Approximation and Representation

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3 Polynomials

In the previous chapter, we spent a considerable amount of time developing Fourier series partly in the hope that, by the end, their existence and convergence properties felt not as if they were handed from on high, but instead emerge as natural consequences of a function's *underlying structure*. What we can learn from the Master theorem is that there isn't *a single unifying theory* of which functions are "Fourier-anenable" — indeed, a function seems to *accumulate* desirable Fourier-analytic properties as it is endowed with greater and greater regularity, but that regularity comes in a variety of different flavors: absolute integrability L^1, square-integrability L^2, k-times continuous differentiability C^k, bounded variation BV, Dini conditions to establish regularity at particular points in the domain, etc.

So, taking a step back, what is the *common theme* among these conditions? Indeed, each characterizes a very particular kind of regularity on behalf of a function f, but what we can understand is that the conditions elucidate something about how *cooperative* f is as a target for Fourier approximation. What we understand about the procedure from a bird's-eye perspective is: given a function $f\colon [-\pi,\pi] \to \mathbb{R}$, we want to approximate $f(x)$ pointwise for $x \in [-\ell,\ell]$ without ever "plugging in $x \mapsto f(x)$ explicitly." To this end, we take the trigonometric functions

$$\mathcal{T} = \{1,\ \cos(t),\ \sin(t),\ \cos(2t),\ \sin(2t),\ \dots\} = \{1,\ \cos(nt),\ \sin(nt)\}_{n=1}^{\infty},$$

and we approximate $f(x)$ by a **linear combination** of members of $\mathcal{T}$ evaluated at $t = x$:

$$S_N f(x) = \frac{a_0}{2} + \sum_{n=1}^{N} [a_n \cos(nx) + b_n \sin(nx)], \qquad x \in [-\pi,\pi],$$

where the Fourier coefficients are

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\,dt, \quad a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\cos(nt)\,dt, \quad b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\sin(nt)\,dt.$$

In the limit as $N \to \infty$, the partial sums $S_N f \to \mathcal{S}f$, and $\mathcal{S}f(x) = f(x)$ pointwise for points of continuity $x \in [-\pi,\pi]$ — or to the midpoint of the left- and right-sided limits $\mathcal{S}f(x) = \frac{f(x^+)+f(x^-)}{2}$ at points of discontinuity — so long as the target function f was "sufficiently regular." What we mean by sufficiently regular, of course, came down to what question we're wanting to answer, and this is what led us to characterize properties of the Fourier expansion by function class (as in the master theorem).

Now, however, we want to turn our attention to the other half of the assumption: why did we imagine this approximation of functions $f\colon [-\pi,\pi] \to \mathbb{R}$ by members of $\mathcal{T}$ *could* be appropriate in the first place? We've alluded to the answer in passing, but here, it will prove central to our development and, hopefully, will elucidate just how powerful an *appropriately chosen basis* is for productive analysis.

3.1 The Practitioner's Parables

From Fourier to Regression: An Overview

We begin our examination here, which will ultimately spann this and the following chapter, within the same well-trodden territory of Fourier series. Immediately, we want to direct our attention toward a luxury we often take for granted in analysis: namely, in constructing the Fourier representation $\mathcal{S}f$, we've assumed thus far that the *entire function* $f\colon [-\pi,\pi] \to \mathbb{R}$ is accessible to us.

Concretely, let us stop to consider how we *really* compute the integral defining the constant Fourier coefficient,

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\,dt.$$

What we're tempted to believe is that "an integral" unilaterally directs us to apply the Fundamental Theorem of Calculus:

$$\int_{-\pi}^{\pi} f(t)\,dt \stackrel{?}{=} F(\pi) - F(-\pi) \quad \text{for some antiderivative } F \text{ satisfying } F'(t) = f(t),\ t \in [-\pi,\pi].$$

However, what's clear is that when we strip away all of the ambient assumptions on the functional form of f, what remains in the notation $\int_{-\pi}^{\pi} f(t)\,dt$ is a much more concrete directive: "form the Riemann sums $\sum_{\mathcal{P}} f(c_i)\Delta t_i$ over finer and finer partitions $\mathcal{P} = \{-\pi = t_0, t_1, \dots, t_k = \pi\}$ of $[-\pi,\pi]$, and verify that the sums stabilize" — where, to be clear, we require a very particular form of stability for the integral to make sense. In particular, the Riemann integral demands that the sums converge to the same value over *every* choice of partition $\mathcal{P}$ and over *every* choice of sample points c_i within each subinterval $[t_{i-1}, t_i]$ — only then is the notation $\int f(t)\,dt$ well-defined.

What an integral quietly presupposes, then, is that we possess an *infinite observational budget* for querying f. That is, in order to integrate a function *absent* sufficient regularity (or absent our ability to verify its presence), we're effectively being directed to query f at arbitrary points $c_i \in [t_{i-1}, t_i]$, where the bin-widths $\Delta t_i = t_i - t_{i-1} \to 0$ as we let the mesh of our partition $\|\mathcal{P}\| \to 0$.

What we understand is that, in "theoretical" settings, the functions of interest are of a *known form.* We're *given* $f(t) = t^2$ for $t \in [-\pi,\pi]$, at which point we can ascertain the function's behavior directly — we can derive $f'(t) = 2t$ at every $t \in [-\pi,\pi]$, that $f''(t) = 2$ a constant for every $t \in [-\pi,\pi]$, that all higher order derivatives $f^{(k)}(t), k \geq 3$ vanish on $[-\pi,\pi]$, and so on. Most importantly, once we possess such an exact functional form, the observational budget *is* infinite, because the formula *is* the oracle: hand me any $t \in [-\pi,\pi]$ and I hand you back $f(t) = t^2$. There is no gap between "what f does" and "what we know about f."

But, in practice, we almost never know the form of a function of interest — in this light, let us *flip* the perspective and see what falls out. Concretely, suppose that we do not possess the functional form of f — instead, what we have is a finite sample

$$\mathcal{D} = \{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\}.$$

Here, we suppose that the **data** $\mathcal{D}$ has been sorted in terms of the $x_i \in [-\pi, \pi]$ — i.e., $-\pi \leq x_1 < x_2 < \cdots < x_n \leq \pi$ — and that y_i is the "evaluation of interest" at location x_i. Perhaps we sampled a signal (y) at regular time intervals (x); perhaps we measured temperature (y) at n points along a rod (x); perhaps we recorded a patient's response (y) at each of n clinic visits indexed by date (x). In each case, we *believe* there is some underlying process f generating the measurements y_i according to the corresponding "location" x_i, but that the exact form of the **data generating process** (DGP) — i.e., the functional form of f — eludes us.

Notice what we *do* have, though. The ordered sample points $x_1 < x_2 < \cdots < x_n$ carve $[-\pi, \pi]$ into subintervals $[x_{i-1}, x_i]$ — they are, in effect, a bona fide partition of $[-\pi, \pi]$ (more precisely, of $[x_1, x_n]$ where x_1 is close to $-\pi$ and x_n is close to π). At each sample point x_i, we observe a sample on the associated y_i. Supposing, then, that we agree to imagine that the y_i are a result of such a functional relationship with f — suppose, for instance, that $y_i = f(x_i)$ *exactly* for some function $f\colon [-\pi, \pi] \to \mathbb{R}$ — then the sum

$$\sum_{i=1}^{n} y_i \Delta x_i := \sum_{i=1}^{n} f(x_i) \Delta x_i, \qquad \Delta x_i := x_i - x_{i-1},$$

is a *perfectly legitimate* Riemann sum when the partition $\mathcal{P}$ is taken to be the x-coordinates in our given data.

What is different in this situation, however, is that we're *stuck* with the partition our data provides. The mesh $\|\mathcal{P}\| = \max_i \Delta x_i$ is whatever it happens to be — we cannot send it to zero. Even if we observed perfectly even bin widths $\Delta x_i = 2\pi/n$ for every i, we would *still be stuck* with the given partition, unable to refine the Riemann sum $\sum_{i=1}^{n} f(x_i)\Delta x_i$ without collecting more data.

Moreover, we've quietly been supposing the best possible case: that $y_i = f(x_i)$ *exactly*. In practice, this is almost never true. What we *actually* observe, should it really be the case that a DGP generates y_i as a function of the x_i, are the *noisy* observations $y_i \approx f(x_i)$. In practice, we specify such a DGP by supposing

$$y_i = f(x_i) + \epsilon_i, \qquad i = 1, \ldots, n,$$

where the **errors** $\epsilon_i = y_i - f(x_i)$ serve as a catch-all bucket absorbing everything our model cannot capture — we might imagine the errors accumulate from, among other sources, measurement error, environmental variation, the influence of other unobserved variables driving the DGP, or simply the inadequacy of whatever form we have assumed for f.

This is the setting we'll inhabit, in various guises, for much of this book: we have n noisy observations from a process we don't fully understand, and we want to recover as much of the underlying structure as we can. What changes, of course, is whether we *acknowledge* this noisy structure and, if we do, how we *utilize* it. Indeed, in this chapter, we disregard noise *almost entirely* — what we'll find, however, is that its presence is *simply unavoidable*, even in the best case imaginable. What we're forced to reckon with, ultimately, is the interplay between what we *hope to learn*, and what our finitely many data are *actually able to teach us* about the process we suppose to be producing it.

Here, we give an initial statement of a familiar framework we'll inhabit through most of this chapter, though perhaps in a less canonically familiar setting.

Definition 3.1.1 (The Linear Model (v1))

Suppose we observe the **data**

$$\mathcal{D} = \{(x_1, y_1), \dots, x_n, y_n)\},$$

where $n \in \mathbb{N}$ is a known **sample size**, and where we suppose that the relationship between **outcome** y_i and **predictor** x_i pairs, (x_i, y_i), is "well-described" by a **linear combination** of $p = k+1$ known functions $\phi_0, \phi_1, \dots, \phi_k$ of the predictor x; i.e., that

$$y_i = \beta_0\phi_0(x_i) + \beta_1\phi_1(x_i) + \cdots + \beta_k\phi_k(x_i) + \epsilon_i, \qquad i = 1, \dots, n,$$

for some $\beta_0, \beta_1, \dots, \beta_k \in \mathbb{R}$.

The p functions $\phi_0, \phi_1, \dots, \phi_k$ are called the **basis functions** (or **regressors**; here, we're imagining that "the same predictor is being reused in various functional forms across the model").

The scalars $\beta_0, \beta_1, \dots, \beta_k$ are the **coefficients** (or **model parameters**) which are typically the unknown quantities of interest in a "regression problem."

The ϵ_i are perceived to be **random errors**, comprising "everything the model can't (or won't) explain" after accounting for the functions ϕ_j of our predictor x. (Occasionally, we perceive these errors to be **normally distributed** — but we're quick to clarify that this is certainly *not* a requirement for the linear model to make sense.

If we assemble everything in **matrix notation**:

$$\mathbf{\Phi} = \underbrace{\begin{pmatrix} \phi_0(x_1) & \phi_1(x_1) & \cdots & \phi_k(x_1) \\ \phi_0(x_2) & \phi_1(x_2) & \cdots & \phi_k(x_2) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_0(x_n) & \phi_1(x_n) & \cdots & \phi_k(x_n) \end{pmatrix}}_{p=k+1 \text{ columns}} \in \mathbb{R}^{n\times p},$$

$$\mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} \in \mathbb{R}^n, \qquad \boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{pmatrix} \in \mathbb{R}^p, \qquad \boldsymbol{\epsilon} = \begin{pmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{pmatrix} \in \mathbb{R}^n,$$

then we can equivalently write the system of n equations defining the linear model as

$$\mathbf{y} = \mathbf{\Phi}\boldsymbol{\beta} + \boldsymbol{\epsilon}.$$

We call $\mathbf{y}$ the **outcome vector**, $\mathbf{\Phi}$ the **design matrix** (or **model matrix**), $\boldsymbol{\beta}$ the **coefficient vector** (or **parameter vector**), and $\boldsymbol{\epsilon}$ the **error vector**.

Remark.

The distinction that p, the *number of model parameters*, equals $k+1$, is something of a customary choice of notation. What we'll find is that, most often, the **constant term** $\beta_0 \in \boldsymbol{\beta}$ is endowed with some "distinguishing properties" that warrant its special treatment. In particular, most often, the linear model specifies $\phi_0 = \mathbf{1}$ the constant function equal to 1 for any x, so that the leading column in $\mathbf{\Phi}$ is just a vector of 1s. In this way, the $\beta_0\phi_0(x_i)$ term is usually just $\beta_0 \cdot 1 = \beta_0$ for each $i = 1, \dots, n$. Combined with the rest of the predictors, what we imagine the model doing is "estimating the baseline mean β_0 across the full vector $\mathbf{y}$, and then modeling *conditional deviations* from the global mean of the y's given the ith participant's predictor was observed to be $x = x_i$.

In essence, we call $p = k+1$ so that p can denote "the number of *p*arameters we need to estimate, *including* the baseline mean," while k denotes "the number of relationships we've specified as driving conditional deviations *away from* the baseline mean."

Almost immediately, one raises a qualm: "what if the relationship is *really nonlinear*?" The answer is that it almost certainly *is* nonlinear — but the linear model accomodates this just fine. The word "linear" in "linear model" does *not* refer to the relationship between y and x — it refers to the relationship between y and the *coefficients* β_j. Imagine the situation when we want to model a vector of $n = 3$ outcomes y_i as conditional deviations from a global mean driven by some observed predictor x_i and its associated quadratic x_i^2 (where, again, we assume that $x_1 < x_2 < x_3$ are ordered and distinct). Then, in the notation above, we'd have

$$\phi_0(x_i) = 1, \quad \phi_1(x_i) = x_i, \quad \phi_2(x_i) = x_i^2, \qquad i = 1, 2, 3,$$

and we'd write the model whose system of equations is described by

$$\begin{aligned} y_i &= \phi_0(x_i)\beta_0 + \phi_1(x_i)\beta_1 + \phi_2(x_i)\beta_2 + \epsilon_i \\ &= 1 \cdot \beta_0 + x_i \cdot \beta_1 + x_i^2 \cdot \beta_2 + \epsilon_i \\ &= \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \epsilon_i, \qquad\qquad i = 1, 2, 3, \end{aligned}$$

as

$$\underbrace{\begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix}}_{\mathbf{y}} = \underbrace{\begin{pmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ 1 & x_3 & x_3^2 \end{pmatrix}}_{\mathbf{\Phi}} \underbrace{\begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{pmatrix}}_{\boldsymbol{\beta}} + \underbrace{\begin{pmatrix} \epsilon_1 \\ \epsilon_2 \\ \epsilon_3 \end{pmatrix}}_{\boldsymbol{\epsilon}}.$$

What we understand, then, is that the relationship between y and x as specified is *certainly nonlinear* — we've explicitly modeled y_i as a function including a quadratic term, x_i^2. What makes it a "linear model" is that, once we imagine (x_1, x_2, x_3) as "a list of numbers," then so too is (x_1^2, x_2^2, x_3^2) "just a list of numbers." How we've *obtained* these numbers is quite immaterial to the model itself — it doesn't really care whether you imagine x_i^2 as "deriving from x_i" or as "a completely new number" (at least, not on its face — we will see that it *really does* care, it just won't tell you so.)

An Initial Model Fit

What makes this immateriality concrete is how we proceed to *estimate* the unknown quantities in the model. To start, suppose we're working in the *noiseless* case, so that $\epsilon_1 = \epsilon_2 = \epsilon_3 = 0$ (equivalently, $\boldsymbol{\epsilon} = \mathbf{0}$ — we use the symbol $\mathbf{0}$ to denote a vector of zeros in $\mathbb{R}^3$.) When $\boldsymbol{\epsilon} = \mathbf{0}$, the system in the linear model becomes

$$\mathbf{y} = \mathbf{\Phi}\boldsymbol{\beta}.$$

This is not yet a *statistical* problem — it's an *algebraic* one. Concretely, realize that if we write the noiseless system out as

$$\begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \beta_0 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + \beta_1 \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} + \beta_2 \begin{pmatrix} x_1^2 \\ x_2^2 \\ x_3^2 \end{pmatrix},$$

then the question is stated in terms of whether there exists a vector $\boldsymbol{\beta} \in \mathbb{R}^3$ such that $\mathbf{y}$ can be expressed as a **linear combination** of the columns of $\mathbf{\Phi}$. The questions, then, are twofold:

whether such a solution vector β *exists* and, if it does, whether it can be *computed.* What we understand is that verifying existence explicitly at the beginning is almost always faster than "letting unsolvability emerge as an artifact of a solution-finding algorithm."

For instance, suppose that we'd only sampled $n = 2$ observations (instead of $n = 3$). Then, the system of equations could be written explicitly as

$$y_1 = \beta_0 + \beta_1 x_1 + \beta_2 x_1^2,$$
$$y_2 = \beta_0 + \beta_1 x_2 + \beta_2 x_2^2.$$

Suppose we'd found a solution vector $\beta^\star$. Then, as a consequence of the *linearity of the model*, it would follow that if we subtract the equations above and plug in the solution, we would have

$$y_1 - y_2 = \beta_1^\star(x_1 - x_2) + \beta_2^\star(x_1^2 - x_2^2) = (\beta_1^\star + \beta_2^\star(x_1 + x_2))(x_1 - x_2).$$

In particular, divide both sides by $x_1 - x_2$ to form the **first-order divided difference**, given by

$$\frac{y_1 - y_2}{x_1 - x_2} = \beta_1^\star + \beta_2^\star(x_1 + x_2).$$

To see how this looks in action, suppose that we'd observed the values $(x_1, y_1) = (0, 2)$ and $(x_2, y_2) = (1, 3)$. We compute

$$\frac{y_1 - y_2}{x_1 - x_2} = 1, \quad x_1 + x_2 = 1,$$

so that the first-order divided difference condition reduces to

$$\beta_1^\star + \beta_2^\star = 1.$$

This is one equation in two unknowns — in particular, what we observe upon rearrangment is that

$$\beta_1^\star = 1 - \beta_2^\star.$$

Read this to mean: For any choice of $\beta_2^\star$, we can set $\beta_1^\star = 1 - \beta_2^\star$, and then recover $\beta_0^\star$ from either of the original equations — say the first:

$$\beta_0^\star = y_1 - \beta_1^\star x_1 - \beta_2^\star x_1^2 = 2 - (1 - \beta_2^\star)(0) - \beta_2^\star(0)^2 = 2.$$

What we can say, then, is that when we observe the data

$$\mathcal{D} = \{(0, 2), (1, 3)\},$$

the full family of solutions to the noiseless linear system

$$y_1 = \beta_0 + \beta_1 x_1 + \beta_2 x_1^2,$$
$$y_2 = \beta_0 + \beta_1 x_2 + \beta_2 x_2^2,$$

is given by

$$(\beta_0^\star,\ \beta_1^\star,\ \beta_2^\star) = (2,\ 1 - \beta_2^\star,\ \beta_2^\star), \qquad \beta_2^\star \in \mathbb{R}.$$

Let us take a moment to *intuitively* restate the model's reasoning as reflected by the derivations above. First of all, we observe that the original parameterization of the β vector was specified in terms of three distinct components: an intercept β_0, a linear-term slope β_1, and a quadratic-term slope β_2. Nevertheless, in our *solution* $\beta^\star$ based on the observed data $\mathcal{D}$, we recognize that only *one* parameter varies independently — namely, the quadratic slope $\beta_2^\star \in \mathbb{R}$. In particular, once we set *any* $\beta_2^\star \in \mathbb{R}$, we're instructed to take as the solution's linear slope $\beta_1^\star = 1 - \beta_2^\star$, and for any such pair of slopes $(\beta_1^\star = 1 - \beta_2^\star, \beta_2^\star)$, taking our intercept to be $\beta_0^\star = 2$ yields a solution vector $(\beta_0^\star, \beta_1^\star, \beta_2^\star) = (2, 1 - \beta_2^\star, \beta_2^\star)$ solving $y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2$, $i = 1, 2$ under our observed data $\mathcal{D}$: in particular, plugging in $(x_1, y_1) = (0, 2)$ and $(x_2, y_2) = (1, 3)$, this says that the solution vector $\beta^\star$ described above satisfies

$$\begin{aligned} 2 &= \beta_0^\star + \beta_1^\star(0) + \beta_2^\star(0)^2 = \beta_0^\star, \\ 3 &= \beta_0^\star + \beta_1^\star(1) + \beta_2^\star(1)^2 = \beta_0^\star + \beta_1^\star + \beta_2^\star. \end{aligned}$$

And, of course, this is reasonable in light of the given *noiseless model* — the noiseless model instructs us to take our observed data $\mathcal{D}$ as *gospel*. What the model is literally telling us is that "if $\beta^\star$ solves the system and you've observed a sample point $x_i = 0$ in $\mathcal{D}$, then according to the noiseless specification, I'm *forced* to believe its corresponding y_i characterizes the truth. Since you've told me that $y_i = 2$ at $x_i = 0$, and since you've not given me any reason to doubt the exactness of this correspondence, I'll *force* the solution to pass through $(x, y) = (0, 2)$ no matter what by setting the y-intercept of any solution accordingly." And indeed, $\beta_0^\star$ here *really is* "just the y-intercept" — i.e., when $x_i = 0$, the model predicts $y_i = \beta_0^\star$ exactly.

From here, what we understand is that, once the model sets $\beta_0^\star = 2$ in order for the quadratic $y = \beta_0^\star + \beta_1^\star x + \beta_2^\star x^2$ to pass through the point $(x, y) = (0, 2)$, it treats setting $(\beta_1^\star, \beta_2^\star)$ like a balancing act: if $\beta_2^\star = 1$, then we obtain $(\beta_0^\star, \beta_1^\star, \beta_2^\star) = (2, 0, 1)$, corresponding to the quadratic model

$$f(x) = x^2 + 2.$$

At the point $x = 0$, the curve *predicts* $f(x) = 0^2 + 2 = 2$, verifying that the observed sample point $(x, y) = (0, 2) \in \mathcal{D}$ falls on the curve. Similarly, at $x = 1$, the model predicts $f(x) = 1^2 + 2 = 3$, so that the other sample point $(x, y) = (1, 3) \in \mathcal{D}$ also falls on the curve.

But, in the above, we set $\beta_1^\star = 1 - \beta_2^\star$ according to the constraint $\beta_1^\star + \beta_2^\star = 1$, but this wasn't our only recourse — indeed, we can certainly ask what happens if we set $\beta_2^\star = 1 - \beta_1^\star$ and set the linear coefficient $\beta_1^\star = 1$? Then the quadratic coefficient $\beta_2^\star = 1 - \beta_1^\star = 0$ vanishes, we obtain a solution $(\beta_0^\star, \beta_1^\star, \beta_2^\star) = (2, 1, 0)$, and we obtain the linear model

$$f(x) = x + 2.$$

Again, one can verify the model conforms with the two points in our given data $\mathcal{D}$, as do, for instance,

$$f(x) = \frac{1}{2}x^2 + \frac{1}{2}x + 2; \qquad f(x) = 100x^2 - 99x + 2; \qquad f(x) = -\pi x^2 + (1 + \pi)x + 2.$$

Again, *all* of these curves agree with the data $\mathcal{D}$; what changes is how they behave *away* from our observed data. To see just how dramatically, evaluate a few of these curves at $x = 3$: for instance, at $\beta^\star = (2, 1, 0)$, we have

$$f(x) = x + 2 \implies f(3) = 3 + 2 = 5,$$

while at $\beta^\star = (2, -99, 100)$,

$$f(x) = 100x^2 - 99x + 2 \implies f(3) = 900 - 297 + 2 = 605.$$

At $x = 0$ and $x = 1$, every member of the family agrees — they must, by construction. But two units to the right of the data, the predictions range from 5 to 605, and there is *nothing* in the model telling us which to prefer. And, ultimately, this is a price we're forced to pay if we choose to use *this model* to model *this data*.

Let us pause for a moment, though, and really think about what we're trying to do here. Given only two data points — $\mathcal{D} = \{(0,2),(1,3)\}$ — we somewhat absentmindedly "went along" with the quadratic model we'd proposed *assuming* we'd possess $n = 3$ data points. The problem is that we were given only $n = 2$ data points; we simply never had enough information to justify our model in the first place. We say the model is **overparameterized** relative to what we could hope for the model to reasonably learn.

Concretely, given only two data points, the seemingly most natural choice would be to model their relationship is not a *quadratic*; it is simply a *line*. Intuitively, what we understand is that the segment connecting the two data points $(0,2)$ and $(1,3)$ represents a fairly natural choice for predicting outputs $\hat{y}$ at levels of the predictor lying inside the range of our observed data, $\hat{x} \in [0,1]$. In such a case, if we *truly don't know* "what's going on" in the regime where the predictor x takes values in $[0,1]$ strictly, such a **linear interpolation** would seem to "minimize the level of undue influence we're able to exert on the resultant predictions." Having then enforced linearity *inside* the range of our observed predictors $[0,1]$ — and absent any reason to believe otherwise — the most reasonable predictions *outside* $[0,1]$ would, intuitively, seem to be **linear extrapolations** based on the data-derived segment connecting $(0,2)$ and $(1,3)$. In this way, what we might assert is that the "most appropriate model" in this case is the linear one

$$f(x) = 2 + x, \qquad x \in \mathbb{R}.$$

Remark.

Of course, we still came to the conclusion that the linear model was "most appropriate" even though we started in the broader class of quadratic models. Yet, we can't help but imagine how much simpler life would have been if we'd *started* by only considering the class of linear models — i.e., those of the form

$$y = \beta_0 + \beta_1 x.$$

Such a move reduces what was a fairly intensive logical argument to simple algebra. Our hope, quite often, is to reduce a complex problem's structure to something we can solve algorithmically.

What changes when we add a third data point, then, is that we permit the model to discover *curvature*. Suppose we observe the additional sample $(x_i, y_i) = (3,8)$, so that our full dataset becomes

$$\mathcal{D}_3 = \{(0,2),(1,3),(3,8)\},$$

where we've indexed the data by the sample size $n = 3$.

At this point, one observes that the prediction yielded by our previous linear model, $f(3) = 5$, undershoots the observed $y_3 = 8$ at the same level of the predictor. What we choose to do with this information is, of course, up to us — nevertheless, if we're supposing that the data has been generated perfectly in accordance with the rest of the process, we really should ask ourselves how we intend to proceed. On the one hand, when we only had $n = 2$ samples, the linear model $f(x) = x+2$ proved to be "best" at capturing our original data, supposing our goal is to minimize the amount of influence we're exerting on the model's predictions.

On the other hand, now that we have $n = 3$ samples, it appears as if the *class* of linear models is inadequate — try as we might, we'll *never* find a line that goes through all three points in $\mathcal{D}_3$. So, suppose we were to go back to the original class of quadratic noiseless models given by

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2, \qquad i = 1,2,3.$$

Plugging in the observed data gives

$$\begin{aligned}2 &= \beta_0\\ 3 &= \beta_0 + \beta_1 + \beta_2\\ 8 &= \beta_0 + 3\beta_1 + 9\beta_2,\end{aligned}$$

which, from $\beta_0 = 2$, immediately gives $\beta_1 + \beta_2 = 1$, the same constraint from the original data, *and a second constraint*, $3\beta_1 + 9\beta_2 = 6$. One finds from our previous elimination that the noiseless quadratic model corresponding to our data $\mathcal{D}_3$ is

$$p_3(x) = 2 + \frac{1}{2}x + \frac{1}{2}x^2,$$

where we've again indexed the model f_n by the sample size $n = 3$.

One again verifies that the model p_3 predicts $\mathcal{D}_3$ perfectly. Furthermore, this is the simplest polynomial function capable of perfectly fitting the data — we could also do so with a class of cubic polynomials, but barring any reason to believe such a structure *should* exist, the quadratic class is perfectly capable of describing the data, and it does so in a *simpler way*. (We tend to prefer **parsimonious** models — that is, within a group of models which perform comparably on our observed data, we tend to prefer those which minimize model complexity). We can verify that

$$p_3(4) = 2 + 2 + 8 = 12,$$

while we also note that our previous linear model, $p_2(x) := 2 + x$, would have predicted

$$p_2(4) = 2 + 4 = 6.$$

One might observe this and begin to think that the models p_2 and p_3 are in conflict — recall that the prediction by model p_2 at $x = 3$ disagreed with our third sample point $(x_3, y_3) = (3, 8)$ (where we've set the quadratic model in order for $p_3(3) = 8$) by $|p_3(3) - p_2(3)| = |8 - 5| = 3$ units. Now, at $x = 4$, we observe that the predictions by the two models disagree by a more substantial $|p_3(4) - p_2(4)| = |12 - 6| = 6$ units. Suppose, then, that observe a fourth data point $(x_4, y_4) = (4, 6)$ — where, again, we assume that the new sample point has been generated perfectly, in accordance with the existing data — and we update our data

$$\mathcal{D}_4 = \{(0, 2), (1, 3), (3, 8), (4, 6)\}.$$

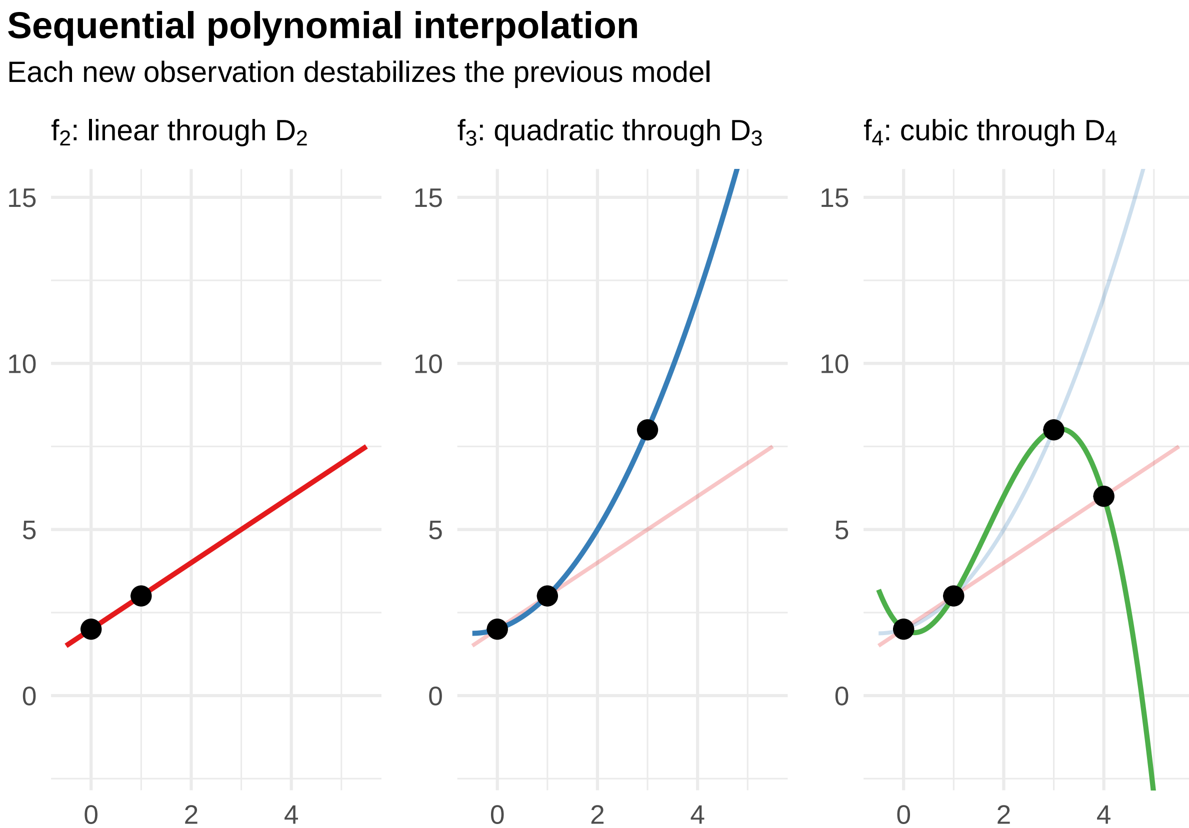
Notice that the fourth sample *agrees* with the prediction given by the linear model p_2: $|y_4 - p_2(4)| = |6-6| = 0$. On the other hand, y_4 clearly disagrees with the quadratic model's prediction at the same level of the predictor: $|y_4 - p_3(4)| = |6 - 12| = 6$.

Furthermore, what we observe is that, just as the class of linear models was insufficient for characterizing the three samples in $\mathcal{D}_3$, the class of quadratic models $f(x_i) = \beta_0 + \beta_1 x_i + \beta_2 x_i^2$ is insufficient for completely characterizing the four samples in $\mathcal{D}_4$ — in particular, one can verify there is *no* quadratic that passes through all four points of $\mathcal{D}_4$ simultaneously. If we trust the trend, it would seem as if a *cubic* model would be sufficient; i.e., if we set

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \beta_3 x_i^3, \qquad i = 1, 2, 3, 4,$$

then we'd be able to solve for a unique $\beta^\star = (\beta_0^\star, \beta_1^\star, \beta_2^\star, \beta_3^\star)$ such that the sample points $(x_i, y_i) \in \mathcal{D}_4$ are all generated by the model, and it would be the lowest order polynomial capable of passing through all four points in $\mathcal{D}_4$ uniquely. Indeed, this is true; one obtains (after a somewhat more involved elimination than before)

$$p_4(x) = 2 - x + \frac{5}{2}x^2 - \frac{1}{2}x^3$$



Looking back on the story thus far, we have now built three successive models, each tailored to its generation of data:

$$p_2(x) = 2 + x, \qquad p_3(x) = 2 + \frac{1}{2}x + \frac{1}{2}x^2, \qquad p_4(x) = 2 - x + \frac{5}{2}x^2 - \frac{1}{2}x^3.$$

Each is the *lowest-degree polynomial* that passes through all available data perfectly. Each was, at the time of its construction, the parsimonious choice. And each was *invalidated* by the next observation: p_2 predicted $p_2(3) = 5$ but we got $(x_3, y_3) = (3, 8)$; p_3 predicted $p_3(4) = 12$ but we got $(x_4, y_4) = (4, 6)$. Right now, p_4 is perfect at fitting $\mathcal{D}_4$, but what right have we earned to be *confident* in how p_4 will perform on a future sample?

Very little. And the situation is, on reflection, *even worse* than it first appears. Look at the trajectory of the coefficients across the three models:

	β_0	β_1	β_2	β_3
p_2	2	1	—	—
p_3	2	1/2	1/2	—
p_4	2	−1	5/2	−1/2

The intercept $\beta_0 = 2$ has been stable — this is entirely a consequence (a gift, even) of having observed the sample point $(x_1, y_1) = (0, 2)$. But the linear coefficient β_1 went from 1 down to 1/2 then even more to -1 — it does not seem that the coefficient is stabilizing as we observe more samples, but rather oscillating quite wildly in an effort to catch up with the new information. Each new observation isn't *refining* the previous model; just *destabilizing* it.

Once again, we stress that, here, we've been working with *perfect data*. We can't pin the problems on "measurement error" or "randomness" — the appropriate models always return $f_n(x_i) = y_i$ exactly. It would seem somewhat paradoxical; something about what we're doing is simply *wrong*, but it would appear as though we're making a principled move at every step along the way.

3.1.1 Chasing the Dragon

A Setup For Disaster

Still, all of this would still seem perfectly reasonable supposing the data were *truly generated by a polynomial.* Concretely, supposing the DGP really were a fixed polynomial of degree $d > 0$, say

$$f(x) = \beta_0 + \beta_1 x + \beta_2 x^2 + \cdots + \beta_d x^d,$$

then as soon as the sample size $n > d$, the table above would begin to stabilize: the extra data points would be consistent with the existing model, no new terms would be needed, and things would start to settle down. This is fine — and, if we're so lucky as to receive such a finitely parameterized polynomial DGP, we should accept the interpolation strategy as perfectly legitimate.

But it would already seem quite wishful thinking to imagine more complex processes would be characterized by a polynomial of finite-degree. Yet, what we *might hope* is for there to be a saving grace hidden in plain sight: suppose, for instance, that we were to observe samples drawn from an exponential DGP — say, one of the form

$$g(x) := e^{-x^2}, \qquad x \in \mathbb{R}.$$

Of course, we recall the Taylor series for e^x is just

$$e^x \sim 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \cdots = \sum_{n=0}^{\infty} \frac{x^n}{n!};$$

plugging $x \mapsto -x^2$ into this expression, we can write the Taylor series for g as

$$e^{-x^2} \sim 1 - x^2 + \frac{1}{2}x^4 - \frac{1}{6}x^6 + \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{n!}, \qquad \forall x \in \mathbb{R}.$$

We would then perhaps hope that, as our sample size $n \to \infty$, the polynomial interpolants g_n constructed in the manner described above would, term by term, begin to resemble the Taylor coefficients: $\beta_0 = 1, \beta_1 = 1, \beta_2 = \frac{1}{2}, \beta_3 = \frac{1}{6}$, and so on. That is, we'd hope that, eventually, our strategy of interpolating the data would succeed under an exponential DGP — and, indeed, in *this case*, it's a reasonable wish. The reason why is that the function $g(x) = e^{-x^2}$ is **analytic** (or **entire**): i.e., its Taylor series converges for all $x \in \mathbb{R}$.

But we know that e^x is special, particularly with regards to its Taylor series. By contrast, suppose that we observed data generated by

$$h(x) := \frac{1}{1 + x^2}, \qquad x \in \mathbb{R}.$$

We can recall (or verify, as appropriate) that the Taylor series for $\frac{1}{1-x}$ which converges for $|x| < 1$ is

$$\frac{1}{1-x} \sim 1 + x + x^2 + x^3 + \cdots = \sum_{n=0}^{\infty} x^n,$$

and realizing that, equivalently,

$$h(x) = \frac{1}{1-(-x^2)}, \qquad |-x^2| < 1 \implies |x| < 1,$$

we can form a Taylor series for $h(x)$, convergent for x within the radius of convergence $R = 1$ — i.e., for $x \in (-1, 1)$ — as

$$h(x) = \frac{1}{1+x^2} \sim 1 - x^2 + x^4 - x^6 + \cdots = \sum_{n=0}^{\infty} (-1)^n x^{2n}, \qquad |x| < 1.$$

When we juxtapose the Taylor series — namely, the above with

$$g(x) = e^{-x^2} \sim 1 - x^2 + \frac{1}{2}x^4 - \frac{1}{6}x^6 + \cdots = \sum_{n=0}^{\infty} \frac{1}{n!}(-1)^n x^{2n}, \qquad \forall x \in \mathbb{R},$$

what we observe is that the two are astonishingly similar in form. Nevertheless, it is just as immediate *why* the former series for $h(x)$ — which, for convenience, we defined for *all* $x \in \mathbb{R}$ (even the divergent x), as

$$\mathcal{T}[h(x)] := \sum_{n=0}^{\infty} (-1)^n x^{2n}, \qquad x \in \mathbb{R}.$$

In particular, we observe that $\mathcal{T}[h(x)]$ diverges for $|x| \geq 1$: in such a case, $\mathcal{T}[h]$ is an alternating series whose terms are strictly increasing in magnitude whenever $|x| \geq 1$:

$$\forall n \geq 1: \quad |(-1)^n x^{2n}| = |x|^{2n} \cdot 1 \leq |x|^{2n} \cdot |x| = |(-1)^{n+1} x^{2n+1}|.$$

The terms are oscillating more and more wildly for larger $|x| \geq 1$ — at the ends, for instance, one may verify that each $x = \pm 1$ give the harmonic series.

It is just as clear why the series

$$\mathcal{T}[g(x)] := \sum_{n=0}^{\infty} \frac{1}{n!}(-1)^n x^{2n}, \qquad x \in \mathbb{R},$$

does not suffer such issues: its scaling factor of $1/n!$ ensures that, at some point, the successive multiplications by x^{2n} for $|x| \geq 1$ are eventually overtaken by the ever-growing factors of $n \to \infty$ in $1/n!$. "Even though $|x|^{2n} = |x| \cdot |x| \cdots |x|$ may have $|x|$ large enough that it beats $n! = n \cdot (n-1) \cdots 2 \cdot 1$ for a while, eventually $n!$ is going to catch up; and, when it does, it's going to stay ahead forever after."

The Gaussian and The Cauchy

We've deliberately been somewhat illusory about the whole affair, but what we've really done here has a salient punchline that brings us back to the previous discussion: imagine that we take h and g as defined above to be the Data Generating Processes underlying some data,

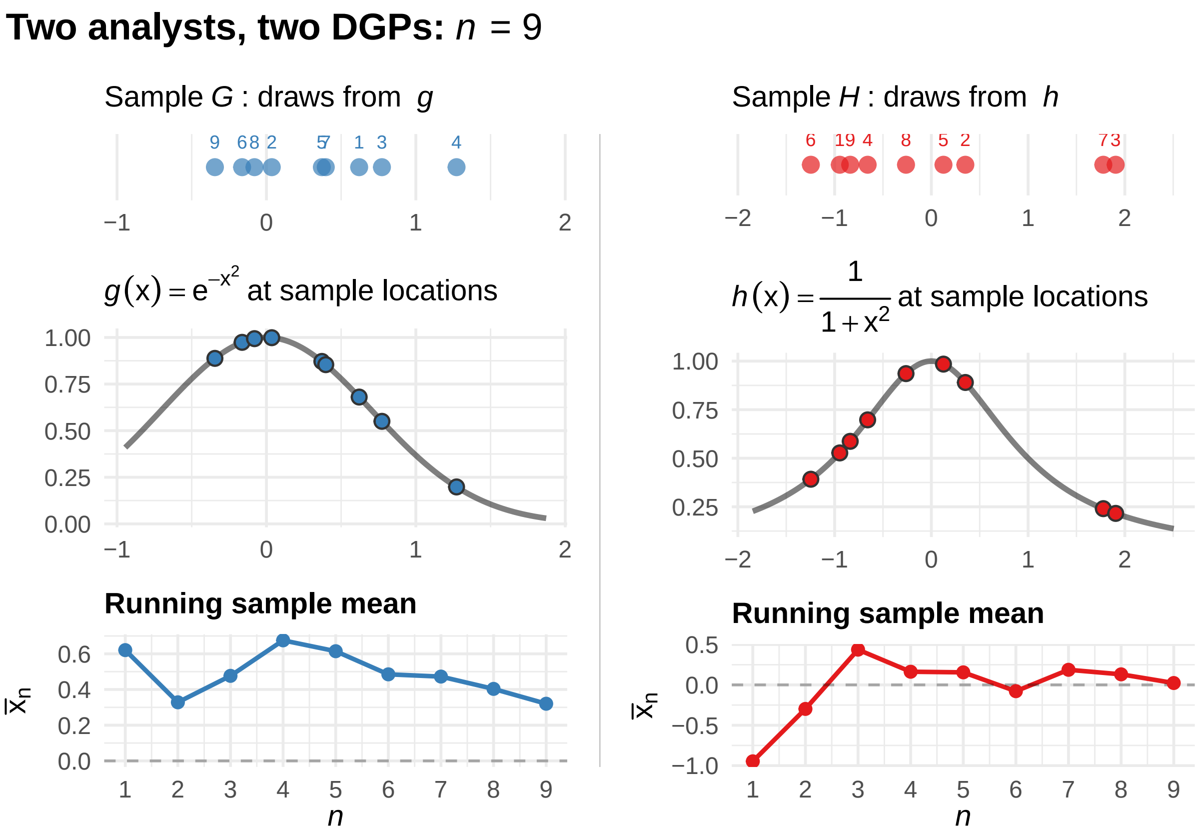
$$\mathcal{G} = \{(x_i, g(x_i)\}_{i=1}^{m}, \qquad \mathcal{H} = \{(x_j, h(x_j)\}_{j=1}^{n}.$$

We again suppose that the observed "y_i" are generated perfectly in accordance with the DGPs as defined above. Two analysts are given data $\mathcal{G}$ and $\mathcal{H}$ respectively, but neither knows the precise functional forms of g or h, and they're asked to reverse-engineer the functional relationship to the best of their ability — largely the same setup as before, only now, *we're* dictating the DGP so that we may learn from the cautionary tale of the analyst of $\mathcal{H}$.

Remark.

In an analogous, more realistic setting, all the analysts would observe are the raw draws x_i/x_j — the density evaluations $g(x_i)/h(x_j)$ would be *unobservable*, and we'd hope to infer the approximate shape of the density by "how many draws we observed near a given x." Here, we *give* these evaluations to the analysts; it is a more relaxed setting than what we'd encounter in practice. As we will see, it leads to catastrophe regardless.

We imagine that, as the data "continue to roll in," the analysts repeatedly recompute the sample means of their observed data in an effort to summarize the center of location of the data. The plot below relays their computations at each sample size n.



Concretely, rather than imagining $g(x) = e^{-x^2}$ and $h(x) = \frac{1}{1+x^2}$ as "functions to be evaluated at chosen input locations $x \in \mathbb{R}$," we imagine that we the plot above depicts samples *drawn from* these curves in accordance with the DGP itself. That is, again, we might imagine going into the populations characterizing $\mathcal{G}/\mathcal{H}$ and drawing sample points x_i/x_j from each population, where we interpret the evaluations $g(x_i)/h(x_j)$ as *unobserved* quantities that are "driving the likelihood" we observe a given x_i/x_j.

To really make this "likelihood" intuition concrete, we could imagine *normalizing* g and h: indeed, one verifies (quite instructively, if you haven't seen it before; see the exercises at the end of the chapter for a pair of guided derivations) that

$$g(x) \geq 0 \quad \text{and} \quad h(x) \geq 0, \qquad \forall x \in \mathbb{R},$$

that

$$\int_{-\infty}^{\infty} g(x)\,dx = \int_{-\infty}^{\infty} e^{-x^2}\,dx = \sqrt{\pi},$$

and that

$$\int_{-\infty}^{\infty} h(x)\,dx = \int_{-\infty}^{\infty} \frac{1}{1+x^2}\,dx = \pi.$$

In particular, then, g and h can each be normalized to form a probability density function. For simplicity, we overwrite the previous unnormalized DGPs by the new, "practically equivalent" functions

$$g(x) = \frac{1}{\sqrt{\pi}} e^{-x^2}, \qquad x \in \mathbb{R},$$

and

$$h(x) = \frac{1}{\pi}\frac{1}{1+x^2}, \qquad x \in \mathbb{R}.$$

Though we haven't formally defined it in the case when the samples are *infinitely supported* (i.e., when we can observe $x \in \mathbb{R}$ anywhere in the entire real line rather than from strictly within a bounded interval $x \in [a, b]$), it would seem as if, upon this normalization, each g and h have been endowed with a probabilistic interpretation: we observe $g(x) \geq 0$ and $h(x) \geq 0$ for any $x \in \mathbb{R}$, and the integral of the newly normalized densities over $\mathbb{R}$ each equal 1. Indeed, even if things are a bit hand-wavy at this point, we can assure you that this *is valid* — i.e., we really can imagine the normalized g and h as valid probability densities.

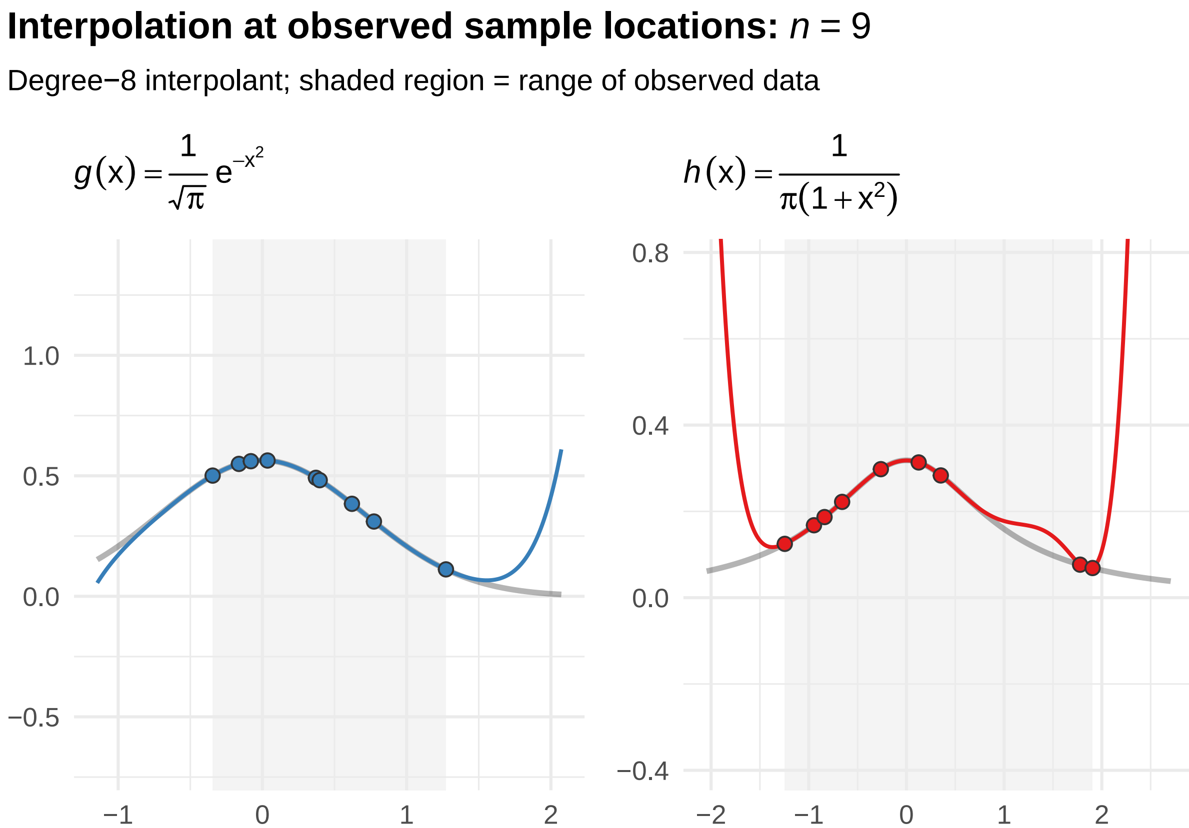
But, again, the analysts don't know any of this — *all* they see are the scatterplots of the data. Really, they were *always* "imagining the situation probabilistically," even when draming g and h as deterministic functions to be queried at particular inputs. The normalization just makes their intuition explicit: the data points are drawn from a distribution, and the shape of that density determines what "typical" data looks like.

So, at $n = 9$, both analysts see roughly the same thing: a cluster of observations near the origin, roughly symmetric, and both with sample means hovering around zero: i.e., under the observed data with $m = n = 9$,

$$\frac{1}{m}\sum_{i=1}^{m} x_i = \frac{1}{n}\sum_{j=1}^{n} x_j \approx 0.$$

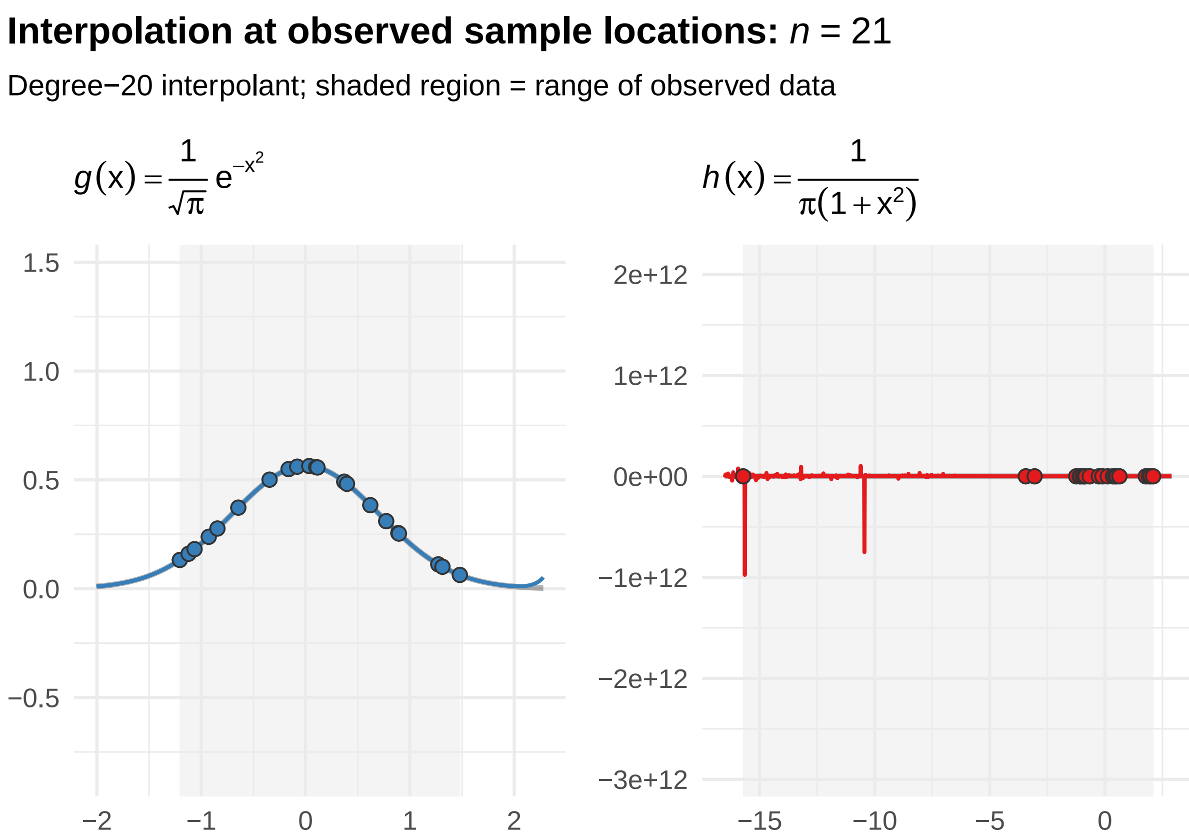
Nothing has yet emerged to suggest the data differ in a meaningful way. By almost every reasonable metric, the data *are* the same. What we imagine is that the analysts, noticing the structural similarities, might opt to pursue similar modeling strategies.

Suppose, then, that they each decide to pursue the strategy of interpolating the data by polynomials, the same strategy we illustrated in motivating the current discussion. They obtain two polynomials, and they plot each polynomial curve as a function of x, overlaying the original data points in $\mathcal{G}/\mathcal{H}$ on top for comparison. What they observe, then, is something like the following.



The "correct" density lines in dark gray here are overlaid for our understanding, but again, it is worth re-emphasizing that the entirety of what the analysts know is subsumed by the plots and curves in blue and red, respectively. And, at this point, based on what we see in the shaded region corresponding to the range of predictors actually represented in our observed data, it seems as if both analysts would be reassured they are on the right track.

It is really only when the second batch of data rolls in that things begin to unravel for the analyst of $\mathcal{H}$. We suppose that after receiving a second wave of $n_2 = m_2 = 12$ samples, the analysts simply roll them in with their existing data comprised of $n_1 = m_1 = 9$ samples. Together, the combined data $\mathcal{G} = \{(x_i, g(x_i))\}_{i=1}^{m}$ and $\mathcal{H} = \{(x_j, h(x_j))\}_{j=1}^{n}$ consists of $n = m = 21$ total sample points. Again, the data is perfect, no noise, yada yada — they proceed to re-fit the interpolating polynomials, at which point they observe the following.



The analyst of $\mathcal{G}$ is, deservedly, quite thrilled. The degree-20 interpolant has essentially merged with the true curve: each wave of data has only seemed to improve the fit.

The analyst of $\mathcal{H}$ is, also deservedly, quite appalled. One of the twelve new observations landed at $x \approx -15$ — far from the cluster near the origin that had characterized the first batch. What results is a polynomial whose behavior is necessarily so erratic as to describe a scattering of points $(x_j, h(x_j))$ which is seemingly devoid of structural integrity in the presence of this "seeming outlier." Of course, as the oracles, we know that at inputs $x \approx -15$, the DGP h yields

$$h(-15) = \frac{1}{\pi} \frac{1}{1 + (-15)^2} \approx 0.0014,$$

and that we may (somewhat loosely at this point) "interpret this evaluation probabilistically."

We quickly contrast this, for instance, with the competing DGP g's evaluation of the same "probability" (which, to be somewhat more precise about the interpretation, could be viewed as "how often the DGP based on g produces sample points $x_i \in \mathcal{G}$ near $x = -15$"):

$$g(-15) = \frac{1}{\sqrt{\pi}} e^{-(-15)^2} \approx 1.084 \times 10^{-98}.$$

That is: while the Cauchy density h regards an observation near $x = -15$ as uncommon, it is still *entirely legitimate* — roughly a 1-in-700 event, and this is something we could have anticipated observing on a sample of size $n = 21$ fairly often under the same DGP. On the other hand, the Gaussian density g regards the same observation near $x = -15$ as, for all practical purposes, *impossible*. This is not a colloquial use of the word impossible — it is *emphatic*, and one should quickly observe that the reciprocal of the number of atoms in the observable universe is, on the lower end, 10^{-83} — which is to say that $g(-15) \approx 10^{-98}$ is *incomprehensibly small*. We can remain, for all practical purposes, absolutely certain that the analyst of $\mathcal{G}$ will never receive a similarly destabilizing data point as that which has already plagued $\mathcal{H}$, and not because she is fortunate — but because her DGP g is *structurally incapable* of producing one.

Yet, of course, these analysts remain unaware of their folly. Suppose the analyst of $\mathcal{H}$, furious over the "contaminated data," rushes to the experimental team and insists that they've really messed it up this time — "we were rolling *just fine* at $n = 9$ — what happened, guys?"

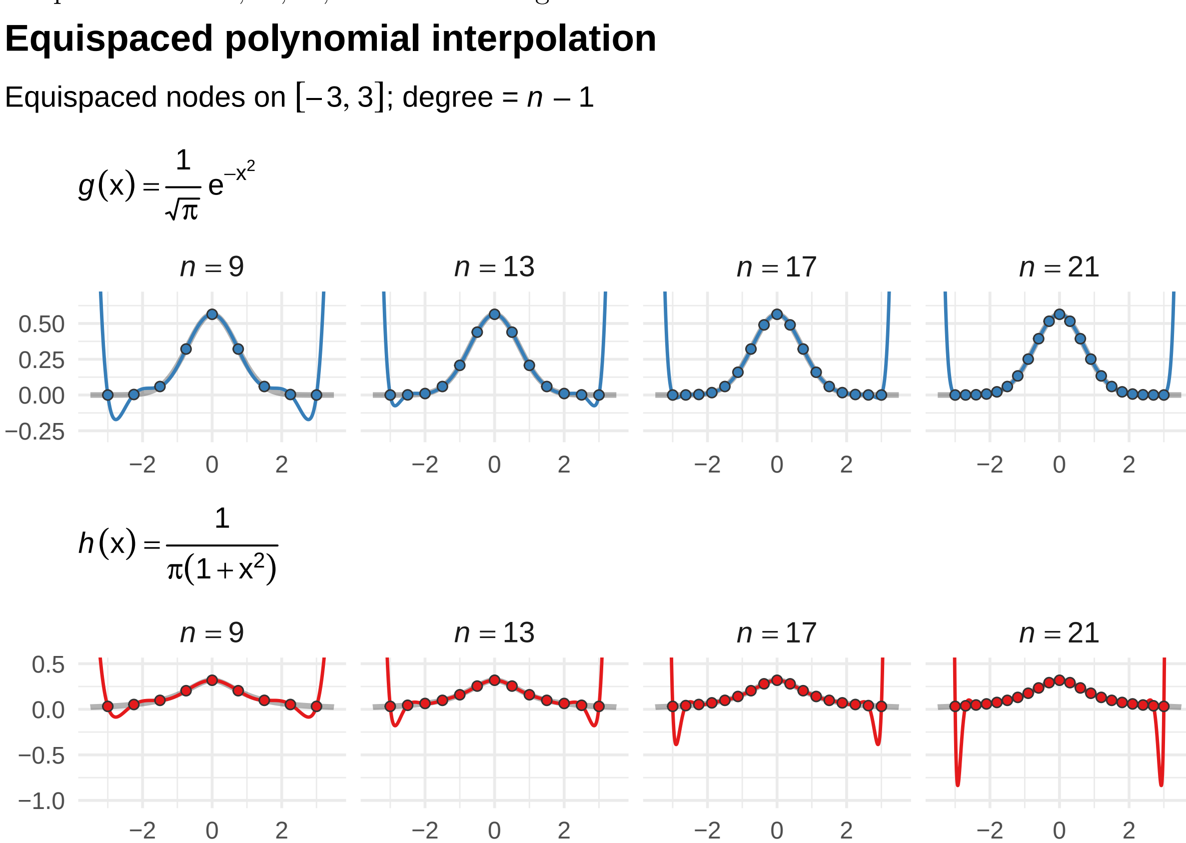
And the truth is that he *isn't* crazy — but *no* amount of data the team could convince him otherwise. Indeed, suppose we, acting as benevolent oracles, granted each analyst data access to some new data:

$$\mathcal{G}_n = \{(x_i, g(x_i))\}_{i=1}^n, \quad \mathcal{H}_n = \{(x_i, h(x_i))\}_{i=1}^n,$$

where the x_i we're evaluating each of the functions g and h at are *identical* — in particular, both analysts receive data generated from the *same* set of equispaced nodes

$$x_1 < x_2 < \cdots < x_n, \quad \text{uniformly spaced on } [-3, 3].$$

In effect, what we're doing is "circumventing the DGP and allowing the analysts to see how their polynomial approximations would work deployed on *uniform partitions* of $[-3, 3]$." What we deliver the analysts, and their corresponding polynomial approximations, are reflected at four sample sizes $n = 9, 13, 17$, and 21 in the figure below.



Chasing the Dragon

It is strange indeed that these DGPs, each capable of producing data which is so ostensibly similar to the other, exhibit such strikingly different behavior in their respective polynomial interpolants. These "fangs" we observe in the plot of the Cauchy DGPs data emerge gradually as we continue to increase the sample size n. What is somewhat interesting to observe is that, at $n = 9$, both the Gaussian and Cauchy DGPs produce data which are "mildly fanged" — yet, the Gaussian's fangs "curl in" as n increases. It would seem as if, the more points we supply, the more violently the Cauchy polynomial coefficients must thrash around in order to "hit all of the points in the data" — which, again, would seem quite strange, considering the curves are *so ostensibly similar*. Why is the Cauchy DGP just "a bit too unwieldy" for our approximating polynomial to properly interpolate its points?

The answer reveals itself all at once when we look at the actual coefficients comprising each of these polynomials. First, remind yourself once more of the Taylor series corresponding to (the unnormalized) g and h:

$$g(x) = e^{-x^2} \sim 1 - x^2 + \frac{1}{2}x^4 - \frac{1}{6}x^6 + \cdots = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!} x^{2k}, \qquad x \in \mathbb{R}.$$

and

$$h(x) = \frac{1}{1+x^2} \sim 1 - x^2 + x^4 - x^6 + \cdots = \sum_{k=0}^{\infty} (-1)^k x^{2k}, \qquad |x| < 1,$$

Now, look at the table below.

Interpolating polynomials for $g(x) = e^{-x^2}$ vs. Taylor series $T[g]$

	$T[g]$	p_5	p_9	p_{13}	p_{21}
x^0	1	1.0000	1.0000	1.0000	1.0000
x^2	-1	−0.4931	−0.9473	−0.9977	−1.0000
x^4	$\frac{1}{2}$	0.0424	0.3549	0.4859	0.5000
x^6	$-\frac{1}{6} \approx -0.1667$	—	−0.0558	−0.1434	−0.1666
x^8	$\frac{1}{24} \approx 0.0417$	—	0.0030	0.0254	0.0415
x^{10}	$-\frac{1}{120} \approx -0.0083$	—	—	−0.0024	−0.0082
x^{12}	$\frac{1}{720} \approx 0.0014$	—	—	0.0001	0.0013

Interpolating polynomials for $h(x) = \dfrac{1}{1+x^2}$ vs. Taylor series $T[h]$

	$T[h]$	p_5	p_9	p_{13}	p_{21}
x^0	$+1$	1.0000	1.0000	1.0000	1.0000
x^2	-1	−0.3769	−0.8127	−0.9570	−0.9983
x^4	$+1$	0.0308	0.3386	0.7007	0.9694
x^6	-1	—	−0.0581	−0.3077	−0.8327
x^8	$+1$	—	0.0032	0.0718	0.5540
x^{10}	-1	—	—	−0.0081	−0.2589
x^{12}	$+1$	—	—	0.0003	0.0806

In both cases, the interpolating polynomial's coefficients are *converging to the Taylor coefficients* of the respective DGP. For the Gaussian DGP g, this is good news: the Taylor coefficients

$$c_{2k} = \frac{(-1)^k}{k!} \to 0 \quad \text{as} \quad k \to \infty,$$

and, with a rate of decay behaving like $1/k!$, these coefficients will eventually dominate any fixed evaluation of x^{2k} at a given $x \in \mathbb{R}$. What we interpret, then, is that the interpolant's convergence to the true Gaussian is a direct consequence of the Taylor series' convergence: each additional term the interpolant learns contributes a mere $|c_{2k}| = 1/k!$ in magnitude to the overall approximation — a factor shrinking so rapidly that the polynomial can "afford to incorporate it into the existing approximation without destabilizing the existing fit elsewhere." The true coefficient of the x^{12} term is $c_{12} = 1/720 \approx 0.0014$ — a correction so small that the decision to include or exclude it only affects the approximation's behavior at fairly large x, and certainly not within a neighborhood close to 0, say $x \in [-3, 3]$.

For the Cauchy DGP h, the interpolating coefficients are converging *just as faithfully* — but, here, the Taylor coefficients are

$$c_{2k} = (-1)^k,$$

which alternate in sign and remain at unit magnitude $|c_{2k}| = 1$ forever. By $n = 21$, the coefficient of x^4 has reached 0.97, headed for $+1$; the coefficient of x^6 sits at -0.83, headed for -1; that of x^8 at 0.55 is headed for $+1$, and so on. Each new term the interpolant of h seeks to learn is *just as large* as the last.

Furthermore, the series $\sum(-1)^k x^{2k}$ *diverges* for $|x| \geq 1$ — which is to say, the vast majority of the interval $[-3, 3]$. The "fangs at the boundary" are not an artifact of the procedure; they exhibit a polynomial trying desperately to conform to the rules at a smattering of locations it wasn't equipped to handle, that the approximation simply isn't realistic anywhere.

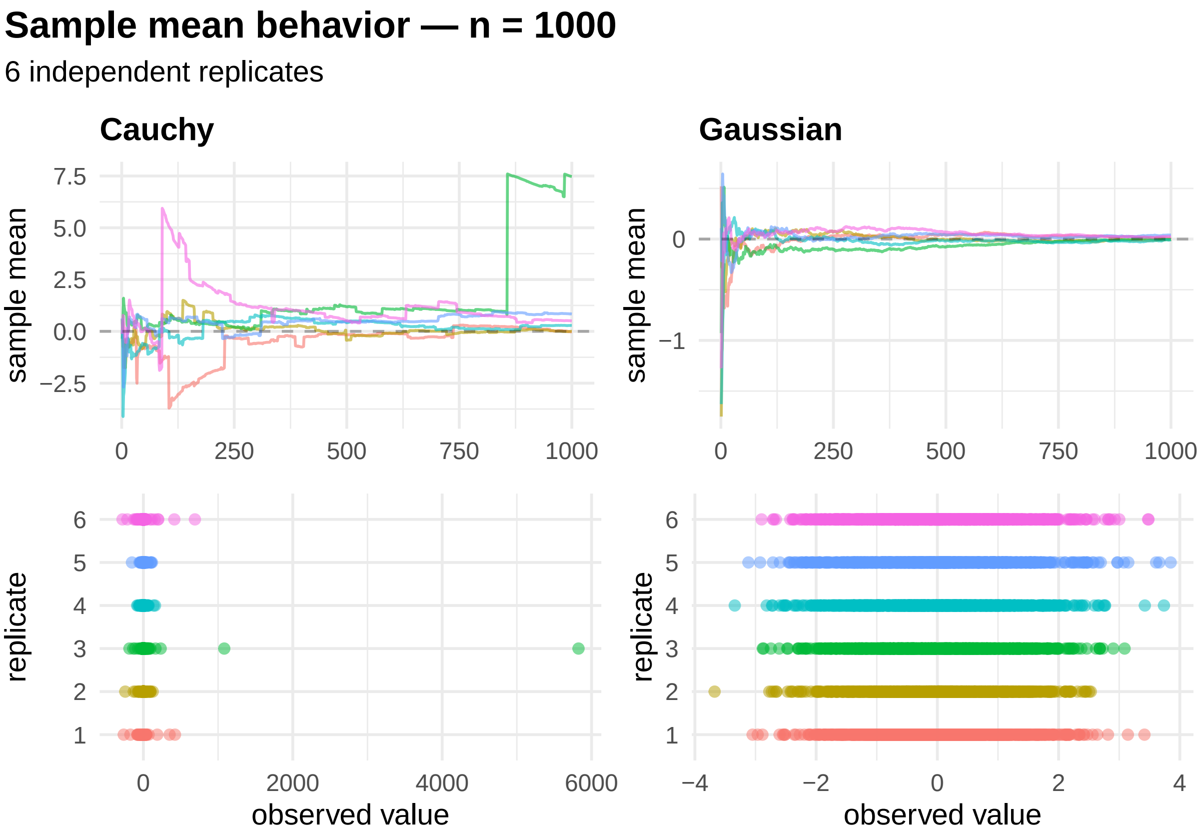
Taming the Dragon

This, then, would seem to mark the end of the parable; an unsatisfying end, to be sure, but an instructive one nonetheless. Indeed, what have we learned from the analysts of $\mathcal{G}$ and $\mathcal{H}$? That *memorization* is not the same as *learning*. Sometimes, we happen upon a data generating process which is well-behaved, and others we're forced to reckon with a process which seems purposefully adversarial — and what we've observed here is that the line separating the situations is far too fine for comfort. We're forced to exercise due diligence in understanding the mechanics of the Cauchy's elusive behavior; indeed, we imagine that if we weren't equipped to handle a function as simple as $\frac{1}{1+x^2}$, we'd remain quite hopeless in modeling the complexities of data generating processes one encounters in practice.

But we also stress that, here, *diligence itself* is a bit of a dragon. What we mean is that, no matter how many times we've recalled the parable's lessons, we find it much harder to willingly "throw away information" than it is to try and squeeze every last drop of it from the given data. It is not a failure to compromise in such situations; after recognizing analytic pursuits aren't working, such compromise *is* diligence exemplified in its purest form. The real struggle as we see it is discerning when matters are "truly hopeless," and those where some other recourse at our disposal should we just avert our gaze to the horizon. Indeed, we reassure you now that it is most often the latter case which is closer to reality — but, more aptly, reality tends to live somewhere in the middle. Inhabiting this middle with *confidence* is, in effect, the data analyst's eternal pursuit.

Looking ahead, we choose to close with an intuitive gesture towards something we'll strive to make precise moving forward. Concretely, one way we can understand the parable is through the lens of **moments**. Indeed, we're able to verify is that the Cauchy distribution has no finite mean: $\int_{-\infty}^{\infty} |x| \cdot h(x)\, dx = \infty$. More dramatically, it has no finite moments of *any* order — $\mathbb{E}[X^{2k}] = \infty$ for every $k \geq 1$. The law of large numbers, which guarantees $\overline{x}_n \to \mu$ for the Gaussian, simply does not apply for the Cauchy. The reason is not pathological — it's because the Cauchy's tails are heavy enough that a single sufficiently extreme observation can drag the running average by an arbitrarily large amount, and such observations arrive often enough that the average never recovers.

What we can seek to understand eventually is that the reasons that the Taylor coefficients of h refuse to decay and that the moments of the corresponding distribution refuse to exist are, in effect, *the same*. We'll make this connection precise when we introduce the Fourier transform and its statistical analogue, the characteristic function; for now, we leave the parable with the following figure, whose idiosyncrasies we encourage you to attempt to decipher according to the intuition developed here.



3.2 The Importance of a Basis

3.2.1 Polynomials

A Brief Review

Let us now attempt to reconcile what we learned from the parable with a presentation minding the rigor of the matter a bit more forthrightly. At this point, we encourage you to remind yourself of the definitions we presented at the beginning of this chapter — in particular, we recall that we began by investigating a noiseless system of equations taking the form

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2, \qquad i = 1, \dots, n.$$

Consider now a generalization of the basic form above: we might write it as

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \cdots + \beta_k x_i^k, \qquad i = 1, \dots, n.$$

To better facilitate the generalization, we recognize that, if we define the monomials

$$\phi_0(x) := 1, \qquad \phi_1(x) := x, \qquad \phi_2(x) := x^2, \quad \dots, \quad \phi_k(x) := x^k, \quad \dots,$$

then by defining the **design matrix** (or **model matrix**)

$$\Phi := \begin{pmatrix} \phi_0(x_1) & \phi_1(x_1) & \cdots & \phi_k(x_1) \\ \phi_0(x_2) & \phi_1(x_2) & \cdots & \phi_k(x_2) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_0(x_n) & \phi_1(x_n) & \cdots & \phi_k(x_n) \end{pmatrix} \in \mathbb{R}^{n \times p}, \qquad p := k + 1,$$

and similarly stacking the rest of the model in vector form:

$$y := \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} \in \mathbb{R}^n, \qquad \beta := \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{pmatrix} \in \mathbb{R}^p,$$

then we can write the noiseless linear model in the compressed matrix form given by

$$y = \Phi \beta.$$

What we observed in the first act of the parable were a series of data,

$$\mathcal{D}_2 := \{(0,2),(1,3)\}, \quad \mathcal{D}_3 := \{(0,2),(1,3),(3,8)\}, \quad \mathcal{D}_4 := \{(0,2),(1,3),(3,8),(4,6)\},$$

and, having been told these data were generated in accordance with a *perfect* DGP, we proceeded to fit a polynomial model that **interpolated** the samples $(x_i, y_i) \in \mathcal{D}_k$ exactly. In particular, then, what we stated was that the "polynomials of lowest degree interpolating the data" were given by

$$p_2(x) := 2 + x, \qquad p_3(x) := 2 + \frac{1}{2}x + \frac{1}{2}x^2, \qquad p_4(x) = 2 - x + \frac{5}{2}x^2 - \frac{1}{2}x^3$$

Furthermore, what we seemed to discover, at least implicitly, was that these interpolating-polynomials-of-lowest-degree were *unique* — we emphasize "of-lowest-degree" since, if we admitted a class of polynomials one degree higher for each data set, our solution class suddenly becomes *infinite*. This was what we found initially, when we attempted to fit a quadratic model containing $p = 3$ monomial terms ϕ_0, ϕ_1, ϕ_2 to our original data $\mathcal{D}_2$ containing only $n = 2$ samples.

Notice, importantly, that the class of solutions became infinite as soon as the number of monomials p exceeded the number of samples in our data n. In matrix notation, this would correspond with a design matrix Φ of the form

$$\Phi = \begin{pmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \end{pmatrix} \in \mathbb{R}^{2\times 3};$$

observe that this matrix is *wider than it is tall*. Such a "wide matrix" should be compared with, say,

$$\Phi = \begin{pmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ 1 & x_3 & x_3^2 \\ 1 & x_4 & x_4^2 \end{pmatrix} \in \mathbb{R}^{4\times 3},$$

i.e., a matrix which is *taller than it is wide*. Such a "tall matrix" would coincide with, for instance, our attempt to fit a quadratic model in $p = 3$ monomial terms to interpolate the data containing $n = 4$ samples $\mathcal{D}_4$.

Now, what we observe is that neither of these cases — the "wide" matrix ($n < p$) nor the "tall" matrix ($n > p$) — coincides with a model that actually came out of the parable. Indeed, we *gestured* toward the possibility, but what we found was that the quadratic model on $\mathcal{D}_2$ produced an *infinite number* of solutions — at $n = 2$ and $p = 3$ the quadratic class is too flexible to *uniquely* identify a solution $\beta \in \mathbb{R}^3$. Yet, at $n = 4$ and $p = 3$, the quadratic class isn't flexible enough to *produce* a valid solution $\beta \in \mathbb{R}^3$ at all — i.e., a solution *doesn't exist*.

Instead, when we fit p_2 to $\mathcal{D}_2$ at $n = 2$, we used a *linear* model comprised of $p = 2$ monomial terms $\phi_0(x) = 1, \phi_1(x) = x$, and we claimed the solution $\beta = (2, 1)^\top \in \mathbb{R}^2$ was *unique*. When we fit p_3 to $\mathcal{D}_3$ at $n = 3$, we used a *quadratic* model comprised of $p = 3$ monomial terms $\phi_0(x) = 1, \phi_1(x) = x, \phi_2(x) = x^2$, and we claimed the solution $\beta = (2, 1/2, 1/2)^\top \in \mathbb{R}^3$ was unique. It would appear that, in order to obtain a solution to the noiseless system $\beta \in \mathbb{R}^p$ which *exists and is unique* given a sample of size n, we would, in general, require the columns of Φ be comprised of exactly $p = n$ monomial terms $\phi_0, \phi_1, \dots, \phi_{n-1}$.

Indeed, at least in the non-trivial case, we have the following result.

Theorem 3.2.1 (Polynomial Interpolation)

Suppose $x_1, x_2, \dots, x_n \in \mathbb{R}$ be *distinct* real numbers ($x_1 \neq x_2 \neq \cdots \neq x_n$), and let $y_1, y_2, \dots, y_n \in \mathbb{R}$ be arbitrary. Then, there *exists a unique* polynomial

$$p \in \Pi_{n-1} := \{\text{polynomials of degree at most } n-1\} := \left\{ \sum_{j=0}^{n-1} a_j x^j : \quad a_0, \dots, a_{n-1} \in \mathbb{R} \right\},$$

such that

$$p(x_i) := a_0 + a_1 x_i + a_2 x_i^2 + \cdots + a_{n-1} x_i^{n-1} = y_i, \qquad i = 1, \dots, n.$$

Proof:

The proof asserts both (i) *existence* of a polynomial $p \in \Pi_{n-1}$ such that $p(x_1) = y_1, \dots, p(x_n) = y_n$, and that (ii) this polynomial p is *unique*. Such a proof requires us to show both parts separately. Here, we focus on the proof of uniqueness — indeed, we defer the existence argument to a *constructive one* presented in the subsequent material.

So, to show uniqueness, we often proceed by supposing the converse and deriving a contradiction. Namely, here, begin by supposing that $p \in \Pi_{n-1}$ and $q \in \Pi_{n-1}$ were *distinct* polynomials (i.e., at least one of their coefficients disagree) of degree at most $n-1$ satisfying $p(x_i) = y_i$ and $q(x_i) = y_i$ for all $i = 1, \dots, n$. Then, since the right sides would be equal, we subtract each side to force the requirement that

$$r(x_i) := p(x_i) - q(x_i) = 0, \qquad i = 1, \dots, n.$$

On the other hand, when we add or subtract polynomials $p, q \in \Pi_{n-1}$, we're certainly not increasing the degree of the result (e.g., $\underbrace{(3x+5) - (x-2)}_{p(x)-q(x)} = \underbrace{2x+7}_{r(x)}$ — we'll formalize this in a tick). Indeed,

$$p, q \in \Pi_{n-1} \implies r := p - q \in \Pi_{n-1};$$

i.e., if p and q are each of degree at most $n-1$, then so too is $r = p - q$ at most of degree $n-1$ (where, quite possibly, r's degree could shrink as a result of the subtraction).

What the constraint $r(x_i) = 0$, $i = 1, \dots, n$ says in this light is that r, a polynomial of degree at most $n-1$, must vanish at n distinct points $x_1, \dots, x_n$. Again deferring the rigorous discussion, we can imagine that at $n = 2$, $r \in \Pi_{n-1}$ is a polynomial of degree $n-1 = 1$ (i.e., a line), and we're constraining this line to vanish at exactly $n = 2$ points (x_1, y_1) and (x_2, y_2) — it is, of course, *impossible* to do this without setting the line perfectly on the x-axis (i.e., setting r to be the constant line $r(x) = 0$ for all x).

In general, the same sort of argument shows that in order for $r \in \Pi_{n-1}$ to satisfy n constraints, it must be that r is the **zero** (or **trivial**) polynomial,

$$r \equiv 0 \in \Pi_{n-1}.$$

Finally, since $0 \equiv r = p - q$, we have $p = q \in \Pi_{n-1}$, which establishes the claimed uniqueness.

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The proof above leaned on two assertions we haven't yet formally justified: that (i) Π_{n-1} is "closed under subtraction" — that $p, q \in \Pi_{n-1}$ implies $p - q \in \Pi_{n-1}$ — and that (ii) a polynomial of degree at most $n-1$ which vanishes at n distinct points *must* be the zero polynomial. Both of these are consequences of the *algebraic structure* of polynomials, in a manner we now seek to make precise.

Theorem 3.2.2 (Properties of Polynomials)

Let Π_k denote the set of all polynomials of degree at most k with real coefficients. Then:

(a) (Closure under Linear Combinations) If $p, q \in \Pi_k$ and $\alpha, \beta \in \mathbb{R}$, then

$$\alpha p + \beta q \in \Pi_k.$$

(b) (Root Bound) If $0 \neq p \in \Pi_k$ is not the zero polynomial, then p has at most k real-valued roots.

In establishing the proof of the root bound, we require the following lemma (whose proof, though certainly instructive, we ultimately defer to the reader — noting, perhaps, that we will establish essentially the same intuition therein while proving the root bound.)

Lemma 3.2.3 (Polynomial Factor)

If $p \in \Pi_k$ and $p(a) = 0$ for some $a \in \mathbb{R}$, then there exists a polynomial $q \in \Pi_{k-1}$ such that

$$p(x) = (x-a)q(x), \qquad x \in \mathbb{R}.$$

Proof:

(a) To establish the closure of Π_k, suppose that $p, q \in \Pi_k$ are typical members such that

$$p(x) = \sum_{j=0}^{k} a_j x^j, \qquad q(x) = \sum_{j=0}^{k} b_j x^j, \qquad x \in \mathbb{R}.$$

Then, by defining the addition $p + q$ by

$$p(x) + q(x) := \sum_{j=0}^{k} (a_j + b_j) x^j, \qquad x \in \mathbb{R},$$

and multiplication of p and q by scalars $\alpha, \beta \in \mathbb{R}$ respectively by

$$\alpha p(x) := \sum_{j=0}^{k} (\alpha a_j) x^j, \qquad \beta q(x) := \sum_{j=0}^{k} (\beta b_j) x^j,$$

what we verify is that

$$\alpha p(x) + \beta q(x) := \sum_{j=0}^{k} (\alpha a_j + \beta b_j) x^j$$

is a polynomial of degree k, so that $\alpha p + \beta q \in \Pi_k$. (The key point here is that we verify the coefficients of the linear combination remain valid and, hence, constitute a well-defined polynomial $\alpha p + \beta q \in \Pi_k$ simply by inheriting it from the real coefficients in the summations, $\alpha a_j + \beta b_j \in \mathbb{R}$.)

(b) For the Root Bound, let $0 \neq p \in \Pi_k$ be a nontrivial, degree-k polynomial of the form

$$p(x) = c_0 + c_1 x + c_2 x^2 + \cdots + c_k x^k.$$

Then, if p has no real roots, we're done (verify why this is obvious!); otherwise, let a be a root of p. By the Polynomial Factor Lemma, we have

$$p(x) = (x-a)q(x), \qquad q \in \Pi_{k-1}.$$

If $q \equiv 0$, then $p(x) = (x-a) \cdot 0 = 0$ for all x, implying $p \equiv 0 \in \Pi_k$ is the trivial polynomial — a contradiction to our original assumption. In particular, it must be that $q \in \Pi_{k-1}$ is nontrivial.

In particular, then: if q has no real roots, we're done (and p has one root, $x = a$). If not and $b \neq a$ is a root of q, then

$$p(b) = \underbrace{(b-a)}_{\neq 0} \underbrace{q(b)}_{=0} = 0,$$

so that b is a root of $p \in \Pi_k$ as well. In particular, we may write

$$p(x) = (x-a)(x-b)r(x), \qquad r \in \Pi_{k-2}.$$

We repeat the process of reapplying the Polynomial Factor Lemma until the resultant polynomial on the right has no remaining roots (in which case we conclude it has as many roots as the number of times we were able to apply the Lemma), or until it the polynomial becomes identically a constant $c \in \Pi_0$ after k iterations. In the latter, most extreme case, we ultimately obtain a representation for p of the form

$$p(x) = c(x-a_1)(x-a_2)\cdots(x-a_k).$$

This polynomial has exactly k roots, and is structurally incapable of possessing more. Any other, less extreme possibility terminates when the representation on the right has *fewer* than k roots; in particular, it must be that $p \in \Pi_k$ has at most k roots.

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Notice what happened in the proof of the root bound: in the "most extreme case," we started with a polynomial of the form

$$p(x) = c_0 + c_1 x + \cdots + c_k x^k,$$

and we ended with a *new representation* for the *same polynomial*,

$$p(x) = c(x-a_1)(x-a_2)\cdots(x-a_k).$$

What we also seemed to indicate was that, whenever we found a polynomial $p \in \Pi_k$ had less than k roots (say it has $j \leq k$), our repeated applications of the Factor Lemma would proceed and terminate according to

$$\begin{aligned}
&(0) \qquad & p(x) &= \sum_{i=0}^{k} c_i x^i, & 0 &\not\equiv p \in \Pi_k;\\
&(1) \qquad & p(x) &= (x-a_1)r_1(x), & 0 &\not\equiv r_1 \in \Pi_{k-1};\\
&(2) \qquad & p(x) &= (x-a_1)(x-a_2)r_2(x), & 0 &\not\equiv r_2 \in \Pi_{k-2};\\
& & &\vdots & &
\end{aligned}$$

$$(j) \qquad p(x) = (x - a_1)(x - a_2) \cdots (x - a_j) r_j(x), \qquad 0 \not\equiv r_j \in \Pi_{k-j}.$$

At each line of this process, the object on the left — $p(x)$ — has not changed. Indeed, the polynomial p is a function: a rule that accepts as input a real number $x \in \mathbb{R}$ and returns a real number, $p(x) \in \mathbb{R}$. Two expressions that dictate equivalent rules — i.e., that return the same value at every $x \in \mathbb{R}$ — are *not different polynomials*; they are *different descriptions* of the *same polynomial*.

That is, in the steps of the algorithm outlined above, what we observe is that at step (0), p is expressed as a linear combination of the functions $\{1, x, x^2, \ldots, x^k\}$ weighted by real numbers $\{c_0, c_1, c_2, \ldots, c_k\}$. On the other hand, at step (j), p is expressed as a product of j linear factors $(x - a_1)(x - a_2) \cdots (x - a_j)$ and a residual polynomial r_j, where the numbers $\{a_1, \ldots, a_j\}$ are its roots. The weights c_i and the roots a_i are, in general, completely different numbers, and neither set of numbers "is the polynomial"; they are what we obtain when we *describe* the polynomial in a *particular language*. It just so happens that, here, the monomial expression at step (0) and the factored expression at step (j) describe the same rule p by characterizing two seemingly distinct structural properties of the original polynomial — at step (0), "we know how much each monomial $x_i \in \{1, x, \ldots, x^k\}$ comprises the whole polynomial p by looking at the coefficient c_i"; at step (j), "we know where the real roots $\{a_1, a_2, \ldots a_j\}$ of p fall in $\mathbb{R}$."

This distinction — between the polynomial p and a particular representation — is not merely philosophical. Indeed, what we understand is that if we recall, for instance, the polynomials from the parable

$$p_2(x) := 2 + x, \qquad p_3(x) := 2 + \frac{1}{2}x + \frac{1}{2}x^2, \qquad p_4(x) = 2 - x + \frac{5}{2}x^2 - \frac{1}{2}x^3,$$

stack the model coefficients in a table as follows:

	β_0	β_1	β_2	β_3
p_2	2	1	—	—
p_3	2	1/2	1/2	—
p_4	2	−1	5/2	−1/2

Reading down the columns of the coefficients, it is only the intercept $\beta_0 = 2$ which remains stable across the three models — in particular, the linear β_1 and quadratic β_2 terms seem to shift, even going so far as changing signs: the linear coefficient attached to β_1 fluctuates from 1 in p_2 to -1 in p_4.

It is genuinely curious, then, that the coefficient $\beta_0 = 2$ — which, recall, we deduced directly from the common sample point $(0, 2)$ present across each of the datasets $\mathcal{D}_2, \mathcal{D}_3, \mathcal{D}_4$ — is the sole estimate that remains stable. Indeed, the reason is visible in the design matrix itself. For the model p_2 given data $\mathcal{D}_2 = \{(0, 2), (1, 3)\}$, the system determining the model, $\mathbf{y} = \mathbf{\Phi B}$, was

$$\begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix}.$$

The system is lower triangular, and we can observe directly that the top row identifies $\beta_0 = 2$ directly. How we obtain β_1, on the other hand, requires us to have identified that $\beta_0 = 2$ from the first constraint, at which point we deduce from the second row that

$$3 = \beta_0 + \beta_1 \implies \beta_1 = 3 - 2 = 1.$$

Still, this deduction was fairly straightforward: it was apparent from the **triangular** form of the matrix $\Phi = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$ that we could "compute the β_i coefficients from the top down." It was straightforward because the first row told us precisely what β_0 was, automatically forcing the value of the only unknown in the second row β_1.

Compare this now to the corresponding system for p_3 on $\mathcal{D}_3 = \{(0,2),(1,3),(3,8)\}$:

$$\begin{pmatrix} 2 \\ 3 \\ 8 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 3 & 9 \end{pmatrix} \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{pmatrix}.$$

Again, we're essentially handed $\beta_0 = 2$ for free in the first row because the first sample point has $x_1 = 0$, at which each monomial $\phi_1(x) = x, \phi_2(x) = x^2, \ldots$ evaluates to zero. However, what happens in the following rows is immediately less clear — indeed, the second and third rows describe the equations

$$\begin{aligned} \beta_0 + \beta_1 + \beta_2 &= 3, \\ \beta_0 + 3\beta_1 + 9\beta_2 &= 8. \end{aligned}$$

Nothing is *wrong* with these equations; they're just **dense**. Even if we plug in what we know from the first row, $\beta_0 = 2$, and reduce appropriately, we're still left with

$$\begin{aligned} \beta_1 + \beta_2 &= 1, \\ 3\beta_1 + 9\beta_2 &= 6, \end{aligned}$$

which isn't quite helpful precisely because it doesn't tell us what β_1 and β_2 *are*, only what they must *satisfy* in order to form a valid solution to the original system. This is, in effect, what we intend to underscore when we distinguish objects from their representations — i.e., it is clear that the dense representation "works" insofar as characterizing the quadratic model p_3 on $\mathcal{D}_3$, but it does little to reveal the model's *underlying structure*, whatever it may be.

So, to reiterate, if we look to the original system on $\mathcal{D}_3$,

$$\begin{pmatrix} 2 \\ 3 \\ 8 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 3 & 9 \end{pmatrix} \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{pmatrix},$$

one deduces according to the logic deployed above that a solution $\beta \in \mathbb{R}^3$ must satisfy

$$\begin{aligned} \beta_0 &= 2 \\ \beta_1 + \beta_2 &= 1, \\ 3\beta_1 + 9\beta_2 &= 6. \end{aligned}$$

Again, our "issue" with this deduction isn't that it's "wrong," just that it's quite verbose — it tells us exactly what β_0 is, but only seems to "describe" what β_1 and β_2 are.

What we understand from a first course in linear algebra is that, given two equations in two unknowns, we're often able to employ **substitution** or **elimination** to reduce the system into a more explicit form. Indeed, from

$$\beta_1 + \beta_2 = 1,$$

$$3\beta_1 + 9\beta_2 = 6,$$

we can rearrange $\beta_2 = 1 - \beta_1$, plug in

$$3\beta_1 + 9(1-\beta_1) = 6 \implies \beta_1 = \frac{1}{2},$$

and from the rearrangement obtain

$$\beta_2 = 1 - \beta_1 = \frac{1}{2}.$$

In particular, any solution $\beta \in \mathbb{R}^3$ solving the original system is forced to satisfy

$$\begin{aligned}\beta_0 &= 2\\ \beta_1 &= \frac{1}{2},\\ \beta_2 &= \frac{1}{2},\end{aligned}$$

which is of the explicit form we desired.

But what can we say about the process in general? Indeed, let us take a step back and perform the same series of operations on the general form of the quadratic model given $n = 3$ samples, which is

$$\begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ 1 & x_3 & x_3^2 \end{pmatrix} \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{pmatrix}.$$

If we go to row-reduce the system in the usual way, we might start by realizing that if we subtract row 1 from rows 2 and 3, we obtain

$$\begin{pmatrix} y_1 \\ y_2 - y_1 \\ y_3 - y_1 \end{pmatrix} = \begin{pmatrix} 1 & x_1 & x_1^2 \\ 0 & x_2 - x_1 & x_2^2 - x_1^2 \\ 0 & x_3 - x_1 & x_3^2 - x_1^2 \end{pmatrix} \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{pmatrix}.$$

We observe that in coefficient matrix, there are common factors of $x_2 - x_1$ in the second row (since $x_2^2 - x_1^2 = (x_2 - x_1)(x_2 + x_1)$) and $x_3 - x_1$ in the third. Supposing as we have been that the nodes x_i are distinct, we may divide by $x_2 - x_1 \neq 0$ and $x_3 - x_1 \neq 0$ to obtain

$$\begin{pmatrix} y_1 \\ \dfrac{y_2 - y_1}{x_2 - x_1} \\ \dfrac{y_3 - y_1}{x_3 - x_1} \end{pmatrix} = \begin{pmatrix} 1 & x_1 & x_1^2 \\ 0 & 1 & x_1 + x_2 \\ 0 & 1 & x_1 + x_3 \end{pmatrix} \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{pmatrix}$$

Now, if we subtract row 2 from row 3 and divide it by $(x_3 - x_2)$, we obtain

$$\begin{pmatrix} y_1 \\ \dfrac{y_2 - y_1}{x_2 - x_1} \\ \dfrac{\dfrac{y_3 - y_1}{x_3 - x_1} - \dfrac{y_2 - y_1}{x_2 - x_1}}{x_3 - x_2} \end{pmatrix} = \begin{pmatrix} 1 & x_1 & x_1^2 \\ 0 & 1 & x_1 + x_2 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{pmatrix}$$

At this point, observe that the coefficient matrix on the right is **upper triangular** — ones on the diagonal, arbitrary entries above the diagonal, and zeros below it. On the left-side, we observe a new kind of structure emerging from the row operations. The first entry is y_1, which is just querying the value of p at the first node x_1. However, the second row is no longer describing a constraint on the position of the polynomial's nodes — i.e., that "at $x = x_2$, $p(x_2) = y_2$" — instead, it constrains

$$\frac{y_2 - y_1}{x_2 - x_1};$$

this is no longer a positional constraint, but rather one controlling the *rate of change* of the polynomial evaluations $y_1 = f(x_1)$ and $y_2 = f(x_2)$ as x moves from x_1 to x_2. We define

$$[y_1, y_2] := \frac{y_2 - y_1}{x_2 - x_1}$$

to be the **first divided difference** of the data $\mathcal{D}_3$, and one immediately recognizes that the first divided difference is precisely the slope of the secant line connecting the points (x_1, y_1) with (x_2, y_2). "The first row of this new system constrains the position of the first node $y_1 = f(x_1)$; the second row looks at the position of that first node $(x_1, f(x_1))$, observes where $(x_2, f(x_2))$ is supposed to be, and forces the slope of the linear component β_1 to be such that the line through (x_1, y_1) also passes through (x_2, y_2)." In other words, we've "exchanged" a constraint on *where the polynomial goes* for a constraint on *how fast it gets there*. The original row 2 said "$p(x_2) = y_2$" — a positional constraint. The new row 2 says "the average rate of change of p moving from x_1 to x_2 equals the first divided difference $[y_1, y_2]$."

The third entry on the left admits an analogous interpretation. Define the **second divided difference** by

$$[y_1, y_2, y_3] := \frac{[y_1, y_3] - [y_1, y_2]}{x_3 - x_2} = \frac{\dfrac{y_3 - y_1}{x_3 - x_1} - \dfrac{y_2 - y_1}{x_2 - x_1}}{x_3 - x_2},$$

and observe that this is a *difference of slopes* normalized by the node spacing between x_2 and x_3. That is, $[y_1, y_2, y_3]$ characterizes "how the linear rate of change must, well, *itself* change" in order for the model to pass through all three points — i.e., it characterizes a **discrete curvature**. As the first divided difference is to a "discrete first derivative," the second divided difference is to a "second discrete derivative," computed directly from the observed data.

In this language of divided differences, what the new system reads is

$$\begin{pmatrix} [y_1] \\ [y_1, y_2] \\ [y_1, y_2, y_3] \end{pmatrix} = \begin{pmatrix} 1 & x_1 & x_1^2 \\ 0 & 1 & x_1 + x_2 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{pmatrix}.$$

(We may understand $[y_1] = y_1$ as a **zeroth divided difference** if we're so inclined).

Viewed as a description of the solution's underlying mechanism, this formulation of the system is much clearer than our initial one. Again, the reason is because this new system is *triangular* — indeed, if one "starts at the bottom row," we start by reading off

$$\beta_2 = [y_1, y_2, y_3]$$

directly. Then, we observe that

$$\beta_1 + (x_1 + x_2)\beta_2 = [y_1, y_2] \implies \beta_1 = [y_1, y_2] - (x_1 + x_2)[y_1, y_2, y_3].$$

Finally, after some simplification, we obtain

$$\beta_0 + x_1\beta_1 + x_1^2\beta_2 = y_1 \implies \beta_0 = y_1 - x_1[y_1, y_2] + x_1x_2[y_1, y_2, y_3]$$

Together, the solution in this language of divided differences is

$$\begin{pmatrix}\beta_0\\ \beta_1\\ \beta_2\end{pmatrix} = \begin{pmatrix}[y_1] - x_1[y_1, y_2] + x_1x_2[y_1, y_2, y_3]\\ [y_1, y_2] - (x_1 + x_2)[y_1, y_2, y_3]\\ [y_1, y_2, y_3]\end{pmatrix}.$$

Indeed, one can verify that these formulas applied to the data $\mathcal{D}_3 = \{(0, 2), (1, 3), (3, 8)\}$ yields the familiar solution

$$\begin{pmatrix}\beta_0\\ \beta_1\\ \beta_2\end{pmatrix} = \begin{pmatrix}2\\ 1/2\\ 1/2\end{pmatrix}.$$

But what we understand from the general formulation is that, really, it was "quite lucky" that the constraint on β_0 reduced to the mere identification $\beta_0 = 2$. The reason for this reduction is just as apparent from the form above — the top row involves a linear combination of divided differences, where all orders besides the zeroth $[y_1]$ are weighted by x_1, which happened to coincide with the data point $(x_1, y_1) = (0, 2)$. Since $x_1 = 0$, the whole thing reduces to $\beta_0 = [y_1] = y_1 = 2$. In general, though, it is really the *quadratic* coefficient which we're able to "read off" in this manner — indeed, we compute the second divided difference y_1, y_2, y_3 directly from the raw data, and we identify this quantity precisely as the quadratic coefficient β_2. We then back-substitute this second order difference in the row constraining $\beta_1 = [y_1, y_2] - (x_1 + x_2)[y_1, y_2, y_3]$, computing the first divided difference $[y_1, y_2]$ along the way to identifying the linear slope β_1. Only then, having already computed the first and second divided differences/β_2 and β_1, do we move to the row constraining $\beta_0 = [y_1] - x_1[y_1, y_2] + x_1x_2[y_1, y_2, y_3]$, where we plug everything in and solve for the intercept β_0.

Changing the Questions

So again, returning to the original form of the system for $\mathcal{D}_3$ given by

$$\begin{pmatrix}2\\ 3\\ 8\end{pmatrix} = \begin{pmatrix}1 & 0 & 0\\ 1 & 1 & 1\\ 1 & 3 & 9\end{pmatrix}\begin{pmatrix}\beta_0\\ \beta_1\\ \beta_2\end{pmatrix},$$

what we understand is that, *mechanically*, row reduction transformed the original system — three *distinct* positional constraints on p that "at $x = x_i$, $f(x_i) = y_i$" — into a new system asking *hierarchical* constraints: a value, a slope running from that value, and the curvature of that same slope.

The way we got from the original system to the hierarchical one was via row operations on the system — namely, we transitioned from

$$\begin{pmatrix}y_1\\ y_2\\ y_3\end{pmatrix} = \begin{pmatrix}1 & x_1 & x_1^2\\ 1 & x_2 & x_2^2\\ 1 & x_3 & x_3^2\end{pmatrix}\begin{pmatrix}\beta_0\\ \beta_1\\ \beta_2\end{pmatrix} \longmapsto \begin{pmatrix}[y_1]\\ [y_1, y_2]\\ [y_1, y_2, y_3]\end{pmatrix} = \begin{pmatrix}1 & x_1 & x_1^2\\ 0 & 1 & x_1 + x_2\\ 0 & 0 & 1\end{pmatrix}\begin{pmatrix}\beta_0\\ \beta_1\\ \beta_2\end{pmatrix}.$$

To make this a bit more rigorous, observe that if we retrace our steps through the row operations, then if the original system on the left above is

$$y = \Phi\beta,$$

then, in lieu of the "row operations" we conducted mechanically, perhaps we could hope to define a matrix L such that

$$Ly = L\Phi\beta := \Psi\beta, \qquad \Psi := L\Phi.$$

That is, in a more expanded notation, perhaps we could hope to find a matrix $L = \big((\ell_{ij})\big)$ such that when we left-multiply the system as

$$\begin{pmatrix} \ell_{11} & \ell_{12} & \ell_{13} \\ \ell_{21} & \ell_{22} & \ell_{23} \\ \ell_{31} & \ell_{32} & \ell_{33} \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} \ell_{11} & \ell_{12} & \ell_{13} \\ \ell_{21} & \ell_{22} & \ell_{23} \\ \ell_{31} & \ell_{32} & \ell_{33} \end{pmatrix} \begin{pmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ 1 & x_3 & x_3^2 \end{pmatrix} \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{pmatrix},$$

the numbers ℓ_{ij} are chosen such that

$$\begin{pmatrix} \ell_{11} & \ell_{12} & \ell_{13} \\ \ell_{21} & \ell_{22} & \ell_{23} \\ \ell_{31} & \ell_{32} & \ell_{33} \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} [y_1] \\ [y_1, y_2] \\ [y_1, y_2, y_3] \end{pmatrix},$$

and

$$\begin{pmatrix} \ell_{11} & \ell_{12} & \ell_{13} \\ \ell_{21} & \ell_{22} & \ell_{23} \\ \ell_{31} & \ell_{32} & \ell_{33} \end{pmatrix} \begin{pmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ 1 & x_3 & x_3^2 \end{pmatrix} = \begin{pmatrix} 1 & x_1 & x_1^2 \\ 0 & 1 & x_1 + x_2 \\ 0 & 0 & 1 \end{pmatrix}.$$

In effect, we're hoping to "turn row operations into matrix multiplication"; it is perhaps "how we'd tell the computer to perform the analogous operations."

Indeed, one can verify that the matrix

$$L := \begin{pmatrix} 1 & 0 & 0 \\ \dfrac{-1}{x_2 - x_1} & \dfrac{1}{x_2 - x_1} & 0 \\ \dfrac{1}{(x_2 - x_1)(x_3 - x_1)} & \dfrac{-1}{(x_2 - x_1)(x_3 - x_2)} & \dfrac{1}{(x_3 - x_1)(x_3 - x_2)} \end{pmatrix}$$

accomplishes precisely this. The numbers themselves aren't nearly as important as the *structure* of L. Indeed, we observe that the entries above the diagonal vanish, $\ell_{12} = \ell_{13} = \ell_{23} = 0$ — i.e., the matrix L is lower triangular. And, when we apply the lower triangular L to the dense matrix by mapping $\Phi \mapsto L\Phi$, we obtain our desired upper triangular matrix,

$$\begin{pmatrix} 1 & 0 & 0 \\ \dfrac{-1}{x_2 - x_1} & \dfrac{1}{x_2 - x_1} & 0 \\ \dfrac{1}{(x_2 - x_1)(x_3 - x_1)} & \dfrac{-1}{(x_2 - x_1)(x_3 - x_2)} & \dfrac{1}{(x_3 - x_1)(x_3 - x_2)} \end{pmatrix} \begin{pmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ 1 & x_3 & x_3^2 \end{pmatrix} = \begin{pmatrix} 1 & x_1 & x_1^2 \\ 0 & 1 & x_1 + x_2 \\ 0 & 0 & 1 \end{pmatrix}.$$

Similarly, when we apply L to the data vector $\mathbf{y}$, we obtain

$$\begin{pmatrix} 1 & 0 & 0 \\ \dfrac{-1}{x_2 - x_1} & \dfrac{1}{x_2 - x_1} & 0 \\ \dfrac{1}{(x_2 - x_1)(x_3 - x_1)} & \dfrac{-1}{(x_2 - x_1)(x_3 - x_2)} & \dfrac{1}{(x_3 - x_1)(x_3 - x_2)} \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} [y_1] \\ [y_1, y_2] \\ [y_1, y_2, y_3] \end{pmatrix}.$$

Thus, in its entirety, the row-reduced system reads

$$L\mathbf{y} = L\Phi\beta, \qquad \text{i.e.,} \qquad \begin{pmatrix} [y_1] \\ [y_1, y_2] \\ [y_1, y_2, y_3] \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & x_1 & x_1^2 \\ 0 & 1 & x_1 + x_2 \\ 0 & 0 & 1 \end{pmatrix}}_{U := L\Phi} \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{pmatrix}.$$

Taking stock of the whole procedure, we observe that in introducing the matrix L, it acted on the *left* of both $\mathbf{y}$ and Φ — transforming the data into divided differences and the design matrix into upper triangular form — while leaving the unknowns β entirely untouched. What is crucial about the interpretation is that the numbers β_0, β_1, and β_2 still measure the influence of the monomials $\phi_0(x) = 1$, $\phi_1(x) = x$ and $\phi_2(x) = x^2$; yet, what is just as apparent is that the *questions* we're asking to ascertain these influences have fundamentally changed. Again, the original system asks three positional questions — "where does p go at x_1, x_2, and x_3?" — while the row-reduced system asks three hierarchical constraints — "where does p go at x_1; given this value, how does the slope of p change to attain another value at x_2; given this value and slope, how does the curvature of p change to attain a third value at x_3?"

We stress that the reduced form of the system we derived in the last ribbon *really is* an improvement insofar as understanding the mechanism driving the solution. When we levy a new positional constraint on the polynomial, it isn't really that the coefficients β_i shift according to the "location of the new node," but rather according to "how aggressively we must contort our polynomial to hit the new node, given it must still hit all of the previous nodes." Nevertheless, what we observe is that, in spite of our careful consideration of the *questions* we're asking, the *language of our solution* isn't rendered any cleaner for it.

The divided differences — the hierarchical answers to these new questions — sit on the *left-hand side* of the reduced system. They describe the data $\mathbf{y}$. But the unknowns on the right are still $\beta_0, \beta_1, \beta_2$ — the monomial coefficients. These were the original curiosity — we were wondering why, in the table

	β_0	β_1	β_2	β_3
p_2	2	1	—	—
p_3	2	1/2	1/2	—
p_4	2	-1	5/2	$-1/2$

the other coefficients continued to bounce around while β_0 remained constant. At this point, we've revealed this to be a "lucky coincidence" — a consequence of the sample point $(x_1, y_1) = (0, 2)$ appearing in every dataset. But the coefficients are "actually" being computed via, e.g.,

$$\begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix} = \begin{pmatrix} [y_1] - x_1[y_1, y_2] \\ [y_1, y_2] \end{pmatrix}, \qquad \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{pmatrix} = \begin{pmatrix} [y_1] - x_1[y_1, y_2] + x_1 x_2 [y_1, y_2, y_3] \\ [y_1, y_2] - (x_1 + x_2)[y_1, y_2, y_3] \\ [y_1, y_2, y_3] \end{pmatrix},$$

and so on. And what is quite unfortunate about the form of these solutions is that they're *constantly shifting* to account for the newest higher-order monomial — indeed, we see the same "basic structure" in both of the solutions written above, with β_1 a function of the first divided difference $[y_1, y_2]$ in both cases, and with β_0 a function of $[y_1]$ and $[y_1, y_2]$ in both cases. When we add on a cubic β_3 term, we imagine the solution would look something like

$$\begin{pmatrix}\beta_0\\ \beta_1\\ \beta_2\\ \beta_3\end{pmatrix} = \begin{pmatrix} \underbrace{[y_1] - x_1[y_1,y_2] + x_1x_2[y_1,y_2,y_3]}_{\text{same structure}} + \underbrace{w_0(x_1,x_2,x_3)[y_1,y_2,y_3,y_4]}_{\text{correction}} \\ \underbrace{[y_1,y_2] - (x_1+x_2)[y_1,y_2,y_3]}_{\text{same structure}} + \underbrace{w_1(x_1,x_2,x_3)[y_1,y_2,y_3,y_4]}_{\text{correction}} \\ \underbrace{[y_1,y_2,y_3]}_{\text{same structure}} + \underbrace{w_2(x_1,x_2,x_3)[y_1,y_2,y_3,y_4]}_{\text{correction}} \\ \underbrace{[y_1,y_2,y_3,y_4]}_{\text{new structure}} \end{pmatrix},$$

which we interpret as "given a new sample (x_4, y_4), takes the basic structure of $(\beta_0, \beta_1, \beta_2)$ derived from $\mathcal{D}_3$ and introduces correction terms: combinations of the new highest order divided difference $[y_1, y_2, y_3, y_4]$ weighted by functions w_j of the sample nodes we've already modeled, x_1, x_2, x_3."

Even though we've asked better questions — ones suggesting a concrete order in which to query the data (first compute $[y_1, y_2, y_3, y_4]$, then $[y_1, y_2, y_3]$, and so on) — we still have to *recompute every coefficient* any time we introduce a new monomial term. In this sense, the lower-order β_i coefficients are quite unstable.

Yet we can also observe a "kernel of stability" characterizing the coordinates β_i in the solution above. Indeed, we perceive each row as "clean signal plus a contamination":

$$\begin{aligned}
\beta_3 &= \underbrace{[y_1,y_2,y_3,y_4]}_{\text{clean, stable}} \\
\beta_2 &= \underbrace{[y_1,y_2,y_3]}_{\text{stable}} + \underbrace{w_2(x_1,x_2,x_3)[y_1,y_2,y_3,y_4]}_{\text{one correction}} \\
\beta_1 &= \underbrace{[y_1,y_2]}_{\text{stable}} - \underbrace{(x_1+x_2)[y_1,y_2,y_3] + w_1(x_1,x_2,x_3)[y_1,y_2,y_3,y_4]}_{\text{two cumulative corrections}} \\
\beta_0 &= \underbrace{[y_1]}_{\text{stable}} - \underbrace{x_1[y_1,y_2] + x_1x_2[y_1,y_2,y_3] + w_0(x_1,x_2,x_3)[y_1,y_2,y_3,y_4]}_{\text{three cumulative corrections}}.
\end{aligned}$$

Supposing we were to identify each β_i by its "most distinguishable atom," we would almost certainly point to the stable ith order divided difference. The corrections — the terms involving higher-order divided differences, weighted by functions of the nodes — are what separate β_j from the "clean" quantity $[y_1, \dots, y_{j+1}]$. As we continue adding more data, we're forced to add additional monomial terms, and we necessarily induce more corrections on the lowest-order coefficients. That is, we'll never stop re-contaminating the lower-order coefficients — again, *this is why* we observe instability in the non-constant coefficients

	β_1	β_2	β_3
p_2	1	—	—
p_3	1/2	1/2	—
p_4	−1	5/2	−1/2

But this raises an important, devastatingly obvious question: if every β_j is "really" the jth divided difference $[y_1, \dots, y_{j+1}]$ plus "corrections" that exist only to satisfy the *monomial representation*, then why are we insisting on the monomial representation at all?

Indeed, we recall what the polynomial interpolation theorem actually promised: the existence and uniqueness of a *polynomial* $p \in \Pi_{n-1}$ satisfying $p(x_i) = y_i$. Again, at its most basal level, a polynomial is "just a function," so it is "just a rule that assigns inputs $x \in \mathbb{R}$ to outputs $p(x) \in \mathbb{R}$." Intuitively, if we could "list every $x \in \mathbb{R}$ alongside its evaluation $p(x) \in \mathbb{R}$," *never* dictating a "functional form" for the polynomial, and the table would *still characterize* "the same polynomial p." It is immaterial to the polynomial *how* we "come up with this number $p(x)$" — just that we do, and that it is right.

In the same way that a table of evaluations could act as a representation for the polynomial p, its representation in terms of monomials is just that: *a representation*, and nothing more. The subtlety is that, this whole time, we've been operating tacitly under the definition we gave of a polynomial $p \in \Pi_k$ *as a linear combination of monomials*: indeed, recall we *defined* Π_{n-1} to be "the class of polynomials of degree at most $n-1$," where in particular,

$$p \in \Pi_{n-1} \implies \forall x \in \mathbb{R}\colon\ p(x) = \sum_{j=0}^{n-1} a_j x^j \quad \text{for some fixed} \quad a_0, a_1, \dots, a_{n-1} \in \mathbb{R}.$$

That is, the monomial representation of a polynomial $p \in \Pi_{n-1}$ isn't merely a convenient choice — it is, at least as we've set things up so far, baked into the *definition*. So, when we've been saying "p is a polynomial of degree at most $n-1$," what we've *meant every time* was precisely that "p admits an expansion $\sum a_j x^j$ with real coefficients." This seems to be directly at odds with what we just said though, which is that, in general, a polynomial is "just a class of structured rules" assigning real inputs x to real outputs $p(x)$.

As it turns out, neither perspective is wrong — they both gesture towards a common ground. Indeed, when we defined Π_{n-1}, the class of polynomials of degree at most $n-1$, as $\left\{\sum_{j=0}^{n-1} a_j x^j\colon\ a_0, \dots, a_{n-1} \in \mathbb{R}\right\}$, we were making two claims at once, and it is worth separating them. The first claim is about *membership*: what does it mean for a function to be "a polynomial of degree at most $n-1$?" This question is answered directly by the monomial representation: a polynomial p is of degree $n-1$ if and only if it can be written $\sum_{j=0}^{n-1} a_j x^j$.

The second claim, which we haven't stated but which we've silently assumed, is about *identity*, and it is what we've spent the last several pages attempting to dismantle. Recall the order of events leading us here: we were given n data points, and, largely based on preexisting intuition, we decided the class of polynomials Π_{n-1} was suitable for interpolating those data points. We then *told you* "how to construct a polynomial" as an expansion in terms of monomials, and from that moment forward, the problem of "finding the polynomial" became synonymous with "finding the monomial coefficients β." But these are *not* the same task! The first asks for a function — a rule, as in our first perspective above. The second asks for a *description* of that function in a *particular language* — the language of monomials. The interpolation theorem guarantees the uniqueness *of the function*. It says *nothing* about privileging any particular description of it.

Formally, what reconciles the two perspectives is the observation that the monomial expansion is *sufficient for membership* — if a function admits a monomial expansion, then it is a polynomial — but it is not *necessary for computation*. In fact, we *already know* a way to do this — given a polynomial in monomial form with j real roots, we can apply the Polynomial Factor Lemma j times, and each iteration will give an *equivalent computational formula* for evaluating $p(x)$ without ever having to evaluate its original monomial form. Similarly, if we're given a polynomial with j real roots, we can "unfactor" it to return an equivalent monomial representation.

It should stand to reason, then, that if we're *given* the factored form of a polynomial and we're *only interested* in evaluating it, then we shouldn't need to redundantly "unfactor" it back into monomial form just to verify it is a polynomial — the factored form should stand as a *certificate of membership* on its own. And if the factored form can certify membership, then so too can *any* representation that can be converted back to monomials when verification is required. The question, then, is not "which representation is a polynomial?" but "which representation is *useful*?"

Changing the Language

Let us pursue such a change of representation and see what emerges. In the previous ribbon, we identified that for the polynomials

$$p_2(x) = \beta_0 + \beta_1 x, \qquad p_3(x) = \beta_0 + \beta_1 x + \beta_2 x^2,$$

their respective coefficients β — which each exist and are unique, given $n = 2$ and $n = 3$ noiseless samples on distinct values of the predictors x_i collected in data $\mathcal{D}_2$ and $\mathcal{D}_3$ — are

$$\begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix} = \begin{pmatrix} [y_1] - x_1[y_1, y_2] \\ [y_1, y_2] \end{pmatrix}, \qquad \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{pmatrix} = \begin{pmatrix} [y_1] - x_1[y_1, y_2] + x_1 x_2 [y_1, y_2, y_3] \\ [y_1, y_2] - (x_1 + x_2)[y_1, y_2, y_3] \\ [y_1, y_2, y_3] \end{pmatrix},$$

where

$$[y_1] = y_1, \qquad [y_1, y_2] = \frac{y_2 - y_1}{x_2 - x_1}, \qquad \text{and} \qquad [y_1, y_2, y_3] = \frac{\dfrac{y_3 - y_2}{x_3 - x_2} - \dfrac{y_2 - y_1}{x_2 - x_1}}{x_3 - x_1} = \frac{[y_2, y_3] - [y_1, y_2]}{x_3 - x_1}$$

are the zeroth, first, and second divided differences, respectively.

Now, notice that, in our recapitulation, we subconsciously made a move that gestures toward the broader point of the matter: when we listed the divided differences $[y_1]$, $[y_1, y_2]$, and $[y_1, y_2, y_3]$ at the bottom, we listed them *once* — even though they appear in *both* solutions. The divided differences are not "the $n = 2$ divided differences" and "the $n = 3$ divided differences"; they are *the same quantities*, computed from the same data, serving as common ingredients in both formulas. What changes between the two columns is not the divided differences, but the *corrections around them* — the $-x_1[y_1, y_2]$ in $\beta_0^{(2)}$ (with the superscript denoting the model p_2) becomes $-x_1[y_1, y_2] + x_1 x_2 [y_1, y_2, y_3]$ in $\beta_0^{(3)}$, and so on. But again, it's not that the corrections change entirely erratically — they maintain all previous corrections, just "appending a new one" for the newest higher order term.

But let us observe just one thing. Working within the linear model

$$\begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix} = \begin{pmatrix} [y_1] - x_1[y_1, y_2] \\ [y_1, y_2] \end{pmatrix},$$

we have that

$$\beta_1 = [y_1, y_2] \implies \beta_0 = [y_1] - x_1[y_1, y_2] = y_1 - \beta_1 x_1.$$

In particular, we can plug the expression for β_0 back in to the general form of the (linear) polynomial model p_2:

$$p_2(x) = \beta_0 + \beta_1 x = (y_1 - \beta_1 x_1) + \beta_1 x = y_1 + [y_1, y_2](x - x_1).$$

That is,

$$p_2(x) = y_1 + \frac{y_2 - y_1}{x_2 - x_1}(x - x_1).$$

Supposing we were to write

$$m := [y_1, y_2] = \frac{y_2 - y_1}{x_2 - x_1},$$

this reads

$$p_2(x) = y_1 + m(x - x_1).$$

We've just derived the **point-slope form** of a line through (x_1, y_1) with slope $m = \frac{y_2 - y_1}{x_2 - x_1}$ (set so that it also falls on the point (x_2, y_2)). Observe that the monomial coefficients β_0 and β_1 have dissolved — what remains is the data point y_1, the divided difference/linear slope m, and the shifted input variable $(x - x_1)$.

One is often taught the point-slope form of a line in elementary algebra. On the other hand, one is almost never taught the analogous "point-slope-curvature form of a parabola"; yet, the same procedure that led us to point-slope form produces it as well, as we will see below. And indeed, we're led to similar analogues for higher order derivatives, each with a resemblant form to the well-known point-slope. But, rather than state these piecemeal, let us attempt to understand the procedure from another familiar light: namely, the Polynomial Factor Lemma.

Suppose we restate the goal of fitting the linear model above as follows. First, we seek a polynomial $p_2 \in \Pi_1$, the class of linear models, satisfying $p_2(x_1) = y_1$ and $p_2(x_2) = y_2$. Imagine then that we somewhat arbitrarily constrain the line $p_2 \in \Pi_1$ to satisfy $p_2(x_1) = y_1$. Define

$$r_1(x) = p_2(x) - y_1.$$

By construction, we have

$$r_1(x_1) = p_2(x_1) - y_1 = y_1 - y_1 = 0,$$

so that, by the factor lemma, there exists a (constant) polynomial $c_1 \in \Pi_0$ such that

$$r_1(x) = c_1(x - x_1), \qquad \text{i.e.,} \qquad p_2(x) = y_1 + c_1(x - x_1).$$

We're quick to notice the structural parallel with the proof we gave of the polynomial root bound. There, we were given a polynomial p with a *known* root at $x = a$, and the factor lemma told us $p(x) = (x - a)q(x)$ — in other words, we found that

$$p \text{ vanishes at } x = a \iff (x - a) \text{ factors } p.$$

Here, the situation is reversed, but the *logic* remains identical. Indeed, at the start, we *don't* have a polynomial with a known root — neither x_1 nor x_2 are *roots* of p_2, they are *constraints* (or, more aptly, points which are involved in the constraints that $p_2(x_1) = y_1$ and $p_2(x_2) = y_2$). Yet, these constraints are *equations* — which means they can be *rearranged to equal zero.* This is why we define $r_1 := p_2 - y_1$ to measure the residual between y_1 and p_2's output at a point $x \in \mathbb{R}$; it is explicitly so that

$$r_1 \text{ vanishes at } x = x_1 \iff (x - x_1) \text{ factors } r_1,$$

and we obtain

$$r_1(x) = c_1(x - x_1) \qquad (\text{since } (x - x_1) \text{ factors } r_1),$$

which, upon plugging in $r_1 = p_2 - y_1$ and isolating p_2, gives the at this point familiar equation

$$p_2(x) = y_1 + c_1(x - x_1).$$

On the other hand, we've perhaps created enough distance between ourselves and the matter that you've forgotten we know what c_1 is — it is the first divided difference

$$c_1 = m = [y_1, y_2] = \frac{y_2 - y_1}{x_2 - x_1}.$$

This is forced just by comparing the forms of the representations of p_2 — yet, we could have also derived this explicitly using the other constraint $p_2(x_2) = y_2$:

$$y_2 = p_2(x_2) = y_1 + c_1(x_2 - x_1) \iff c_1 = \frac{y_2 - y_1}{x_2 - x_1}.$$

Furthermore, we simply note that haven't heard from our monomial coordinates β_j in a while. This is precisely the point.

Now, we repeat the construction one level higher — i.e., we want a factored form for the quadratic model $p_3 \in \Pi_2$ given samples $i = 1, 2$ as above and the new $(x_3, y_3) = (3, 8)$. We *immediately* notice that, crucially, whatever $p_3 \in \Pi_2$ is, it *must agree* with p_2 at x_1 and x_2. For this reason, we may *start* by defining

$$s(x) := p_3(x) - p_2(x).$$

Since $s = p_3 - p_2 \in \Pi_2$ (cf. Polynomial closure under linear combinations), and

$$s(x_1) = p_3(x_1) - p_2(x_1) = y_1 - y_1 = 0, \qquad s(x_2) = p_3(x_2) - p_2(x_2) = y_2 - y_2 = 0,$$

the situation begins to look analogous to the proof of the root bound yet again. Indeed, s is a polynomial of degree at most 2 with two known roots x_1 and x_2, we apply the factor lemma twice, once to obtain

$$s(x) = (x - x_1)t(x), \qquad t \in \Pi_1,$$

and again (after perhaps noting that $t(x_2) = \frac{s(x_2)}{x_2 - x_1} = 0$ since $x_1 \neq x_2$), we obtain

$$s(x) = (x - x_1)(x - x_2)c_2, \qquad c_2 \in \Pi_0 \equiv \mathbb{R}.$$

Plugging back into $p_3 = p_2 + s$:

$$p_3(x) = p_2(x) + c_2(x - x_1)(x - x_2) = y_1 + [y_1, y_2](x - x_1) + c_2(x - x_1)(x - x_2).$$

This is the "point-slope-curvature form" we alluded to earlier. Indeed, y_1 is the point, $[y_1, y_2]$ is the discrete analogue to the slope, and we can recover the discrete analogue to the curvature, c_2, by utilizing the last constraint $p_3(x_3) = y_3$:

$$y_3 = p_3(x_3) = y_1 + [y_1, y_2](x_3 - x_1) + c_2(x_3 - x_1)(x_3 - x_2),$$

from which we obtain

$$c_2 = \frac{y_3 - y_1 - [y_1, y_2](x_3 - x_1)}{(x_3 - x_1)(x_3 - x_2)} = \frac{\dfrac{y_3 - y_1}{x_3 - x_1} - [y_1, y_2]}{x_3 - x_2} = \frac{[y_1, y_3] - [y_1, y_2]}{x_3 - x_2};$$

i.e., c_2 is the second divided difference $c_2 = [y_1, y_2, y_3]$, and p_3 becomes

$$p_3(x) = y_1 + [y_1, y_2](x - x_1) + [y_1, y_2, y_3](x - x_1)(x - x_2).$$

The pattern is now fully visible, and it generalizes without modification. To construct $p_4 \in \Pi_3$ through all four points of $\mathcal{D}_4$: define $u(x) := p_4(x) - p_3(x)$, observe that u vanishes at x_1, x_2, and x_3 (since p_3 and p_4 agree on the data they share), apply the factor lemma three times to obtain $u(x) = c_3(x-x_1)(x-x_2)(x-x_3)$, and impost $p_4(x_4) = y_4$ to determine $c_3 = [y_1, y_2, y_3, y_4]$. At each stage, the correction $p_n - p_{n-1}$ must vanish at all previous nodes x_i — the factor lemma leaves no alternative but the product, $\prod_{i=1}^{n-1}(x - x_i)$ — and the newest data $p_n(x_n) = y_n$ determines *only* the newest coefficient, $c_n = [y_1, \dots, y_{n-1}, y_n]$. Indeed, information propagates *forward* through the coefficients (divided differences) c_j; this seems useful, and at this point, we'd like to articulate precisely why it is.

Remark.

We note that the Newton construction here has completed the deferred proof of existence we gestured toward in the Polynomial Interpolation theorem. Indeed, uniqueness was established earlier via the root bound, and the Newton interpolant

$$p(x) = \sum_{j=0}^{n-1} [y_1, \dots, y_{j+1}] \prod_{i=1}^{j} (x - x_i) \in \Pi_{n-1}$$

is the explicit construction of a polynomial $p \in \Pi_{n-1}$ satisfying $p(x_i) = y_i$ for each $i = 1, \dots, n$.

Developing a New Basis

We have, over the course, of several ribbons, arrived at two distinct representations of the same interpolating polynomial — one in terms of monomials, and the other in terms of factored products — and we have observed, both concretely and in general, that the latter factored form is *stable under data augmentation*, while the monomial form is not. We now seek to formalize these notions.

Definition 3.2.4 (Basis for Π_k)

A collection of $k+1$ polynomials

$$\mathcal{B} = \{\psi_0, \psi_1, \dots, \psi_k\} \subset \Pi_k$$

is called a **basis** for Π_k if every polynomial $p \in \Pi_k$ can be written as a linear combination of the form

$$p(x) = c_0\psi_0(x) + c_1\psi_1(x) + \dots + c_k\psi_k(x) = \sum_{j=0}^{k} c_j \psi_j(x)$$

for a *unique* choice of coefficients $(c_0, c_1, \dots, c_k) \in \mathbb{R}^{k+1}$.

That a polynomial $p \in \Pi_k$'s representation under a given basis *exists* (for every $p \in \Pi_k$) and that it is *unique* (no two distinct coefficient vectors under a fixed basis product the same polynomial) are both essential: the first ensures the basis $\mathcal{B}$ is expressive enough to describe all of Π_k, and the second ensures that the coefficients under $\mathcal{B}$ are well-defined and unambiguous.

Indeed, there are two bases we've met so far for the class of polynomials.

Theorem 3.2.5 (Two Bases for Π_k)

Let

$$\Pi_k := \{\text{polynomials of degree at most } k\}.$$

Define the set

$$\mathcal{M}_k = \{1, x, x^2, \dots, x^k\},$$

and given fixed, distinct nodes $x_1, \dots, x_k \in \mathbb{R}$ with $x_1 \neq \cdots \neq x_k$, define

$$\mathcal{N}_k := \left\{ 1, (x - x_1), (x - x_1)(x - x_2), \dots, \prod_{i=1}^{k}(x - x_i) \right\}.$$

Then both $\mathcal{M}_k$ and $\mathcal{N}_k$ are bases of Π_k, the class of polynomials of degree at most k. In particular, we call $\mathcal{M}_k$ the **monomial basis** of Π_k, and $\mathcal{N}_k$ the **Newton basis** of Π_k.

Proof:

To show that $\mathcal{M}_k$ and $\mathcal{N}_k$ are bases of Π_k, we need to show that (i) every polynomial admits a representation under $\mathcal{M}_k/\mathcal{N}_k$ (existence), and (ii) this representation is unambiguous (uniqueness). That this follows for the monomial basis $\mathcal{M}_k$ is essentially definitional: every $p \in \Pi_k$ admits a monomial expansion $\sum c_j x^j$ just by our definition of Π_k (showing existence (i)), and that this representation is unique follows by the root bound: if $p(x) = \sum a_j x^j$ and $p(x) = \sum b_j x^j$ for all $x \in \mathbb{R}$, then subtracting both sides gives

$$\sum_{j=0}^{k}(a_j - b_j)x^j = 0 \quad \forall x \in \mathbb{R} \iff a_j = b_j, \quad j = 0, 1, \dots, k.$$

This means that two representations of p under $\mathcal{M}_k$ must have the same coordinates $a_j = b_j$ for every j; i.e., the coordinates are equal, and the expansion is unique.

That $\mathcal{N}_k$ is a basis follows from the iterative construction developed in the previous ribbon. We encourage the reader to reproduce the proof of existence. That a polynomial $p \in \Pi_k$'s representation under the Newton basis is unique, then, follows by observing that if

$$\sum_{j=0}^{k} c_j \psi_j(x_i) = \sum_{j=0}^{k} d_j \psi_j(x_i), \qquad i = 1, \dots, k+1,$$

then, going term by term:

(i) At x_1: every Newton basis function besides ψ_0 vanishes since $\psi_j(x_1) = 0$ for $j \geq 1$. The equation for $i = 1$ reduces to $c_0 = d_0$.

(ii) At x_2: $\psi_j(x_2) = 0$ for $j \geq 2$, so the equation reduces to $c_0 + c_1(x_2 - x_1) = d_0 + d_1(x_2 - x_1)$. We already know that $c_0 = d_0$, so this simplifies to $c_1(x_2 - x_1) = d_1(x_2 - x_1)$. Since $x_2 \neq x_1$, we conclude $c_1 = d_1$.

(iii) At x_3: The same logic forces $c_2 = d_2$, using only that $c_0 = d_0$, $c_1 = d_1$, and $(x_3 - x_1)(x_3 - x_2) \neq 0$.

The pattern continues: at each node x_{j+1}, all Newton basis functions of index $> j$ vanish, all coefficients of index $< j$ have already been shown to agree, and the jth coefficient is isolated on both sides — and they're forced to agree, since $\prod_{i=1}^{j}(x_{j+1} - x_i) \neq 0$ whenever the x_i are distinct.

$\|$

Remark.

It is worth noting that the two uniqueness arguments above are genuinely different in character. The reason why is, essentially, that we stated "what the zero polynomial $0 \in \Pi_k$ is" in terms of the monomial basis: in particular,

$$p \equiv 0 \in \Pi_k \iff \text{inside } p(x) = \sum_{j=0}^{k} a_k x^k, \quad a_0 = a_1 = \cdots = a_k = 0.$$

That is, we've *seemingly defined* the zero polynomial in Π_k relative to the monomial basis, requiring its coordinates in this basis to vanish in order to *be* the zero polynomial. But this goes the *other way* as well: if we know a polynomial *is* the zero polynomial, then its monomial coefficients a_j must vanish for all j. This is why when we deduce $p(x) = \sum a_j x^j = \sum b_j x^j$ and rearrange to $\sum (a_j - b_j) x^j = 0$, we can conclude immediately that each term's coefficients $a_j = b_j$.

On the other hand, in the Newton basis, if we started in a similar way by saying $p(x) = \sum c_j \psi_j(x) = \sum d_j \psi_j(x)$ and subtract $\sum (c_j - d_j)\psi_j(x) = 0$, we can still conclude that their difference is identically zero, but we can *not* use this to immediately conclude $c_j = d_j$ for all j since the ψ_j are not monomials, and we only know the correspondence between "zero polynomial $\iff$ vanishing coordinates" under the monomial basis. Knowing that $\sum (c_j - d_j)\psi_j(x) = 0$ tells us the expression on the left, *when expanded in monomials*, has all zero coefficients. And we could have proceeded by "re-expanding" the expression on the left in terms of monomials (by expanding the ϕ_j and collecting terms), showing it forces the same equivalence; but the node-by-node argument facilitates the conclusion directly, without ever converting to monomials.

What the node-by-node argument above reveals, though, is something we should recognize: it is precisely *forward substitution* through a lower-triangular system. Indeed, to establish uniqueness of representation under the Newton basis, we argued in a sequential manner: at x_1, only ψ_0 contributed, forcing $c_0 = d_0$. At x_2, only ψ_0 and ψ_1 contributed, and with $c_0 = d_0$ from the first round, we force $c_1 = d_1$. Each step relies on the previous ones, and no step could be disturbed by a later one.

This is the *triangular structure* we observed concretely in the parable, now appearing in the abstract. Though we've done so previously, we take the opportunity to once more reformalize what will be a critical notion moving forward.

Definition 3.2.6 (Design Matrix Relative to a Basis)

Given a basis $\mathcal{B} = \{\psi_0, \psi_1, \dots, \psi_k\}$ of Π_k, suppose the nodes $x_1, \dots, x_n$ are distinct. Then, the

design matrix of $\mathcal{B}$ at the nodes $x_1, \dots, x_n$ is

$$\Psi_{\mathcal{B}} := \begin{pmatrix} \psi_0(x_1) & \psi_1(x_1) & \cdots & \psi_k(x_1) \\ \psi_0(x_2) & \psi_1(x_2) & \cdots & \psi_k(x_2) \\ \vdots & \vdots & \ddots & \vdots \\ \psi_0(x_n) & \psi_1(x_n) & \cdots & \psi_k(x_n) \end{pmatrix} \in \mathbb{R}^{n \times (k+1)},$$

Remark.

Here, we index the first basis function as ψ_0; this indexing is nothing more than a notational choice, and we certainly feel free to re-index the first basis member by, say ψ_1, starting at the subscript $j = 1$ instead of $j = 0$, if it serves to clarify matters intuitively.

In the square case where $n = k + 1$, the interpolation problem $p(x_i) = y_i$, $i = 1, \dots, n$ becomes the linear system

$$\mathbf{y} = \Psi \mathbf{c},$$

where $\mathbf{c} = (c_0, \dots, c_k)$ are the coordinates of $p(x)$ in the basis $\mathcal{B}$. This is, for instance, how we proceeded in deriving the Newton coefficients $c_j = [y_1, \dots, y_j]$ — indeed, under the Newton basis $\mathcal{N}_k = \{1, (x - x_1), \dots, \prod_{i=1}^{k}(x - x_i)\}$, the design matrix is

$$\Psi_{\mathcal{N}_k} = \begin{pmatrix} 1 & 0 & 0 & \cdots & 0 \\ 1 & (x_2 - x_1) & 0 & \cdots & 0 \\ 1 & (x_3 - x_1) & (x_3 - x_1)(x_3 - x_2) & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & (x_n - x_1) & \cdots & \cdots & \prod_{i=1}^{k}(x_n - x_i) \end{pmatrix} \in \mathbb{R}^{n \times (k+1)}.$$

When we set

$$\mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}, \qquad \mathbf{c} = \begin{pmatrix} c_0 \\ c_1 \\ \vdots \\ c_k \end{pmatrix},$$

this leads to the system

$$\begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & \cdots & 0 \\ 1 & (x_2 - x_1) & 0 & \cdots & 0 \\ 1 & (x_3 - x_1) & (x_3 - x_1)(x_3 - x_2) & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & (x_n - x_1) & \cdots & \cdots & \prod_{i=1}^{k}(x_n - x_i) \end{pmatrix} \begin{pmatrix} c_0 \\ c_1 \\ \vdots \\ c_k \end{pmatrix},$$

where we can sequentially solve for the "point-slope-higher derivative" forms by

$$c_0 = y_1 = [y_1]; \qquad c_0 + c_1(x_2 - x_1) = y_2 \implies c_1 = \frac{y_2 - y_1}{x_2 - x_1} = [y_1, y_2];$$

and so on.

Now, by contrast, the monomial basis $\mathcal{M}_k = \{1, x, x^2, \dots, x^k\}$ yields the following design:

$$\Phi := \Psi_{\mathcal{M}_k} = \begin{pmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^k \\ 1 & x_2 & x_2^2 & \cdots & x_2^k \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^k \end{pmatrix} \in \mathbb{R}^{n\times(k+1)}.$$

We call Φ the **Vandermonde matrix**. The Vandermonde matrix is generally **dense** — i.e., the entries of a dense matrix are not equipped with much, if any, identifiable structure, in contrast with, say, a triangular matrix like the design yielded by the Newton basis.

At this point, we have seen a few matrix structures: the monomial basis $\mathcal{M}_k$ yielding the dense design $\Psi_{\mathcal{M}_k}$, and the Newton basis $\mathcal{N}_k$ yielding a lower-triangular design $\Psi_{\mathcal{N}_k}$. We ultimately prefer the latter triangular structure in $\Psi_{\mathcal{N}_k}$ as produced by the Newton basis precisely because it is *stable under data augmentation*, enabling us to solve the system via *forward-substitution* without destabilizing coefficients derived from previous waves of data. Recall, we arrived at the Newton basis through extensive polynomial reasoning: residuals, the factor lemma, divided differences, back-substitution formulas rearranged and simplified. This was productive, but it was also laborious — and it was specific to the Newton basis.

Now that we have the formal language, we can reverse the process that produced Newton: i.e., instead of deriving a basis through polynomial reasoning and *observing* the corresponding matrix structure, we can *postulate* a desirable matrix structure and reverse-engineer a basis producing it as its design. The factor lemma, which did all the heavy lifting for Newton, turns out to do the heavy lifting here as well.

Example 3.2.7 (Identity Structure):

First, suppose we ask: is there a basis $\mathcal{L}_n = \{\ell_1, \ell_2, \dots, \ell_n\}$ for Π_{n-1} whose design matrix at the nodes $x_1, \dots, x_n$ is the **identity matrix**

$$\Psi_{\mathcal{L}_n} = I_n := \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix} = \operatorname{diag}(1, 1, \dots, 1) \in \mathbb{R}^{n\times n}?$$

(Observe that we index the basis functions ℓ_j by the sample size n as opposed to prescribing their own index k.)

Given data $\mathcal{D}_n = \{(x_i, y_i)\}_{i=1}^n$, the coefficients $\mathbf{c} = (c_1, c_2, \dots, c_n)^\top$ characterizing the solution under the identity design, should such a set of coefficients exist, would be forced to satisfy $\mathbf{y} = \Psi_{\mathcal{L}_n}\mathbf{c}$, or

$$\begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{pmatrix},$$

which would hold if and only if

$$c_j = y_j, \qquad j = 1, 2, \dots, n.$$

So, if we were able to identify functions $\ell_j \in \mathcal{L}_n$ such that

$$\Psi_{\mathcal{L}_n} = \begin{pmatrix} \ell_1(x_1) & \ell_2(x_1) & \cdots & \ell_n(x_1) \\ \ell_1(x_2) & \ell_2(x_2) & \cdots & \ell_n(x_2) \\ \vdots & \vdots & \ddots & \vdots \\ \ell_1(x_n) & \ell_2(x_n) & \cdots & \ell_n(x_n) \end{pmatrix} \stackrel{?}{=} \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix} = I_n,$$

what we would have is that the unique polynomial p of degree at most $n-1$ interpolating the data $\mathcal{D}_n$ admits a representation under the basis $\mathcal{L}_n$ given by

$$p(x) = \sum_{j=1}^{n} c_j \ell_j(x) = y_1 \ell_1(x) + y_2 \ell_2(x) + \cdots + y_n \ell_n(x).$$

(Verify that p would interpolate $\mathcal{D}_n$ since $\ell_j(x_j) = 1$ for all j and $\ell_j(x_i) = 0$ for $i \neq j$ by construction.)

The only question, then, is whether we *can* identify such functions ℓ_j such that

$$\Psi_{\mathcal{L}_n} = I_n \iff \ell_j(x_i) = \delta_{ij} := \begin{cases} 1 & i = j, \\ 0 & i \neq j. \end{cases}$$

Indeed we can, and the factor lemma tells us how. Fix a node x_j. The function ℓ_j must vanish at every node x_i with $i \neq j$ — that is, ℓ_j must have $n-1$ prescribed roots (notice how this differs from the triangular Newton basis — here, *every* basis function ℓ_j inherits all $n-1$ roots from the data, not just the preceding $j-1$.)

Since $\ell_j \in \Pi_{n-1}$ is a polynomial of degree at most $n-1$, the root bound tells us this is the most roots it can possibly have without vanishing identically to the zero polynomial $0 \in \Pi_{n-1}$. The factor lemma, applied $n-1$ times (exactly as in the proof of the root bound), forces

$$\ell_j(x) = c \prod_{\substack{i=1 \\ i \neq j}}^{n} (x - x_i)$$

for some constant $c \in \mathbb{R}$. The product on the right already accounts for $n-1$ factors, placing ℓ_j at degree exactly $n-1$. What remains a free parameter is the constant c, which we can determine using the last nontrivial constraint that $\ell_j(x_j) = 1$:

$$\ell_j(x_j) = c \prod_{\substack{i=1 \\ i \neq j}}^{n} (x_j - x_i) = 1 \implies c = \frac{1}{\prod_{\substack{i=1 \\ i \neq j}}^{n} (x_j - x_i)},$$

which yields our final expression for the basis functions $\ell_j \in \mathcal{L}_n$:

$$\ell_j(x) = \prod_{\substack{i=1 \\ i \neq j}}^{n} \frac{x - x_i}{x_j - x_i}, \qquad j = 1, 2, \dots, n.$$

Indeed, we have just derived a new basis for Π_{n-1}:

Theorem 3.2.8 (Lagrange Basis)

Given distinct nodes $x_1, \dots, x_n \in \mathbb{R}$, the **Lagrange basis** for Π_{n-1} relative to these nodes is the collection

$$\mathcal{L}_n = \{\ell_1, \ell_2, \dots, \ell_n\},$$

where

$$\ell_j(x) := \prod_{\substack{i=1 \\ i \neq j}}^{n} \frac{x - x_i}{x_j - x_i}, \qquad j = 1, 2, \dots, n.$$

The design matrix of $\mathcal{L}_n$ at the nodes is the identity matrix, $\Psi_{\mathcal{L}_n} = I_n$.

(We defer the formal proof that $\mathcal{L}_n$ defines a formal basis of Π_{n-1} to an exercise. Indeed, it is entirely analogous to how we argued for the Vandermonde and Newton bases.)

The Lagrange basis is, in a sense, the maximally convenient representation for *solving* the interpolation problem: the design matrix $\Psi_{\mathcal{L}_n}$ is the identity I_n, so the coefficients c_j are just the data y_j, and there is nothing to compute. But the computational convenience comes at an administrative cost: the Lagrange basis is, in another sense, maximally *inconvenient* under *data augmentation*. The identity matrix has no structure that extends as new data arrives: whenever we receive a new x_{n+1}, every existing basis function ℓ_j — defined relative to a product involving *every node* in the data — must be redefined. Adding one data point changes every basis function — just like how, in the monomial basis, every coefficient *besides* the constant $\beta_0 = 2$ shifted around as we added new data. The constancy of the intercept $\beta_0 = 2$ under the monomial basis was an artifact of the sample point $(x_1, y_1) = (0, 2)$ present in every sample; the difference is that Lagrange is immune to even such "lucky cases of coefficient stability" emerging as artifacts of happenstance. (In some ways, this is still better than the "unpredictable instability" we encountered under the monomial basis; at least under the Lagrange basis, we *know* the coefficients will be *unstable everywhere, every time.*)

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We now consider yet another structure well worth our exploration.

Example 3.2.9 (Orthogonal Structure):

Suppose we ask: is there a basis $\mathcal{O}_n = \{\omega_1, \omega_2, \ldots, \omega_n\}$ for Π_{n-1} whose design matrix at the nodes $x_1, \ldots, x_n$ has *orthogonal* columns? That is, for which

$$\Psi_{\mathcal{O}_n}^\top \Psi_{\mathcal{O}_n} = \text{diagonal?}$$

Denoting the jth column of $\Psi_{\mathcal{O}_n}$ by $\boldsymbol{\omega}_j = (\omega_j(x_1), \ldots, \omega_j(x_n))^\top \in \mathbb{R}^n$, the condition is

$$\boldsymbol{\omega}_j^\top \boldsymbol{\omega}_m = \sum_{i=1}^{n} \omega_j(x_i)\, \omega_m(x_i) = 0, \qquad j \neq m :$$

distinct basis functions, when evaluated at the nodes and stacked into column vectors, must be orthogonal in $\mathbb{R}^n$. If this held, then each coefficient c_j could be extracted independently, via

$$c_j = \frac{\boldsymbol{\omega}_j^\top \mathbf{y}}{\boldsymbol{\omega}_j^\top \boldsymbol{\omega}_j} = \frac{\displaystyle\sum_{i=1}^{n} \omega_j(x_i)\, y_i}{\displaystyle\sum_{i=1}^{n} \omega_j(x_i)^2}.$$

No system to solve. No substitution of any kind, neither forward nor backward. Each coefficient is an inner product of the data with the jth basis function's column, normalized by the basis function's squared norm. And — unlike the Lagrange basis, where the coefficients were also immediately available — no coefficient depends on any other coefficient's value.

Can we construct such a basis? Indeed, the procedure is classical: suppose, for instance, we take the monomial basis $\mathcal{M} = \{1, x, x^2, \ldots\}$ and seek to **orthogonalize** them with respect to the discrete inner product

$$\langle f, g\rangle_n := \sum_{i=1}^{n} f(x_i)\, g(x_i).$$

(One should take care to verify that $\langle \cdot, \cdot \rangle_n$ constitutes a valid inner product on the space of real functions defined on the finitely many nodes $\{x_1, \dots, x_n\}$.)

Start with $\omega_1(x) = 1$. Then define $\omega_2(x) = x - \operatorname{proj}_{\omega_1}(x)$, where $\operatorname{proj}_{\omega_1}(x) = \frac{\langle x, \omega_1\rangle_n}{\langle \omega_1, \omega_1\rangle_n} \cdot \omega_1(x) = \frac{\sum x_i}{n}$ is the projection of x onto the constant function at the nodes — i.e., the sample mean $\overline{x}$. So $\omega_2(x) = x - \overline{x}$: the "centered" linear term. Continue: $\omega_3(x) = x^2 - \operatorname{proj}_{\omega_1}(x^2) - \operatorname{proj}_{\omega_2}(x^2)$, which subtracts from x^2 its components along the constant and centered-linear directions. At each step, the new basis function is the "residual" of the next monomial after removing everything that correlates with the previously constructed functions.

This is the **Gram-Schmidt process**, applied to the monomial sequence $\{1, x, x^2, \dots, x^{n-1}\}$ under the discrete inner product $\langle \cdot, \cdot \rangle_n$. The resulting basis $\mathcal{O}_n = \{\omega_1, \dots, \omega_n\}$ consists of polynomials that are orthogonal when evaluated at the nodes — and one can verify that the design matrix $\Psi_{\mathcal{O}_n}$ has orthogonal columns by construction.

‖

Before we conclude the ribbon, we briefly recall our throughline: indeed, the last example should be contrasted with something the reader who has worked through Chapter 2 will recognize immediately. The coefficient formula

$$c_j = \frac{\displaystyle\sum_{i=1}^{n} \omega_j(x_i)\, y_i}{\displaystyle\sum_{i=1}^{n} \omega_j(x_i)^2}$$

is, coefficient for coefficient, the discrete analogue of

$$a_n = \frac{\langle f, \cos(nx)\rangle}{\|\cos(nx)\|^2} = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos(nx)\, dx$$

— the Fourier coefficient formula. And indeed, at equispaced nodes $x_i = -\pi + 2\pi i/N$, the trigonometric functions $\mathcal{T} = \{1, \cos(x), \sin(x), \cos(2x), \dots\}$ produce a design matrix whose columns satisfy exactly the orthogonality condition we postulated:

$$\sum_{i=1}^{N} \cos(m x_i)\cos(n x_i) = 0, \qquad m \neq n.$$

The trigonometric basis achieves this without any Gram-Schmidt computation — the orthogonality is *intrinsic* to the functions, a consequence of the periodicity and symmetry of sines and cosines at equispaced points. This is the discrete manifestation of the continuous orthogonality relations

$$\int_{-\pi}^{\pi} \cos(mx)\cos(nx)\, dx = 0, \qquad m \neq n,$$

that were the foundation of the entire Fourier development in the previous chapter. The data we're considering here in $\mathcal{D}_k$ are *not* equispaced, though; can you figure out why this is an issue for the Fourier basis?

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Assessing Basis Behavior

To close the current development, we revisit the example from the first half of the parable and demonstrate the behavior of solutions under several bases of Π_{n-1}. Indeed, recall that in the running example, we received three waves of data

$$\mathcal{D}_2 = \{(0,2),(1,3)\}, \qquad \mathcal{D}_3 = \{(0,2),(1,3),(3,8)\}, \qquad \mathcal{D}_4 = \{(0,2),(1,3),(3,8),(4,6)\},$$

and computed — in the monomial basis $\mathcal{M}_k = \{1, x, x^2, \dots, x^{k-1}\}$ for $k = 2,3,4$ — the interpolating polynomials

$$p_2^{(\mathcal{M})}(x) = 2 + x, \qquad p_3^{(\mathcal{M})}(x) = 2 + \frac{1}{2}x + \frac{1}{2}x^2, \qquad p_4^{(\mathcal{M})}(x) = 2 - x + \frac{5}{2}x^2 - \frac{1}{2}x^3.$$

These polynomials are the objects of interest, and we seek to verify that every basis we consider produces exactly these same three polynomials — thereby demonstrating the *uniqueness* component of the Polynomial Interpolation Theorem.

Example 3.2.10 (Newton Basis):

To start, we take as our basis of Π_{n-1} the Newton basis

$$\mathcal{N}_{n-1} = \left\{ 1, (x-x_1), (x-x_1)(x-x_2), \dots, \prod_{i=1}^{n-1}(x-x_i) \right\}.$$

Under $\mathcal{N}_{n-1}$, the interpolating polynomial is

$$p_{n-1}^{(\mathcal{N})}(x) = c_0 \cdot 1 + c_1(x-x_1) + c_2(x-x_1)(x-x_2) + \dots + c_{n-1}\prod_{i=1}^{n-1}(x-x_i).$$

We know that the coefficients c_j are just the divided differences

$$c_j = [y_1, \dots, y_{j+1}], \qquad j = 0, 1, \dots, n-1,$$

as derived in the previous ribbon. Hence, for the interpolating polynomial of $\mathcal{D}_2$ with $(x_1, y_1) = (0,2)$ and $(x_2, y_2) = (1,3)$, we obtain

$$c_0 = [y_1] = y_1 = 2, \qquad c_1 = [y_1, y_2] = \frac{y_2 - y_1}{x_2 - x_1} = \frac{3-2}{1-0} = 1.$$

We thus assemble

$$p_2^{(\mathcal{N})}(x) = c_0 \cdot 1 + c_1(x - x_1) = 2 \cdot 1 + 1 \cdot (x - 0) = 2 + x.$$

This is, of course, the *same representation* as what we recovered under the monomial basis $\mathcal{M}$. The difference is that, now, suppose we go to fit the interpolating polynomial through $\mathcal{D}_3$ with the same data for $j = 1, 2$ and the new sample $(x_3, y_3) = (3, 8)$. Then, the interpolating polynomial $p_3^{(\mathcal{N})}$ under the Newton basis *retains these coefficients* from $p_2^{(\mathcal{N})}$:

$$c_0 = [y_1] = 2, \qquad c_1 = [y_1, y_2] = 1,$$

and we only have to compute the new coefficient

$$c_2 = [y_1, y_2, y_3] = \frac{[y_2, y_3] - [y_1, y_2]}{x_3 - x_1} = \frac{\frac{5}{2} - 1}{3 - 0} = \frac{1}{2}$$

after computing $[y_2, y_3] = \frac{8-3}{3-1} = \frac{5}{2}$.

Thus, at $n = 3$, the polynomial p_3 represented under the Newton basis is

$$p_3^{(\mathcal{N})}(x) = c_0 + c_1(x - x_1) + c_2(x - x_1)(x - x_2) = 2 + x + \frac{1}{2}x(x-1).$$

We can verify upon expanding the factor of $\frac{1}{2}x(x-1)$ that this agrees with the monomial basis $p_3^{(\mathcal{M})}(x) = 2 + \frac{1}{2}x + \frac{1}{2}x^2$.

We can verify that for the model $p_4^{(\mathcal{N})}$, we obtain c_0, c_1, and c_2 as above, along with

$$c_3 = [y_1, y_2, y_3, y_4] = -\frac{1}{2},$$

yielding

$$p_4^{(\mathcal{N})} = 2 + x + \frac{1}{2}x(x-1) - \frac{1}{2}x(x-1)(x-3).$$

In particular, we reveal that once again, Newton leaves the earlier coefficients untouched; data augmentation under $\mathcal{N}$ injects the newest contribution only into the newest coefficient. Gathering the four coefficients into a table, we observe

$$p^{(\mathcal{N})}(x) = \sum_{j=0}^{k} c_j\, \psi_j(x)$$

c	c_0	c_1	c_2	c_3
$\mathcal{D}_2$	2	1	—	—
$\mathcal{D}_3$	2	1	$\frac{1}{2}$	—
$\mathcal{D}_4$	2	1	$\frac{1}{2}$	$-\frac{1}{2}$

ψ	ψ_0	ψ_1	ψ_2	ψ_3
$\mathcal{D}_2$	1	x	—	—
$\mathcal{D}_3$	1	x	$x(x-1)$	—
$\mathcal{D}_4$	1	x	$x(x-1)$	$x(x-1)(x-3)$

We're quick to contrast this with the monomial coefficient table we computed earlier:

$$p^{(\mathcal{M})}(x) = \sum_{j=0}^{k} \beta_j\, \phi_j(x)$$

β	β_0	β_1	β_2	β_3
$\mathcal{D}_2$	2	1	—	—
$\mathcal{D}_3$	2	$\frac{1}{2}$	$\frac{1}{2}$	—
$\mathcal{D}_4$	2	-1	$\frac{5}{2}$	$-\frac{1}{2}$

ϕ	ϕ_0	ϕ_1	ϕ_2	ϕ_3
$\mathcal{D}_2$	1	x	—	—
$\mathcal{D}_3$	1	x	x^2	—
$\mathcal{D}_4$	1	x	x^2	x^3

Indeed, the Newton basis reveals a sense of *stability* among its estimated coefficients c_j absent from the monomial basis's estimated β_j, all while retaining the stability of the basis functions ψ_j themselves.

Example 3.2.11 (Lagrange Basis):

In the Lagrange basis, the situation is quite a bit different. Recall that the Lagrange basis $\mathcal{L}_n = \{\ell_1, \dots, \ell_n\}$ at nodes $x_1, \dots, x_n$ consists of the basis functions

$$\ell_j(x) = \prod_{\substack{i=1 \\ i \neq j}}^{n} \frac{x - x_i}{x_j - x_i}, \qquad j = 1, \dots, n.$$

The coefficients c_j in the expansion

$$p^{(\mathcal{L})}(x) = c_1 \ell_1(x) + \cdots + c_n \ell_n(x)$$

are simply the data values,

$$c_j = y_j, \qquad j = 1, \dots, n.$$

For the data $\mathcal{D}_2 = \{(0, 2), (1, 3)\}$, the Lagrange basis at the nodes $x_1 = 0$ and $x_2 = 1$ are

$$\ell_1^{(2)}(x) = \prod_{i \neq 1} \frac{x - x_i}{x_j - x_i} = \frac{x - x_2}{x_1 - x_2} = \frac{x - 1}{0 - 1} = 1 - x,$$

and

$$\ell_2^{(2)}(x) = \prod_{i \neq 2} \frac{x - x_i}{x_j - x_i} = \frac{x - x_1}{x_2 - x_1} = \frac{x - 0}{1 - 0} = x.$$

With $c_1 = y_1 = 2$ and $c_2 = y_2 = 3$, we obtain the Lagrange representation of p_2,

$$p_2^{(\mathcal{L})}(x) = y_1 \ell_1^{(2)}(x) + y_2 \ell_2^{(2)}(x) = 2(1 - x) + 3(x) = 2 + x,$$

again recovering our familiar form.

Yet, the situation changes completely when we move to augment the data. Indeed, considering $\mathcal{D}_3$ consisting of the *same initial samples* for $j = 1, 2$ along with a new sample $(x_3, y_3) = (3, 8)$, the Lagrange basis at nodes $(x_1, x_2, x_3) = (0, 1, 3)$ becomes

$$\begin{aligned}
\ell_1^{(3)}(x) &= \frac{(x-1)(x-3)}{(0-1)(0-3)} = \frac{(x-1)(x-3)}{3}, \\
\ell_2^{(3)}(x) &= \frac{(x-0)(x-3)}{(1-0)(1-3)} = -\frac{x(x-3)}{2}, \\
\ell_3^{(3)}(x) &= \frac{(x-0)(x-1)}{(3-0)(3-1)} = \frac{x(x-1)}{6}.
\end{aligned}$$

With $(c_1, c_2, c_3) = (y_1, y_2, y_3) = (2, 3, 8)$, the Lagrange representation of p_3 is

$$p_3^{(\mathcal{L})}(x) = 2 \cdot \frac{(x-1)(x-3)}{3} - 3 \cdot \frac{x(x-3)}{2} + 8 \cdot \frac{x(x-1)}{6}.$$

One can verify that the coefficients and basis functions associated with the polynomial p under the Lagrange basis develop as indicated in the table below:

$$p^{(\mathcal{L})}(x) = \sum_{j=1}^{n} c_j\,\ell_j(x)$$

c	c_1	c_2	c_3	c_4
$\mathcal{D}_2$	2	3	—	—
$\mathcal{D}_3$	2	3	8	—
$\mathcal{D}_4$	2	3	8	6

ℓ	ℓ_1	ℓ_2	ℓ_3	ℓ_4
$\mathcal{D}_2$	$1-x$	x	—	—
$\mathcal{D}_3$	$\dfrac{(x-1)(x-3)}{3}$	$-\dfrac{x(x-3)}{2}$	$\dfrac{x(x-1)}{6}$	—
$\mathcal{D}_4$	$-\dfrac{(x-1)(x-3)(x-4)}{12}$	$\dfrac{x(x-3)(x-4)}{6}$	$-\dfrac{x(x-1)(x-4)}{6}$	$\dfrac{x(x-1)(x-3)}{12}$

What we observe is that, even though the coefficients c_j are *extremely stable* — indeed, they require no computation at all — the basis functions ℓ_j are *extremely unstable*, changing dramatically with every new sample. Compare with the Newton basis!

Example 3.2.12 (Orthogonal):

Finally, consider the orthogonal basis at the nodes, constructed by applying Gram-Schmidt to the monomial basis $\{1, x, x^2, \ldots\}$ with respect to the discrete inner product

$$\langle f, g\rangle_n := \sum_{i=1}^{n} f(x_i)g(x_i).$$

Since the inner product itself depends on the node set, we anticipate that *both* the basis functions *and* the coefficients shift under data augmentation. Let us verify this explicitly.

To work in the orthogonal Gram-Schmidt basis, we start by defining the constant basis function

$$\omega_1(x) := 1,$$

and then proceed by iteratively constructing the basis functions

$$\omega_2(x) := x - \mathrm{proj}_{\omega_1}(x), \qquad \omega_3(x) := x^2 - \mathrm{proj}_{\omega_1}(x^2) - \mathrm{proj}_{\omega_2}(x^2),$$

and, in general,

$$\omega_1(x) = 1; \qquad \omega_j(x) = x^{j-1} - \sum_{i=1}^{j-1} \mathrm{proj}_{\omega_i}(x^{j-1}), \qquad j = 2, \ldots, n.$$

where we define the **projection** of x^j onto ω_i by

$$\mathrm{proj}_{\omega_i}(x^j) = \frac{\langle x^j, \omega_i\rangle_n}{\langle \omega_i, \omega_i\rangle_n} \cdot \omega_i(x).$$

Now, for the given data $\mathcal{D}_2$ with nodes $(x_1, x_2) = (0, 1)$, we start by setting $\omega_1^{(2)}(x) = 1$, and then compute

$$\omega_2^{(2)}(x) = x - \mathrm{proj}_{\omega_1}(x) = x - \frac{\langle x, \omega_1\rangle_n}{\langle \omega_1, \omega_1\rangle_n} \cdot \omega_1(x)$$

Compute

$$\langle x, \omega_1 \rangle_n = \langle x, 1 \rangle_n = \sum_{i=1}^{n} x_i \cdot 1 = 0 + 1 = 1,$$

and (for $\mathcal{D}_2$ with $n = 2$)

$$\langle \omega_1, \omega_1 \rangle_n = \langle 1, 1 \rangle_n = \sum_{i=1}^{n} 1 = n = 2.$$

Thus $\operatorname{proj}_{\omega_1}(x) = \frac{1}{2}$. Together with $\omega_1(x) = 1$ for every x, this gives

$$\omega_2^{(2)}(x) = x - \frac{1}{2}.$$

So, under the orthogonal Gram-Schmidt basis, we have

$$p_2^{(\mathcal{O})}(x) = c_1 \omega_1^{(2)}(x) + c_2 \omega_2^{(2)}(x),$$

where the coefficients c_j are computed via the inner-product formulas

$$c_j = \frac{\langle \mathbf{y}, \boldsymbol{\omega}_j \rangle_n}{\langle \boldsymbol{\omega}_j, \boldsymbol{\omega}_j \rangle_n} = \frac{\sum_{i=1}^{n} y_i \omega_j(x_i)}{\sum_{i=1}^{n} \omega_j(x_i)^2}.$$

We compute

$$c_1 = \frac{\langle \mathbf{y}, \boldsymbol{\omega}_1^{(2)} \rangle}{\langle \boldsymbol{\omega}_1^{(2)}, \boldsymbol{\omega}_1^{(2)} \rangle} = \frac{y_1 \cdot 1 + y_2 \cdot 1}{1^2 + 1^2} = \frac{2+3}{2} = \frac{5}{2},$$

and

$$c_2 = \frac{\langle \mathbf{y}, \boldsymbol{\omega}_2^{(2)} \rangle}{\langle \boldsymbol{\omega}_2^{(2)}, \boldsymbol{\omega}_2^{(2)} \rangle} = \frac{y_1 \cdot (0 - \frac{1}{2}) + y_2 \cdot (1 - \frac{1}{2})}{(-\frac{1}{2})^2 + (\frac{1}{2})^2} = \frac{-1 + \frac{3}{2}}{\frac{1}{2}} = 1.$$

Assembling,

$$p_2^{(\mathcal{O})}(x) = \frac{5}{2} \cdot 1 + 1 \cdot \left(x - \frac{1}{2} \right) = \frac{5}{2} + x - \frac{1}{2} = 2 + x,$$

which, at this point, is our familiar friend.

We won't illustrate this process for the other datasets we've considered, but one should verify that the tables of coefficients and basis functions under the orthogonal Gram-Schmidt basis $\mathcal{O}_n$ develop as follows as we augment the data $\mathcal{D}_k$:

$$p^{(\mathcal{O})}(x) = \sum_{j=1}^{n} c_j \, \omega_j(x)$$

c	c_1	c_2	c_3	c_4
$\mathcal{D}_2$	$\frac{5}{2}$	1	—	—
$\mathcal{D}_3$	$\frac{13}{3}$	$\frac{29}{14}$	$\frac{1}{2}$	—
$\mathcal{D}_4$	$\frac{19}{4}$	$\frac{13}{10}$	$-\frac{1}{2}$	$-\frac{1}{2}$

ω	ω_1	ω_2	ω_3	ω_4
$\mathcal{D}_2$	1	$x - \frac{1}{2}$	—	—
$\mathcal{D}_3$	1	$x - \frac{4}{3}$	$x^2 - \frac{22}{7}x + \frac{6}{7}$	—
$\mathcal{D}_4$	1	$x - 2$	$x^2 - 4x + \frac{3}{2}$	$x^3 - 6x^2 + \frac{43}{5}x - \frac{6}{5}$

Notice that *nothing here* is stable — neither the basis functions nor the coefficients. One might reasonably wonder whether the orthogonal basis offers any advantage at all. It does, and it is a subtle one: each coefficient c_j is computed by a *single inner product* of the data with $\boldsymbol{\omega}_j$, independently of every other coefficient — there is no system to solve, forward or backward, and no coefficient depends on the value of any other. This is a virtue the other bases do not share: the monomial requires a dense solve, Newton requires forward substitution, and Lagrange (though it demands no solve) depends on every node x_i to assemble any single basis function. Whether this independence is worth the cost of rebuilding both the basis and the coefficients at every augmentation depends, as we shall see, on what one intends to do with them.

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This concludes our development of the polynomial interpolation problem. We summarize the taxonomy of bases and their associated design matrix structures in the following table:

Basis of Π_{n-1}	**Design Structure**	**Solution Method**	**Behavior Under Data Augment**
Monomial $\mathcal{M}_{n-1}$	Dense (Vandermonde)	Back-substitution	Unstable; dense
Newton $\mathcal{N}_{n-1}$	Lower triangular	Forward substitution	Stable; triangular
Lagrange $\mathcal{L}_{n-1}$	Identity	Immediate (read off values)	Unstable; redefines every ℓ_j
Orthogonal $\mathcal{O}_{n-1}$	Orthogonal: $\Psi^\top\Psi = \text{diag}$	Inner products	Unstable; redefines every ω_j

Every basis in this table represents the same polynomial — the unique $p \in \Pi_{n-1}$ satisfying $p(x_i) = y_i$. They differ only in the language of representation, and each language carries its own structural virtues and costs. The question "which representation is useful?" — which we raised at the close of the previous green ribbon — now has a nuanced answer: it depends on what you need. Stability under augmentation favors Newton. Immediate coefficients favor Lagrange. Independent coefficient extraction favors orthogonal bases. And the trigonometric basis, alone among those we have seen, provides orthogonality as an intrinsic property of the functions rather than an artifact of the data.

But we have, in all of this, been operating within the confines of a single assumption: that we have n data points, and we fit a polynomial of degree exactly $n-1$ passing through each one. The polynomial interpolation theorem guarantees this is always possible and always unique. What we have *not* addressed is the question that the parable's second act posed most urgently: given that the interpolating polynomial passes through the data perfectly, does it behave well *between* the data points? The coefficient instability we resolved was a problem of *representation*; the Cauchy analyst's catastrophe was a problem of *approximation*. These are not the same problem, and the tools we have developed so far — bases, design matrices, triangularity, orthogonality — do not, by themselves, resolve the second. For that, we will need to leave the finite-dimensional world of Π_{n-1} and ask what happens when the function we are trying to approximate is not itself a polynomial — when the DGP lives in a space that Π_{n-1}, no matter how large n, can only approximate.

3.3 Fitting Points or Functions?

Reopening the Parable

So, at this point what we understand is that if we're given a dataset consisting of finitely many sample points, say $\mathcal{D}_n = \{(x_1, y_1), \dots, (x_n, y_n)\}$ for real valued predictor nodes $x_i \in \mathbb{R}$ and outcome data $y_i \in \mathbb{R}$, we're always able to identify a polynomial of degree at most $n-1$, $p \in \Pi_{n-1}$, such that p interpolates the data $\mathcal{D}_n$ exactly — i.e.,

$$p(x_i) = y_i, \qquad i = 1, \dots, n.$$

The existence and uniqueness of this interpolating polynomial p is guaranteed by the apt Polynomial Interpolation. We may intuit the explicit construction of the unique interpolating polynomial p straightforwardly in terms of the Newton basis, $\mathcal{N}_{n-1} = \left\{1, (x - x_1), (x - x_1)(x - x_2), \dots, \prod_{i=1}^{n-1}(x - x_i)\right\}$, and the uniqueness guaranteed by the interpolation theorem assures us that our choice of representative basis $\mathcal{N}_{n-1}$ over, say, the monomial basis $\mathcal{M}_{n-1} = \{1, x, x^2, \dots, x^{n-1}\}$, is unimportant if our concern is merely in evaluations of $p(x)$. The reason we may prefer the Newton basis is that its coefficients and basis functions are \emph{stable under data augmentation. That is, unlike the monomial basis — whose expansion in terms of basis functions $p^{(\mathcal{M})}(x) = \beta_0 + \beta_1 x + \dots + \beta_{n-1}x^{n-1}$ sees the coefficients $\beta_0, \dots, \beta_{n-1}$ fluctuate with the addition of a new basis function x^n with coefficient β_n — the Newton basis expansion $p^{(\mathcal{N})}(x) = c_0 + c_1(x - x_1) + \dots + c_{n-1}\prod_{i=1}^{n-1}(x - x_i)$ has coefficients $c_0, \dots, c_{n-1}$ which *remain fixed* when we add a new data point x_n with basis function $\prod_{i=1}^{n}(x - x_i)$.

This is all well and good, and we're better off for having taken the time to understand it. But now, let us take a moment to recall the parable's setup. In the first half, we began by issuing some fairly arbitrary data $\mathcal{D}_2 = \{(0, 2), (1, 3)\}$ and simply explored what it would mean for "a quadratic $p \in \Pi_2$ to interpolate that data." After some reflection, we realized that the class of quadratic interpolants was simply too large to identify a unique interpolating quadratic — both $2 + x^2$ and $2 + x$ are in the class Π_2, and both interpolate $\mathcal{D}_2$ perfectly. The solution, then, was to simply *reduce the size* of the class of interpolants under consideration; namely, by considering polynomials $p \in \Pi_1$, the class of linear interpolants (i.e., lines $p(x) = mx + b$). The line $p(x) = 2+x$ is the unique line interpolating the data in $\mathcal{D}_2$, and at the time, we implicitly took uniqueness over a reduced class as a "good enough reason" to prefer the line over the quadratic for interpolating $\mathcal{D}_2$.

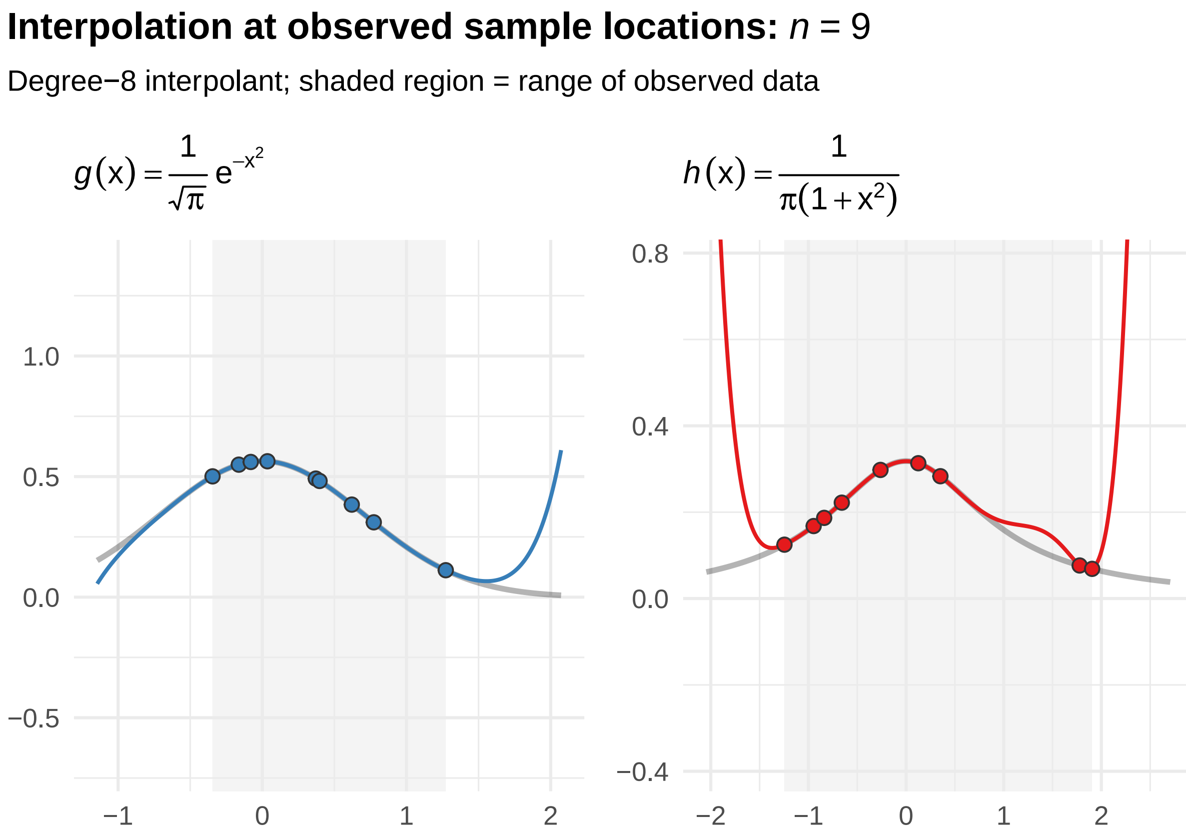
Yet, what happened when we appended a new — again, mostly contrived — third sample $(x_3, y_3) = (3, 8)$ to the existing data was a bit discomforting: we realized that *no* linear polynomial could interpolate the new data $\mathcal{D}_3 = \{(0, 2), (1, 3), (3, 8)\}$, and were we to still insist on interpolating the data exactly, we would need to consider enlarging the class of models from the linear polynomials Π_1 to the quadratic polynomials Π_2 after all. On the one hand, it is intuitively clear that this *had to happen* — we're not magicians, nor fortune-tellers, and it wouldn't make sense to chastise ourselves for being wrong about something we *didn't know*. And, again, the data we generated in the first half of the parable were, for all intents and purposes, picked because "they look nice" and "they didn't break anything in the example we had in mind" — which is to say, they were *not* generated by a "true DGP."

But this is what led us to the second half of the parable, and it is the half we've yet to truly reckon with as of this point. Recall that we defined two data generating processes g (the Gaussian) and h (the Cauchy) by their normalized functional forms

$$g(x) = \frac{1}{\sqrt{\pi}} e^{-x^2}, \qquad h(x) = \frac{1}{\pi} \frac{1}{1 + x^2},$$

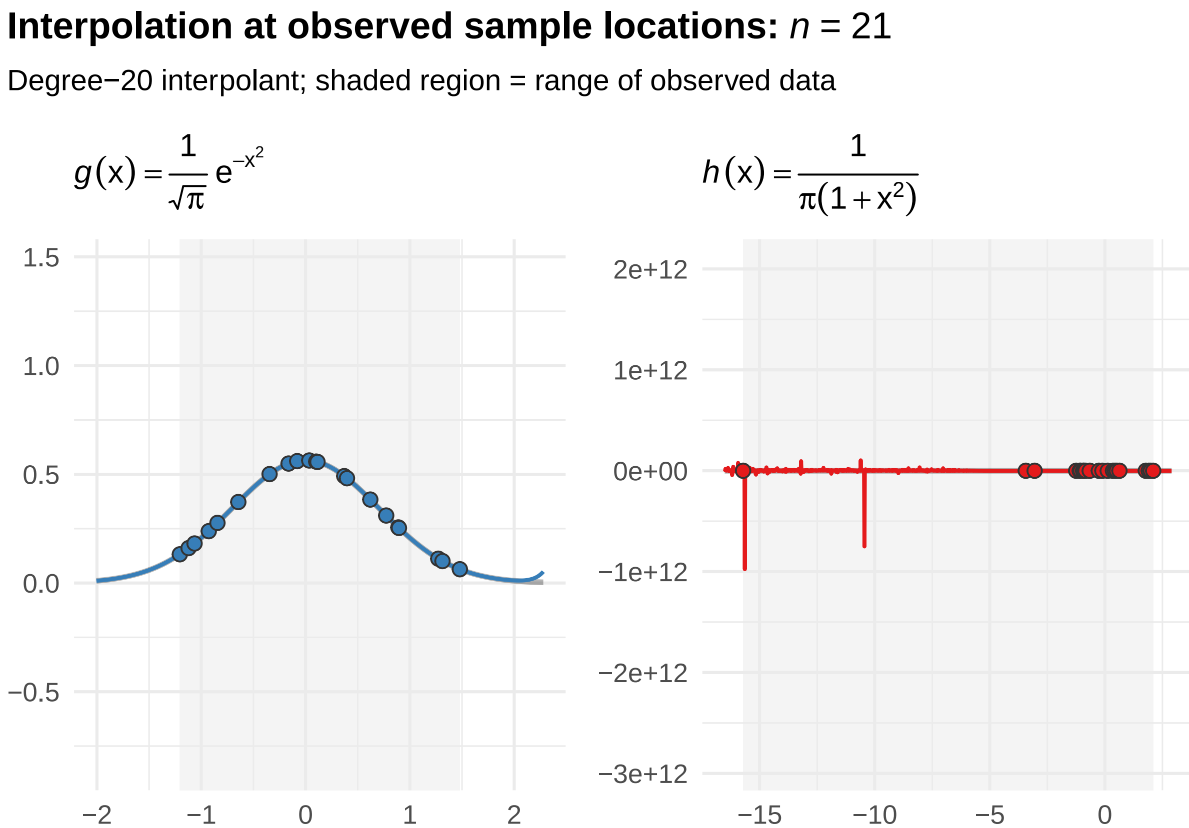
generated datasets $\mathcal{G}_n \sim g$ and $\mathcal{H}_n \sim h$, and assigned each to one of two competing analysts. We informed them that the data they'd receive was generated perfectly in accordance with their respective processes — essentially, so that no questions of "data quality" or "measurement error" could save the analysts from what was to follow.

At $n = 9$, the data $\mathcal{G}_n$ and $\mathcal{H}_n$, whose samples were randomly generated by the processes g and h respectively, looked "fairly similar" when placed on a scatterplot. Since the data was "generated perfectly," the analysts supposed that fitting an interpolating polynomial through the data — which we now know is guaranteed to exist and be unique *for each fixed dataset* by the Interpolation Theorem — couldn't be the worst idea, and they proceeded to do so. We recall that they observed the following:



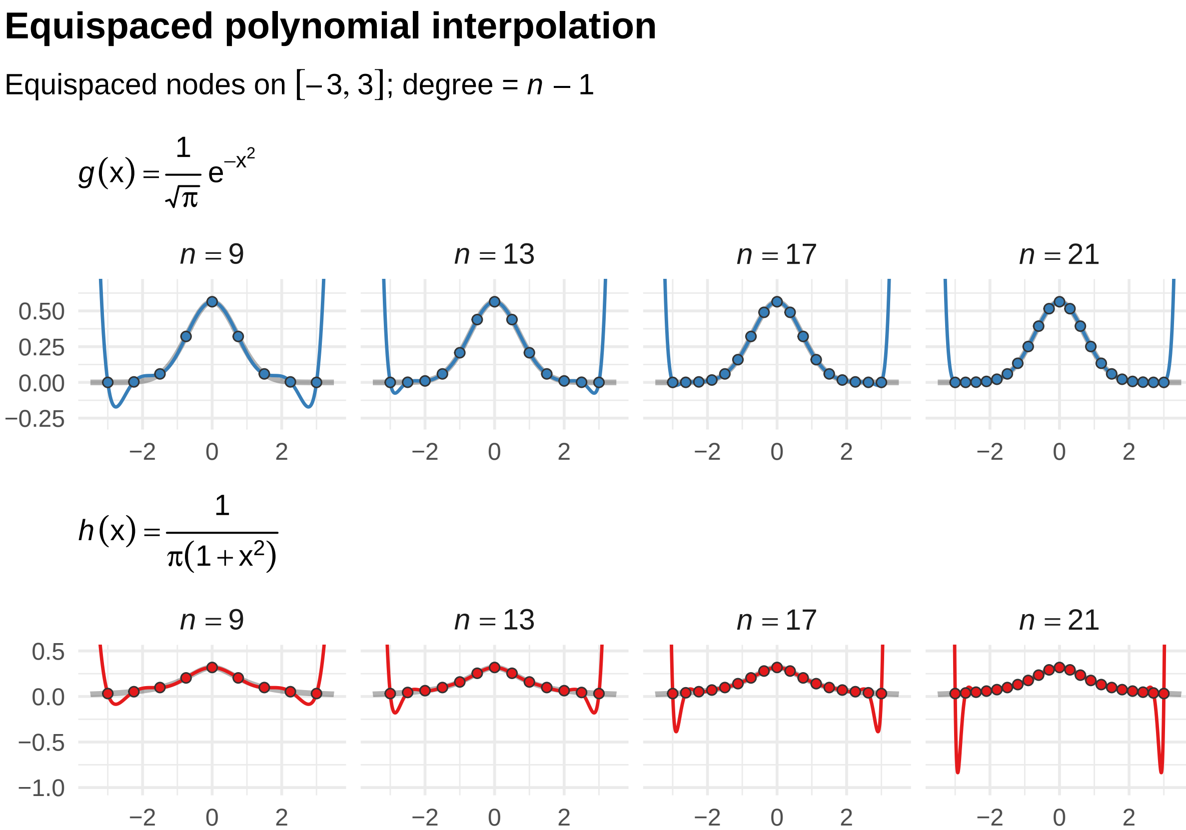
Indeed, these curves "look about the same." There isn't a reason to believe that the one on the left is "proper" and the on on the right "isn't" — and indeed, if the analyst of $\mathcal{H}_9$ was to estimate $h(1.5)$ by the interpolating polynomial $p_9^{(h)}(1.5)$ depicted in the graph above, her answer wouldn't raise eyebrows relative to "how wrong she could have been," say, by just "guessing blindly."

But then, the analysts were cursed by *more data*. In particular, at $n = 21$ (consisting of the first $n_1 = 9$ samples, appended by a new batch of $n_2 = 12$ samples), the analysts observed the following interpolating polynomials:



Again the interpolation theorem has *done its job*: it has found an interpolating polynomial capable of bouncing between the outlier generated by the Cauchy density at $x \approx -15$ and the bulk of the samples nearer $x \approx 0$ — but this interpolation is, for all intents and purposes, *completely unusable* for prediction on $h(x)$. We can no longer say that $h(-10)$ is "reasonably approximated" by $p_{21}^{(h)}(-10)$, nor anywhere else — it is a terrible, ugly thing, and we did not discredit the analyst of $\mathcal{H}_{21}$ into thinking she'd believe otherwise.

But, really: what is she to do? "Trim the outlier"? *The data is perfect!* To "trim the outlier" would, in a very precise way, be to discard some kernel of *what h really is*, which is a machine capable of generating large outliers at $x \approx -15$ at small samples of even size $n = 21$. Again, the analysts don't know the functional forms; all they have is the data. As the oracles, we're even able to compute that the probability of the Gaussian DGP g generating such a sample at $x \approx -15$ is on the order of 10^{-98} — negligible compared to the analogous probability of roughly $1/700$ under the Cauchy DGP h. And this all seems to reinforce the notion that it would be *entirely improper* for the analyst to "pretend that data point didn't exist." We can even take it one step further and realize that the problem persists even when we bestow the analysts with data generated on equispaced nodes, where we observe the "Cauchy's fangs" emerge for higher and higher n:



Theoretically, we traced our inability to approximate h back to its Taylor coefficients: those of g decayed like $1/k!$ (that is, quite rapidly), while those of h retained a constant contribution across all terms in its Taylor expansion — i.e., h's Taylor coefficients *don't* decay. New observations always have the potential to completely destabilize what was "beginning to look like" a decent approximation — and what is perhaps scariest is that the "looks like" here is *always* acting to deceive us.

One might wonder, then, whether the issue lay with the polynomial itself as a tool for *approximating the functions* g and h. We stress that this is an entirely different question from what the Polynomial Interpolation Theorem addresses: that result, in part, tells us that polynomials of degree $n-1$ or higher are always capable of *interpolating the points* $(x_1, y_1), \ldots, (x_n, y_n)$ — but it says nothing about whether the resulting polynomial bears any resemblance to a function from which those points may have been drawn. This distinction — between "interpolating points" and "approximating functions" — is what implicitly separated the arbitrarily generated data of the parable's first half from the data generated by the functional forms of the Gaussian and Cauchy DGPs in its second. And one might assume, based on what we've observed thus far — namely, that h isn't well-approximated by its Taylor series — that the answer is *no*: that some functions simply elude approximation by polynomials, with the Cauchy h serving as the face of such a class.

What is genuinely surprising to us, then, is that such an ostensibly reasonable conjecture turns out to be *dead wrong*.

3.3.1 Weierstrass's Game

We begin the section by stating its central result, due to German mathematician Karl Weierstrass (with whom we will become intimately familiar) in 1885, in its standard form.

Theorem 3.3.1 (Weierstrass Approximation Theorem)

Let $f\colon [a,b] \to \mathbb{R}$ be a continuous function on a closed and bounded interval $[a,b] \subset \mathbb{R}$. Then for any tolerance $\epsilon > 0$, there exists a *finite* degree $N = N(\epsilon) \in \mathbb{N}$ such that some polynomial $p \in \Pi_N$ uniformly approximates f on $[a,b]$; that is,

$$\sup_{x\in[a,b]} |f(x) - p(x)| := \|f - p\|_\infty < \epsilon.$$

Before we discuss with what we can *do* with the theorem, we want to linger on its ramifications for just a moment, particularly with regards to the parable. The statement of the Weierstrass Approximation Theorem is short, but what it asserts is quite remarkable. Indeed, the hypothesis on the target $f\colon [a,b] \to \mathbb{R}$ is essentially as weak as we could ask of any function: we require only that f be *continuous* (not differentiable, nor analytic, nor smooth in any sense beyond the barest sense), and we're guaranteed the existence of a class of polynomials Π_N containing a member p approximating f *uniformly well across all of* $[a,b]$ *simultaneously.* What the theorem *doesn't* tell us is, well, how to *find* the integer N, nor the class Π_N, and especially not the polynomial $p \in \Pi_N$ itself. Indeed, the Weierstrass theorem asserts *existence* of these objects, and nothing more.

Remark.

The reader should compare Weierstrass's theorem here with one due to Fejér (Theorem 2.3.10) as stated Chapter 2. What you may notice is that the theorems address similar questions — and indeed, they have similar answers. We ask you to simply stash this observation away for the time being — we'll address it come the intermission below.

We're *immediately* struck by the impossibility of the approximating polynomials being produced by the Taylor series of the target function f. We've already observed a function — the Cauchy density h — with a Taylor series converging only within a fixed radius $|x| < 1$. Relative to its Taylor polynomials, we can thus intuit our inability to find an $N = N(\epsilon)$ and a $p \in \Pi_N$ such that p and h are uniformly close across, say, all of $x \in [-3,3]$. Yet, what Weierstrass guarantees is that a *polynomial* — something of *precisely* the same form as the Taylor series, "$a_0+a_1x+a_2x^2+\cdots+a_Nx^N$ for some N and a_j" — produces such a uniformly approximating polynomial of *finite* degree for *every* continuous function — *including the Cauchy* and, as we'll find in our development, an even more exotic function, one which we find illuminates just how weak a condition mere continuity really is.

Of course, the theorem makes specific demands of the domain on which we hope to approximate f by a polynomial p. In particular, the requirement that $[a, b]$ be a closed and bounded interval is essential, and we cannot simply "wave it away." Indeed, suppose we were to instead ask whether there were a polynomial p such that $\sup_{x\in\mathbb{R}} |h(x) - p(x)| < \epsilon$ for our Cauchy density h on *all* of $\mathbb{R}$. We wouldn't have to look very hard to find such an attempt structurally hopeless: $h(x) \to 0$ as $|x| \to \infty$, while *any* nonconstant polynomial p has $|p(x)| \to \infty$ as $|x| \to \infty$, and the difference $|h(x)-p(x)|$ blows up to $+\infty$ along with p in either direction. The constant polynomial $p \equiv 0$ fares no better — it misses $h(0) = 1/\pi$ by a fixed amount and never recovers. Polynomials and decaying functions are simply incompatible "at infinity," and no choice of N nor $p \in \Pi_N$ can rescue the situation. Yet — and this is the entire force of the theorem — on *any* closed and bounded interval $[a, b]$, no matter how small & negative we take a nor large & positive we take b — the Weierstrass theorem applies in full, and a polynomial of finite degree approximating h to within ϵ *exists*.

Perhaps most importantly for the parable we've been developing, however, is somethign else that the Weierstrass theorem *doesn't* tell us: namely, that the approximating polynomial $p \in \Pi_N$ is anything like the interpolating polynomials our analysts have been constructing. The interpolating polynomial $p_{n-1} \in \Pi_{n-1}$ is the unique element satisfying $p_{n-1}(x_i) = f(x_i)$ at n specified nodes; the Weierstrass polynomial $p \in \Pi_N$, by contrast, is some element of *some* class Π_N satisfying $|f(x) - p(x)| < \epsilon$ at *every* location $x \in [a, b]$. These are answers to fundamentally different questions, and the polynomials answering them are, in general, fundamentally different objects. The Weierstrass polynomial owes the data values $f(x_i)$ *nothing*: it need not pass through any of them exactly, nor even particularly closely at any individual point — it need only stay within ϵ of f uniformly. The interpolating polynomial, conversely, is forced to agree with f precisely at the nodes, and it spends every degree of freedom it has doing so.

Applied to our analyst of $\mathscr{H}_n$ on, say, $[-3, 3]$, then, what the Weierstrass theorem tells us is that for any tolerance $\epsilon > 0$ we care to specify — be it 10^{-3}, 10^{-10}, or any other — there exists a polynomial p such that

$$\sup_{x\in[-3,3]} |h(x) - p(x)| < \epsilon.$$

The polynomial *exists*. It is sitting somewhere in the union $\bigcup_{N=0}^{\infty} \Pi_N$, waiting to be found. The catastrophic interpolating polynomials we constructed in the parable's second half are not it; the "fangs" the analyst observed at the boundary are not what a good polynomial approximation to h on $[-3, 3]$ has to look like at all. There is a polynomial that approximates h to graphical accuracy on the entire interval, and another that does so to within any tolerance we may demand — and the Weierstrass theorem assures us of as much, in spite of the analyst's best efforts to convince herself otherwise.

But the analyst, equipped now with the theorem and her catastrophic interpolant, would be entitled to ask the obvious question: *where is it?* The Weierstrass theorem guarantees the polynomial exists, but it offers her no recipe for constructing one, no bound on the degree she'd need, and no indication of whether her interpolating strategy could be salvaged or whether something altogether different is required. All she is certain is that *if* there is a method that can identify a representative polynomial p within the (nonempty!) set of good approximants to h guaranteed by Weierstrass, she would very much like to understand what it is, and whether it'd help in salvaging her analysis.

Weierstrass By Construction: A Naive Attempt

So, if it weren't already clear, we're *not* going to "prove the Weierstrass Approximation Theorem" — at least, not in the abstract. Indeed, what we seek to do is *explicitly construct* this uniformly approximating polynomial, which would, of course, suffice for proving the original theorem anyway. Of course, this is easier said than done — we speculate there is a reason Karl Weierstrass *wasn't* more explicit in his original statement, and it isn't for a lack of trying. Indeed, we will see that it is *simply hard* to identify the "right" polynomial — to bury the lede just a bit, all we will say now is that this polynomial cannot be constructed using any of the methods we've considered thus far *in Chapter 3*.

Nevertheless, let's try anyway, just to see what happens when we do. Concretely, let us suppose for the duration of this setup that we have an adversary, say Weierstrass himself, and he and we are to play a game. The rules are as follows: we commit, in advance, to a *strategy* for constructing polynomials on $[-1, 1]$. Once we've committed, Weierstrass will hand us a function $f \in C[0, 1]$ and a tolerance $\epsilon > 0$ of his choosing. We win if our strategy produces a polynomial satisfying

$$\|f - p\|_\infty := \sup_{x \in [0,1]} |f(x) - p(x)| < \epsilon;$$

otherwise, we lose.

Now, at this point, the only tools we have at our disposal are the ones this chapter has developed: polynomial bases, interpolation, and the design matrices that accompany them. Weierstrass, for his part, knows this — and, to make things interesting, he offers a concession: he will evaluate f for us at the $n + 1$ equispaced points $0, \frac{1}{n}, \frac{2}{n}, \dots, 1$, and hand us the resulting data

$$\mathcal{D}_n = \left\{ \left(\frac{k}{n},\ f\left(\frac{k}{n}\right) \right) \right\}_{k=0}^{n}$$

We may use this data however we like, and we are free to request n as large or as small as we wish. The catch, of course, is that Weierstrass chooses f *after* seeing our strategy — and he will choose adversarially.

Round 1

We thus begin our time with Weierstrass by examining the tools at our disposal and asking which, if any, have the "best chance" of defeating Weierstrass out the gate. Given the data $\mathcal{D}_n$ consisting of $n+1$ samples on equispaced nodes $x_0 = 0, x_1 = 1/n, x_2 = 2/n, \dots, x_n = 1$ with corresponding evaluations $f(0), f(1/n), f(2/n), \dots, f(1)$, we might start by picking our favorite basis from the previous section, adapting it to $\mathcal{D}_n$, and taking as our construction p the unique interpolating polynomial of degree n as guaranteed by the interpolation theorem.

But which basis should we choose? Does it matter? We've seen that the monomial, Newton, Lagrange, and orthogonal bases all produce the *same* interpolating polynomial (again guaranteed by the interpolation theorem). So any basis we choose will yield the same p — the basis is a matter of *representation*, not of substance. So let us simply choose whichever representation is most *revealing* for the task at hand; for reasons that will become clear momentarily, we choose the Lagrange basis

$$\mathcal{L}_n[f(x)] = \sum_{k=0}^{n} f\left(\frac{k}{n}\right) \ell_k(x), \qquad \ell_k(x) = \prod_{\substack{j=0 \\ j \neq k}}^{n} \frac{x - x_j}{x_k - x_j}.$$

We choose this representation not because it is computationally convenient (the Newton basis would be better for such a purpose) — rather, we choose it because it makes the *structure* of the interpolant maximally transparent. Look at the formula: the interpolating polynomial $\mathcal{L}_n[f]$ is a sum of the basis functions ℓ_k, each weighted directly by the evaluations $c_k = f(k/n)$ as given by Weierstrass himself.

In fact, the Lagrange representation reveals something else, something we really ought to have noticed earlier: the basis functions ℓ_k satisfy a **partition of unity**. Indeed, one should verify that, for *every* $x \in [0,1]$ (not just at the nodes $k/n \in \mathcal{D}_n$ where $\sum \ell_k(x_j) = 1$ is trivial!), we have

$$\sum_{k=0}^{n} \ell_k(x) = \sum_{k=0}^{n} \left(\prod_{\substack{j=0 \\ j\neq k}}^{n} \frac{x - x_j}{x_k - x_j} \right) = 1.$$

(Hint: check this by verifying that if $f \equiv 1$ is the constant function on $[0,1]$ then f is its own interpolant; in particular, use the fact that $\mathcal{L}_n[f(x)] = 1$ for every $x \in [0,1]$.)

So, the interpolant $\mathcal{L}_n[f(x)] = \sum f(k/n)\ell_k(x)$ is a sum of the data values $f(k/n)$ weighted by functions $\ell_k(x)$ that *sum to* 1 *for every* x. Consider, then, how this does and does not resemble some special brands of "linear combination." For concreteness, we restate definitions of some familiar, and perhaps one unfamiliar, kinds of combinations.

Definition 3.3.2 (Special Linear Combinations)

Suppose that $x_0, x_1, \dots, x_n \in \mathbb{R}$ are fixed real numbers.

(a) Recall that a **linear combination** of the given $x_0, x_1, \dots, x_n \in \mathbb{R}$ is an expression of the form

$$\sum_{k=0}^{n} w_k x_k : \quad w_k \in \mathbb{R}. \qquad \text{(Linear Combination)}$$

No restrictions are placed on the weights w_k other than they be real numbers.

(b) If in the definition of a linear combination, we impose that the weights **sum to unity**:

$$\sum_{k=0}^{n} w_k = 1,$$

then we obtain an **affine combination**:

$$\sum_{k=0}^{n} w_k x_k : \quad w_k \in \mathbb{R} \quad \text{s.t.} \quad \sum_{k=0}^{n} w_k = 1. \qquad \text{(Affine Combination)}$$

(c) If in the definition of an affine combination, we even further restrict each weight in the partition of unity to be **nonnegative**:

$$w_k \geq 0, \qquad \sum_{k=0}^{n} w_k = 1,$$

then we obtain a **convex combination**:

$$\sum_{k=0}^{n} w_k x_k : \quad w_k \in \mathbb{R} \quad \text{s.t.} \quad w_k \geq 0, \quad \sum_{k=0}^{n} w_k = 1. \qquad \text{(Convex Combination)}$$

We've already observed that the Lagrange basis produces a partition of unity in its weights:

$$\sum_{k=0}^{n} \ell_k(x) = 1, \qquad \forall x \in [0,1].$$

Thus, the Lagrange interpolant

$$\mathcal{L}_n[f(x)] = \sum_{k=0}^{n} f\left(\frac{k}{n}\right) \ell_k(x)$$

is an affine combination of the data values $f(0), f(1/n), \dots f(1)$; the "weights" $\ell_k(x)$ sum to 1, and the "coefficients" are the data values $f(k/n)$ themselves.

But is $\mathcal{L}_n[f]$ a *convex* combination? Indeed, this would require, in addition to the ℓ_k serving as a partition of unity for every $x \in [0,1]$, that each $\ell_k(x) \geq 0$ be nonnegative for every $x \in [0,1]$. Unfortunately, we can immediately verify that the answer is *no* — and, moreover, the reason it is "no" is not an accident of our data, but a structural feature of the Lagrange basis deployed on *any* data. The issue is that each ℓ_k is a polynomial of degree n with exactly n roots — one at each node x_j with $j \neq k$ — and a polynomial of degree n with n distinct real roots *must change sign* at each root. (Intuitively, each root is *simple* — the factor $(x - x_j)$ appears exactly once in the product defining ℓ_k — and a polynomial that arrives at a simple root from the positive side must emerge negative on the other. It cannot "touch the axis and bounce back" as it would at a repeated root like $(x - x_j)^2$; it *crosses through*, and the crossing flips its sign.) To verify this explicitly, one may check that for $n = 2$, the equispaced nodes on $[0,1]$ are $x_0 = 0, x_1 = 1/2, x_2 = 1$, and the Lagrange basis functions are

$$\ell_0(x) = 2(x - \tfrac{1}{2})(x-1), \qquad \ell_1(x) = -4x(x-1), \qquad \ell_2(x) = 2x(x - \tfrac{1}{2}).$$

At $x = 1/4$, for instance, we thus have

$$\ell_0(1/4) = \frac{3}{8}, \qquad \ell_1(1/4) = \frac{3}{4}, \qquad \ell_2(1/4) = -\frac{1}{8}.$$

So, in the expansion

$$\mathcal{L}_2[f(1/4)] = \frac{3}{8} f(0) + \frac{3}{4} f(1/2) - \frac{1}{8} f(1),$$

we can observe that the weights *do* form a partition of unity: $\sum \ell_k(1/4) = \frac{3}{8} + \frac{3}{4} - \frac{1}{8} = 1$. Yet, clearly $\ell_2(1/4) = -\frac{1}{8} < 0$, so that $\mathcal{L}_n[f(1/4)]$ is *not a convex* combination of the data $f(x_k)$; it is merely an *affine* combination.

At this point, we acknowledge the obvious: why does any of this matter in defeating Weierstrass? The answer is that affine and convex combinations carry *fundamentally different guarantees* about the structure of their outputs. Indeed, recall from Chapter 2 (Definition 2.4.5 and the surrounding discussion) that a convex combination of values $x_0, x_1, \dots, x_n \in \mathbb{R}$ is always "self-contained" — in particular, if the coefficients $c_k \geq 0$ are nonnegative and form a partition of unity $\sum_k c_k = 1$, we have

$$\min_k x_k \leq \sum_{k=0}^{n} c_k x_k \leq \max_k x_k.$$

This is decidedly *not* a property harbored by affine combinations. (To see this explicitly: in the expression for $\mathcal{L}_2[f(1/4)]$ above, imagine that $0 \leq f(0) \leq f(1/2) << f(1)$ — i.e., $f(1)$ is a large positive number weighted by the negative coefficient $-1/8$ in $\mathcal{L}_2[f(1/4)]$. Then $\mathcal{L}_2[f(1/4)]$ can certainly be negative, even if we also had $f(x) \geq 0$ nonnegative for all $x \in [0,1]$. Convince yourself that this simply cannot happen when the coefficients are strictly nonnegative!)

What this means for our game against Weierstrass, then, is immediate: the Lagrange interpolant, being merely an *affine* combination of the data values $f(k/n)$, is free to produce values $\mathcal{L}[f(x)]$ *outside the range* of f. If $f\colon [0,1] \to [0,1]$, for instance, there is nothing in the structure of $\mathcal{L}_n[f]$ preventing it from evaluating to -3, $+47$, or any other number it pleases — the negative weights in the Lagrange basis permit the interpolant to escape the **convex hull** of the data entirely. (This is precisely what the "fangs" in the equispaced Cauchy interpolant *are*: the function $h(x) = 1/\pi(1+x^2)$ takes values in $[0, 1/\pi]$, and yet the Lagrange interpolant at equispaced nodes produces values of much larger magnitude near the boundary.)

We wouldn't appear to have much confidence in our interpolating polynomial $\mathcal{L}_n[f]$, if for no other reason than we *already know* its defects: it simply doesn't work for approximating the Cauchy density on equispaced nodes. Nevertheless, it is what we have, and so we submit it: our Round 1 strategy is to take as our approximant

$$p(x) = \mathcal{L}_n[f(x)],$$

the Lagrange interpolant at equispaced nodes.

Weierstrass, who has been watching patiently, hands us $f = h$, the Cauchy density rescaled to $[0,1]$. We already know what happens; we concede the first round to Weierstrass.

Round 2

Before we can regroup, a voice from the gallery interrupts — immediately, we recognize our peer as none other than great Russian mathematician Pafnuty Lvovich Chebyshev. Chebyshev, a household name due to his own concentration inequality (cf. Chapter 2), happens to have spent the better part of the 1850s thinking about precisely the kind of approximation problem we've just lost. Indeed, he and Weierstrass have played this game before — and our new friend is happy to offer insights he's garnered from rounds past, should they aid in our attempts to take Weierstrass down once and for all.

Chebyshev begins by asking us to look more carefully at the Lagrange interpolant. "You know it fails for the Cauchy density," he says, "but do you know *why*?" We gesture vaguely at the fanga, the sign-changing basis functions, the affine-not-convex structure we just diagnosed. "Yes, yes," he retorts, "but there is something more precise." He directs our attention back to the Lagrange formula and asks us to consider the following.

Chebyshev: Fix any $x \in [0,1]$ that is *not* one of the given nodes $x_k = k/n$, and ask: 'how far away is your approximation from the truth?' Define the **interpolation error**

$$e_n(x) := f(x) - \mathcal{L}_n[f(x)].$$

By the interpolation theorem,

$$e_n(x_k) = f(x_k) - \mathcal{L}_n[f(x_k)] = 0, \qquad k = 0, 1, \dots, n,$$

so the error function e_n vanishes at all $n+1$ nodes $e_k = k/n$, $k = 0, 1, \dots, n$.

"Now," Chebyshev continues, "suppose for the moment that the function Weierstrass will choose is sufficiently smooth." We may linger, wondering whether this is an appropriate assumption — we take care to observe that Weierstrass did *not* dictate any smoothness conditions on the functions he's allowed to choose — yet, lacking any other suitable recourse, we choose to follow Chebyshev's lead.

In such a case that f has $n+1$ continuous derivatives on $[0,1]$, Chebyshev asks us to consider the auxilliary function

$$\varphi(t) := \underbrace{f(t) - \mathcal{L}_n[f(t)]}_{=e_n(t)} - \lambda \prod_{k=0}^{n} (t - x_k)$$

where λ is a constant chosen so that φ vanishes at some fixed point $x^\star \in [0,1]$ distinct from the nodes $x_0, x_1, \ldots, x_n$. "This may look like a lot of nonsense," he continues, "but just trust me."

Remark.

The reader is encouraged to verify that such a λ exists and is unique — indeed, since $\mathcal{L}_n[f(x_k)] = f(x_k)$ at every node and $x \mapsto \prod_{k=0}^{n}(x_k - x_k) = 0$ at every node, we have $\varphi(x_k) = 0$ for $k = 0, 1, \ldots, n$ automatically. We then choose λ to make $\varphi(x^\star) = 0$ as well, which requires

$$\lambda = \frac{f(x^\star) - \mathcal{L}_n[f(x^\star)]}{\prod_{k=0}^{n}(x^\star - x_k)},$$

where the denominator $\prod(x^\star - x_k) \neq 0$ since $x^\star$ is not one of the original nodes.

"So," Chebyshev says, "φ vanishes at $n+2$ distinct points in $[0,1]$: the $n+1$ nodes $x_0, x_1, \ldots, x_n$, and the additional, distinct point $x^\star \in [0,1]$ we constructed."

"Oh," he hesitates, "and I suppose I should ask — are you familiar with Rolle's Theorem?"

Theorem 3.3.3 (Rolle's Theorem)

Let $\varphi\colon [a,b] \to \mathbb{R}$ be continuous on $[a,b]$ and differentiable on (a,b), with

$$\varphi(a) = \varphi(b) = 0.$$

Then, there exists at least one point $c \in (a,b)$ such that

$$\varphi'(c) = 0.$$

Proof:

Consider two cases. First, suppose

$$f(x) = 0 \quad \text{for all} \quad x \in [a,b].$$

In this case, any value $c \in [a,b]$ can serve as the c guaranteed by the theorem, as the function is constant on $[a,b]$, whence its derivative is zero.

Second, suppose

$$f(x) \neq 0 \quad \text{for some} \quad x \in (a,b).$$

Clearly f attains a minimum and maximum somewhere in $[a,b]$ (cf. the **Extreme Value Theorem**). Recall by our hypothesis that $f(a) = f(b) = 0$, and that in this case, $f(x) \neq 0$

for some $x \in (a, b)$. Thus, f will have either a positive absolute maximum value at some $c_+ \in (a, b)$, a negative absolute minimum value c_- in (a, b), or both.

Take c to be either c_+ or c_- (or either one), depending on which we've identified.

Then, the open interval (a, b) contains c, and either

- $f(c) = f(c_+) \geq f(x)$ for all $x \in (a, b)$, or
- $f(c) = f(c_-) \leq f(x)$ for all $x \in (a, b)$.

Either way, this means f has a local extremum at c. Since f is differentiable on (a, b) and $c \in (a, b)$ strictly, $f'(c)$ exists, and we conclude that at this point c,

$$f'(c) = 0.$$

‖

"Good," says Chebyshev. "Now, recall: φ vanishes at $n+2$ distinct points in $x_0, x_1, \ldots, x_n$, and our chosen root $x^\star$. Supposing that $x_j < x^\star < x_{j+1}$ and $0 \leq x_0 < x_1 < \cdots < x^\star < \cdots < x_n \leq 1$, we can clearly apply Rolle's Theorem to φ on each subinterval $[x_i, x_{i+1}]$. In particular, since we've constructed φ such that

$$\varphi(x_0) = \varphi(x_1) = \cdots = \varphi(x^\star) = \cdots = \varphi(x_n) = 0,$$

and since we're supposing f (and hence φ) is continuous and "sufficiently smooth" — which we now restate formally as requiring Weierstrass's chosen function f be at least $n+1$ times differentiable on $(0, 1)$ — we can apply Rolle's theorem to find points in each interval $c_i \in (x_i, x_{i+1})$ (along with our extra root $c^\star \in (x^\star, x_{j+1})$) such that

$$\varphi'(c_0) = \varphi'(c_1) = \cdots = \varphi'(c^\star) = \cdots = \varphi'(c_{n-1}) = 0.$$

That is, by one application of Rolle's Theorem, φ' has at least $n+1$ zeros $c_0, c_1, \ldots, c_j, c^\star, c_{j+1}, \ldots, c_{n-1}$ in $(0, 1)$."

Chebyshev pauses. "Do you see the pattern? Apply Rolle's theorem again — this time to φ', which vanishes at $n+1$ distinct points — and conclude that φ'' has at least n zeros. Apply it again: φ''' has at least $n-1$ zeros." We nod somewhat approvingly; "after $n+1$ applications," Chebyshev continues, "we will conclude that $\varphi^{(n+1)}$ has at least one zero in $(0, 1)$. Suppose ξ denotes this lone guaranteed zero of $\varphi^{(n+1)}$."

To compute $\varphi^{(n+1)}$ explicitly, recall the definition:

$$\varphi(t) := \underbrace{f(t) - \mathcal{L}_n[f(t)]}_{= e_n(t)} - \lambda \prod_{k=0}^{n} (t - x_k)$$

Differentiating $n+1$ times: the interpolant $\mathcal{L}_n[f]$ is a polynomial of degree at most n, so its $(n+1)$th derivative vanishes identically. The product $\prod_{k=0}^{n}(t - x_k)$ is a **monic polynomial** of degree $n+1$ — that is, its leading term in the monomial basis is t^{n+1} — so its $(n+1)$th derivative is the *constant value* $(n+1)!$. (Verify this explicitly!) Thus,

$$\varphi^{(n+1)}(t) = f^{(n+1)}(t) - 0 - \lambda \cdot (n+1)!.$$

Substituting in our derived constraint $\varphi^{(n+1)}(\xi) = 0$ and solving for λ:

$$\lambda = \frac{f^{(n+1)}(\xi)}{(n+1)!}.$$

"Now substitute back." Recall that λ was defined so that our chosen $x^\star$ was a root satisfying $\varphi(x^\star) = 0$, which gave us

$$\lambda = \frac{f(x^\star) - \mathcal{L}_n[f(x^\star)]}{\prod_{k=0}^n (x^\star - x_k)}.$$

We now have two representations for λ, which we can equate and rearrange:

$$\underbrace{\frac{f^{(n+1)}(\xi)}{(n+1)!} = \frac{f(x^\star) - \mathcal{L}_n[f(x^\star)]}{\prod_{k=0}^n (x^\star - x_k)}}_{\text{Both} = \lambda} \implies f(x^\star) - \mathcal{L}_n[f(x^\star)] = \frac{f^{(n+1)}(\xi)}{(n+1)!}\prod_{k=0}^n (x^\star - x_k).$$

"But the $x^\star \in [0,1]$ you chose was arbitrary," Chebyshev observes. "It can truly be anything you'd like — that is, we may replace $x^\star$ by any $x \in [0,1]$, understanding that the intermediate point $\xi = \xi(x) \in (0,1)$ may depend on our choice. What I can tell you for certain, then, is the following:"

Theorem 3.3.4 (Interpolation Error)

Suppose f has $n+1$ continuous derivatives on $[a,b]$, and let $\mathcal{L}_n[f]$ denote the Lagrange interpolant at nodes $x_0, x_1, \dots, x_n \in [a,b]$. Then for every $x \in [a,b]$, there exists $\xi_x \in (a,b)$ such that

$$f(x) - \mathcal{L}_n[f(x)] = \frac{f^{(n+1)}(x)}{(n+1)!}\prod_{k=0}^n (x - x_k).$$

Chebyshev glares — perhaps we look impatient. He told us this was leading somewhere, but at this point even Weierstrass would seem to be nodding off. "Please stick with me" Chebyshev beckons — "I've been playing Karl's game longer than you've been alive, so do forgive me if I'm a bit terse as I'm trying to catch you up."

Perhaps sensing the building anxiety amongst us and the peanut gallery, Chebyshev pleads: "*Just look* at what the Interpolation Error theorem is telling you. Two factors," he says, pointing to the right hand side formula therein. "The left one,

$$\frac{f^{(n+1)}(\xi)}{(n+1)!},$$

is controlled *entirely* by the function f of Weierstrass's choice. It is, in effect, the $(n+1)$th Taylor coefficient of f — its *regularity* at the $(n+1)$th derivative level, divided by a scaling factor $(n+1)!$."

"The right one,

$$\omega_{n+1}(x) := \prod_{k=0}^n (x - x_k),$$

is a polynomial of degree $n+1$ determined *entirely* by the nodes $x_0, x_1, \dots, x_n$. It has nothing whatsoever to do with f." Chebyshev taps the board. "Everything about Weierstrass's chosen function is trapped inside the left factor of my Interpolation Error formula. But let me ask: do you understand *how* we control the right factor, ω_{n+1}?"

We pause and stare at the formula above. Indeed, it may not be entirely clear what Chebyshev's point is if all we're thinking about is *changing basis*. But what we know from the Interpolation theorem is that this choice of representative basis is immaterial to our ability to approximate Weierstrass's function given only information on the $n+1$ equispaced samples $x_0 = 0, x_1 = 1/n, x_2 = 2/n, \dots, x_n = 1$ on $[0,1]$. This is to say that *once we've fixed* the $n+1$ data points, *any* basis of Π_{n+1} will yield the same interpolating polynomial.

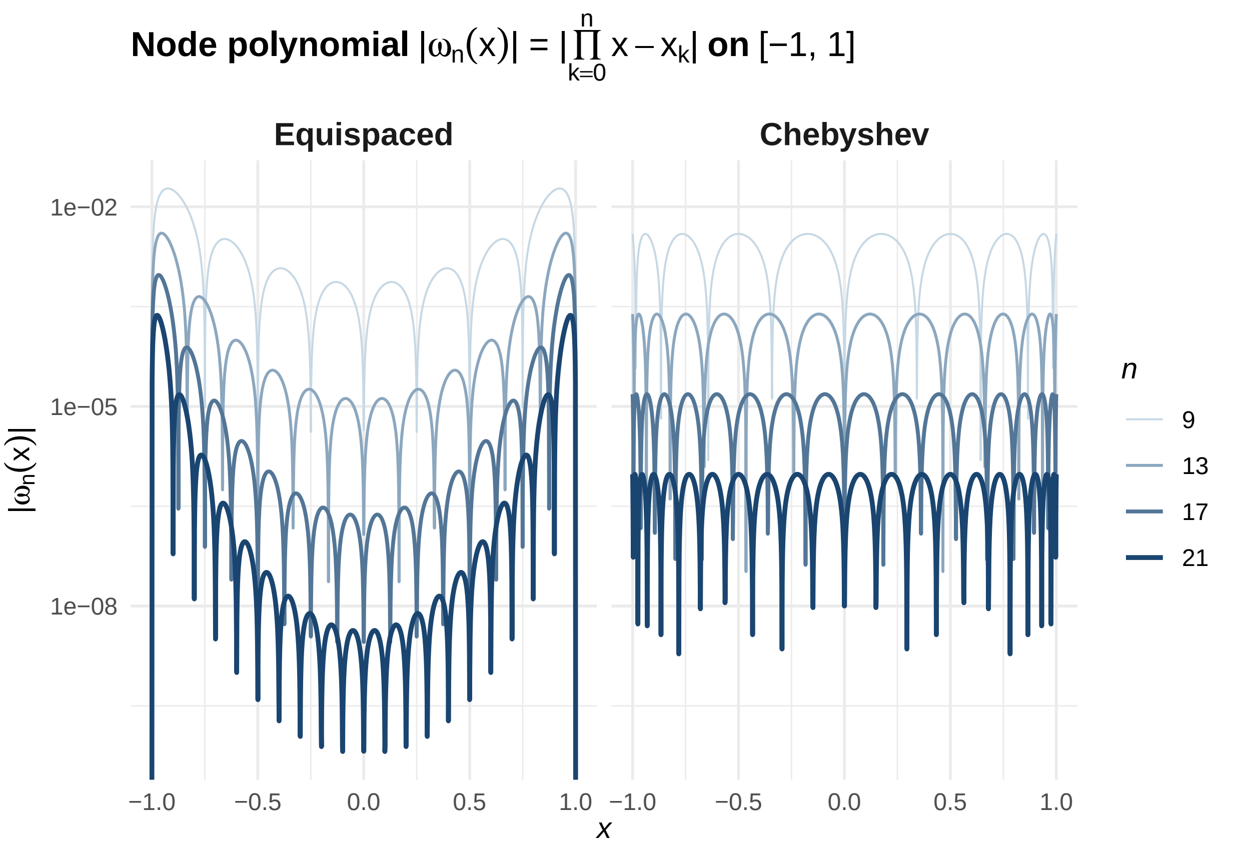
But what Chebyshev is saying is that, really, if we insist that our strategy is based on *interpolating* the points we're given, then Weierstrass has been loading the deck from the start by making them equispaced. "Look at the formula again," continues Chebyshev: "the formula reads

$$f(x) - \mathcal{L}_n[f(x)] = \underbrace{\frac{f^{(n+1)}(\xi)}{(n+1)!}}_{\text{controlled by Weierstrass}} \times \underbrace{\prod_{k=0}^{n}(x-x_k)}_{\omega_{n+1}(x)}.$$

The node polynomial ω_{n+1} is built from the nodes x_k that Weierstrass hands over and nothing else. It is a monic polynomial of degree $n+1$. Its shape, its size, its peaks — all are determined by *where you put the nodes.* It is the one piece of this formula that we haven't tried to touch," he grins, "and I have a proposition, if Karl would care to indulge in a modification to his usual game of three-card monte."

Weierstrass, startled by all this attention, rubs his eyes to think. Finally, he retorts: "It's anything for you Pafnuty," he grins, "as long as I still get to see the nodes you chose *before* I hand you my function."

Chebyshev, a bit surprised by the concession, gestures somewhat excitedly towards the board. "Before," he begins, "Weierstrass handed you $x_k = k/n$, and you accepted. Do you know what ω_{n+1} looks like at equispaced nodes? Let me show you:"

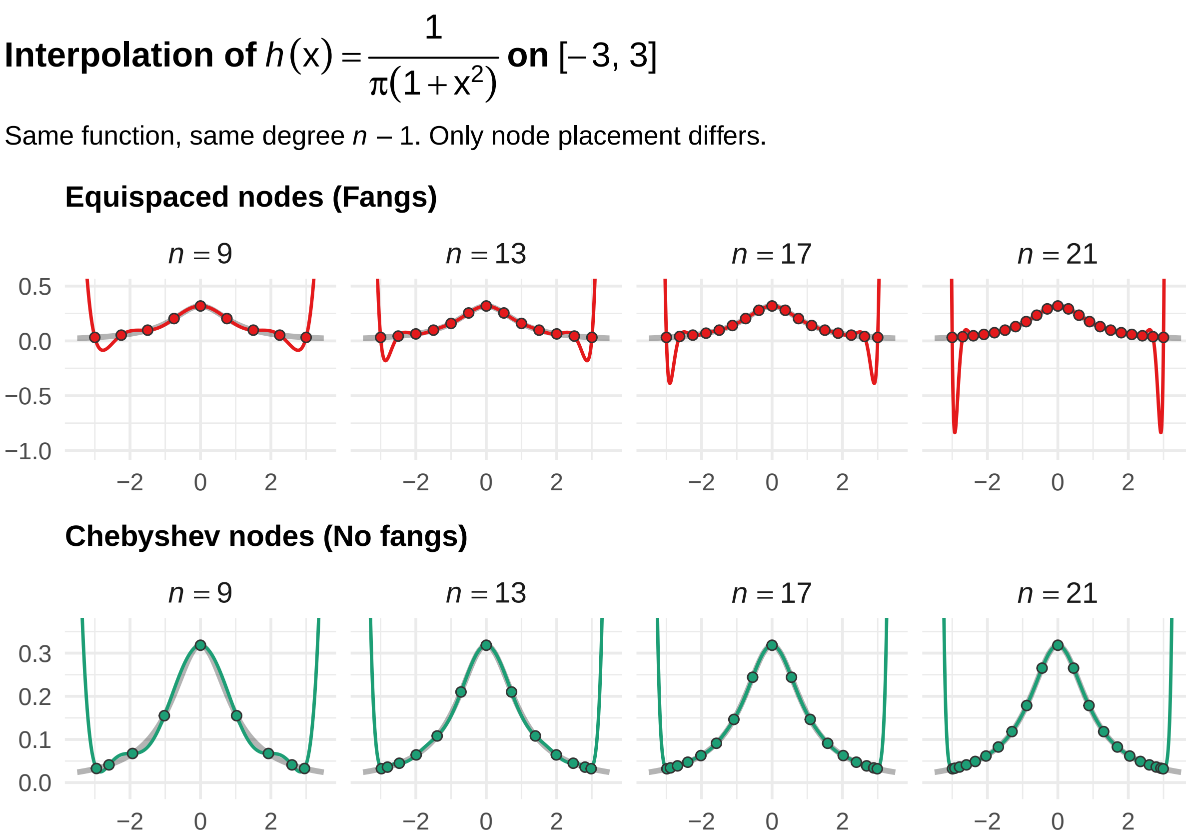


"Without loss of generality," Chebyshev concedes, "I'm working on $[-1, 1]$ — the Chebyshev polynomials live here naturally, and everything transfers to an arbitrary interval $[a, b]$ by the substitution

$$x \mapsto \tfrac{a+b}{2} + \tfrac{b-a}{2}x.$$

With all this in mind, look at the board. What is it telling you? As n grows, *both* panels see $|\omega_{n+1}|$ shrink — but they shrink in *very* different ways. On the equispaced side, the polynomial collapses rapidly further near the middle (the **interior**) of $[-1, 1]$ — you can see the darker curves well below the lighter ones for x near 0 — but the peaks near the **boundary** $x \approx \pm 1$ barely budge. Adding more equispaced nodes helps where the approximation *was already decent*, and does *essentially nothing* where it was catastrophic — recall the Cauchy's fangs, for instance."

Chebyshev taps the right panel. "Now look at mine. The peaks shrink **uniformly** — boundary, interior, everywhere. We control the polynomial *globally*, and we can tame the Cauchy. Take a look at what happens:"



The contrast is immediate. The same function h, the same polynomial degree, the same Lagrange interpolator — and yet the Chebyshev-node interpolants converge visibly toward h as n grows, while the equispaced interpolants tear themselves apart at the boundary. The *only* difference, then, is in the placement of the nodes.

"You see?" Chebyshev perks up. "Your analyst's fangs were *never* intrinsic to the Cauchy density — they were an artifact of the *equispaced nodes* you gave her. The node polynomial ω_{n+1} refused to shrink along the boundary, and that's where the polynomial interpolant *needed the most information* to accurately reproduce the Cauchy's decay. My nodes fix this by clustering near the boundary; precisely where the equispaced nodes leave us most exposed." He pauses. "And this isn't a trick or heuristic. It is *optimal*; no other function-naïve node placement can beat it."

Speaking on our behalf, suppose we ask whether all of this can be made precise. "Indeed," exclaims a beckoning Chebyshev — "let's return to the board."

Definition 3.3.5 (Chebyshev Polynomial)

A nth **Chebyshev Polynomial** is the function $T_n \colon [-1,1] \to \mathbb{R}$ defined by

$$T_n(x) := \cos(n \arccos(x)), \qquad x \in [-1,1].$$

"Now hang on just a second," we cry — "that is *not* a polynomial!" Chebyshev, exasperated, hangs his head. "They most certainly *are* polynomials: just compute

$$T_0(x) = \cos(0) = 1, \qquad T_1(x) = \cos(\arccos(x)) = x,$$

and if you don't believe me for the higher degrees, use the cosine product-to-sum formula (cf. Theorem 2.2.8 (P1))

$$\cos A \cos B = \frac{\cos(A+B) + \cos(A-B)}{2}$$

applied to $A = \theta$ and $B = n\theta$ and rearrange to obtain

$$\cos((n+1)\theta) = 2\cos(\theta)\cos(n\theta) - \cos((n-1)\theta).$$

Make the substitution $\theta = \arccos x$, so that $x = \cos\theta$, and you obtain

$$\cos((n+1)\arccos x) = 2\underbrace{\cos(\arccos x)}_{=x}\cos(n\arccos x) - \cos((n-1)\arccos x).$$

This looks messy, but it contains *several* terms of our desired form

$$T_k(x) = \cos(k \arccos x);$$

in particular, on switching out the notation, we observe the recurrence

$$T_{n+1}(x) = 2xT_n(x) - T_{n-1}(x).$$

Since $T_0 = 1$ and $T_1 = x$ are polynomials, induction gives that T_n is a polynomial of degree n for every $n \geq 0$. Verify for yourselves:"

$$\begin{aligned}
&T_0(x) = 1, \qquad T_1(x) = x,\\
&T_2(x) = 2xT_1(x) - T_0(x) = 2x^2 - 1,\\
&T_3(x) = 2xT_2(x) - T_1(x) = 4x^3 - 3x,\\
&T_4(x) = 2xT_3(x) - T_2(x) = 8x^4 - 8x^2 + 1,
\end{aligned}$$

where we observe that $T_0 \in \Pi_0 \equiv \mathbb{R}$, $T_1 \in \Pi_1$, and in general, $T_k \in \Pi_k$ the class of polynomials of degree (at most) k.

"Furthermore," Chebyshev adds, "the leading coefficient of T_n is 2^{n-1} for $n \geq 1$ — you can read it off from the recurrence, since each application of $T_{n+1} = 2xT_n - T_{n-1}$ doubles the leading coefficient. So, if you want one of my polynomials to have a leading coefficient of 1 — i.e., if you want the **monic** Chebyshev polynomial of degree n — it is simply $T_n/2^{n-1}$." (Verify this explicitly for a few of the polynomials we computed above; e.g., $T_2/2$ has leading coefficient 1 on the x^2 term.)

"Good," says Chebyshev. "Now, the reason I bring this up is that, originally, I posed the following question: among all monic polynomials of degree $n+1$ on $[-1,1]$, which one has the *smallest possible peak*? That is: I set out to identify

$$\min_{\substack{q \,\text{monic} \\ deg\, q = n+1}} \max_{x \in [-1,1]} |q(x)|.$$

What I ended up finding was, of course, that the **minimax** polynomial satisfying the above was the monic Chebyshev polynomial $T_{n+1}/2^n$."

Remark.

Chebyshev has no problem understanding his minimax notation, but especially at a glance, we might find it a bit cumbersome. "It's a minimum, *of* a maximum? How's that matter?" We encourage one to read this notation "from the outside in, then back out, then back in once more." Of course, we elaborate.

First, look to the min and the class it ranges over: here, the monic polynomials q of degree exactly n. So the object we're looking to find is some particular monic $q \in \Pi_n$. *Second*, look to the max: for any fixed candidate q, the expression $\max_{x\in[-1,1]} |q(x)|$ is just a single number — the largest value $|q|$ attains on $[-1,1]$. Different candidates q produce different numbers, and the min asks us to find the candidate whose number is smallest. *Third*, "back out": once we've found the candidate $q^\star$ achieving the minimum, we ask whether we're *certain* no other monic polynomial in Π_n could do better.

To see what this is really asking, let us work at the simplest nontrivial degree: $n = 2$. We're looking for the monic quadratic $q(x) = x^2 + bx + c$ on $[-1,1]$ whose peak $\max_{[-1,1]} |q|$ is as small as possible. An obvious first candidate is $q_1(x) = x^2$:

$$\max_{x\in[-1,1]} |q_1(x)| = \max_{x\in[-1,1]} |x^2| = 1.$$

Can we do better? Notice that q_1 hits its peak at the boundary $x = \pm 1$ and its minimum value of 0 at $x = 0$. The "budget" is lopsided: all the height is at the edges, the interior is wasted. What if we shift the parabola downward to redistribute? Consider

$$q_2(x) = x^2 - \frac{1}{2}.$$

This is still monic and still degree 2. Its values at the key points are

$$q_2(\pm 1) = 1 - \frac{1}{2} = \frac{1}{2}, \qquad q_2(0) = -\frac{1}{2},$$

so that $\max_{[-1,1]} |q_2(x)| = 1/2$. We've *halved* the peak just by subtracting a constant.

Now we "back out": can we do *even* better? What if we shift further — say, $q_3(x) = x^2 - 0.6$? Then $q_3(\pm 1) = 0.4$ (smaller at the boundary), but $q_3(0) = -0.6$ (larger in the interior). The max is now 0.6, which is *worse*. Shifting up instead — $q_4(x) = x^2 - 0.3$ — gives $q_4(\pm 1) = 0.7$ and $q_4(0) = -0.3$; the max is 0.7, also worse.

What happened with $q_2 = x^2 - 1/2$ is that we chose the constant so that the magnitude at the boundary $|q_2(\pm 1)| = 1/2$ *exactly equals* the magnitude at the interior extremum $|q_2(0)| = 1/2$. The polynomial **equioscillates**: it alternates between $+1/2$ and $-1/2$, and any attempt to lower one peak necessarily raises another. There is no slack left to exploit. This is what makes q_2 optimal.

And indeed: $q_2(x) = x^2 - 1/2 = (2x^2 - 1)/2 = T_2(x)/2$, the monic Chebyshev polynomial of degree 2, with peak exactly $1/2 = 2^{1-2} = 2^{-1}$, validating the theorem for $n = 2$. The minimax result says this equioscillation characterizes the optimum at *every* degree: the monic $T_n/2^{n-1}$ is the unique monic polynomial of degree n that equioscillates between $\pm 2^{1-n}$ on $[-1, 1]$, and any monic polynomial that fails to equioscillate has a strictly larger peak.

"Now," Chebyshev continues", "the optimal nodes. If I want the node polynomial ω_{n+1} — a degree-$(n+1)$ polynomial, mind you — to equal the minimax polynomial in that class — which we now know is $T_{n+1}/2^n$ — then I certainly need its roots to be the same as T_{n+1}. And where does $T_{n+1}(x) = \cos((n+1)\arccos x)$ vanish?" He looks at us expectantly. "A cosine vanishes at odd multiples of $\pi/2$: that is,

$$\cos\alpha = 0 \iff \alpha = \frac{(2k-1)\pi}{2}, \quad k = 1, 2, 3, \dots$$

In particular, letting $\alpha = (n+1)\arccos x$ in the above shows that

$$\cos((n+1)\arccos x) = 0 \iff (n+1)\arccos x = \frac{(2k-1)\pi}{2} \iff x = \cos\left(\frac{(2k-1)\pi}{2(n+1)}\right), \quad k \in \mathbb{N}.$$

I believe, then, that I've earned the right to present you with a formal definition:"

Definition 3.3.6 (Chebyshev Nodes)

The **Chebyshev nodes** of order $n+1$ on $[-1, 1]$ are the $n+1$ roots of the Chebyshev polynomial T_{n+1}:

$$x_k = \cos\left(\frac{(2k-1)\pi}{2(n+1)}\right), \qquad k = 1, 2, \dots, n+1.$$

Furthermore, on a general interval $[a, b]$, the Chebyshev nodes may be obtained by the affine map

$$x \mapsto \tfrac{a+b}{2} + \tfrac{b-a}{2}x.$$

"Notice where they sit," Chebyshev says, gesturing at the board. "Near $x = \pm 1$, the nodes cluster together; near $x = 0$, they spread apart. This is the opposite of what your intuition wants — you'd think uniform spacing should be 'fairest' — but the clustering is precisely what I just showed you a moment ago in the Cauchy figure a moment ago. What we can imagine is that the equispaced node polynomial refused to shrink near the boundary because there weren't enough nodes to keep the factors $(x - x_k)$ small. My nodes *compensate* for this by placing more nodes near the edges, so that each factor $(x - x_k)$ 'stays small' where it matters most, and the product $\omega_{n+1} = \prod(x - x_k)$ shrinks uniformly rather than concentrating nearer the interior."

With the definition and minimax theorem now in hand, Chebyshev formally states his guarantee:

Theorem 3.3.7 (Chebyshev Minimax)

Among all monic polynomials of degree n on $[-1, 1]$, the monic Chebyshev polynomial

$$q^\star(x) := \frac{T_n(x)}{2^{n-1}} = \frac{1}{2^{n-1}} \cos(n \arccos x)$$

achieves the smallest possible peak:

$$\min_{\substack{q \text{ monic} \\ \deg q = n}} \max_{x \in [-1,1]} |q(x)| = \max_{x \in [-1,1]} \left| \frac{T_n(x)}{2^{n-1}} \right| = \frac{1}{2^{n-1}},$$

and no other monic polynomial of degree n achieves this value — that is, $q^\star = T_n/2^{n-1}$ is the *unique minimax solution* over the class of degree n polynomials.

"And so," Chebyshev says, "let me finish what I started. Recall my interpolation error formula:

$$f(x) - \mathcal{L}_n[f(x)] = \frac{f^{(n+1)}(\xi_x)}{(n+1)!} \cdot \omega_{n+1}(x).$$

The right factor $\omega_{n+1}(x) = \prod_{k=0}^{n}(x - x_k)$ is a monic polynomial of degree $n+1$ determined entirely by the nodes. At equispaced nodes, this factor grows near the boundary and refuses to shrink no matter how large n becomes — you saw as much in the figure. At *my* nodes, $\omega_{n+1} = T_{n+1}/2^n$, and the minimax theorem gives

$$\max_{x \in [-1,1]} \left| \omega_{n+1}(x) \right| = \frac{1}{2^n}.$$

Exponentially small in n, mind you. So, the interpolation error satisfies

$$\max_{x \in [-1,1]} \left| f(x) - \mathcal{L}_n[f(x)] \right| \leq \frac{\max \left| f^{(n+1)} \right|}{(n+1)!} \cdot \frac{1}{2^n},$$

where I've substituted the evaluation $|f^{(n+1)}(\xi_x)|$ with the maximum $|f^{(n+1)}|$ attains over $[-1, 1]$.

"Now," Chebyshev continues, "let me revisit your two DGPs with this bound in hand. For the Gaussian,

$$g(x) = \frac{1}{\sqrt{\pi}} e^{-x^2},$$

the factor to the left of $1/2^n$ decays like

$$\frac{\max |g^{(n+1)}|}{(n+1)!} \sim O\left(\frac{1}{n!}\right);$$

indeed, we didn't even *need* my nodes for the Gaussian; the factorial decay inherent to the Taylor expansion of the Gaussian kernel e^{-x^2} overcomes any deficits we incur by haphazard node placement."

"On the other hand, for the Cauchy density

$$h(x) = \frac{1}{\pi(1+x^2)},$$

the left factor satisfies

$$\frac{\max |h^{(n+1)}|}{(n+1)!} = O(1);$$

i.e., it *doesn't decay*. We don't get any help from the Cauchy, but it doesn't hurt us that badly — in particular, it doesn't blow up as $n \to \infty$. Under equispaced nodes, we inherit *no additional help*, and the fangs emerge; by instead *strategically placing* our nodes, we're able to overwhelm the constancy of the left factor by an exponentially decaying right factor, which is why

$$\max_{x\in[-1,1]} |h(x) - \mathcal{L}_n[h(x)]| \leq \underbrace{O(1)}_{\text{Cauchy's (ir-)regularity}} \times \underbrace{\frac{1}{2^n}}_{\text{Chebyshev's nodes}} \overset{n\to\infty}{\longrightarrow} 0,$$

so the fangs disappear." Chebyshev taps the interpolation figure once more for emphasis. "Same Lagrange interpolation, same Cauchy density, same polynomial degree. The *only* change is at the nodes, and the convergence is exponential."

He pauses, then somewhat hastily adds: "Really, on your analysts' interval $[-3, 3]$, the affine map stretches the scaling a bit, but the same principle applies. My nodes cluster where the equispaced polynomial was most exposed, and the improved interpolant converges. What you saw in the figure, then, is precisely the figure in action."

Chebyshev allows himself a bow. We agree: he's earned it.

Weierstrass, on the other hand, does not applaud.

"Very nice, Pafnuty," Weierstrass smirks. He reaches into his coat and produces a new function (or, more aptly, *class* of functions, indexed by parameters $a \in (0,1)$ and $b \in 2\mathbb{Z}+1$, i.e., the odd integers):

$$W_{a,b}(x) := \sum_{k=0}^{\infty} a^k \cos(b^k \pi x), \qquad 0 < a < 1, \quad b \text{ odd}, \quad ab > 1 + \tfrac{3}{2}\pi.$$

"I've been working on this one," Weierstrass slinks back in his chair, "just for a couple of wise guys like you."

We stare at the board. It is a sum of cosines — each term $a^k \cos(b^k \pi x)$ oscillating at frequency b^k, weighted by the geometrically decaying (since $a \in (0,1)$) amplitude, a^k.

Weierstrass, sensing our unease, takes his time. "Before I explain what this function *does*," he says, "let me first ensure you that it *exists*. You're looking at an infinite series, and infinite series require justification." He writes:

Theorem 3.3.8 (Weierstrass M-Test)

Let $\{f_k\}_{k=0}^{\infty}$ be a sequence of functions on a common domain $D \subseteq \mathbb{R}$, and suppose there exists a sequence of constants $M_k \geq 0$ such that

$$|f_k(x)| \leq M_k, \quad \text{for all } x \in D, \quad k = 0, 1, 2, \dots,$$

with

$$\sum_{k=0}^{\infty} M_k < \infty.$$

Then, the series

$$\sum_{k=0}^{\infty} f_k(x) \text{ converges,}$$

and moreover, this convergence is **uniform** on D.

Furthermore, if each f_k is continuous on D, then the limit of the series $\sum_{k=0}^{\infty} f_k$ is continuous on D.

Proof:

Here, we want to show that $\sum f_k$ converges uniformly on D whenever each term f_k is uniformly bounded for $x \in D$; i.e., if $\{M_k\}$ is a sequence of constants such that $|f_k(x)| \leq M_k$ for all $x \in D$ at each $k = 0, 1, 2, ...$, and if the series of the bounding factors $\sum M_k$ converges, then so too does the series $\sum f_k$ whose terms f_k are bounded by the individual M_k.

Recall that saying a series "converges uniformly" precisely means that the partial sums of the series,

$$S_N(x) = \sum_{k=0}^{N} f_k(x),$$

converge uniformly *as a sequence*. And one way to interpret uniform convergence of a sequence S_n is: for every $\epsilon > 0$, there exists a threshold $N \in \mathbb{N}$ such that for all $n \geq N$,

$$\sup_{x \in D} |S_n(x) - \mathcal{S}(x)| < \epsilon,$$

where $\mathcal{S}$ is the limit. The supremum over $x \in D$ is what makes it uniform — N works for *every* x at once.

The only problem, of course, is that we don't have a proposed limit $\mathcal{S}$. Instead, we leverage the Cauchy criterion. Indeed, each pointwise evaluation $x \mapsto S_n(x)$ at a fixed n is just a real number, $S_n(x) \in \mathbb{R}$, and we know that the real line is *complete*. A sequence of real numbers converges if and only if it is *Cauchy*: that is, if for every $\epsilon > 0$, there exists a threshold N such that for all $n > m \geq N$,

$$|S_n(x) - S_m(x)| < \epsilon.$$

Now, notice that $S_n(x) - S_m(x)$ is just the "tail chunk" of the series from term $m + 1$ to term n:

$$S_n(x) - S_m(x) = \sum_{k=m+1}^{n} f_k(x).$$

Bounding this in absolute value,

$$|S_n(x) - S_m(x)| = \left| \sum_{k=m+1}^{n} f_k(x) \right| \leq \sum_{k=m+1}^{n} |f_k(x)| \leq \sum_{k=m+1}^{n} M_k,$$

where the first inequality is the triangle inequality and the second is the hypothesis that $|f_k(x)| \leq M_k$.

Since $\sum_{k=0}^{\infty} M_k < \infty$ by assumption, the tails of the series vanish: given $\epsilon > 0$, we can choose N such that $\sum_{k=N+1}^{\infty} M_k < \epsilon$. Then, for all $n > m \geq N$,

$$|S_n(x) - S_m(x)| \leq \sum_{k=m+1}^{n} M_k \leq \sum_{k=N+1}^{\infty} M_k < \epsilon.$$

The key observation is that the right-hand side, $\sum_{k=N+1}^{\infty} M_k$, *does not depend on the input* $x \in D$. In particular, the same threshold N works for every $x \in D$ simultaneously. So the partial sums are *uniformly* Cauchy on D: for each fixed x, completeness of $\mathbb{R}$ gives convergence of $S_n(x)$ to some limit $\mathcal{S}(x)$, and the x-independence of N guarantees this convergence is uniform on D.

For the continuity claim: suppose each f_k is continuous on D, and fix a point $x_0 \in D$. We want to show the limit $\mathcal{S}$ is continuous at x_0; if we can, that x_0 is arbitrary will establish the continuity of $\mathcal{S}$ on all of D.

So, for any $\epsilon > 0$, choose N large enough so that

$$\sup_{x \in D} |\mathcal{S}(x) - S_N(x)| < \frac{\epsilon}{3}$$

(possible since $S_N \to \mathcal{S}$ uniformly on D.) Since S_N is a finite sum of continuous functions f_k, S_N is continuous at x_0: there exists $\delta > 0$ such that

$$|x - x_0| < \delta \implies |S_N(x) - S_N(x_0)| < \frac{\epsilon}{3}.$$

Now, set $N = N(\epsilon)$ in order for the first sup bound to hold, and then set $\delta = \delta(\epsilon, N)$ so that the second bound holds. Since at this N we have $|\mathcal{S}(x) - S_N(x)| < \epsilon/3$ for all $x \in D$, it holds at both points x and x_0 we set so that $|x - x_0| < \delta$ — in particular, this forces

$$|x - x_0| < \delta \implies |\mathcal{S}(x) - \mathcal{S}(x_0)| \leq \underbrace{|\mathcal{S}(x) - S_N(x)|}_{<\epsilon/3} + \underbrace{|S_N(x) - S_N(x_0)|}_{<\epsilon/3} + \underbrace{|\mathcal{S}(x_0) - S_N(x_0)|}_{<\epsilon/3} < \epsilon.$$

at the given N and δ. Since $x_0 \in D$ was arbitrary, such an N and δ can be set for any $\epsilon > 0$ to force continuity of the limit $\mathcal{S}$ in a neighborhood about any point $x \in D$, and we conclude that $\mathcal{S} = \sum f_k$ is continuous on D whenever the f_k are continuous on D.

$\|$

"Good," says Weierstrass. "Now let me apply this to my function. Each term in the series defining $W_{a,b}$ is

$$f_k(x) = a^k \cos(b^k \pi x), \quad 0 < a < 1, \quad b \text{ odd}, \quad ab > 1 + \tfrac{3}{2}\pi.$$

Now, each of the cosine contributions has magnitude $|\cos(b^k \pi x)| \leq 1$ uniformly in x, so we may establish the term-wise bound

$$|f_k(x)| = |a^k \cos(b^k \pi x)| \leq a^k := M_k, \quad \forall x \in \mathbb{R}.$$

Since $0 < a < 1$, the bounding constants $M_k = a^k$ form a convergent geometric series:

$$\sum_{k=0}^{\infty} M_k = \sum_{k=0}^{\infty} a^k = \frac{1}{1-a} < \infty.$$

Thus, by the M-test, we conclude

$$W_{a,b}(x) = \sum_{k=0}^{\infty} a^k \cos(b^k \pi x) \text{ converges uniformly in } x \in \mathbb{R}.$$

Furthermore, since each partial sum is a finite sum of cosines — cosines are continuous — the M-test's continuity guarantee ensures continuity: $W_{a,b} \in C(\mathbb{R})$."

Remark.

Right quick, we want to demonstrate what the M-test is protecting us from. As such, consider the sequence

$$f_k(x) = x^k, \quad x \in [0, 1].$$

Each f_k is continuous, yet the sequence converges pointwise to the *discontinuous* function

$$f(x) = \begin{cases} 0 & 0 \le x < 1, \\ 1 & x = 1. \end{cases}$$

Why doesn't the M-test apply here? Each f_k is continuous, yes, but we *can not* bound the terms of the sequence by constants whose infinite series converges. Indeed, because

$$\sup_{x \in [0,1]} x^k = 1, \quad \forall k \in \mathbb{N},$$

the task of setting the constants M_k such that

$$M_k \ge |x^k|, \quad \forall x \in [0, 1] \iff M_k \ge \sup_{x \in [0,1]} |x^k| = \sup_{x \in [0,1]} |x|^k = 1, \quad k = 0, 1, 2, ...$$

This makes $\sum M_k = \infty$, and we cannot proceed.

"So," continues Weierstrass, "$W_{a,b}$ exists, converges uniformly, and is continuous on all of $\mathbb{R}$. Now, let me explain what makes it special."

He taps the kth term. "Consider the anatomy of $f_k(x) = a^k \cos(b^k \pi x)$. It has two moving parts: an *amplitude* a^k, which decays geometrically in k since $0 < a < 1$, and a *frequency* b^k, which grows geometrically in k since $b \ge 3$. The explicit form of f_k for several k are

$$\begin{aligned} k = 0: &\qquad f_0(x) = \cos(\pi x), \\ k = 1: &\qquad f_1(x) = a \cos(b \pi x), \\ k = 2: &\qquad f_2(x) = a^2 \cos(b^2 \pi x), \end{aligned}$$

and so on. At $k = 0$, f_0 is just a cosine, but at $k = 1$, we observe b oscillations of amplitude a. At $k = 2$, a growing number b^2 of oscillations of shrinking amplitude a^2. At $k = 10$, b^{10} oscillations — and for a particular value, say $b = 13$, this results in roughly 10^{11} oscillations of f_{10} at a tiny amplitude of a^{10}. The function $W_{a,b}$ is the superposition of *all of these simultaneously*: oscillation riding on faster oscillation, ad infinitum."

"The question," Weierstrass continues, "is which force wins. The decay of the amplitudes a_k wants to make $W_{a,b}$ smooth — if the high-frequency terms $\cos(b^k \pi x)$ were negligible, the function would inherit the smoothness of its low-frequency terms, and you could differentiate term-by-term without consequence. Yet the growth of the frequencies b^k wants to make $W_{a,b}$ rough — each successive cosine oscillates b times faster than the last, introducing structure at finer and finer scale."

"Oh," he pauses — "and I suppose I haven't told you the best part. Have you tried to differentiate $W_{a,b}$?"

We haven't, but Weierstrass is already at the board. "Suppose, for the sake of argument, that you *could* differentiate $W_{a,b}$ term by term. The derivative of the kth term is

$$f_k'(x) = -\pi(ab)^k \sin(b^k \pi x),$$

which has amplitude $\pi(ab)^k$. Now, we force $ab > 1 + \frac{3}{2}\pi$ by hypothesis, so the derivative amplitudes $(ab)^k$ *grow geometrically* in k. The series of derivatives,

$$\sum_{k=0}^{\infty} f_k'(x) = -\pi \sum_{k=0}^{\infty} (ab)^k \sin(b^k \pi x),$$

has terms whose amplitudes $(ab)^k \to \infty$. The M-test, which saved us a moment ago, now tells us *nothing* — the bounding constants $M_k = \pi(ab)^k$ form a *divergent* geometric series."

"Thus," he lingers, "you simply *cannot* justify differentiating f_k under the sum. But this failure is not merely technical — it is not just that we lack the tools to *justify* the differentiation, it is that the derivative *genuinely does not exist*."

Remark.

The proof that $W'_{a,b}(x)$ fails to exist at *every* $x \in \mathbb{R}$ is more delicate than the convergence argument, and we will not reproduce it — the interested reader may consult Hardy's 1916 treatment, or the proof linked here. What we *can* say, however, is that the intuition is exactly as described: the amplitudes a^k decay fast enough to guarantee convergence of $\sum f_k$ (continuity), but not fast enough to guarantee convergence of $\sum f_k'$ (non-differentiability). The amplitudes a^k and frequencies b^k are in a tug of war, and the condition $ab > 1 + \frac{3}{2}\pi$ ensures that the frequencies "win at the level of derivatives," while "losing at the level of the function itself."

"There is just one more thing I'd like you to observe about my function before we return to the game," Weierstrass presses. "You know that $W_{a,b}$ is continuous everywhere, differentiable nowhere, and built from cosines. But look at what happens when we rearrange my series:

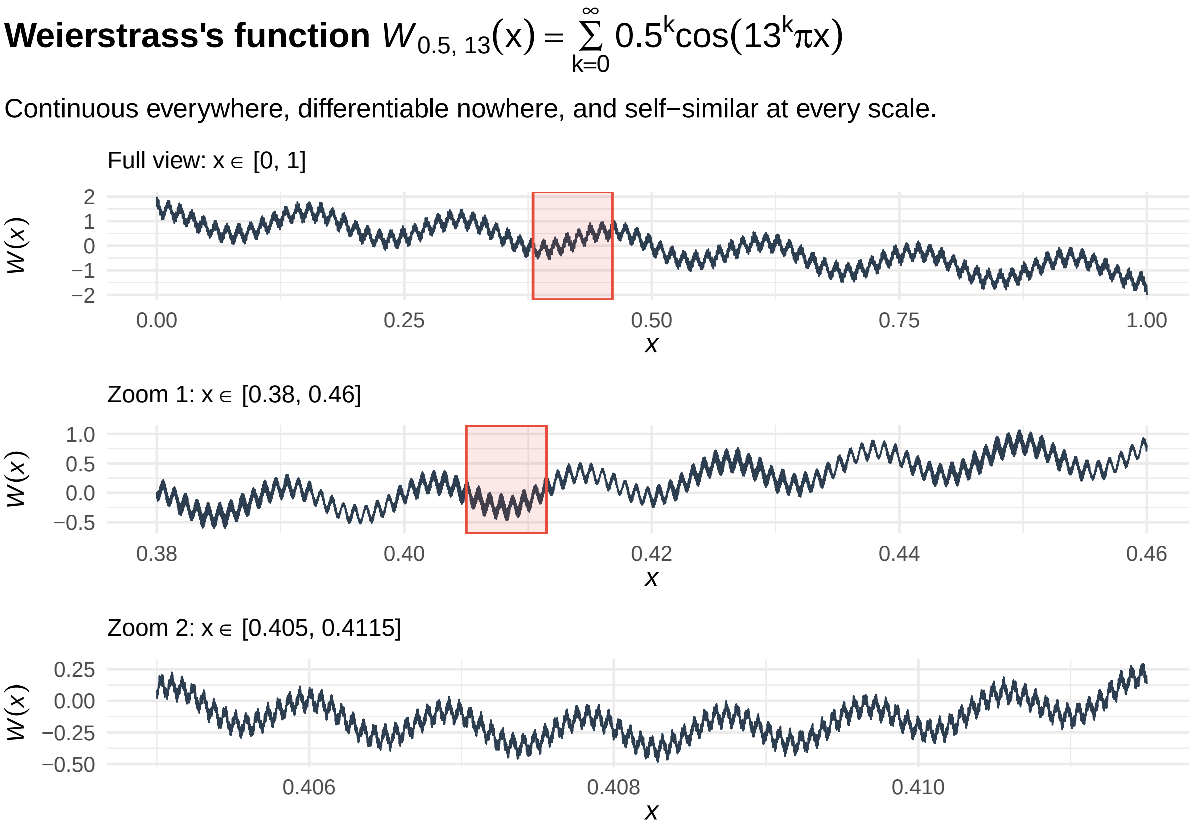
$$W_{a,b}(x) = \cos(\pi x) + a \sum_{k=0}^{\infty} a^k \cos(b^{k+1}\pi x) = \cos(\pi x) + aW_{a,b}(bx).$$

What we can observe in

$$W_{a,b}(x) = \cos(\pi x) + aW_{a,b}(bx)$$

is that $W_{a,b}$ satisfies a **functional equation**: up to a single smooth cosine, $W_{a,b}(x)$ is just a rescaled copy of itself, evaluated at b times the input $x \mapsto W_{a,b}(bx)$, and scaled by a in amplitude."

"Visually," Weierstrass puffs, now vigorously scrawling at the white board, "you might imagine this means the plot of my function looks something like this:"



Weierstrass wipes away the sweat from his brow. “So you see,” he manages, “*If* you zoom in on any patch of the graph by a factor of b horizontally and stretch by $1/a$ vertically, and you see *the same function* — not approximately, but *exactly*.” He taps the board. “This is the **self-similarity** of $W_{a,b}$. There is no scale at which the function ‘looks smooth.’ The roughness you see at one magnification reappears, identically, at the next. It is not a surface feature that resolves itself should we ‘zoom in enough’; it *is* the function.”

Weierstrass constructed $W_{a,b}$ in 1872, thirteen years before the approximation theorem bearing his name. At the time, many mathematicians — Ampère among them — had conjectured that every continuous function must be differentiable “almost everywhere,” or at least at “most points.” Weierstrass’s construction demolished this belief: $W_{a,b}$ is differentiable at *no* points, and it is not pathological or artificial — it is a sum of cosines, among the most natural building blocks in all of analysis.

We now return to the game. Recalling Chebyshev’s interpolation error formula reads

$$f(x) - \mathcal{L}_n[f(x)] = \frac{f^{(n+1)}(\xi_x)}{(n+1)!} \cdot \omega_{n+1}(x),$$

the left factor requires $f^{(n+1)}(\xi)$. But $W_{a,b}$ has no first derivative, let alone an $(n+1)$th. The expression $W_{a,b}^{(n+1)}(\xi)$ is not merely large — it is *undefined.* This is not a situation where we can “manipulate ω_{n+1} to overwhelm the left factor,” since it simply doesn’t exist in this case. The error formula simply does not apply.

The entire framework Chebyshev developed — the decomposition into “DGP regularity” and “node geometry,” the minimax optimization, the Chebyshev nodes — requires that f have enough derivatives for the formula to *parse.* For $W_{a,b}$, it does not parse. The error formula is a *conditional* guarantee: *if* f is “smooth enough,” *then* Chebyshev’s nodes give exponential convergence. $W_{a,b}$

does not satisfy this condition, so choosing better nodes can't provide enough help to force anything better.

But, to be clear, this is not just a failure of Chebyshev's brand of interpolation — *any* method whose convergence guarantees depend on a smooth target f will fail for $W_{a,b}$ for precisely the same reason: $W_{a,b}$ isn't smooth."

And yet, we emphasize that the Weierstrass Approximation Theorem *still applies* to $W_{a,b}$! Indeed, $W_{a,b}$ is continuous on $[0,1]$, so for every $\epsilon > 0$, a polynomial p with $\|W_{a,b} - p\|_\infty < \epsilon$ is guaranteed by the theorem. Again, the polynomial is guaranteed, but Chebyshev's tools cannot find it — not because the tools themselves are flawed, but because their hypotheses are not met by our proposed target $W_{a,b}$.

If it wasn't clear by this point, we've *lost again*. It is unfortunate: even though Chebyshev helped us beat the Cauchy, Weierstrass's weapon $W_{a,b}$ was simply too wild for the same tools to correct.

Intermission

Chebyshev, a bit dismayed by the whole affair, stares at the board. "Hold on just a moment," he mutters to himself. He scrawls for a moment, turning red in the face — "in fact," he says, more slowly now, as though working it out for himself, "suppose we started with my own polynomials $T_n(x) = \cos(n \arccos x))$. The substitution $x = \cos\theta$ turns this into

$$T_n(x) = \cos(n \arccos x) \implies T_n(\theta) = \cos(n\theta).$$

And that," his expression drops, "is a Fourier cosine function."

He paces for a minute before walking back to the board. "And I can even take this further. Suppose we had a polynomial p expressed in *my* basis — that is, as a linear combination of Chebyshev polynomials

$$p(x) = \sum_{k=0}^{n} c_k T_k(x), \qquad x \in [-1,1].$$

This is a polynomial of degree n on $[-1,1]$. Make the substitution $x = \cos\theta$, so that $\theta = \arccos x$ and θ ranges over $[0,\pi]$ as x ranges over $[-1,1]$. Under this substitution, every Chebyshev polynomial becomes a cosine:

$$T_k(x) = T_k(\cos\theta) = \cos(k\theta).$$

So, the polynomial p, re-expressed as a function of $\theta \in [0,\pi]$, becomes

$$p(\cos\theta) = \sum_{k=0}^{n} c_k \cos(k\theta), \qquad \theta \in [0,\pi].$$

Chebyshev stops and stares at what he's written. He compares

$$W_{a,b}(x) = \sum_{k=0}^{\infty} a^k \cos(b^k \pi x)$$

with his own

$$p(\cos\theta) = \sum_{k=0}^{n} c_k \cos(k\theta).$$

"The coefficients c_k in my Chebyshev expansion," he says slowly, "are the Fourier cosine coefficients of the function $\theta \mapsto p(\cos\theta)$ on $[0,\pi]$. These are not analogous. They are not 'similar in spirit.' They are the *same computation* in different coordinates."

He turns to look at Weierstrass. “Karl, the tool that defeated me — your $W_{a,b}$ — is a sum of cosines. The tool I used to fight back — my T_n — *is* a cosine. The Fourier cosine basis on $[0, \pi]$ and the Chebyshev polynomial basis on $[-1, 1]$ are the same basis, connected by $x = \cos\theta$. I have been doing Fourier analysis this entire time. I just didn't know it.”

Nobody says anything for a moment. Then Chebyshev sits down.

Remark.

We linger on this connection between Chebyshev polynomials and Fourier series, because we think the reader should too. What Chebyshev has just observed is not a curiosity or a notational coincidence. It is a structural identity: polynomial approximation on $[-1, 1]$ in the Chebyshev basis *is* Fourier cosine approximation on $[0, \pi]$, full stop. The change of variables $x = \cos\theta$ converts one into the other, and the conversion is exact — no approximation, no loss, no residual. Everything the reader learned about Fourier series in Chapter 2 — orthogonality, convergence rates, the Dirichlet and Fejér kernels, the role of smoothness in determining coefficient decay — has a polynomial translation on $[-1, 1]$, and vice versa.

But then this observation raises a question that, once noticed, is difficult to set aside. Indeed, we recall a remark we made near the statement of Weierstrass's theorem — there, we hearkened *all* the way back to chapter 2, where we *already proved* that continuous functions f on compact intervals $[a, b]$ (here, you could rescale the function affinely such that it is defined on $[-\pi, \pi]$) can be uniformly approximated — not by polynomials, but by the **Cesàro means** of their Fourier series:

$$\sigma_N f(x) = \frac{1}{N+1} \sum_{j=0}^{N} S_j f(x),$$

where

$$S_j f(x) = \frac{a_0}{2} + \sum_{k=1}^{j} [a_k \cos(kx) + b_k \sin(kx)],$$

$$a_0 = \frac{1}{\pi} \int_{\pi}^{\pi} f(x)\, dx, \qquad a_k = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(kx)\, dx, \qquad b_k = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(kx)\, dx.$$

Fejér's theorem (Theorem 2.3.10) guarantees that $\sigma_N f \rightrightarrows f$ on $[a, b]$ — i.e., that $\sigma_N f$ converges uniformly to f on $[a, b]$, which is precisely the same as saying

$$\|\sigma_N f - f\|_\infty \to 0 \quad \text{as} \quad N \to \infty.$$

or, stated more formally, that given $\epsilon > 0$, we can choose $N = N(\epsilon) \in \mathbb{N}$ based *purely* on ϵ such that

$$n \geq N \implies \|\sigma_N f - f\|_\infty = \sup_{x \in [a,b]} \left|\sigma_n[f(x)] - f(x)\right| < \epsilon$$

To be clear — what this the Cesàro means satisfy *almost everything* we're looking for in a candidate to beat Weierstrass at his own game. Indeed, they *must* be able to beat $W_{a,b}$

So we have to ask: if Fejér already solved the problem back in Chapter 2, why are we talking about it again?

The answer is subtle, but that subtlety is, our view, the single most underserved sentiment in all of mathematics: it concerns the *complexity* of the approximating objects. It is, quite fundamentally, the same distinction we made in Chapter 1: these Cesàro means σ_N are comprised of a finite number of Fourier partial sums S_n, each of which are comprised of sums of sines and cosines $\sin(nx)$ and $\cos(nx)$ weighted by integrals of evaluations of our function f weighted by sines and cosines in the coefficients a_k and b_k.

All of this might sound like a mouthful. And maybe it should — because at this point, we *haven't even remembered what a cosine is.* Again, forget the calculator — how would Euclid have computed a Fourier series?

He couldn't have. Because $\cos(kx)$ is *not* a finite arithmetic object. It is an infinite power series:

$$\cos(kx) = 1 - \frac{(kx)^2}{2!} + \frac{(kx)^4}{4!} - \frac{(kx)^6}{6!} + \cdots = \sum_{m=0}^{\infty} \frac{(-1)^m (kx)^{2m}}{(2m)!}.$$

Every cosine in $\sigma_N f$ — and there are finitely many of them, one for each harmonic $k = 0, 1, \ldots, N$ — *hides* an infinite series. But look at the mth term of the cosine Taylor series:

$$\underbrace{\frac{(-1)^m k^{2m}}{(2m)!}}_{A_m} x^{2m} := A_m x^{2m}.$$

This connects back to Chapter 2, where we observed that $\cos(\cdot)$ is an **even** function — indeed, we observe that the Taylor expansion contains only members of the form

$$\mathcal{C}_m := \{1, x^2, x^4, \ldots, x^{2m}\}.$$

What we might intuitively imagine, then, is that when we add in the analogous $\sin(\cdot)$ terms,

$$\sin(kx) = x - \frac{(kx)^3}{3!} + \frac{(kx)^5}{5!} - \frac{(kx)^7}{7!} + \cdots = \sum_{m=0}^{\infty} \frac{(-1)^m (kx)^{2m+1}}{(2m+1)!},$$

and, again looking to the mth term, setting

$$\underbrace{\frac{(-1)^m k^{2m+1}}{(2m+1)!}}_{B_m} x^{2m+1} := B_m x^{2m+1},$$

that what we observe is the *odd* monomials filling in:

$$\mathcal{S}_m := \{x, x^3, x^5, \ldots, x^{2m+1}\}.$$

Together, $\mathcal{C}_m \cup \mathcal{S}_m = \{1, x, x^2, x^3, x^4, \ldots\}$ — but this is just the monomial basis $\mathcal{M}$. The cosines, expanded in powers of x, contribute only the even monomials x^0, x^2, $x^4 \ldots$; the sines contribute only the odd monomials $x^1, x^3, x^5 \ldots$ Neither family alone spans the space of polynomials. You need *both* the fill out the monomial basis — and this is the *reason* a Fourier series "requires both," and why Cesàro means are then able to approximate *any* continuous function — not *just* the even or odd ones.

The even/odd decomposition of f in Chapter 2 — which, at the time, seemed like a *symmetry property* of trigonometric series — is, viewed through the monomial lens, a statement about *which powers of x* each family of transcendental functions harbors in the representation *we actually compute.*

So, what we observe in composite is that, when we *really* compute

$$\sigma_N f(x) = \frac{a_0}{2} + \sum_{k=1}^{N} \underbrace{\left(1 - \frac{k}{N+1}\right)}_{:= w_k} \Big[a_k \cos(kx) + b_k \sin(kx)\Big],$$

— where the

$$w_k := 1 - \frac{k}{N+1} = \frac{(N+1)-k}{N+1}, \qquad k = 1, \dots, N,$$

are the Cesàro weights that taper the Fourier coefficients toward zero — we are *really* computing

$$\begin{aligned}
\sigma_N f(x) &= \frac{a_0}{2} + \sum_{k=1}^{N} w_k \left[a_k \sum_{m=0}^{\infty} \frac{(-1)^m k^{2m}}{(2m)!} x^{2m} \; + \; b_k \sum_{m=0}^{\infty} \frac{(-1)^m k^{2m+1}}{(2m+1)!} x^{2m+1}\right] \\
&= \frac{a_0}{2} + \sum_{k=1}^{N} w_k \sum_{m=0}^{\infty} (-1)^m \left[\frac{a_k k^{2m}}{(2m)!}\, x^{2m} \; + \; \frac{b_k k^{2m+1}}{(2m+1)!}\, x^{2m+1}\right].
\end{aligned}$$

The outer sum over k is finite (only N terms). The inner sum over m is infinite (each cosine and sine unpacks into infinitely many monomials). If we now swap the order of summation — collecting by monomial degree rather than by frequency — we obtain

$$\begin{aligned}
\sigma_N f(x) &= \frac{a_0}{2} \; + \; \sum_{m=0}^{\infty} (-1)^m \underbrace{\left[\sum_{k=1}^{N} \frac{w_k\, a_k\, k^{2m}}{(2m)!}\right]}_{\text{even coefficient } \alpha_{2m}} x^{2m} \; + \; \sum_{m=0}^{\infty} (-1)^m \underbrace{\left[\sum_{k=1}^{N} \frac{w_k\, b_k\, k^{2m+1}}{(2m+1)!}\right]}_{\text{odd coefficient } \alpha_{2m+1}} x^{2m+1} \\
&= \sum_{\ell=0}^{\infty} \alpha_\ell\, x^\ell,
\end{aligned}$$

where the coefficient α_ℓ at each monomial degree ℓ is a finite sum (over $k = 1, \dots, N$) of Fourier coefficients weighted by powers of the frequencies:

$$\alpha_\ell = \begin{cases} \dfrac{a_0}{2} + \displaystyle\sum_{k=1}^{N} \frac{(-1)^m\, w_k\, a_k\, k^{2m}}{(2m)!} & \ell = 2m \text{ (even)}, \\ \displaystyle\sum_{k=1}^{N} \frac{(-1)^m\, w_k\, b_k\, k^{2m+1}}{(2m+1)!} & \ell = 2m+1 \text{ (odd)}. \end{cases}$$

We encourage you to now entirely discard the explicit forms of the α_ℓ (while perhaps choosing to remember that each is computable, in the Chapter 1 "sense of Euclid"), and instead look to the general form of

$$\sigma_N f(x) = \sum_{\ell=0}^{\infty} \alpha_\ell\, x^\ell$$

The re-indexing in terms of ℓ might read like notational slight of hand, but it isn't — all we did was swap a finite sum (over *frequencies* $k = 1, \dots, N$) past an infinite sum (over *monomial degrees* $m = 0, 1, 2, \dots$), and this is always when the outer sum is finite. What the re-indexing *reveals*, however, is that the Cesàro mean was "never really comprised N functions" in the sense Weierstrass demands — it was always a single infinite power series $\sum_{\ell=0}^{\infty} \alpha_\ell x^\ell$, whose coefficients

α_ℓ are finite, computable numbers determined by the Fourier data. The trigonometric representation fragmented this series into "N parallel channels organized by frequency"; the monomial representation "reassembles it into one infinite series organized by degree."

And now the crucial observation: since $\sigma_N f \rightrightarrows f$ *anyway*, even in spite of the fact that it was "always infinite," what we can deduce is that the *tail* of the infinite series $\sum \alpha_\ell x^\ell$ is negligible. Indeed: for any $\epsilon > 0$, there exists a finite M such that

$$\left\| \sum_{\ell=0}^{\infty} \alpha_\ell x^\ell - \sum_{\ell=0}^{M} \alpha_\ell x^\ell \right\|_\infty = \left\| \sum_{\ell=M+1}^{\infty} \alpha_\ell x^\ell \right\|_\infty < \epsilon.$$

The truncated series

$$p_M(x) := \sum_{\ell=0}^{M} \alpha_\ell x^\ell$$

is a polynomial — an element of Π_M, finite in the Euclidean sense — and it satisfies

$$\|f - p_M\|_\infty \le \underbrace{\|f - \sigma_N f\|_\infty}_{<\epsilon\,\text{(Fejér)}} + \underbrace{\|\sigma_N f - p_M\|_\infty}_{<\epsilon\,\text{(tail truncation)}} < 2\epsilon.$$

The whole of the peanut gallery seem to let out a collective gasp as we scrawl in our last ϵ. Funnily enough, we don't even appear to realize what we've done — but Chebyshev, seemingly revived from his stuper, leaps straight to the board.

"Do you realize what you've just done?" Chebyshev says, barely containing himself. "You've *proved it*. The Weierstrass Approximation Theorem. You just proved it — right there, on the board — using Fejér's theorem from Chapter 2 and nothing else."

We stare at our own work. He's right.

- Step 1: Fejér guarantees $\|f - \sigma_N f\|_\infty < \epsilon$.
- Step 2: the Cesàro mean is a power series whose tail is negligible, so truncation at degree M costs at most ϵ.
- Step 3: the triangle inequality. The polynomial $p_M \in \Pi_M$ exists and uniformly approximates f to within 2ϵ.

That's Weierstrass's theorem in a nutshell. No really — we just proved it.

Weierstrass, who has been watching from his chair, begins to applaud — slowly, and with a smile that suggests he is not entirely surprised. "Congratulations," he says. "So you've found a polynomial. Now, why don't you just tell try and me: what *is* it?"

We look at our polynomial. Its coefficients are:

$$\alpha_\ell = \sum_{k=1}^{N} \frac{(\text{Cesàro weight})_k \times (\text{Fourier coefficient})_k \times k^\ell}{\ell!}.$$

Suppose we're wondering what the degree M of the polynomial p_M we found is. Immediately, we realize that we *don't* know — it depends on how fast the tail decays, which depends on N, which depends on ϵ, and the relationship between M, N, and ϵ is some function we never bothered to compute.

And the notation we used above is deliberately quite illusory — but, in practice, it matters *what* the coefficients α_ℓ are explicitly. So, what are they? Finite sums of Fourier integrals weighted by factorial-suppressed frequency powers — computable in principle, yet completely unilluminating in practice.

Is the polynomial a convex combination of the data values? This mattered for the whole probabilistic interpretation of the Fejér kernel — it was part of why it was "intuitively appealing" to us at the time. And again, the answer is a resounding "No" — there is no reason the polynomial should be convex, and in general it isn't.

Can our "Cesàro polynomial p_M" overshoot the range of f? Certainly. Does it have any structural properties at all? None that we can see.

And, perhaps most damning of all for p_M: the coefficients α_ℓ are built from the Fourier coefficients a_k and b_k, which are *integrals*:

$$a_k = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos(kx)\,dx, \qquad b_k = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin(kx)\,dx.$$

Weierstrass, on the other hand — he handed us *point evaluations*: $f(0), f(1/n), f(2/n), \ldots, f(1)$. He *explicitly did not* hand us the ability to integrate f against cosines over $[-\pi, \pi]$. Our proof establishes that the polynomial exists, yes — but the polynomial itself requires information about f that the game never provided. We proved the theorem while violating the rules.

Remark.

The reader who was "only interested" in our game to obtain a constructive proof of Weierstrass's theorem may, thus, feel free to leave. Yet we would encourage you to continue — because we believe the "proof as given" is scarcely illuminating and, moreover, perhaps even obstructive. We do not *feel* the theorem yet, and that is what we aim to do by the end here.

Chebyshev, sensing our deflation, sits back down. "Don't be too hard on yourselves," he says quietly. "You've done something real. The theorem is proved. The polynomial *exists*. But" — and here he pauses — "existence is not construction, and the polynomial you found is, if I may be frank, terrible. It has no closed form. It has no structure. It requires global information about f — integrals against cosines — that no finite set of point evaluations can provide. And it inherits nothing about the geometry of f: it is not a convex combination of anything, it has no reason to stay within the range of f, and its rate of convergence is buried under two layers of uncontrolled approximation."

He looks at the table on the board.

	Ch. 2 (Fourier Series)	**Ch. 3 (Polynomials)**
"Hard" method	Dirichlet partial sums $S_N f$	Lagrange interpolation $\mathcal{L}_n[f]$
"Soft" method	Cesàro means $\sigma_N f$	???
Complexity	Finite $\times$ infinite	Finite $\times$ finite

"You've shown that the bottom-right entry *exists*," Chebyshev says, "by extracting it from the bottom-left. But the extraction destroyed everything that made the Cesàro means beautiful: the nonnegativity, the convexity, the closed form, the probabilistic interpretation and, perhaps most of all, the clean proof. What's sitting in that bottom-right cell *is* a polynomial, yes — but it is a polynomial discovered by hiring an orchestra, and then sending everyone home except the pianist." He pauses. "Perhaps it is better," he ponder, "to walk directly to the piano."

We agree. The question has shifted. It is no longer "does a uniformly approximating polynomial exist?" — we have just answered that, and the answer is yes. The question is now: can someone hand us a polynomial that is *good*? One with a closed form, built from the equispaced data

$f(0), f(1/n), \dots, f(1)$ and nothing else, whose weights are nonneg, whose convergence proof uses only continuity, and whose structure reveals *why* it works rather than merely *that* it works?

Perhaps we open the floor to questions — our peanut gallery is, after all, not short on talent. Fejér, whose theorem we just used as the load-bearing wall of our proof, shifts uncomfortably in his seat. Runge, whose own name adorns the phenomenon that started this whole mess, winces sympathetically. Faber, who could have told Chebyshev from the start that no node sequence would work for every continuous function, excuses himself entirely.

Yet, none of them can tell us what to build. Fejér's construction was the one we just gutted for parts. Runge identified the disease but not the cure. Faber proved the impossibility but offered no alternative. And Chebyshev — Chebyshev gave us the Fourier connection that made the whole argument possible, and he will, as we are about to discover, give the next man through the door the single most important tool in his proof. But the construction itself is not his to make.

So perhaps we'd have given in and accepted the ugly polynomial as the best we could do; at least, had we not acknowledged a chime from a man who, perhaps unrightfully, we imagine was sat somewhere near the back.

"May I?"

Round 3

We turn. Sergei Natanovich Bernstein has been watching from the back since the beginning.

Bernstein was born in 1880 into a Jewish family in Odessa — the same Russian Empire that Chebyshev (1821–1894) knew his entire life. His mother sent him to Paris, where he studied at the Sorbonne and, in 1904, at the age of twenty-four, solved Hilbert's 19th problem on the analytic nature of solutions to elliptic partial differential equations. It was the kind of debut that should have opened every door in European mathematics.

It did not open the doors in Russia. The Russian Empire did not recognize foreign doctoral degrees for the purpose of university appointment, so Bernstein — having already solved one of Hilbert's twenty-three problems — was required to defend a *second* doctoral thesis at Kharkov in 1913, on the very subject we are now discussing: the approximation of continuous functions by polynomials. The work that concerns us here, published in a two-page paper in the *Communications of the Kharkov Mathematical Society* in 1912, sits between these two dissertations. It is, by any reasonable measure, among the most consequential two pages in the history of analysis.

The years that followed were not kind. Bernstein axiomatized probability theory in 1917 — the first such axiomatization, predating Kolmogorov's measure-theoretic foundations by sixteen years — but his algebraic approach was eventually superseded. He survived the political purges that swept Soviet academia in the 1930s, was forced from Kharkov to the Academy of Sciences in Leningrad, and fled to Kazakhstan during the siege of Leningrad in 1941. He returned after the war, spent his remaining decades at the Steklov Institute in Moscow, and devoted the final years of his life to a labor that, given our narrative, borders on poetic: editing and annotating the collected works of Chebyshev.

The man whose inequality Bernstein is about to borrow spent his last years in Bernstein's care. We imagine the incarnation before us now is the 1912 version: thirty-two years old, eight years past the Sorbonne, still young enough to see something in the equispaced data that everyone else has overlooked. His training is in partial differential equations, but his instincts — as we are about to discover — are those of a probabilist. He does not introduce himself with a theorem or a formula. He asks a question.

"Give me back the equispaced data."

We — and Chebyshev, and Weierstrass himself — stare. The *equispaced* data? The data from Round 1? The data that Chebyshev just spent the entirety of Round 2 explaining was the problem?

"Yes," Bernstein says. "The evaluations $f(0), f(1/n), f(2/n), \dots, f(1)$. I won't be needing different nodes." He pauses, then turns to Chebyshev. "In fact — Pafnuty, if I may — I'd like to borrow your inequality."

Chebyshev, bewildered, nods.

Bernstein walks to the board. "You've been trying to build a polynomial that *passes through* the data," he says. "I'm going to build one that *averages* it." He writes:

Definition 3.3.9 (Bernstein Polynomial)

Let $f\colon [0,1] \to \mathbb{R}$ be any function. Then, the nth **Bernstein polynomial** of f is

$$B_n[f(x)] := \sum_{k=0}^{n} f\left(\frac{k}{n}\right) b_{n,k}(x), \qquad b_{n,k}(x) := \binom{n}{k} x^k (1-x)^{n-k}.$$

The functions $b_{0,n}, b_{1,n}, \dots, b_{n,n}$ are the **Bernestein basis polynomials** of degree n.

"Same data as Lagrange's basis," he says, tapping the $f(k/n)$ terms, "that you used in round 1: $f(0), f(1/n), \dots, f(1)$. I am not asking Weierstrass for anything he hasn't already given us — no integrals, no inner product, and no other information about f other than the $n+1$ point evaluations $f(k/n)$, $k = 0, 1, \dots, n$."

"But," we object, "those are not Lagrange basis functions. $b_{n,k}(x) \neq \ell_k(x)$, but the terms in $\mathcal{L}_n[f(x)]$ are $f(k/n)\ell_k(x)$, and those of $B_n[f(x)]$ are $f(k/n)b_{n,k}(x)$. These are *decisively different objects*, Bernstein — we've attached the *same coefficients* to a *different set of basis functions*."

"Precisely," Bernstein agrees. "And that is the entire point. Let me show you what my basis functions $b_{n,k}$ do that Lagrange's ℓ_k cannot." He writes two properties on the board:

Theorem 3.3.10 (Properties of $b_{n,k}$)

For a fixed $n \in \mathbb{N}$, let $b_{n,k}$, $k = 0, 1, \dots, n$ be the kth Bernstein basis function

$$b_{n,k}(x) = \binom{n}{k} x^k (1-x)^{n-k}.$$

Then $b_{n,k}$ satisfies:

(i) **(Nonnegativity)** For every $x \in [0,1]$ and every $k \in \{0, 1, \dots, n\}$,

$$b_{n,k}(x) \geq 0.$$

(ii) **(Partition of Unity)** For every $x \in [0,1]$,

$$\sum_{k=0}^{n} b_{n,k}(x) = 1.$$

Proof:

Nonnegativity follows by the nonnegativity of the binomial coefficient

$$\binom{n}{k} = \frac{n!}{k!(n-k)!} \geq 0$$

and from the fact that $x \in [0,1]$ satisfies $x^k \geq 0$ for every k.

That the weights $b_{n,k}$ form a partition of unity follows by the Binomial theorem:

$$(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^k b^{n-k},$$

we have

$$\sum_{k=0}^{n} b_{n,k}(x) = \sum_{k=0}^{n} \binom{n}{k} x^k (1-x)^{n-k} = (x + (1-x))^n = 1^n = 1.$$

‖

"And from these two properties alone," Bernstein observes, "we can deduce the property you mentioned earlier:"

Theorem 3.3.11 (Convexity of $B_n[f(x)]$)

For any function $f\colon [0,1] \to [m, M]$ bounded between m and M, the Bernstein polynomial $B_n f$ satisfies

$$m \leq B_n[f(x)] \leq M, \qquad \text{For all } x \in [0,1].$$

That is: $B_n[f(x)]$ is a **convex combination** of the data values $f(0), f(1/n), \ldots, f(1)$ at every point $x \in [0,1]$, and thus $B_n f$ inherits the range of f.

Proof:

Since $b_{n,k}(x) \geq 0$ and $\sum_{k=0}^{n} b_{n,k}(x) = 1$, we have

$$B_n[f(x)] = \sum_{k=0}^{n} f\left(\frac{k}{n}\right) b_{n,k}(x) \geq \sum_{k=0}^{n} m \cdot b_{n,k}(x) = m \sum_{k=0}^{n} b_{n,k}(x) = m \cdot 1 = m,$$

and entirely analogously, we obtain the upper bound

$$B_n[f(x)] \leq M \cdot 1 = M.$$

‖

"Recall your Cauchy's fangs," Bernstein says; "how, even though the DGP $h = \frac{1}{\pi(1+x^2)}$ took values in $[0, 1/\pi]$ exclusively, your interpolating polynomial deployed on equispaced nodes in $[-3, 3]$ developed those trademark fangs. Realize, then, that *my* polynomial is *structurally incapable* of the same error — the polynomial $B_n[f(x)]$ *cannot overshoot*. $B_n f$ will *never* develop fangs — it simply cannot, for any function, no matter its behavior."

"But," Chebyshev interjects from his seat — "$B_n[f(x)]$ doesn't pass through the data. Indeed, $\mathcal{L}_n[f(x)]$ passes through the data by construction, but we can observe that

$$B_n[f(x_k)] \neq f(x_k)$$

in general. YOur basis functions satisfy $b_{n,k}(j/n) \neq \delta_{kj}$ — the design matrix of your basis at the equispaced nodes is *not* the identity I_k. You've *abandoned interpolation*."

"Not entirely," counters Bernstein, "but *mostly*, yes; and that is *also* the point. Your diagnosis in item (ii) of the intermission was correct: interpolation consumes every degree of freedom on forcing $\mathcal{L}_n f$ to satisfy the $(n+1)$ *local* constraints $\mathcal{L}_n[f(x_k)] = f(x_k)$. But in turn, you leave *nothing* to control the polynomial's *global* behavior away from the given nodes. My polynomial doesn't pass through the data in general — instead, it *averages* them, weighted by the $b_{n,k}$, and the weights shift smoothly as x moves across $[0,1]$. At a given node $x_j = j/n$, the binomial weight $b_{n,k}(j/n)$ is largest at $k = j$ and decays as k moves away — so $B_n[f(j/n)]$ is a weighted average of *all* the data values, concentrated near $f(j/n)$ but drawing from its neighbors. The node doesn't 'own' its data value exclusively; it shares it with the neighborhood."

He pauses. "There is one exception, actually, and it is worth noting: at the *endpoints* $x = 0$ and $x = 1$, the binomial degenerates. $\mathrm{Bin}(n, 0)$ places all its weight on $k = 0$; $\mathrm{Bin}(n, 1)$ places all its weight on $k = n$. So $B_n[f(0)] = f(0)$ and $B_n[f(1)] = f(1)$ exactly — my polynomial *does* interpolate at the boundary, and it does so for free, without being asked. The endpoints are the only points where no averaging occurs." He glances at the interpolation figures still on the board. "It is perhaps not lost on you that the boundary is where your fangs lived."

"But then," we press, "how do you know the approximation *actually* converges anywhere other than the endpoints? If $B_n f$ doesn't even agree with f at the nodes, why should we believe it agrees with f *anywhere*, much less *uniformly everywhere*?"

Bernstein smiles. "Pafnuty, this is where I'll put your inequality to use." He turns back to the board. "Fix a point $x \in [0,1]$ and look at the weight

$$b_{n,k}(x) = \binom{n}{k} x^k (1-x)^{n-k}.$$

Does the expression look familiar to anyone?"

Chebyshev's eyes widen. Indeed, $b_{n,k}$ is nothing more than the **probability mass function** of a Binomial(n, x) random variable.

Definition 3.3.12 (Binomial Random Variable)

Suppose that for some fixed $p \in [0,1]$, a random variable X taking values in the dichotomous support $\mathcal{X} = \{0,1\}$ is distributed according to the law

$$\mathbb{P}(X = 1) = p, \qquad \mathbb{P}(X = 0) = 1 - p.$$

We say that X is a **Bernoulli** random variable with parameter $p \in [0,1]$, and we write

$$X \sim \mathrm{Bernoulli}(p),$$

where we can define the probability mass function of X in shorthand

$$p_X(x) := \mathbb{P}(X = x) = p^x (1-p)^{1-x}, \qquad x \in \{0,1\}.$$

(verify the equivalence to the law described by $\mathbb{P}(X=1)=p$, $\mathbb{P}(X=0)=1-p$!) Moreover, one verifies that the mean and variance of X are given by

$$\mathbb{E}[X] = \sum_{x} x p_X(x) = 0 p_X(0) + 1 p_X(1) = p,$$

and

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = p - p^2 = p(1-p).$$

Then, suppose we observe n such Bernoulli(p) trials sequentially, where the outcome of the kth trial bears no influence on the $(k+1)$th or any other (i.e., where the successive realizations $X_1, \dots, X_n$ of the Bernoulli(p) RV are **independent**). If we then define S_n to be the sum of these $X_i \overset{ind}{\sim}$ Bernoulli(p) — i.e., if

$$S_n := \sum_{i=1}^{n} X_i,$$

then we say that S_n is a **Binomial** random variable with parameters $n \in \mathbb{N}$ and $p \in [0,1]$, and we write

$$S_n \sim \mathrm{Binomial}(n,p), \qquad \Big(\text{or } S_n \sim \mathrm{Bin}(n,p)\Big)$$

where we can define the probability mass function of S_n accordingly as

$$p_{S_n}(x) = \mathbb{P}(S_n = x) = \binom{n}{x} p^x (1-p)^{n-x}, \qquad x \in \{0,1,\dots,n\}.$$

One can verify similarly to our computation above that the mean and variance of $S_n \sim \mathrm{Bin}(n,p)$ are

$$\mathbb{E}[S_n] = np, \qquad \mathrm{Var}(S_n) = np(1-p).$$

"So," Bernstein continues, "at a fixed $x \in [0,1]$, the Bernstein basis function $b_{n,k}(x)$ is *precisely* the probability that a Binomial(n,x) random variable S_n equals k:

$$b_{n,k}(x) = \mathbb{P}(S_n = k) = \binom{n}{k} x^k (1-x)^{n-k}, \qquad k \in \{0,1,\dots,n\}.$$

This is clearly not an analogy; it is a literal identification! The Bernstein basis function $b_{n,k}$ at the point x *is* the binomial probability mass function with parameter $p = x$." He lets this sit for a moment. "And so, my polynomial becomes

$$B_n[f(x)] = \sum_{k=0}^{n} f\left(\frac{k}{n}\right) \underbrace{\binom{n}{k} x^k (1-x)^{n-k}}_{=\mathbb{P}(S_n=k)} = \sum_{k=0}^{n} f\left(\frac{k}{n}\right) \mathbb{P}(S_n = k) = \mathbb{E}\left[f\left(\frac{S_n}{n}\right)\right],$$

where the last equality is simply the definition of the expected value of a function of a discrete random variable: if S_n takes values $k \in \{0,1,\dots,n\}$, then

$$\mathbb{E}[g(S_n)] := \sum_{k=0}^{n} g(k)\mathbb{P}(S_n = k).$$

"Thus," he pauses, "The Bernstein polynomial of x, $B_n[f(x)]$, is the **expected value** of f evaluated at the random variable S_n/n, where $S_n \sim \mathrm{Bin}(n,x)$. The polynomial that looked like an algebraic construction — a sum of binomial coefficients $\binom{n}{k}$ times powers of x and $1-x$ — is, secretly, an expectation."

The room has gone quiet; at this point, we're all invested in Bernstein's lecture. He continues, "so, we can see that by somewhat arbitrarily defining a random variable $S_n \sim \mathrm{Bin}(n, x)$ with the 'success probability' set to the point of evaluation $x \in [0, 1]$ itself, we obtain a remarkable clean expression:

$$B_n[f(x)] = \mathbb{E}\left[f\left(\frac{S_n}{n}\right) \,\middle|\, S_n \sim \mathrm{Bin}(n, x) \right].$$

"Now," he continues, "I prove the uniform convergence of $B_n f \rightrightarrows f$."

Theorem 3.3.13 (Bernstein's Theorem (Weierstrass by Construction))

Let $f\colon [0, 1] \to \mathbb{R}$ be continuous. Define the Bernstein polynomials

$$B_n[f(x)] := \sum_{k=0}^{n} f\left(\frac{k}{n}\right) b_{n,k}(x) = \mathbb{E}\left[f\left(\frac{S_n}{n}\right) \,\middle|\, S_n \sim \mathrm{Bin}(n, x) \right],$$

where the Bernstein basis functions $b_{n,k}$ are

$$b_{n,k}(x) := \binom{n}{k} x^k (1-x)^{n-k} = \mathbb{P}\left(S_n = k \,\middle|\, S_n \sim \mathrm{Bin}(n, x) \right).$$

Then, the Bernstein polynomials $B_n f$ converge uniformly to f on $[0, 1]$:

$$\|f - B_n f\|_\infty = \sup_{x \in [0,1]} |f(x) - B_n[f(x)]| \to 0 \quad \text{as} \quad n \to \infty;$$

in other words,

$$B_n f \rightrightarrows f \quad \text{uniformly on } [0, 1].$$

Proof:

Bernstein turns to the board. "The proof requires three ingredients — I believe you've established all of them previously in Chapter 2." (Of course Bernstein's been reading along!)

"But it's been a minute," he continues, "so allow me to remind you of them."

(i) **(Concentration)** Recall that $S_n \sim \mathrm{Bin}(n, x)$ is a sum of n independent Bernoulli(x) trials. Its mean and variance are

$$\mathbb{E}\left[\frac{S_n}{n}\right] = \frac{1}{n}\mathbb{E}[S_n] = \frac{1}{n}[nx] = x,$$

and

$$\mathrm{Var}\left(\frac{S_n}{n}\right) = \frac{1}{n^2}\,\mathrm{Var}(S_n) = \frac{1}{n^2}(nx(1-x)) = \frac{x(1-x)}{n}.$$

Now — Pafnuty, take note — we may apply Chebyshev's inequality (Theorem 2.4.7): for any $\delta > 0$,

$$\mathbb{P}\left(\left| \frac{S_n}{n} - x \right| > \delta \right) \le \frac{\mathrm{Var}(S_n/n)}{\delta^2} = \frac{x(1-x)}{n\delta^2}.$$

This says that "the probability $\mathbb{P}$ that a realization of the the random quantity S_n/n lands a distance greater than δ from the point of evaluation x, is controlled by the deterministic expression $\frac{x(1-x)}{n\delta^2}$ depending only on the point x. To remove this dependence, we can observe that, since $x(1-x) \leq 1/4$ for $x \in [0,1]$ (verify: the function $x \mapsto x(1-x) = x - x^2$ is a downward facing parabola with maximum $1/4$ at $x = 1/2$), we obtain the **uniform bound**

$$\mathbb{P}\left(\left|\frac{S_n}{n} - x\right| > \delta\right) \leq \frac{1}{4n\delta^2} \overset{n\to\infty}{\longrightarrow} 0.$$

In the language of Definition 2.4.8, this is precisely convergence in probability of S_n/n to x: i.e., we have that

$$\frac{S_n}{n} \overset{P}{\to} x.$$

The sample proportion S_n/n concentrates at the point of evaluation x — and, again, it is crucial here that the bound $1/4n\delta^2$ does not depend on x. The *same* n controls the concentration for *every* $x \in [0,1]$ simultaneously. Intuitively, this does seem to make sense — the point x is playing the role of the success probability, and this says that as $n \to \infty$, the *proportion of successes*, S_n/n, that we observe in the binomial sequence $S_n \sim \mathrm{Bin}(n,x)$ comprised of converges back to the *probability of a success* p as $n \to \infty$ (where, again, here we're identifying the parameter p as the point of evaluation x).

(ii) **(Continuous Mapping)** Since f is continuous on $[0,1]$ (the running assumption), and since we've verified that $S_n/n \overset{P}{\to} x$, the Continuous Mapping Theorem (Lemma 2.4.9(v)) gives

$$f\left(\frac{S_n}{n}\right) \overset{P}{\to} f(x).$$

That is: applying a continuous function f to a sequence of random variables that converges in probability like $S_n/n \overset{P}{\to} x$ preserves the convergence: $f(S_n/n) \overset{P}{\to} f(x)$. Since S_n/n is "getting close on average" to x and since there are "no gaps in the map $x \mapsto f(x)$," the "closeness carries over" without issue.

(Bounded Convergence) Since f is continuous on the *compact* (i.e., closed and bounded) interval $[0,1]$, there exists $M > 0$ such that $|f(t)| \leq M$ for all $t \in [0,1]$. In particular, this means that

$$\left|f\left(\frac{S_n}{n}\right)\right| \leq M \quad \text{for every realization of } S_n,$$

since $S_n/n \in [0,1]$ always. So the sequence $\{f(S_n/n)\}_{n=1}^{\infty}$ is uniformly bounded.

By Bounded Convergence (Lemma 2.4.9(vi)), convergence in probability of a uniformly bounded sequence implies convergence of the expectations:

$$B_n[f(x)] = \mathbb{E}\left[f\left(\frac{S_n}{n}\right) \,\middle|\, S_n \sim \mathrm{Bin}(n,x)\right] \longrightarrow \mathbb{E}[f(x) \mid S_n \sim \mathrm{Bin}(n,x)] = f(x),$$

where the last equality follows since since $f(x)$ is just a constant not involving the random quantity S_n being averaged in the expectation.

"Read that last chain again: $B_n[f(x)] \to f(x)$ at the point x. And since $x \in [0,1]$ was arbitrary, we have that $B_n[f(x)] \to f(x)$ pointwise on $[0,1]$."

He taps the board. "But I said something stronger: that $B_n f \rightrightarrows f$ *uniformly* on $[0,1]$, not merely pointwise. Indeed, this claim follows," gesturing towards the $x(1-x) \le 1/4$ bound, "since the concentration bound $1/4n\delta^2$ was *uniform in x*. We controlled the closeness of S_n/n to x in probability independently of x in establishing (i); and nothing in (ii) nor (iii) re-introduce the dependency. Thus, what we must conclude is that

$$\|f - B_n f\|_\infty = \sup_{x\in[0,1]} \Big| f(x) - B_N[f(x)] \Big| \to 0 \quad \text{as} \quad n \to \infty;$$

that is,

$$B_n f \rightrightarrows f \quad \text{uniformly on } [0,1].$$

$\|$

Bernstein sets down his chalk.

"Let me say that again, so that we are all very clear about what just happened. No Derivatives of f appear anywhere in this proof. No $f^{(n+1)}(\xi_x)$. No Taylor coefficients. No analyticity. No smoothness hypotheses of any kind. Oh, and Weierstrass" he revels, "hand me those evaluations of $W_{a,b}(0), W_{a,b}(1/n), \ldots W_{a,b}(1)$." Weierstrass reluctantly obliges, whereupon Bernstein quickly writes

$$B_n(W_{a,b}; x) = \sum_{k=0}^{n} W_{a,b}\left(\frac{k}{n}\right)\binom{n}{k} x^k (1-x)^{n-k}.$$

"Of course, $W_{a,b}$ is continuous on $[0,1]$, so the theorem applies: $\|W_{a,b} - B_n(W_{a,b})\|_\infty \to 0$ as $n \to \infty$." This nowhere-differentiable $W_{a,b}$ that caused Chebyshev such strife — a Fourier series of cosines at exponentially increasing frequencies — that rendered the entire interpolation program moot — has seemingly just been solved *not* by digging deeper into the deterministic troves of analysis, but by replacing the problem with something different entirely, something *probabilistic*.

And we compare, for a moment, what Bernstein has just produced with our Cesàro solution derived during the intermission. Our Fejér-truncate polynomial exists — we proved it — but it required Fourier integrals that the game never provided. It had no closed form, no attractive structural properties, and a rate of convergence buried under two layes of uncontrolled approximation. Bernstein's polynomial uses the *same equispaced data* Weierstrass handed us all the way back in Round 1: the point evaluations $f(0), f(1/n), \ldots, f(1)$, and nothing more. Its formula sits on a single line; it is a convex combination of the data values at every $x \in [0,1]$, so it inherits the range of f automatically, and its convergence proof *regained a semblance of intuitive simplicity* — it required three steps, Chebyshev's inequality, continuous mapping, and bounded convergence, and each were things we'd developed in a seemingly disparate context back in Chapter 2.

One might imagine that, during the intermission, we proved the theorem by hiring an orchestra and sending everyone home except the pianist. Bernstein walked directly to the piano.

Remark.

The attentive reader may notice something about the *rate* of convergence. The concentration bound was

$$\mathbb{P}\left(\left| \frac{S_n}{n} - x \right| > \delta \right) \le \frac{1}{4n\delta^2},$$

which vanishes like $O(1/n)$ at a fixed δ. One can show (and we leave the details to the exercises) that this yields an approximation rate of

$$\|f - B_n f\|_\infty = O\left(\omega_f\left(\frac{1}{\sqrt{n}}\right)\right),$$

where $\omega_f(\delta) := \sup_{|x-y|<\delta} |f(x) - f(y)|$ is the **modulus of continuity** of f — the worst-case oscillation of f over intervals of width δ.

These rates are, frankly, *terrible*. Chebyshev's interpolant achieved exponential convergence $O(2^{-n})$ for the Cauchy density on $[-1, 1]$ — and Bernstein's polynomial converges at the comparatively glacial rate $O(1/\sqrt{n})$. The tradeoff is stark: Chebyshev's method is fast but conditional (it requires smoothness of f), while Bernstein's method is slow but *unconditional* (it requires only continuity). The Bernstein polynomial pays for its universality with its rate.

And yet: the rate $1/\sqrt{n}$ should ring a bell. It is the *standard deviation* of S_n/n:

$$\mathrm{sd}\left(\frac{S_n}{n}\right) = \sqrt{\frac{x(1-x)}{n}} \leq \frac{1}{2\sqrt{n}} = O\left(\frac{1}{\sqrt{n}}\right).$$

The width of the averaging window that produces $B_n f$ is controlled by the standard deviation of the binomial, which is $O(1/\sqrt{n})$. The rate of polynomial approximation is the rate of probabilistic concentration. Bernstein's proof didn't just produce a polynomial — it revealed that the *speed* of approximation is governed by the *width* of a probability distribution. We know the width. What we don't yet know is the *shape*.

We note this observation, and we set it aside. The proof of uniform convergence did not need the shape — it needed only the *width*, which Chebyshev's inequality provided. The shape is there, but it will demand an explanation that simply cannot be answered using only the tools this chapter has developed. Indeed, answering them will require a way to characterize the *shape* of a distribution, not just its width.

But that is a story for the next chapter.

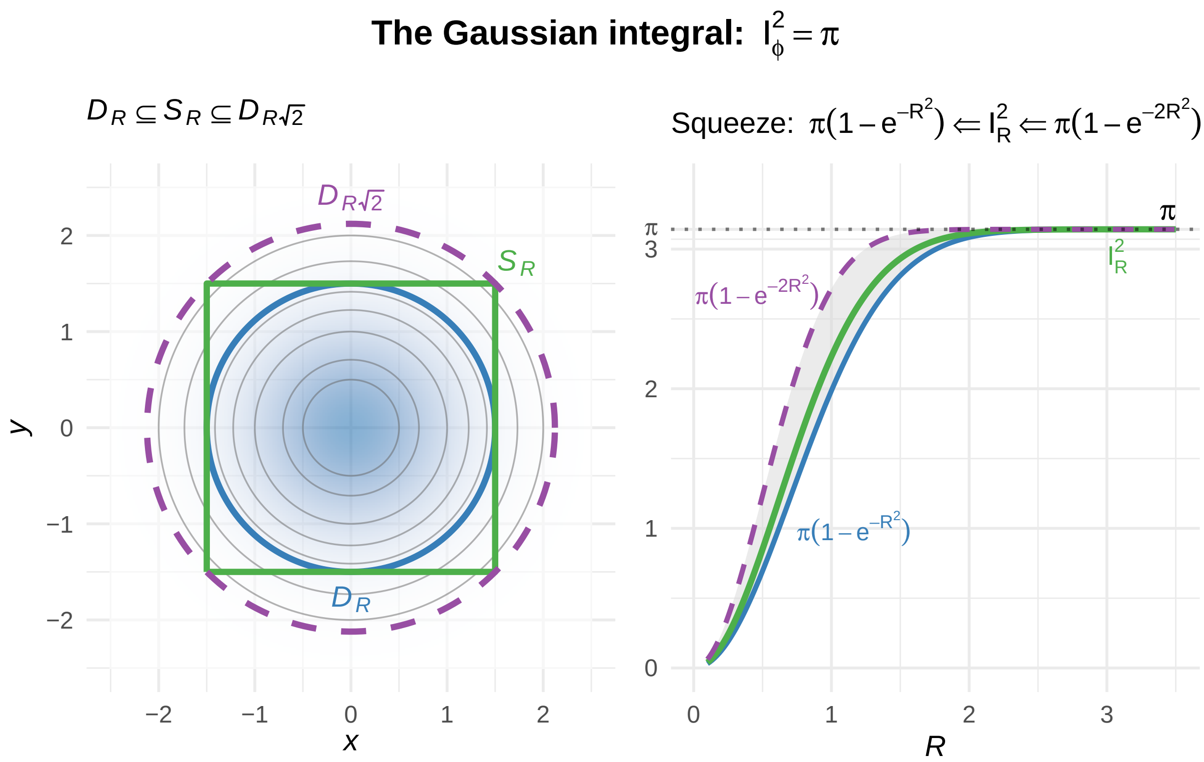
Bernstein allows himself a moment. Then, quietly: "Same equispaced data Weierstrass gave you in Round 1. Same evaluations $f(k/n)$ your Lagrange interpolant choked on. I used them as ingredients in an expected value instead of as hard constraints. The polynomial I built doesn't pass through any of the data points — and it converges uniformly to f for every continuous function on $[0, 1]$. Including," he glances at Weierstrass, "$W_{a,b}$."

He pauses, then adds: "And Pafnuty — thank you for the inequality. I couldn't have done it without you."

Round 3 goes to Bernstein. The Weierstrass Approximation Theorem is proved — by construction, from the data, in two pages and three lines of probability.

3.4 Guided Exercises

3.4.1 Some Important Preliminaries



Exercise 0: The Gaussian Integral

In this exercise, we compute an integral that will accompany us for the remainder of the book — namely, the **Gaussian integral**

$$\int_{-\infty}^{\infty} e^{-x^2}\, dx.$$

The integrand e^{-x^2} is strictly positive and supported on all of $\mathbb{R}$, which forces us to confront — perhaps for the first time — the question of what $\int_{-\infty}^{\infty} f(x)\, dx$ actually *means.*

(a) (Improper integrals of positive functions.) Suppose $f > 0$ is continuous on all of $\mathbb{R}$. When we write

$$\int_{-\infty}^{\infty} f(x)\, dx,$$

we mean the **improper Riemann integral**: the symbols ∞ and $-\infty$ are not elements of $\mathbb{R}$, and the notation is shorthand for a limit. But which limit?

There are (at least) three possible readings:

(i) **(Double limit.)** Take a genuine double limit in a and b simultaneously: we say that

$$\lim_{\substack{a\downarrow -\infty \\ b\uparrow +\infty}} \int_a^b f(x)\, dx = L$$

if, for every $\epsilon > 0$, there exists $N > 0$ such that $a < -N$ and $b > N$ imply

$$\left| \int_a^b f(x)\, dx - L \right| < \epsilon.$$

The key word is "whenever" — a and b are free to approach $-\infty$ and $+\infty$ at completely different rates. The double limit exists only if the integral stabilizes regardless of how a and b wander off to their respective infinities.

(ii) **(Independent limits.)** Split at any fixed $c \in \mathbb{R}$ and take each limit on its own:

$$\int_{-\infty}^{\infty} f(x)\,dx := \lim_{a\downarrow-\infty} \int_a^c f(x)\,dx \;+\; \lim_{b\uparrow+\infty} \int_c^b f(x)\,dx,$$

provided both limits exist and are finite.

(iii) **(Symmetric limit.)** Take a single limit along the symmetric path $[-R, R]$:

$$\int_{-\infty}^{\infty} f(x)\,dx := \lim_{R\to\infty} \int_{-R}^{R} f(x)\,dx.$$

For a general function f, these three readings need not agree. The symmetric limit (iii) can exist even when the independent limits (ii) do not: $\int_{-R}^{R} x\,dx = 0$ for every R, yet

$$\int_0^b x\,dx \to +\infty \quad \text{as} \quad b \uparrow +\infty.$$

(The value produced by (iii) in such cases is called the **Cauchy principal value** — it is "the safest way" to integrate on $\mathbb{R}$, since it gives positive and negative contributions a chance to cancel out. Why might odd functions in general produce "misleading" Cauchy principal values?)

There is a fundamental distinction, though, separating the "misleading" principal values produced by odd functions from those produced by strictly positive ones. For $f > 0$ continuous on all of $\mathbb{R}$, there is simply no negative contribution available to mask a divergence through cancellation. The convergence of the integral, if it occurs, is *genuine* — and, as we now argue, all three readings must agree.

(i) Fix $c = 0$ and define

$$G(a) := \int_a^0 f(x)\,dx, \quad a < 0; \qquad H(b) := \int_0^b f(x)\,dx, \quad b > 0.$$

Argue that, since $f > 0$ on all of $\mathbb{R}$, H is strictly increasing for $b \in (0, \infty)$ and G is strictly increasing as a decreases through $(-\infty, 0)$. (Hint: we are merely widening the window over which we integrate a strictly positive function.)

(ii) Suppose the symmetric limit (iii) converges:

$$\lim_{R\to\infty} \int_{-R}^{R} f(x)\,dx =: L < \infty.$$

Verify that $\int_{-R}^{R} f(x)\,dx = G(-R) + H(R)$. Since both G and H are nonnegative and their sum is bounded above by L, conclude that

$$G(-R) \le L \quad \text{and} \quad H(R) \le L \qquad \text{for all} \quad R > 0.$$

That is: each of G and H is individually bounded above.

(iii) A monotone increasing function that is bounded above converges. (This is the completeness of $\mathbb{R}$ — every bounded, monotone sequence has a limit, a fact we established in Chapter 1.) Conclude that

$$\lim_{a\downarrow-\infty} G(a) \quad \text{and} \quad \lim_{b\uparrow+\infty} H(b) \quad \text{exist and are finite.}$$

(iv) Verify that the independent limits (ii) now converge:

$$\lim_{a\downarrow-\infty} G(a) + \lim_{b\uparrow+\infty} H(b) = L.$$

Explain why the choice of splitting point $c = 0$ was immaterial. (Hint: for any $c \in \mathbb{R}$ and any $a < c < b$, we have $\int_a^c f + \int_c^b f = \int_a^b f$.)

(v) Finally, verify that the double limit (i) converges to the same value L. Write

$$\int_a^b f(x)\,dx = G(a) + H(b)$$

for any $a < 0 < b$. We have shown that $\lim_{a\to-\infty} G(a) =: L_1$ and $\lim_{b\to\infty} H(b) =: L_2$ each exist, with $L_1 + L_2 = L$. Let $\epsilon > 0$. By the convergence of G, there exists N_1 such that $a < -N_1$ implies $|G(a) - L_1| < \epsilon/2$. By the convergence of H, there exists N_2 such that $b > N_2$ implies $|H(b) - L_2| < \epsilon/2$. Set $N = \max(N_1, N_2)$. Then whenever $a < -N$ and $b > N$,

$$\left|\int_a^b f - L\right| = |(G(a) - L_1) + (H(b) - L_2)| \le |G(a) - L_1| + |H(b) - L_2| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon,$$

confirming that the double limit converges to $L = L_1 + L_2$.

Conclude: for $f > 0$ continuous on $\mathbb{R}$, the improper integral either converges under all three definitions simultaneously, or diverges under all three. We are therefore entitled to use whichever reading is most convenient. We say the improper integral **converges** when this is the case, and we adopt the symmetric truncation (iii) throughout.

(b) **(The Gaussian integral: setup.)** Let

$$\phi(x) := e^{-x^2}, \qquad x \in \mathbb{R}.$$

Since $\phi > 0$ on all of $\mathbb{R}$, the theory of part (a) applies: the improper integral $I_Z := \int_{-\infty}^{\infty} e^{-x^2}\,dx$ converges if and only if the symmetric truncations $I_R := \int_{-R}^{R} e^{-x^2}\,dx$ are bounded above.

(i) Verify that $e^{-x^2} \le e^{-|x|+1}$ for $|x| \ge 1$. (Hint: for $x \ge 1$, show that $x^2 \ge x - 1$.)

(ii) Using this bound, show that for $R > 1$,

$$I_R = \int_{-R}^{R} e^{-x^2}\,dx \le \int_{-1}^{1} e^{-x^2}\,dx + 2\int_1^R e^{-x^2}\,dx \le 2e + 2e\int_1^\infty e^{-x}\,dx = 4e.$$

Conclude that I_Z converges and we may compute it via $\lim_{R\to\infty} I_R$.

(c) **(A failed attempt.)** Before pursuing the correct approach, observe that the "obvious" strategy fails.

(i) Using the symmetry $e^{-(-x)^2} = e^{-x^2}$, verify that

$$I_Z = 2\int_0^\infty e^{-x^2}\,dx.$$

(ii) Apply the substitution $u = x^2$ to show that

$$2\int_0^\infty e^{-x^2}\,dx = \int_0^\infty u^{-1/2}\,e^{-u}\,du.$$

Explain why repeatedly applying integration by parts to this integral fails to produce a closed form: the power $u^{-1/2}$ shifts by 1 at each step but never reaches a form whose antiderivative against e^{-u} is elementary.

(d) **(The squaring trick.)** We now pursue the method that works. Define $\phi(y) := e^{-y^2}$ (i.e., relabel the dummy variable), and write

$$I_Z^2 = \left(\int_{-\infty}^{\infty} e^{-x^2}\, dx \right) \cdot \left(\int_{-\infty}^{\infty} e^{-y^2}\, dy \right).$$

Since I_Z converges (part (b)), we have $I_R \to I_Z$ and therefore $I_R^2 \to I_Z^2$.

(i) Show that

$$I_R^2 = \int_{-R}^{R} \int_{-R}^{R} e^{-(x^2+y^2)}\, dx\, dy.$$

(Hint: the inner integral $\int_{-R}^{R} e^{-x^2}\, dx$ is a finite number that does not depend on y, so it factors out of the outer integral as a constant. This is linearity of the Riemann integral on the compact domain $[-R, R]^2$.)

(ii) Examining the integrand, we observe the expression $e^{-(x^2+y^2)}$ depends on the pair (x, y) only through the combination $x^2 + y^2$ — the pair's squared distance from the origin. Indeed, verify that the **level sets** of $e^{-(x^2+y^2)}$ are circles centered at the origin: for any $0 < c < 1$,

$$\left\{ (x, y) \in \mathbb{R}^2 \colon\ e^{-(x^2+y^2)} = c \right\} = \left\{ (x, y) \colon\ x^2 + y^2 = -\ln c \right\},$$

which is a circle of radius

$$r := \sqrt{-\ln c} = \sqrt{\ln(1/c)}$$

(since $0 < c < 1$.) That is, the integrand $e^{-(x^2+y^2)}$ of I_R^2 is **radially symmetric**: it takes the same value at every point on a given circle, and furthermore, is decreasing for $\uparrow r \in (0, \infty)$.

The natural domain for a radially symmetric function is a **disk**: we say that

$$D_\rho := \{(x, y) \colon\ x^2 + y^2 \le \rho^2\}$$

is a disk of radius ρ — intuitively, you might imagine a disk as "a solid circle."

Remark.

More carefully, we note that when we say **circle**, what we really mean is a "hollow circumference," or even just "the boundary of a disk." The word "circle" is much more likely to be "invoked improperly," though we can often best discern "what one really means" simply through context.

To be clear, then: if someone "defines a circle the way we have defined a disk" or vice-versa, *but* are thorough enough in establishing their preferred notation such that there is hardly a chance for confusion, then it is not that they are *wrong*; they are just *different*. We get it; we're different too. (Just as well, if someone "uses our conventions without establishing the notation clearly," we *do* perceive them "to be wrong." There is hardly such a thing as an "established notation" in mathematics — we must *say what we mean* if we want to be understood.)

The issue, though, is that we've defined $I_R^2 = \iint_{-R}^{R} e^{-(x^2+y^2)}\,dxdy$ to integrate over the **square**

$$S_R := [-R, R] \times [-R, R];$$

not a disk. We direct you to the figure at the top of the exercise — right now, "we're the square."

But the figure *also* illuminates our strategy: if we can **squeeze** the square S_R in between two disks, say,

$$D_R := \{(x, y)\colon\ x^2 + y^2 \le R^2\}, \quad \text{and} \quad D_{R\sqrt{2}} := \{(x, y)\colon\ x^2 + y^2 \le 2R^2\},$$

then, since $e^{-(x^2+y^2)} \ge 0$ throughout, integrating over a larger domain can *only increase* the value of the integral:

$$\iint_{D_R} e^{-(x^2+y^2)}\,dxdy \le I_R^2 \le \iint_{D_{R\sqrt{2}}} e^{-(x^2+y^2)}\,dxdy.$$

(i) Verify the containments

$$D_R \subseteq S_R \subseteq D_{R\sqrt{2}}.$$

(Hint: if $x^2 + y^2 \le R^2$, then certainly $|x| \le R$ and $|y| \le R$. Conversely, if $|x| \le R$ and $|y| \le R$, then $x^2 + y^2 \le 2R^2$.)

(e) On a disk, the radial symmetry pays off. We want to evaluate

$$\iint_{D_\rho} e^{-(x^2+y^2)}\,dxdy$$

by exploiting the fact that the integrand depends only on the distance $r = \sqrt{x^2 + y^2}$ from the origin.

(i) Thinking geometrically, we might imagine decomposing D_r into **shells** (annular rings) of radius r and infinitesimal width dr, on each of which the value of $e^{-(x^2+y^2)} = e^{-r^2}$ is roughly constant (essentially a level set). The value of the integrand e^{-r^2} depends only on $r<$ and within a shell of infinitesimal width, r barely changes.

Convince yourself why the area ΔA of such a thin shell at radius r with width Δr is approximately

$$\Delta A \approx \underbrace{2\pi r}_{\text{circumference}} \cdot \Delta r.$$

Yeah — make it feel like a Riemann sum. Here's the cleanup and continuation:

(ii) Conclude that the integral over the disk D_ρ can be approximated by summing the integrand's value on each shell, weighted by the shell's area:

$$\iint_{D_\rho} e^{-(x^2+y^2)}\,dx\,dy \;\approx\; \sum_{\substack{\text{shells} \\ \Delta A \subset D_\rho}} e^{-r^2} \cdot 2\pi r\,\Delta r \;\xrightarrow{\Delta r \downarrow 0}\; \int_0^\rho e^{-r^2} \cdot 2\pi r\,dr.$$

The factor of r in the integrand is not a technicality — it *is* the geometry of the circle. Shells at larger radius have more circumference, hence more area at the same width Δr. A shell at $r = 2$ contributes twice the area of a shell at $r = 1$: the "weight" of each shell grows linearly in r, and this linear growth is the factor the integral has been missing.

(iii) Evaluate:
$$\int_0^\rho 2\pi r\, e^{-r^2}\, dr = 2\pi \cdot \frac{1}{2}\left(1 - e^{-\rho^2}\right) = \pi\left(1 - e^{-\rho^2}\right),$$
using the substitution $u = r^2$, $du = 2r\,dr$. (The substitution $u = r^2$ that failed in part (c) succeeds here because the shell factor r supplies exactly the missing piece of $du = 2r\,dr$. We did not produce this factor by algebraic cleverness — the circumference of a circle at radius r put it there.)

Remark.

The formal name for the shell factor r is the **Jacobian** of the polar coordinate transformation $x = r\cos\theta$, $y = r\sin\theta$. In general, a change of coordinates $(x, y) \mapsto (u, v)$ rescales infinitesimal areas by the factor
$$\left|\det\begin{pmatrix} \partial x/\partial u & \partial x/\partial v \\ \partial y/\partial u & \partial y/\partial v \end{pmatrix}\right|,$$
which, for polar coordinates, evaluates to r. The determinant of this matrix measures how much the area of an infinitesimal patch changes under the coordinate map — the same quantity the shell argument computed by geometry. The Jacobian is the shell area, expressed algebraically.

More broadly: the determinant of a matrix tells you how a linear map *scales area* (or volume/hypervolume, in higher dimensions). The Jacobian matrix of a coordinate transformation is the linear map that best approximates the transformation at each point, and its determinant is how much area is locally stretched or compressed. This is why "determinant of a matrix" in linear algebra and "Jacobian of a change of variables" in calculus are the same concept — both answer the question "how does area change under this map?"

(f) (Assembling the squeeze) Using the disk evaluation from the previous part with $\rho = R$ and $\rho = R\sqrt{2}$, show that
$$\pi\left(1 - e^{-R^2}\right) \le I_R^2 \le \pi\left(1 - e^{-2R^2}\right).$$
Take $R \to \infty$ and conclude:
$$I_Z^2 = \pi, \qquad \text{whence} \qquad I_Z = \int_{-\infty}^{\infty} e^{-x^2}\, dx = \sqrt{\pi}.$$

(g) (The standard normal density) At this point, we've essentially finished what we set out to do: we've shown that the Gaussian integral $I_Z = \int_{-\infty}^{\infty} e^{-x^2}\, dx$ evaluates to $\sqrt{\pi}$. In particular, the function
$$f(x) := \frac{1}{\sqrt{\pi}} e^{-x^2}, \qquad x \in \mathbb{R},$$
is strictly positive and integrates to 1 on $\mathbb{R}$ — though we've extended beyond compact intervals $[a, b]$ to the unbounded $\mathbb{R}$, it would *seem like* f constitutes a perfectly valid probability density function.

Indeed, we tell you now (and demonstrate later) that this is true: f *is* a probability density on $\mathbb{R}$. What is perhaps somewhat peculiar, then, is why we often work with *another* form of the *same* density:

$$f_Z(z) := \frac{1}{\sqrt{2\pi}} e^{-z^2/2}, \qquad z \in \mathbb{R}.$$

Indeed, the probabilist often prefers working with f_Z; let us try to see why.

(i) Verify that $f_Z(z) > 0$ for all $z \in \mathbb{R}$ and that

$$\int_{-\infty}^{\infty} f_Z(z)\, dz = 1.$$

(Hint: substitute $x = z/\sqrt{2}$ and use the result you just derived, $I_Z = \sqrt{\pi}$.)

(ii) Verify that the first moments of f and f_Z each vanish: we have

$$\int_{-\infty}^{\infty} x f(x)\, dx = 0, \qquad \int_{-\infty}^{\infty} z f_Z(z)\, dz = 0.$$

Conclude that if X/Z are random variables distributed according to f and f_Z respectively (i.e., $X \sim f(x)$ and $Z \sim f_Z(z)$), then

$$\mathbb{E}_X[X] = 0, \qquad \text{and} \qquad \mathbb{E}_Z[Z] = 0.$$

(Hint: either recognize the integral is over a symmetric integral and argue by evenness/oddness of the integrands $xf(x)/zf_Z(z)$, or substitute $u = x^2/v = z^2/2$. The former approach is easier.)

(iii) Compute the second moment of the original f directly: using integration by parts with

$$u = x, \qquad dv = xe^{-x^2},$$

show that

$$\int_{-\infty}^{\infty} x^2 e^{-x^2}\, dx = \frac{1}{2}\int_{-\infty}^{\infty} e^{-x^2}\, dx = \frac{\sqrt{\pi}}{2}.$$

Conclude the second moment of a random variable $X \sim f(x)$ is

$$\mathbb{E}_X[X^2] = \frac{1}{\sqrt{\pi}} \cdot \frac{\sqrt{\pi}}{2} = \frac{1}{2}.$$

(iv) Starting from

$$\int_{-\infty}^{\infty} x^2 e^{-x^2}\, dx = \frac{\sqrt{\pi}}{2},$$

again substitute $x = z/\sqrt{2}$ to ultimately verify that

$$\int_{-\infty}^{\infty} z^2 e^{-z^2/2}\, dz = \sqrt{2\pi}.$$

Conclude the second moment of a random variable $Z \sim f_Z(z)$ is

$$\mathbb{E}_Z[Z^2] = \frac{1}{\sqrt{2\pi}} \cdot \sqrt{2\pi} = 1.$$

(v) Recalling the variance identity

$$\mathrm{Var}(U) = \mathbb{E}[U^2] - (\mathbb{E}[U])^2,$$

use the results derived above to conclude that if $X \sim f(x)$ and $Z \sim f_Z(z)$, then

$$\mathrm{Var}(X) = \frac{1}{2}, \qquad \text{while} \qquad \mathrm{Var}(Z) = 1.$$

(vi) Together, we observe that

$$X \sim f(x) = \frac{1}{\sqrt{\pi}} e^{-x^2} \implies \mathbb{E}[X] = 0, \quad \mathrm{Var}(X) = \frac{1}{2},$$

while

$$Z \sim f_Z(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2} \implies \mathbb{E}[Z] = 0, \quad \mathrm{Var}(Z) = 1.$$

(h) (Gaussian scale-family) Both f and f_Z above are members of the more general class of the **Gaussian scale-family** of densities,

$$f_a(x) := \sqrt{\frac{a}{\pi}} e^{-ax^2}, \qquad a > 0,$$

parameterized by a **single parameter** $a > 0$ that "controls the width of the bell."

(i) Show that f_a integrates to 1 for every $a > 0$. (Hint: substitute $u = \sqrt{a}z$ and use $I_Z = \sqrt{\pi}$.)

(ii) Compute

$$\mathrm{Var}(U) = \frac{1}{2a}, \qquad U \sim f_a.$$

(iii) Conclude that the *unique choice* of $a > 0$ yielding $\mathrm{Var}(U) = 1$ is $a = 1/2$. Whence,

$$\mathrm{Var}(U) = 1, \quad U \sim f_a \iff a = 1/2, .$$

so that $U \sim f_{1/2}(x) \equiv f_Z(z)$.

(i) (Gaussian location-family) Observe that for *every* $a > 0$, we have $\mathbb{E}[U] = 0$ whenever $U \sim f_a$. That is: the scale parameter $a > 0$ controls the *width* of the rescaled Gaussian, but *not its center*. As such, given a Gaussian random variable of unit variance

$$Z \sim f_Z(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2},$$

suppose we substitute

$$U := Z + \mu.$$

(i) Verify that

$$\mathbb{E}_U[U] = \mathbb{E}_Z[Z] + \mu = \mu,$$

while

$$\mathrm{Var}_U(U) = \mathrm{Var}_Z(Z) = 1.$$

(Hint: if c is a constant, then $\mathbb{E}[X + c] = \mathbb{E}[X] + c$ by linearity, while $\mathrm{Var}(X + c) = \mathrm{Var}(X)$ by consequence of a cancellation in the definition $\mathrm{Var}(X + c) = \mathbb{E}[(X + c)^2] - (\mathbb{E}[X + c])^2$, as one should verify.)

(ii) Conclude that within the Gaussian **location-family**

$$f_\mu(z) := \frac{1}{\sqrt{2\pi}} e^{-(z-\mu)^2/2}, \qquad \mu \in \mathbb{R}$$

we have

$$\mathbb{E}[U] = \mu, \qquad \operatorname{Var}(U) = 1 \qquad \text{for any} \qquad U \sim f_\mu.$$

(j) (Gaussian location-scale family) It is only natural to *combine* our observations on scale- and location-families to derive a class of even greater generality.

In particular: given $Z \sim f_Z$ — centered with $\mathbb{E}[Z] = 0$ and of unit variance $\operatorname{Var}(Z) = 1$ — define

$$X := \sigma Z + \mu, \qquad \mu \in \mathbb{R}, \qquad \sigma > 0.$$

(i) Verify that

$$\mathbb{E}[X] = \mu \in \mathbb{R}, \qquad \operatorname{Var}(X) = \sigma^2 > 0.$$

(Hint: For the mean, we have $\mathbb{E}[aX + b] = a\mathbb{E}[X] + b$ by linearity. For the variance, use $\operatorname{Var}(aX + b) = a^2 \operatorname{Var}(X)$.)

(ii) Using the change of variables

$$Z = \frac{X - \mu}{\sigma},$$

show that the density of $X = \sigma Z + \mu$ is

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(\frac{-(x-\mu)^2}{2\sigma^2}\right), \qquad \mu \in \mathbb{R}, \qquad \sigma^2 > 0.$$

(Hint: $\mathbb{P}(X \leq x) = \mathbb{P}\left(Z \leq \frac{x-\mu}{\sigma}\right)$. Apply the fundamental theorem of calculus: differentiate $\frac{d}{dx}\mathbb{P}(X \leq x) = f_X(x)$, and apply the chain rule on the right side to obtain

$$f_X(x) = \frac{1}{\sigma} f_Z\left(\frac{x-\mu}{\sigma}\right).$$

)

We now formalize what we've established. In the compact-interval setting, we called a continuous, positive function $w > 0$ on $[a, b]$ with $\int_a^b w = 1$ a probability density. We extend the definition in the obvious way.

Definition 3.4.1 (Probability Density (Positive on $\mathbb{R}$))

A continuous, positive function $f \colon \mathbb{R} \to (0, \infty)$ satisfying

$$\int_{-\infty}^{\infty} f(x)\, dx = 1$$

(in the sense of the improper integral from part (a)) is a **probability density on** $\mathbb{R}$. We write $X \sim f$ to mean that X is a random variable distributed according to f, with CDF

$$F(x) := \mathbb{P}(X \leq x) = \int_{-\infty}^{x} f(t)\, dt, \qquad x \in \mathbb{R}.$$

The density $f_Z(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$ derived above is called the **standard normal** or **standard Gaussian** density. We write $Z \sim f_Z$ or, more commonly,

$$Z \sim N(0,1),$$

to mean Z is a standard normal random variable with CDF

$$F_Z(z) := \mathbb{P}(Z \le z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{z} e^{-t^2/2}\, dt, \qquad z \in \mathbb{R}.$$

More generally, we write

$$X \sim N(\mu, \sigma^2), \qquad \mu \in \mathbb{R}, \qquad \sigma^2 > 0,$$

to mean X is a **normal** (or **Gaussian**) random variable with mean μ and variance σ^2, distributed according to the density

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \qquad x \in \mathbb{R}.$$

The standard normal $Z \sim N(0,1)$ sits at the origin of the parameterization — $\mu = 0$, $\sigma = 1$ — and every other member is obtained from it by $X = \sigma Z + \mu$. Conversely, any $X \sim N(\mu, \sigma^2)$ can be **standardized** via

$$Z = \frac{X-\mu}{\sigma} \sim N(0,1).$$

This invertibility is why we chose $\sigma^2 = 1$ as our reference point: nothing is lost by working with the standard normal, since the general case is always a substitution away.

Exercise 1: The Stieltjes Pairing

In Chapter 2, we introduced the Riemann-Stieltjes integral

$$\int_a^b f(t)\, d\alpha(t)$$

as the limit of sums

$$\sum_{i=0}^{k-1} f(c_i)\Delta\alpha_i, \qquad \Delta\alpha_i := \alpha(t_{i+1}) - \alpha(t_i),$$

where $\{a = t_0, t_1, \dots, t_k = b\}$ partitions $[a,b]$, each $c_i \in [t_i, t_{i+1}]$, and α is an integrator of bounded variation on the same interval. The point of that construction was precisely that integration need not occur only with respect to dt: under the Stieltjes framework, the integrator α may be piecewise monotone, or even a step-function.

In this chapter, we constructed an orthogonal basis for Π_{n-1}, the polynomials of degree at most $n-1$, by applying the Gram-Schmidt procedure to the monomial basis

$$\mathcal{M}_{n-1} = \{1, x, x^2, \dots, x^{n-1}\}$$

under the **discrete inner product** on the data nodes $x_1, \dots, x_n \in \mathcal{D}_n$:

$$\langle f, g\rangle_n := \sum_{i=1}^{n} f(x_i)g(x_i).$$

We now cast this construction into the more general Stieltjes framework.

Throughout, let $\mathcal{X} \subseteq \mathbb{R}$ be a common domain, let

$$\alpha \colon \mathcal{X} \to \mathbb{R}$$

be monotone increasing, and define

$$\langle f, g \rangle_\alpha := \int_{\mathcal{X}} f(t)g(t)\, d\alpha(t)$$

whenever the integral exists. We will call this the **Stieltjes pairing** of f and g induced by α.

(a) Let $(V, \mathbb{R})$ be any vector space of real-valued functions defined on a common domain $\mathcal{X}$ for which $\langle f, g \rangle_\alpha$ exists for all $f, g \in V$. Verify that $\langle \cdot, \cdot \rangle_\alpha \colon V \times V \to \mathbb{R}$ defines a valid **semi-inner product**; i.e., that it is:

(i) **(Symmetric)** For any $f, g \in V$,

$$\langle f, g \rangle_\alpha = \langle g, f \rangle_\alpha.$$

(ii) **(Bilinear)** For any $f, g, h \in V$ and constants $c_1, c_2 \in \mathbb{R}$,

$$\langle c_1 f + c_2 g, h \rangle_\alpha = c_1 \langle f, h \rangle_\alpha + c_2 \langle g, h \rangle_\alpha.$$

(iii) **(Positive Semi-Definite)** For any $f \in V$,

$$\langle f, f \rangle_\alpha = \int_{\mathcal{X}} f(t)^2\, d\alpha(t) \geq 0,$$

and

$$f \equiv 0 \in V \implies \langle f, f \rangle_\alpha = 0 \in \mathbb{R}.$$

Conclude that $\langle \cdot, \cdot \rangle_\alpha$ behaves like an inner product in every respect except possible one: it may fail the **definiteness** condition,

$$\langle f, f \rangle_\alpha = 0 \in \mathbb{R} \overset{???}{\implies} f \equiv 0 \in V.$$

(b) **(Why Definiteness May Fail)** To illustrate, suppose $\mathcal{X} = [-1, 1]$, and let α_0 be the step-function integrator with a single jump of size 1 at $t = 0$:

$$\alpha_0(t) := \begin{cases} 0, & t < 0, \\ 1, & t \geq 0. \end{cases}$$

(i) Consider the case when $V = C([-1, 1])$, the space of continuous functions on $[-1, 1]$. Show that for $f, g \in C([-1, 1])$,

$$\langle f, g \rangle_{\alpha_0} = f(0)g(0).$$

(ii) Identify a nontrivial continuous function $f \not\equiv 0$ on $[-1, 1]$ such that

$$\langle f, f \rangle_{\alpha_0} = 0.$$

(iii) Explain the meaning of this failure in words. (Hint: The integrator α_0 only ”looks at” the value of a function at one point, so on the large space $V = C([-1, 1])$, $\langle \cdot, \cdot \rangle_{\alpha_0}$ cannot distinguish many nonzero functions from zero.)

(c) (Reducing the Space) Now let $x_1, \dots, x_n \in \mathbb{R}$ be distinct nodes, and let α_n be the step-function integrator with positive jumps $w_1, \dots, w_n > 0$ at these nodes. That is,

$$\alpha_n(t) := \sum_{j=1}^{n} w_j \mathbf{1}_{\{x_j \le t\}},$$

where $\mathbf{1}_{\{x_j \le t\}}$ is the indicator taking value 1 whenever the jth data node x_j is smaller than the input $t \in [-1, 1]$.

(i) Show that for any functions $f, g \in V$ (now a general vector space),

$$\langle f, g \rangle_{\alpha_n} = \sum_{j=1}^{n} w_j f(x_j) g(x_j).$$

(ii) Consider the case when $V = C(\mathbb{R})$, the space of continuous functions on $\mathbb{R}$. Verify that $\langle \cdot, \cdot \rangle_{\alpha_n} : V \times V \to \mathbb{R}$ need not be definite — in particular, identify a function $f \in C(\mathbb{R})$ such that

$$\langle f, f \rangle_{\alpha_n} = 0, \quad \text{even though} \quad f \not\equiv 0 \text{ on } \mathbb{R}.$$

(iii) Now restrict the pairing to the polynomial space Π_{n-1}. Prove that if $p \in \Pi_{n-1}$,

$$\langle p, p \rangle_{\alpha_n} = 0 \implies p \equiv 0 \text{ on } \Pi_{n-1}.$$

(Hint: since each $w_j > 0$, the equality

$$\sum_{j=1}^{n} w_j p(x_j)^2 = 0$$

forces $p(x_j) = 0$ for every $j = 1, \dots, n$. Why does a polynomial of degree at most $n-1$ that vanishes at n distinct points have to be the zero polynomial?)

(iv) Conclude that, even though $\langle \cdot, \cdot \rangle_\alpha$ is *not* an inner product on the space $V \times V$ when $V = C(\mathbb{R})$, it *is* an inner product on $V \times V$ when $V = \Pi_{n-1}$.

Remark.

In the parable, we found that, given only $n = 2$ data points $\mathcal{D}_2 = \{(0, 2), (1, 3)\}$, the class of quadratic models was too large to produce a unique interpolating polynomial — for instance, both $x^2 + 2$ and $x + 2$ are polynomials in Π_2 that interpolate $\mathcal{D}_2$ exactly. Uniqueness required *reducing* the model class to the linear Π_1, whereupon the data *could* completely determine the unique interpolating polynomial $x + 2 \in \Pi_1$ of the data $\mathcal{D}_2$.

Compare this with the situation at hand. The step-function integrator α_n with n jumps "sees" exactly n values of any function. On the large space $C(\mathbb{R})$, this information is insufficient to enforce definiteness — many nonzero continuous functions vanish at all n nodes. But on Π_{n-1}, the n node values are exactly enough: a polynomial of degree $n - 1$ that vanishes at n distinct points must be identically zero.

(d) **(Continuous integrators)** We now consider integrators $\alpha\colon \mathcal{X} \to \mathbb{R}$ that are **continuously differentiable** and **strictly increasing** on $\mathcal{X}$. In this case, we have

$$\alpha'(t) > 0, \qquad \forall t \in \mathcal{X},$$

and the Stieltjes integral $\int fg d\alpha$ reduces to the weighted integral,

$$\langle f, g\rangle_\alpha = \int_{\mathcal{X}} f(t)g(t)\,\alpha'(t)\,dt.$$

(i) Show that, unlike the previous case (when α could be, say, a step-function), $\langle\cdot,\cdot\rangle_\alpha\colon V \times V \to \mathbb{R}$ *is* a genuine inner product, *even when* we take $V = C(\mathcal{X})$ to be the space of all continuous functions on $\mathcal{X}$ — not just on polynomial spaces.

(Hint: It is clear that the only thing we need to show here is that definiteness follows from the new condition that α be continuously differentiable. We outline the argument, which proceeds by contradiction. Suppose $f \not\equiv 0$.)

- Since $f \not\equiv 0$, choose $x_0 \in \mathcal{X}$ with $|f(x_0)| = c > 0$.
- Since f is continuous, there exists a δ-neighborhood $N_\delta(x_0) := (x_0 - \delta, x_0 + \delta) \cap \mathcal{X}$ on which $|f(x)| > c/2$, whence $f(x)^2 > c^2/4$.
- Since $w = \alpha' > 0$ is continuous on $\mathcal{X}$, it is bounded below by some $\epsilon > 0$ on $\overline{N_\delta(x_0)}$.
- Assemble:

$$\langle f, f\rangle_\alpha = \int_{\mathcal{X}} f(t)^2\,w(t)\,dt \geq \int_{N_\delta(x_0)} f(t)^2\,w(t)\,dt \geq \frac{c^2\epsilon}{4}\cdot \mathrm{len}(N_\delta(x_0)) > 0.$$

This contradicts $\langle f, f\rangle_\alpha = 0$, so $f \equiv 0$.

Remark.

Compare: the step-function integrator α_n is definite on Π_{n-1} but not on $C(\mathcal{X})$. The strictly increasing, continuously differentiable integrator α is definite on *all* of $C(\mathcal{X})$. The difference is that α_n samples f at finitely many points, while α — being strictly increasing — "sees" f everywhere. A continuous, strictly positive weight leaves no gaps for a nonzero function to hide.

Exercise 2: Chebyshev Orthogonality

In Exercise 1, we established that a continuous, strictly positive weight $w\colon \mathcal{X} \to (0, \infty)$ induces a *genuine* inner product on $C(\mathcal{X})$, the space of continuous functions on $\mathcal{X}$, via the Stieltjes pairing

$$\langle f, g\rangle_w := \int_{\mathcal{X}} f(t)g(t)\,w(t)\,dt.$$

We now use this framework to verify our first concrete orthogonal family of polynomials.

Indeed, the Chebyshev polynomials live something of a double life — in this exercise, we'll examine both. On the one hand, each T_n is a *polynomial* — an element of Π_n, defined on all of $\mathbb{R}$ by the three-term recurrence

$$T_{n+1}(x) = 2xT_n(x) - T_{n-1}(x), \qquad x \in \mathbb{R}, \qquad n = 0, 1, 2, \ldots$$

where $T_0(x) := 1$ and $T_1(x) := x$.

On the other hand, whenever $x \in [-1, 1]$, we may write $x = \cos\theta$ for some $\theta \in [0, \pi]$, and the Intermission's striking identity

$$T_n(\cos\theta) = \cos(n\theta), \qquad \theta \in [0, \pi],$$

reveals that *on this subinterval*, T_n is a disguised Fourier cosine. The polynomial lives on $\mathbb{R}$; the cosine identity lives on $[-1, 1]$.

The exercise that follows takes place entirely on $\mathcal{X} = [-1, 1]$, where both descriptions are available. Indeed, recall from Chapter 2 that the Fourier cosines are **orthogonal** on $[0, \pi]$ under the usual Euclidean inner product, meaning that

$$\langle \cos(m\theta), \cos(n\theta) \rangle_{L^2} := \int_0^\pi \cos(m\theta)\cos(n\theta)\, d\theta = 0, \qquad m \neq n.$$

In light of the identity $T_k(\cos\theta) = \cos(k\theta)$, it is only natural to wonder whether there exists a weight function w such that the Stieltjes pairing of T_n and T_m under $\langle \cdot, \cdot \rangle_w$ are orthogonal for every $n \neq m$?

(a) In light of the cosine orthogonality conditions $\langle \cos(m\theta), \cos(n\theta) \rangle_{L^2} = 0$, one might hope that the "orthogonality of T_m and T_n on $\mathcal{X} = [-1, 1]$" would reduce to nothing more than the analogous *unweighted* condition,

$$\langle T_m(x), T_n(x) \rangle_{L^2} := \int_{-1}^1 T_m(x) T_n(x)\, dx \overset{??}{=} 0, \qquad m \neq n.$$

Under the substitution $x = \cos\theta$ with $\theta \in [0, \pi]$, show that this integral becomes

$$\langle T_m(\cos\theta), T_n(\cos\theta) \rangle_{L^2} = \int_0^\pi \cos(m\theta)\cos(n\theta)\, \sin(\theta)\, d\theta.$$

Compute this integral directly for $m = 0, n = 2$ (Hint: use the product-to-sum identity (P1),

$$\cos A \cos B = \frac{\cos(A - B) + \cos(A + B)}{2},$$

and verify the integral

$$\int_{-1}^1 T_m(x) T_n(x)\, dx = -\frac{2}{3} \neq 0.$$

Conclude that T_m and T_n are *not* orthogonal under the unweighted Euclidean inner product $\langle \cdot, \cdot \rangle_{L^2}$.

(b) The culprit in part (a) is that the **Jacobian** of the substitution $x = \cos\theta$ — i.e., the factor we obtain when we susbtitute $dx = -\sin(\theta)\, d\theta$ — "contaminates" the cosine orthogonality. Really, though, if we could just "get rid of the factor of $\sin\theta$," we *would* reduce the problem back to the known case of cosine orthogonality. In light of this, we reconsider the more general form of the Stieltjes pairings

$$\langle T_m, T_n \rangle_w := \int_{-1}^1 T_m(x) T_n(x)\, w(x)\, dx,$$

where $w \colon (-1, 1) \to (0, \infty)$ is a continuous, strictly positive weight function. Show that under the same substitution $x = \cos\theta$ as part (a), the integral on the right above becomes

$$\int_0^\pi \cos(m\theta)\cos(n\theta) \cdot w(\cos\theta) \sin\theta\, d\theta.$$

(c) Verify that in order to integral above to reduce the above — up to a multiplicative constant — to the Fourier cosine inner product $\int_0^\pi \cos(m\theta)\cos(n\theta)\,d\theta$, the weight function w must satisfy

$$w(\cos\theta)\sin\theta = C, \qquad \theta \in (0,\pi).$$

Using $\sin^2\theta + \cos^2\theta = 1$ and $x = \cos\theta$, conclude that w must take the form

$$w(x) = \frac{C}{\sqrt{1-x^2}}, \qquad x \in (-1,1),$$

for some constant $C > 0$.

(d) We can choose the constant C in one of two ways: fairly arbitrarily, likely just taking $C = 1$ with the resulting*unnormalized* Stieltjes pairing $\langle\cdot,\cdot\rangle_w$ lacking a concrete interpretation; or fairly deliberately, taking C such that w integrates to 1 on $(-1,1)$, so that w is a *probability density* and the resulting*normalized* Stieltjes pairing $\langle\cdot,\cdot\rangle_w$ acquires a *probabilistic* interpretation. Throughout the book, we prefer the second choice for many good reasons, many of which will reveal themselves here.

As such, using the substitution $x = \cos\theta$, show that

$$\int_{-1}^{1} \frac{1}{\sqrt{1-x^2}}\,dx = \int_0^\pi d\theta = \pi,$$

and conclude that the unique choice of the **normalizing constant** $C > 0$ for which $\int_{-1}^{1} w(x)\,dx = 1$ is $C = 1/\pi$.

(e) Thus, by setting

$$w_A(x) := \frac{1}{\pi\sqrt{1-x^2}}, \qquad x \in (-1,1),$$

the weight $w_A \geq 0$ with $\int w_A = 1$ satisfies all of the properties of a probability density function. By definition, the CDF corresponding with w_A (on the nontrivial part of its support) is

$$\alpha_A(x) := \int_{-1}^{x} w_A(t)\,dt = \int_{-1}^{x} \frac{1}{\pi\sqrt{1-t^2}}, \qquad x \in [-1,1].$$

Using the substitution $x = \cos\theta$ one final time, evaluate the integral on the right-hand side above to show that

$$\alpha_A(x) = \frac{1}{2} + \frac{1}{\pi}\arcsin(x), \qquad x \in [-1,1].$$

The probability distribution with PDF w_A and CDF α_A is called the **arcsine distribution**, and we write

$$X \sim \mathrm{Arcsine}(-1,1)$$

to indicate that the random variable X takes on values $x \in [-1,1]$ according to the law

$$\mathbb{P}(X \leq x) := F_A(x) = \begin{cases} 0, & x < -1, \\ \dfrac{1}{2} + \dfrac{1}{\pi}\arcsin(x), & x \in [-1,1], \\ 1, & x \geq 1. \end{cases}$$

(We identify the support $\mathcal{X} = [-1,1]$ with "the part of the domain where the CDF actually changes." This is why we distinguish "a random variable's domain," which is all of $\mathbb{R}$ for any random variable distributed according to any probability law, with "a random variable's support," which changes depending on the "interesting part" of the domain where the nontrivial behavior actually occurs.)

(f) Given a random variable

$$X \sim \text{Arcsine}(-1, 1),$$

the Stieltjes pairing $\langle \cdot, \cdot \rangle_{w_A}$ admits a clean probabilistic interpretation. For any continuous $f, g \in C([-1, 1])$, confirm that

$$\langle f, g \rangle_{w_A} = \mathbb{E}_X[f(X)g(X)],$$

where

$$\mathbb{E}_X[f(X)g(X)] = \int_{\mathbb{R}} f(x)g(x)\, dF_A(x).$$

(Hint: relative to the notation we established above,

$$F_A(x) = \begin{cases} \alpha_A(x), & x \in \mathcal{X} := [-1, 1], \\ \text{const.}, & \text{otherwise,} \end{cases}$$

which implies that

$$\int_{\mathbb{R}} f(x)g(x)\, dF_A(x) = \int_{\mathcal{X}} f(x)g(x)\, d\alpha(x).$$

Rewrite the integrator on the right as

$$d\alpha_A(x) = w_A(x)\, dx, \qquad x \in (-1, 1),$$

and observe the form of the integral after substituting back in.)

(g) Deduce that if f and g have mean zero with respect to this distribution — i.e., if

$$\mathbb{E}_X[f(X)] = \mathbb{E}_X[g(X)] = 0,$$

then

$$\langle f, g \rangle_{w_A} = \text{Cov}(f(X), g(X)),$$

so that orthogonality of f and g under $\langle \cdot, \cdot \rangle_{w_A}$ is precisely the statement that $f(X)$ and $g(X)$ are uncorrelated random variables. ("Two continuous functions f and g on $[-1, 1]$ may be wildly tangled as deterministic objects, yet pass them through the arcsine filter — evaluate at $X \sim \text{Arcsine}(-1, 1)$ — and what survives is only their linear covariation.")

With the arcsine weight w_A now in hand, and the probabilistic reading from part (g) available to us, we return to our computation begun in part (b). Show that, under w_A,

$$\langle T_m, T_n \rangle_{w_A} = \frac{1}{\pi} \int_0^{\pi} \cos(m\theta) \cos(n\theta)\, d\theta.$$

Using the cosine orthogonality evaluations from Chapter 2, verify that

$$\langle T_m, T_n \rangle_{w_A} = \begin{cases} 0, & m \neq n, \\ 1, & m = n = 0, \\ \frac{1}{2}, & m = n \geq 1. \end{cases}$$

Conclude that T_m and T_n are orthogonal under the Stieltjes pairing $\langle \cdot, \cdot \rangle_{w_A}$ induced by the arcsine weight w_A.

Exercise 3: Redeeming the Monomials

In the parable, we convicted the monomial basis. Its Vandermonde design matrix was dense, its coefficients were unstable under data augmentation, and we spent the better part of a chapter constructing alternatives — Newton, Lagrange, orthogonal — that avoided its deficiencies. The prosecution was fair and the verdict was deserved: for *interpolation*, the monomials are genuinely inferior.

But the parable's second act introduced a different operation: Gram-Schmidt orthogonalization under a weighted inner product $\langle \cdot, \cdot \rangle_w$. For *this* task, the defense has evidence to present.

(a) Let $w\colon \mathcal{X} \to (0, \infty)$ be a general, continuous, strictly positive weight function, so that (cf. Exercise 1) the Stieltjes pairing $\langle \cdot, \cdot \rangle_w$ defines a genuine inner product on $C(\mathcal{X})$. Show that whenever the first $2n-1$ moments exist (including the zeroth) and we define

$$\mu_w^{(k)} := \int_{\mathcal{X}} x^k \, w(x) \, dx, \qquad k = 0, 1, 2, \dots, 2n-2,$$

that every inner product between monomials x^j and x^k under $\langle \cdot, \cdot \rangle_w$ is a moment of the form

$$\langle x^j, x^k \rangle_w = \mu_w^{(j+k)}.$$

(Hint: observe that $\mu_w^{(k)} = \langle x^k, 1 \rangle_w = \langle 1, x^k \rangle_w$.)

(b) Define the **Gram matrix** of the monomial basis $\mathcal{M}_{n-1} = \{1, x, \dots, x^{n-1}\}$ induced by the Stieltjes inner product $\langle \cdot, \cdot \rangle_w$ to be

$$G_w^{(\mathcal{M})} := \begin{pmatrix} \langle 1, 1 \rangle_w & \langle 1, x \rangle_w & \langle 1, x^2 \rangle_w & \cdots & \langle 1, x^{n-1} \rangle_w \\ \langle x, 1 \rangle_w & \langle x, x \rangle_w & \langle x, x^2 \rangle_w & \cdots & \langle x, x^{n-1} \rangle_w \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \langle x^{n-1}, 1 \rangle_w & \langle x^{n-1}, x \rangle_w & \langle x^{n-1}, x^2 \rangle_w & \cdots & \langle x^{n-1}, x^{n-1} \rangle_w \end{pmatrix} \in \mathbb{R}^{n \times n}.$$

Using part (a), show that the (j,k)th entry of G_w reduces to

$$(G_w^{(\mathcal{M})})_{jk} = \langle x^j, x^k \rangle_w = \mu_w^{(j+k)}, \qquad 0 \le j, k \le n-1,$$

so that the Gram matrix takes the explicit form

$$G_w^{(\mathcal{M})} = \begin{pmatrix} \mu_w^{(0)} & \mu_w^{(1)} & \mu_w^{(2)} & \cdots & \mu_w^{(n-1)} \\ \mu_w^{(1)} & \mu_w^{(2)} & \mu_w^{(3)} & \cdots & \mu_w^{(n)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \mu_w^{(n-1)} & \mu_w^{(n)} & \mu_w^{(n+1)} & \cdots & \mu_w^{(2n-2)} \end{pmatrix}.$$

(c) Conclude that $G_w^{(\mathcal{M})}$ is a **Hankel matrix**: a matrix whose entries are constant along each **anti-diagonal**. Then, explain why this means the $n \times n$ Gram matrix $G_w^{(\mathcal{M})}$ is governed by only $2n-1$ moments under the monomial basis.

(Hint:) For instance, a 3×3 Hankel matrix of the form above would be

$$\begin{pmatrix} \mu_0 & \mu_1 & \mu_2 \\ \mu_1 & \mu_2 & \mu_3 \\ \mu_2 & \mu_3 & \mu_4 \end{pmatrix}$$

(d) Having observed that the Gram matrix $G_w^{(\mathcal{M})}$ of the monomial basis $\mathcal{M}_{n-1} = \{1, x, \dots, x^{n-1}\}$ under any continuous, strictly positive weight function $w(x) > 0$ is Hankel, we now move to consider the same construction under a different basis.

Suppose $\Psi_n = \{\psi_0, \psi_1, \dots, \psi_{n-1}\}$ is any basis of Π_{n-1}, and define the Gram matrix $G_w^{(\Psi)}$ by $(G_w^{(\Psi)})_{jk} = \langle \psi_j, \psi_k \rangle_w$.

(i) Verify that $G_w^{(\Psi)}$ is symmetric. (Hint: commutativity of multiplication.)

(ii) Concretely: consider the Lagrange basis $\{\ell_0, \ell_1, \ell_2, \ell_3\}$ at distinct nodes x_0, x_1, x_2, x_3. Explain why, in general, $\langle \ell_0, \ell_3 \rangle_w \neq \langle \ell_1, \ell_2 \rangle_w$ even though $0+3 = 1+2$. Conclude that $G_w^{(\mathcal{L})}$ is, in general, *not* Hankel.

(iii) Verify that a general symmetric $n \times n$ matrix requires $n(n+1)/2$ independent entries to specify, while the Hankel matrix $G_w^{(\mathcal{M})}$ requires only $2n-1$ moments. (Hint: symmetry forces $(G)_{jk} = (G)_{kj}$, so we need only count the entries on and above the diagonal: $n + (n-1) + \cdots + 1 = n(n+1)/2$.)

(e) Observe that the dimensions computed in parts (b) and (d) — $2n-1$ moments for the monomial basis's Hankel Gram matrix, $n(n+1)/2$ integrals for a general symmetric Gram matrix — are not coincidences: the number of independent quantities needed to populate a Gram matrix is precisely the dimension of the subspace of $\mathbb{R}^{n\times n}$ it belongs to. With this in mind, verify that the Hankel matrix $G_w^{(\mathcal{M})}$ is symmetric, and explain why, in general,

$$\underbrace{\mathbb{Hk}^n}_{(n\times n)\text{-Hankel matrices}} \subset \underbrace{\mathbb{S}^n}_{(n\times n)\text{-symmetric matrices}} \subset \underbrace{\mathbb{R}^{n\times n}}_{(n\times n)\text{-matrices with real entries}}$$

is a chain of **vector subspaces** with respective dimensions

$$\dim(\mathbb{Hk}^n) = 2n-1, \qquad \dim(\mathbb{S}^n) = \frac{n(n+1)}{2}, \qquad \dim(\mathbb{R}^{n\times n}) = n^2.$$

(Hint: a **vector subspace** of a vector space V is a subset $S \subseteq V$ that is closed under addition ($u, v \in S \implies u+v \in S$) and scalar multiplication ($u \in S$, $\alpha \in \mathbb{R} \implies \alpha u \in S$); alternatively, one may check that

$$u, v \in S \subset V, \quad \alpha, \beta \in \mathbb{R} \implies \alpha u + \beta v \in S. \qquad \text{(Linear Subspace)}$$

Any subspace is itself a proper vector space under the operations and scalar field inherited from its parent.

The **dimension** of a subspace is the number of free parameters needed to specify an arbitrary element — i.e., the number of elements in a basis of a vector space. We also encourage you to appreciate the fact that, currently, the special cases of these classes we're examining, the gram matrices $G_w^{(\mathcal{M})} \in \mathbb{H}\mathbb{k}^n$ and $G_w^{(\Psi)} \in \mathbb{S}^n$, are objects whose dimension happen to precisely correspond with *how many integrals* we have to compute to "write the matrix down." Thus, in this case, "subspace dimension $\equiv$ computational complexity" in a very tangible sense.)

(f) Explicitly compute $\dim(\mathbb{H}\mathbb{k}^n)$, $\dim(\mathbb{S}^n)$, and $\dim(\mathbb{R}^{n\times n})$ for $n = 5$, $n = 20$, and $n = 100$. Comment on the difference in behavior as n grows.

(g) Compare the ratios of

$$\frac{\dim(\mathbb{H}\mathbb{k}^n)}{\dim(\mathbb{R}^{n\times n})} \quad \text{and} \quad \frac{\dim(\mathbb{S}^n)}{\dim(\mathbb{R}^{n\times n})} \quad \text{as} \quad n \to \infty.$$

Verify that while the former ratio tends to 0 as $n \to \infty$, the latter tends to a nonzero quantity. Interpret the difference in **asymptotic behavior** qualitatively — why might we lament the fact that symmetric matrices require us to "compute a number of integrals scaling *quadratically* in n, the number of representatives we include in our basis?"

(h) Fill in the gaps in the proof of the following theorem:

Theorem 3.4.2 (Hankel Gram $\iff$ Monomial Basis)

If a polynomial basis $\Psi = \{\psi_0, \psi_1, \psi_2, \dots\}$ has the property that, for every continuous strictly positive weight w, the Gram matrix $G_w^{(\Psi)}$ is Hankel, then there exist constants $a \neq 0$, b, and $c \neq 0$ such that

$$\psi_k(x) = c\,(ax+b)^k, \qquad k = 0, 1, 2, \dots .$$

In other words, up to normalization and affine change of variable, a polynomial basis whose Gram matrix is Hankel *is* the monomial basis.

Proof: Since $G_w^{(\Psi)}$ is Hankel for every weight w, whenever $j+k = r+s$ we have

$$\langle \psi_j, \psi_k \rangle_w = \langle \psi_r, \psi_s \rangle_w.$$

Equivalently,

$$\int_{\mathcal{X}} \Big(\psi_j(x)\psi_k(x) - \psi_r(x)\psi_s(x)\Big)\, w(x)\, dx = 0 \qquad \text{for every } w.$$

Set

$$g(x) := \psi_j(x)\psi_k(x) - \psi_r(x)\psi_s(x).$$

Explain why the identity

$$\int_{\mathcal{X}} g(x)\, w(x)\, dx = 0 \qquad \text{for every continuous strictly positive weight } w$$

forces $g \equiv 0$. (Hint: imitate the neighborhood argument from Exercise 1(d): if g were nonzero somewhere, continuity would force it to keep one sign on a small interval.)

Now take $r = 0$ and $s = j+k$ to obtain

$$\psi_j(x)\psi_k(x) = \psi_0(x)\psi_{j+k}(x).$$

Deduce that the polynomial $\psi_0 \in \Pi_0 \equiv \mathbb{R}$ is a nonzero constant, say $\psi_0(x) = c$.

Define

$$\phi_k(x) := \frac{\psi_k(x)}{c}.$$

Show that

$$\phi_j(x)\phi_k(x) = \phi_{j+k}(x) \qquad \text{and} \qquad \phi_0(x) = 1.$$

Thus,

$$\phi_k(x) = \phi_1(x)^k \qquad \text{for all } k \geq 0.$$

Finally, use the fact that $\deg \phi_k = k$ to show that ϕ_1 must be a polynomial of degree 1. Conclude

$$\phi_1(x) = ax + b$$

for some $a \neq 0$ and $b \in \mathbb{R}$. Therefore,

$$\psi_k(x) = c\,(ax+b)^k,$$

establishing the claim.

‖

Remark.

The monomial basis is not "good" or "bad." It is bad for one job and uniquely good for another. For interpolation — fitting a polynomial through prescribed data points — the Vandermonde system it produces is dense, ill-conditioned, and unstable under augmentation; the Newton, Lagrange, and orthogonal bases each remedy one or more of these defects. But for orthogonalization under a continuous weight — the operation that produces the classical polynomial families — the monomials possess a structural virtue no alternative can match: their multiplication rule $x^j \cdot x^k = x^{j+k}$ compresses the Gram matrix from $\mathbb{S}^n$ into $\mathbb{Hk}^n$, reducing its specification from $n(n+1)/2$ independent integrals to $2n-1$ moments. Part (h) proved this compression is not merely convenient but *necessary*: any basis achieving it is, up to rescaling, the monomial basis in disguise.

The pipeline we have established at this point is:

$$\textbf{Weight} \xrightarrow{\int x^k w\,dx} \textbf{Moments} \xrightarrow{\mu_w^{(j+k)}} \textbf{Hankel matrix } G_w^{(\mathcal{M})}.$$

Remark.

Of course, the fact that Hankel is the unique $O(n)$-dimensional structure achievable by the monomial basis does not, by itself, rule out the possibility that some *other* polynomial basis Ψ might produce Gram matrices lying in some *different* structured subspace $S \subset \mathbb{S}^n$ with $\dim(S) = O(n)$ — just not Hankel. What we proved in Exercise 3(h) is narrower: if $G_w^{(\Psi)}$ is Hankel under every weight w, then Ψ is monomial. We did *not* prove that Hankel is the only $O(n)$-dimensional subspace of $\mathbb{S}^n$ in which a Gram matrix could land.

That said, the Hankel compression arises from a specific algebraic property of the monomials — the multiplication rule $x^j \cdot x^k = x^{j+k}$, which forces the inner product $\langle x^j, x^k \rangle_w$ to depend only on the *sum* $j+k$. For any other basis $\Psi = \{\psi_i\}$, the product $\psi_j(x)\psi_k(x)$ has no reason to depend on a single scalar function of the indices j and k, and without such a rule, the entries of the Gram matrix have no reason to organize themselves into fewer than $\frac{n(n+1)}{2}$ independent values. No alternative $O(n)$-dimensional structure for polynomial Gram matrices is known, and the algebraic reasons suggest that none should exist — but a formal proof of this stronger claim is beyond our present scope.

Exercise 4: Gram-Schmidt on Gram Matrices

Exercise 3 established that the Gram matrix of the monomial basis $\mathcal{M}_{n-1} = \{1, x, x^2, \dots, x^{n-1}\}$

under a continuous, strictly positive weight $w\colon \mathcal{X} \to (0,\infty)$ is the **Hankel matrix**

$$G_w^{(\mathcal{M})} = \begin{pmatrix} \mu_w^{(0)} & \mu_w^{(1)} & \cdots & \mu_w^{(n-1)} \\ \mu_w^{(1)} & \mu_w^{(2)} & \cdots & \mu_w^{(n)} \\ \vdots & \vdots & \ddots & \vdots \\ \mu_w^{(n-1)} & \mu_w^{(n)} & \cdots & \mu_w^{(2n-2)} \end{pmatrix},$$

governed entirely by the $2n-1$ moments $\mu_w^{(0)}, \mu_w^{(1)}, \dots, \mu_w^{(2n-2)}$. We now show that the orthogonal polynomials produced by Gram-Schmidt can be read off from an appropriate factorization. The key idea is that each step of Gram-Schmidt corresponds to one **Schur complement** on the Gram matrix.

Of course, we begin with a definition.

Definition 3.4.3 (Schur Complement)

Let $G \in \mathbb{R}^{n\times n}$. Suppose that we **block-partition** G as

$$G_{(n\times n)} = \left(\begin{array}{c|c} A_{(n_1\times n_1)} & B_{(n_1\times n_2)} \\ \hline C_{(n_2\times n_1)} & D_{(n_2\times n_2)} \end{array}\right),$$

so that the upper left and lower right blocks, $A_{(n_1\times n_1)}$ and $D_{(n_2\times n_2)}$ are themselves $(n_1 \times n_1)$- and $(n_2 \times n_2)$-square matrices with $n_1 + n_2 = n$, respectively.

Whenever $A \in \mathbb{R}^{n_1\times n_1}$ is invertible, we define the **Schur complement of A in G** to be the quantity

$$S_A \coloneqq D - CA^{-1}B.$$

(a) **(The First Step)** Let $\Phi = \{\phi_0, \phi_1, \dots, \phi_{n-1}\}$ be a set of basis functions and $G_w^{(\Psi)}$ their Gram matrix under a continuous Stieltjes weight $\langle\cdot,\cdot\rangle_w$,

$$G_w^{(\Psi)} = \begin{pmatrix} \langle\psi_0,\psi_0\rangle_w & \langle\psi_0,\psi_1\rangle_w & \cdots & \langle\psi_0,\psi_{n-1}\rangle_w \\ \langle\psi_1,\psi_0\rangle_w & \langle\psi_1,\psi_1\rangle_w & \cdots & \langle\psi_1,\psi_{n-1}\rangle_w \\ \vdots & \vdots & \ddots & \vdots \\ \langle\psi_{n-1},\psi_0\rangle_w & \langle\psi_{n-1},\psi_1\rangle_w & \cdots & \langle\psi_{n-1},\psi_{n-1}\rangle_w \end{pmatrix} \in \mathbb{R}^{n\times n}.$$

Suppose we partition $G_w^{(\Psi)}$ as

$$G_w^{(\Psi)} = \left(\begin{array}{c|c} a_{(1\times 1)} & \mathbf{b}^\top_{(1\times(n-1))} \\ \hline \mathbf{b}_{(n-1)\times 1} & C_{(n-1)\times(n-1)} \end{array}\right).$$

(i) Verify that a block partition of $G_w^{(\Psi)}$ of the form above is obtained by setting

$$a = \langle\psi_0,\psi_0\rangle_w \in \mathbb{R}, \qquad \mathbf{b} = \begin{pmatrix} \langle\psi_1,\psi_0\rangle_w \\ \langle\psi_2,\psi_0\rangle_w \\ \vdots \\ \langle\psi_{n-1},\psi_0\rangle_w \end{pmatrix} \in \mathbb{R}^{n-1},$$

and C to be the lower-right block

$$C = \begin{pmatrix} \langle\psi_1,\psi_1\rangle_w & \langle\psi_1,\psi_2\rangle_w & \cdots & \langle\psi_1,\psi_{n-1}\rangle_w \\ \langle\psi_2,\psi_1\rangle_w & \langle\psi_2,\psi_2\rangle_w & \cdots & \langle\psi_2,\psi_{n-1}\rangle_w \\ \vdots & \vdots & \ddots & \vdots \\ \langle\psi_{n-1},\psi_1\rangle_w & \langle\psi_{n-1},\psi_2\rangle_w & \cdots & \langle\psi_{n-1},\psi_{n-1}\rangle_w \end{pmatrix} \in \mathbb{R}^{(n-1)\times(n-1)},$$

(ii) Show that, whenever $a \neq 0$, the Schur complement of $a = \langle\psi_0,\psi_0\rangle_w > 0$ in G is the $(n-1)\times(n-1)$ matrix

$$S_a = C - \frac{1}{a}\mathbf{b}\mathbf{b}^\top \in \mathbb{R}^{(n-1)\times(n-1)}.$$

In particular, verify that the (j,k)th entry of S_a is

$$(S_a)_{jk} = \langle\psi_j,\psi_k\rangle_w - \frac{\langle\psi_j,\psi_0\rangle_w\langle\psi_0,\psi_k\rangle_w}{\langle\psi_0,\psi_0\rangle_w}, \qquad 1 \le j,k \le n-1.$$

(iii) Verify the factorization

$$G_w^{(\Psi)} = \begin{pmatrix} 1 & \mathbf{0}^\top \\ \mathbf{b}/a & I_{n-1} \end{pmatrix} \begin{pmatrix} a & \mathbf{0}^\top \\ \mathbf{0} & S_a \end{pmatrix} \begin{pmatrix} 1 & \mathbf{b}^\top/a \\ \mathbf{0} & I_{n-1} \end{pmatrix},$$

where I_{n-1} is the $(n-1)\times(n-1)$ identity matrix, and $\mathbf{0}$ is an $(n-1)$-vector of zeros.

(iv) Explain, intuitively, what the factorization has achieved. Before the factorization, the Gram matrix $G_w^{(\Psi)}$ was dense: every entry $\langle\psi_j,\psi_k\rangle_w$ — even for $j,k \ge 1$ — is influenced by ψ_0, because each ψ_j may have some component "pointing in the ψ_0 direction." The factorization separates this influence into three pieces:

- The **ratios** $\mathbf{b}/a$, which record how much of each ψ_j points along ψ_0;
- The **scalar** $a = \langle\psi_0,\psi_0\rangle_w$, which records ψ_0's own size;
- The **residual block** S_a, which contains "any remaining information from $G_w^{(\Psi)}$" after stripping out ψ_0's contribution.

In particular: if ψ_0 were already orthogonal to every other basis function — $\mathbf{b} = \mathbf{0}$ — then $S_a = C$, the outer factors collapse to the identity, and the factorization is trivial. The correction $\frac{1}{a}\mathbf{b}\mathbf{b}^\top$ measures precisely how much ψ_0 is entangled with the rest; removing its influence is what produces the Schur complement S_a.

(b) (The Schur complement is Gram-Schmidt) The factorization of the Gram matrix $G_w^{(\Psi)}$ in part (a) isolated ψ_0's contribution and left behind a block S_a containing whatever information in $\psi_1,\dots,\psi_{n-1}$ is independent of ψ_0. We now seek to make this claim precise.

Look again at the formula

$$(S_a)_{jk} = \langle\psi_j,\psi_k\rangle_w - \frac{\langle\psi_j,\psi_0\rangle_w\langle\psi_0,\psi_k\rangle_w}{\langle\psi_0,\psi_0\rangle_w}.$$

Suppose, then, we were to *define*

$$\tilde{\psi}_j := \psi_j(x) - \frac{\langle\psi_j,\psi_0\rangle_w}{\langle\psi_0,\psi_0\rangle_w}\psi_0(x), \qquad j = 1,2,\dots,n-1.$$

(i) Using bilinearity, expand $\langle\tilde{\psi}_j,\tilde{\psi}_k\rangle_w$ and verify that

$$\langle\tilde{\psi}_j,\tilde{\psi}_k\rangle_w = \langle\psi_j,\psi_k\rangle_w - \frac{\langle\psi_j,\psi_0\rangle_w\langle\psi_0,\psi_k\rangle_w}{\langle\psi_0,\psi_0\rangle_w} = (S_a)_{jk}.$$

(ii) We call $\tilde{\psi}_j$ the (first-round) **residuals** — the part of the jth basis function ψ_j "left over" after accounting for the influence of ψ_0. Indeed, verify the decomposition

$$\psi_j = \underbrace{\frac{\langle \psi_j, \psi_0 \rangle_w}{\langle \psi_0, \psi_0 \rangle_w} \psi_0}_{\text{component along } \psi_0} + \underbrace{\tilde{\psi}_j}_{\text{residual}} ,$$

and confirm:

- The two pieces are **orthogonal** under w:

$$\langle \tilde{\psi}_j, \psi_0 \rangle_w = 0, \qquad j = 1, \dots, n-1.$$

  We also write $\tilde{\psi}_j \perp \psi_0$ under w (or under $\langle \cdot, \cdot \rangle_w$.)
- The first piece is a **scalar multiple of** ψ_0 — i.e., it lives entirely in the one-dimensional subspace *spanned by* ψ_0:

$$\operatorname{span}(\psi_0) := \{x \mapsto c \cdot \psi_0(x) \colon\ c \in \mathbb{R}\} .$$

  (We often suppress the explicit mapping $x \mapsto c \cdot \psi_0(x)$ and write

$$\operatorname{span}(\psi_0) = \{c \cdot \psi_0 \colon\ c \in \mathbb{R}\} .$$

)
- The residual $\tilde{\psi}_j$ lies in the **orthogonal complement** $\operatorname{span}^{\perp}(\psi_0)$ of ψ_0 within $\operatorname{span}(\Psi) := \{c_0\psi_0 + \cdots + c_{n-1}\psi_{n-1} \colon\ c_0, \dots, c_{n-1} \in \mathbb{R}\}$:

$$\operatorname{span}^{\perp}(\psi_0) := \{x \mapsto f(x) \colon f \in \operatorname{span}(\Psi),\ \langle f, \psi_0 \rangle_w = 0\} .$$

  (Similarly to the above, we may perhaps more clearly write this as

$$\operatorname{span}^{\perp}(\psi_0) = \{f \in \operatorname{span}(\Psi) \colon\ \langle f, \psi_0 \rangle_w = 0\} .$$

)
  In words, the orthogonal complement $\operatorname{span}^{\perp}(\psi_0)$ is the subspace of *all* functions $f \in \operatorname{span}(\Psi)$ orthogonal to ψ_0 under w.
- We may equivalently write the decomposition as

$$\psi_j = \underbrace{\operatorname{proj}_{\psi_0}(\psi_j)}_{\text{component along } \psi_0} + \underbrace{\tilde{\psi}_j}_{\text{residual}} ,$$

  where we've defined

$$\operatorname{proj}_f(g) := \frac{\langle g, f \rangle_w}{\langle f, f \rangle_w} f$$

  to be the **projection** of g onto f under $\langle \cdot, \cdot \rangle_w$.
- Show that orthogonality of the two pieces gives a **Pythagorean identity**: for each $j = 1, \dots, n-1$,

$$\|\psi_j\|_w^2 = \|\operatorname{proj}_{\psi_0}(\psi_j)\|_w^2 + \|\tilde{\psi}_j\|_w^2.$$

  Verify the interpretation: the squared norm of ψ_j decomposes into the squared norm of its projection onto ψ_0 plus the squared norm of the residual — the "total size" of ψ_j splits into the part "explained by ψ_0" and the part "independent of ψ_0." (The reader who has encountered ANOVA may recognize this as resembling $\mathrm{SS}_{\text{total}} = \mathrm{SS}_{\text{model}} + \mathrm{SS}_{\text{residual}}$. The connection is exact, not analogous; we will return to this subject in Chapter 6.)

(iii) We recognize that the decomposition above yields an **orthogonal direct sum**: the full n-dimensional space $\operatorname{span}(\Psi)$ splits as

$$\operatorname{span}(\Psi) = \operatorname{span}(\psi_0) \oplus \operatorname{span}^{\perp}(\psi_0),$$

meaning that: if $f \in \operatorname{span}(\Psi)$, then f may be decomposed as

$$f = f_{\parallel} + f_{\perp}, \qquad f_{\parallel} \in \operatorname{span}(\psi_0), \qquad f_{\perp} \in \operatorname{span}^{\perp}(\psi_0).$$

(Hint: If f is itself orthogonal to ψ_0, then $f \in \operatorname{span}^{\perp}(\psi_0)$, so take $f_{\perp} := f$ and $f_{\parallel} := 0$. Argue symmetrically for f parallel to ψ_0. For the general case, set $f_{\parallel} := \frac{\langle f,\psi_0\rangle_w}{\langle \psi_0,\psi_0\rangle_w}\psi_0$ and $f_{\perp} := f - f_{\parallel}$; verify that $f_{\perp} \in \operatorname{span}^{\perp}(\psi_0)$.)

(iv) Verify that this decomposition is **unique** by filling in the gaps of the following argument:

- Suppose $f = g_{\parallel} + g_{\perp}$ is another decomposition with $g_{\parallel} \in \operatorname{span}(\psi_0)$ and $g_{\perp} \in \operatorname{span}^{\perp}(\psi_0)$. Subtracting gives

$$f_{\parallel} - g_{\parallel} = g_{\perp} - f_{\perp}.$$

- Explain why the left side lies in $\operatorname{span}(\psi_0)$ and the right side lies in $\operatorname{span}^{\perp}(\psi_0)$.
- Calling the common value h, conclude that $h = c \cdot \psi_0$ for some $c \in \mathbb{R}$ and simultaneously $\langle h, \psi_0\rangle_w = 0$.
- Deduce $c = 0$, whence $h \equiv 0$, $f_{\parallel} = g_{\parallel}$, and $f_{\perp} = g_{\perp}$.
- Conclude: the two subspaces $\operatorname{span}(\psi_0)$ and $\operatorname{span}^{\perp}(\psi_0)$ intersect only at 0, and their dimensions exhaust the whole space:

$$\underbrace{1}_{\dim \operatorname{span}(\psi_0)} + \underbrace{(n-1)}_{\dim \operatorname{span}^{\perp}(\psi_0)} = \underbrace{n}_{\dim \operatorname{span}(\Psi)}$$

(v) Verify that $\tilde{\Psi} = \{\tilde{\psi}_1, \dots, \tilde{\psi}_{n-1}\}$ is a **basis** for $\operatorname{span}^{\perp}(\psi_0)$. (Hint: $\operatorname{span}^{\perp}(\psi_0)$ has dimension $n-1$; the $n-1$ residuals each lie in it; and any linear dependence $\sum_{j=1}^{n-1} \alpha_j \tilde{\psi}_j = 0$ would give $\sum \alpha_j \psi_j = \big(\sum \alpha_j c_j\big)\psi_0$ where $c_j = \langle \psi_j, \psi_0\rangle_w / \langle \psi_0, \psi_0\rangle_w$, contradicting the linear independence of Ψ.)

Remark.

We have spent the last several parts developing the *geometry* of the decomposition

$$\operatorname{span}(\Psi) = \operatorname{span}(\psi_0) \oplus \operatorname{span}^{\perp}(\psi_0).$$

Let us recall why. In part (a), one round of Schur complementation on the Gram matrix $G_w^{(\Psi)}$ produced the factorization

$$G_w^{(\Psi)} = \begin{pmatrix} 1 & \mathbf{0}^{\top} \\ \mathbf{b}/a & I_{n-1} \end{pmatrix} \begin{pmatrix} a & \mathbf{0}^{\top} \\ \mathbf{0} & S_a \end{pmatrix} \begin{pmatrix} 1 & \mathbf{b}^{\top}/a \\ \mathbf{0} & I_{n-1} \end{pmatrix},$$

with the middle factor **block-diagonal**: in particular, we identify that

$$\begin{pmatrix} a & \mathbf{0}^{\top} \\ \mathbf{0} & S_a \end{pmatrix} = \left(\begin{array}{c|cccc} \|\psi_0\|_w^2 & 0 & 0 & \cdots & 0 \\ \hline 0 & \|\tilde{\psi}_1\|_w^2 & \langle \tilde{\psi}_1, \tilde{\psi}_2\rangle_w & \cdots & \langle \tilde{\psi}_1, \tilde{\psi}_{n-1}\rangle_w \\ 0 & \langle \tilde{\psi}_2, \tilde{\psi}_1\rangle_w & \|\tilde{\psi}_2\|_w^2 & \cdots & \langle \tilde{\psi}_2, \tilde{\psi}_{n-1}\rangle_w \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & \langle \tilde{\psi}_{n-1}, \tilde{\psi}_1\rangle_w & \langle \tilde{\psi}_{n-1}, \tilde{\psi}_2\rangle_w & \cdots & \|\tilde{\psi}_{n-1}\|_w^2 \end{array}\right),$$

where $\tilde{\psi}_j$ is the jth first-round residual,

$$\tilde{\psi}_j := \psi_j(x) - \frac{\langle \psi_j, \psi_0 \rangle_w}{\langle \psi_0, \psi_0 \rangle_w} \psi_0(x), \qquad j = 1, 2, \dots, n-1.$$

The first row and column are clean — ψ_0 has been fully separated — but the residuals $\tilde{\psi}_1, \dots, \tilde{\psi}_{n-1}$ are still entangled with one another inside S_a. To finish, we need to disentangle them too.

But this is exactly what part (b) has licensed us to do. We showed that S_a is itself the Gram matrix of a basis — namely, $\tilde{\Psi} = \{\tilde{\psi}_1, \dots, \tilde{\psi}_{n-1}\}$ is a basis for the orthogonal complement of ψ_0, $\mathrm{span}^{\perp}(\psi_0)$. In turn, we're motivated to apply the same machine to S_a in an iterative fashion — and this is exactly what we proceed to do.

(c) **(Iterating to $LDL^{\top}$)** We now formalize the iteration.

- Initialize
$$\psi_j^{[0]} := \psi_j, \qquad j = 0, 1, \dots, n-1.$$
These are the "zeroth-stage residuals" — nothing has been residualized yet, so these are just the original basis functions.
- At stage $r = 0$, the **pivot** is
$$\omega_0 := \psi_0^{[0]} \equiv \psi_0.$$
The pivot is the basis function whose influence we seek to remove from the remaining members of the original basis Ψ — it will be the first of our new family of orthogonal polynomials. To do so, compute the **first-stage residuals**
$$\psi_j^{[1]} := \psi_j^{[0]} - \mathrm{proj}_{\omega_0}\left(\psi_j^{[0]}\right), \qquad j = 1, \dots, n-1.$$
Finally, record
$$d_0 := \|\omega_0\|_w^2; \qquad \text{and} \qquad \ell_{j,0} := \frac{\langle \psi_j^{[0]}, \omega_0 \rangle_w}{\|\omega_0\|_w^2}, \quad j = 1, \dots, n-1.$$
- At stage $r = 1$, the pivot is
$$\omega_1 := \psi_1^{[1]}.$$
Just as before, we compute the **second-stage residuals**
$$\psi_j^{[2]} := \psi_j^{[1]} - \mathrm{proj}_{\omega_1}\left(\psi_j^{[1]}\right), \qquad j = 2, \dots, n-1,$$
and we record
$$d_1 := \|\omega_1\|_w^2; \qquad \text{and} \qquad \ell_{j,1} := \frac{\langle \psi_j^{[1]}, \omega_1 \rangle_w}{\|\omega_1\|_w^2}, \quad j = 2, \dots, n-1.$$
- In general, at stage r, the pivot is
$$\omega_r := \psi_r^{[r]};$$
this is "what remains of the original rth basis function ψ_r after accounting for the collective influence of $\psi_0, \psi_1, \dots, \psi_{r-1}$." The $(r+1)$**th-stage residuals** are
$$\psi_j^{[r+1]} := \psi_j^{[r]} - \mathrm{proj}_{\omega_r}\left(\psi_j^{[r]}\right), \qquad j = r+1, \dots, n-1,$$

and we record

$$d_r := \|\omega_r\|_w^2; \qquad \text{and} \qquad \ell_{j,r} := \frac{\langle \psi_j^{[r]}, \omega_r \rangle_w}{\|\omega_r\|_w^2}, \quad j = r+1, \dots, n-1.$$

- After $n-1$ stages, set

$$\omega_{n-1} := \psi_{n-1}^{[n-1]}, \qquad \text{and record} \qquad d_{n-1} := \|\omega_{n-1}\|_w^2.$$

At the conclusion of the iteration, we have produced n pivots $\Omega := \{\omega_0, \omega_1, \dots, \omega_{n-1}\}$, together with the scalars $d_0, d_1, \dots, d_{n-1}$ and $\ell_{j,r}$ for $0 \le r < j \le n-1$.

(i) Verify that the pivot $\omega_r = \psi_r^{[r]}$ produced at stage r of the iteration is a polynomial of degree exactly r with leading coefficient 1 — i.e., ω_r is a **monic** polynomial, one whose leading coefficient is exactly x^r. (Hint: at each stage, $\psi_j^{[r+1]} = \psi_j^{[r]} - \operatorname{proj}_{\omega_r}(\psi_j^{[r]})$ subtracts a scalar multiple of ω_r, which has degree $r < j$. So the leading term of $\psi_j^{[r+1]}$ is unchanged from $\psi_j^{[r]}$, which is unchanged from $\psi_j^{[0]} = x^j$. Conclude by induction that $\omega_r = \psi_r^{[r]}$ has leading term x^r.)

(ii) Show that the pivots are pairwise orthogonal:

$$\langle \omega_j, \omega_k \rangle_w = 0, \qquad j \neq k.$$

(Hint: if $j < k$, the component along ω_j was removed at stage j. The projection at each subsequent stage $r > j$ removes the component along ω_r, which is itself orthogonal to ω_j — so the "already removed" component is never reintroduced.)

(iii) Verify that the Pythagorean identity from part (b) replays at every stage: at stage r,

$$\|\psi_j^{[r]}\|_w^2 = \|\operatorname{proj}_{\omega_r}\left(\psi_j^{[r]}\right)\|_w^2 + \|\psi_j^{[r+1]}\|_w^2.$$

(iv) Assemble the matrices

$$L := \begin{pmatrix} 1 & 0 & \cdots & 0 \\ \ell_{1,0} & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ \ell_{n-1,0} & \ell_{n-1,1} & \cdots & 1 \end{pmatrix}, \qquad D := \operatorname{diag}(d_0, d_1, \dots, d_{n-1}).$$

Confirm that L is unit lower triangular and D is diagonal with strictly positive entries. Verify the factorization

$$G_w^{(\Psi)} = LDL^\top.$$

The middle factor is now fully diagonal — every basis function has been peeled off, and the entanglement that was present in $G_w^{(\Psi)}$ has been completely resolved into the projection coefficients $\ell_{j,r}$ (stored in L) and the squared norms $d_r = \|\omega_r\|_w^2$ (stored in D).

(d) (Orthogonal Polynomials from L^{-1})

Let us set aside the factorization for a moment and recall the original goal. We know from Exercise 1 that a continuous, positive weight $w\colon \mathcal{X} \to (0,\infty)$ induces an inner product via its Stieltjes pairing $\langle \cdot, \cdot \rangle_w$:

$$\langle f, g \rangle_w = \int_{\mathcal{X}} f(x) g(x)\, w(x)\, dx, \qquad f, g \in C(\mathcal{X}).$$

What we then *suppose* — perhaps motivated by Chebyshev's polynomials under the arcsine weight — is that there exists a family of **monic orthogonal polynomials** $\Omega = \{\omega_0, \omega_1, ...\}$ under $\langle \cdot, \cdot \rangle_w$ on $\mathcal{X}$ such that, for any pair of distinct basis functions in Ω, we have

$$\omega_j \not\equiv \omega_k \in \Omega \implies \langle \omega_j, \omega_k \rangle_w = 0.$$

But what *are* these polynomials? We cannot simply write them down — their coefficients depend on w, and w may be a complicated, transcendental object. We have no direct access to w's preferred coordinate system. What we *can* do is integrate: for any polynomial p, the number $\int_{\mathcal{X}} p(x)\, w(x)\, dx$ is computable, at least in principle. This is the only channel through which w speaks to us.

In light of this observation, suppose we stack the members of the bases $\mathcal{M}_{n-1}$ and Ω into n-dimensional vectors:

$$\mathbf{m}(x) = \begin{pmatrix} 1 \\ x \\ x^2 \\ \vdots \\ x^{n-1} \end{pmatrix}, \qquad \boldsymbol{\omega}(x) = \begin{pmatrix} \omega_0(x) \\ \omega_1(x) \\ \omega_2(x) \\ \vdots \\ \omega_{n-1}(x) \end{pmatrix}.$$

Now, let us *suppose* there exists a linear map T_w whose matrix "translates the monomials $\mathbf{m}$ we know" to "the language of the orthogonal polynomials $\boldsymbol{\omega}$ we don't": we'd have

$$\boldsymbol{\omega}(x) = T_w\, \mathbf{m}(x).$$

(i) Verify that the structure of $T_w = \Big((t_{jk})\Big) \in \mathbb{R}^{n\times n}$ is determined by the expansions of the ω_j: from $\boldsymbol{\omega}(x) = T_w\, \mathbf{m}(x)$,

$$\omega_j(x) = \sum_{k=0}^{n-1} t_{jk}\, x^k = t_{j0} + t_{j1}\, x + \cdots + t_{j(n-1)}\, x^{n-1}, \qquad j = 0, 1, \dots, n-1,$$

and since each ω_j is monic of degree exactly j (part (c)(i)), we must have $t_{jj} = 1$ and $t_{jk} = 0$ for all $k > j$. Conclude that T_w must be **unit lower triangular**: ones on the diagonal, zeros above, and the unknown coefficients of the orthogonal polynomials sitting below.

(ii) Now translate the orthogonality conditions into a statement about T_w. Using bilinearity, show that

$$\langle \boldsymbol{\omega}, \boldsymbol{\omega}^\top \rangle_w = T_w\, \langle \mathbf{m}, \mathbf{m}^\top \rangle_w\, T_w^\top.$$

Verify that $\langle \mathbf{m}, \mathbf{m}^\top \rangle_w = G_w^{(\mathcal{M})}$, the Gram matrix from Exercise 3 — every entry is a moment. Since the ω_j are pairwise orthogonal (part (c)(ii)), the left side is diagonal: call it $D = \operatorname{diag}(d_0, \dots, d_{n-1})$ with $d_r = \|\omega_r\|_w^2 > 0$. The equation becomes

$$D = T_w\, G_w^{(\mathcal{M})}\, T_w^\top.$$

Read this carefully. On the left: the Gram matrix in w's native coordinates — diagonal, clean, every direction independent. On the right: the same information passed through T_w, which translates from the monomial language we speak into w's language, the one we are trying to learn. The Gram matrix $G_w^{(\mathcal{M})}$ is dense because the monomials are entangled from w's perspective; T_w is the change of coordinates that untangles them.

(iii) Rearrange. Set $L := T_w^{-1}$ and verify:

$$G_w^{(\mathcal{M})} = L\,D\,L^\top.$$

This is the same equation read in the other direction. L translates *from* orthogonal polynomials *back to* monomials: $\mathbf{m}(x) = L\,\boldsymbol{\omega}(x)$. The equation says: start with D — the world as w sees it, diagonal — apply L on both sides — translate back to monomial coordinates — and you recover the dense, entangled G we started with.

We *recognize this*. The factorization $G = LDL^\top$ with L unit lower triangular and D diagonal is exactly what iterated Schur complementation produced in part (c). The algorithm was building L all along, column by column, recording projection coefficients at each stage. It was discovering the translation *back from w's language to ours*, because it worked in the only coordinates available: monomial coordinates.

But is this the *only* such factorization? If two different unit lower triangular matrices could both produce an $LDL^\top$ factorization of $G_w^{(\mathcal{M})}$, then T_w would not be uniquely determined, and the whole enterprise would be ambiguous. We now show that this cannot happen.

(iv) Let $\mathbf{c} = (c_0, \dots, c_{n-1})^\top \neq \mathbf{0}$ and set $p(x) = \mathbf{c}^\top \mathbf{m}(x) = c_0 + c_1 x + \cdots + c_{n-1}x^{n-1}$. Show that

$$\mathbf{c}^\top G_w^{(\mathcal{M})}\,\mathbf{c} = \|p\|_w^2 > 0.$$

The inequality is strict: p is a nonzero polynomial and $w > 0$ on $\mathcal{X}$, so $\int_{\mathcal{X}} p(x)^2\, w(x)\, dx > 0$. A matrix satisfying $\mathbf{c}^\top G\,\mathbf{c} > 0$ for every nonzero $\mathbf{c}$ is called **positive definite**.

(v) Suppose $G = L_1 D_1 L_1^\top = L_2 D_2 L_2^\top$ are two factorizations, each with L_i unit lower triangular and D_i diagonal with positive entries. Set $M := L_2^{-1} L_1$ — also unit lower triangular — and verify that

$$M\,D_1\,M^\top = D_2.$$

Examine the $(1,1)$ entry: since M is unit lower triangular, the first row of M is $(1, 0, \dots, 0)$, so $(MD_1M^\top)_{11} = (D_1)_{11}$. Conclude $(D_1)_{11} = (D_2)_{11}$. Now examine the $(j,1)$ entry for $j > 1$: show that $(MD_1M^\top)_{j1} = M_{j1}(D_1)_{11}$, and since $(D_2)_{j1} = 0$ and $(D_1)_{11} > 0$, conclude $M_{j1} = 0$. Iterate down the diagonal to show $M = I$, whence $L_1 = L_2$ and $D_1 = D_2$.

(vi) The $LDL^\top$ factorization of $G_w^{(\mathcal{M})}$ is unique. The Schur complement procedure produced the only L that works. Conclude:

$$\boldsymbol{\omega}(x) = T_w\,\mathbf{m}(x) = L^{-1}\,\mathbf{m}(x).$$

The translation we sought — from the language we speak to the language w speaks — is *the inverse* of the matrix the algorithm constructed. The procedure discovered $L = T_w^{-1}$ because it worked forward through monomial coordinates, recording at each stage how each monomial decomposes onto orthogonal directions.

(e) At this point, we assert a few actionable steps:

1. **Assemble:** Build the Hankel matrix $G_w^{(\mathcal{M})} = \left(\mu_w^{(j+k)}\right)$ from the moment sequence (Exercise 3).
2. **Factor:** Compute $G_w^{(\mathcal{M})} = LDL^\top$ by repeated Schur complementation — pivot, ratios, Schur complement, repeat.

3. **Apply:** Read off $\boldsymbol{\omega} = L^{-1}\mathbf{m}$.

To demonstrate the procedure concretely, we again consider the arcsine weight $w = w_A$ on $\mathcal{X} = (-1, 1)$ and take $n = 4$, so that $\mathcal{M}_{n-1} = \{1, x, x^2, x^3\}$.

(i) Using the substitution $x = \cos\theta$, show that

$$\mu_{w_A}^{(k)} = \int_{\mathcal{X}} x^k \, w_A(x)\, dx = \frac{1}{\pi}\int_0^{\pi} \cos^k(\theta)\, d\theta.$$

Conclude that the odd moments vanish, $\mu_{w_A}^{(2k+1)} = 0$, while the first several even moments are

$$\mu_{w_A}^{(0)} = 1, \qquad \mu_{w_A}^{(2)} = \frac{1}{2}, \qquad \mu_{w_A}^{(4)} = \frac{3}{8}, \qquad \mu_{w_A}^{(6)} = \frac{5}{16}.$$

(ii) Assemble the Hankel Gram matrix

$$G_{w_A}^{(\mathcal{M}_3)} = \begin{pmatrix} \mu_w^{(0)} & \mu_w^{(1)} & \mu_w^{(2)} & \mu_w^{(3)} \\ \mu_w^{(1)} & \mu_w^{(2)} & \mu_w^{(3)} & \mu_w^{(4)} \\ \mu_w^{(2)} & \mu_w^{(3)} & \mu_w^{(4)} & \mu_w^{(5)} \\ \mu_w^{(3)} & \mu_w^{(4)} & \mu_w^{(5)} & \mu_w^{(6)} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 1/2 & 0 \\ 0 & 1/2 & 0 & 3/8 \\ 1/2 & 0 & 3/8 & 0 \\ 0 & 3/8 & 0 & 5/16 \end{pmatrix}.$$

Confirm the factorization $G_{w_A}^{(\mathcal{M}_3)} = LDL^{\top}$ for lower-triangular L and diagonal D by carrying out three passes of the Schur complement procedure from parts (a)-(f).

We outline the process in its entirety, leaving it to you to fill in the gaps as needed.

- Start by partitioning the original Gram as

$$G_{w_A}^{(\mathcal{M}_3)} = \left(\begin{array}{c|ccc} 1 & 0 & 1/2 & 0 \\ \hline 0 & 1/2 & 0 & 3/8 \\ 1/2 & 0 & 3/8 & 0 \\ 0 & 3/8 & 0 & 5/16 \end{array}\right).$$

- Identify

$$a^{(1)} = 1, \qquad \mathbf{b}^{(1)} = \begin{pmatrix} 0 \\ 1/2 \\ 0 \end{pmatrix},$$

and

$$C^{(1)} = \begin{pmatrix} 1/2 & 0 & 3/8 \\ 0 & 3/8 & 0 \\ 3/8 & 0 & 5/16 \end{pmatrix}.$$

- Compute the Schur complement:

$$\begin{aligned} G^{(1)} &= C^{(1)} - \frac{1}{a^{(1)}}\mathbf{b}^{(1)}(\mathbf{b}^{(1)})^{\top} \\ &= \begin{pmatrix} 1/2 & 0 & 3/8 \\ 0 & 3/8 & 0 \\ 3/8 & 0 & 5/16 \end{pmatrix} - \frac{1}{1}\begin{pmatrix} 0 \\ 1/2 \\ 0 \end{pmatrix}\begin{pmatrix} 0 & 1/2 & 0 \end{pmatrix} \\ &= \begin{pmatrix} 1/2 & 0 & 3/8 \\ 0 & 1/8 & 0 \\ 3/8 & 0 & 5/16 \end{pmatrix}. \end{aligned}$$

- Record

$$d_0 = a^{(1)} = 1, \qquad \ell_0 = \frac{1}{a^{(1)}}\mathbf{b}^{(1)} = \begin{pmatrix} 0 \\ 1/2 \\ 0 \end{pmatrix}.$$

The matrices in the final decomposition will be

$$D = \operatorname{diag}(1, d_1, d_2, d_3), \qquad L = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1/2 & \ell_{12} & 1 & 0 \\ 0 & \ell_{13} & \ell_{23} & 1 \end{pmatrix}.$$

- Repeat: partition

$$G^{(1)} = \left(\begin{array}{c|cc} 1/2 & 0 & 3/8 \\ \hline 0 & 1/8 & 0 \\ 3/8 & 0 & 5/16 \end{array}\right),$$

identify

$$a^{(2)} = 1/2, \qquad \mathbf{b}^{(2)} = \begin{pmatrix} 0 \\ 3/8 \end{pmatrix}, \qquad C^{(2)} = \begin{pmatrix} 1/8 & 0 \\ 0 & 5/16 \end{pmatrix},$$

compute the Schur complement of $a^{(2)}$ in $G^{(1)}$ to find

$$G^{(2)} = C^{(2)} - \frac{1}{a^{(2)}}\mathbf{b}^{(2)}(\mathbf{b}^{(2)})^\top = \begin{pmatrix} 1/8 & 0 \\ 0 & 1/32 \end{pmatrix},$$

record

$$d_1 = a^{(2)} = 1/2, \qquad \ell_1 = \frac{1}{a^{(2)}}\mathbf{b}^{(2)} = \begin{pmatrix} 0 \\ 3/4 \end{pmatrix},$$

and observe the matrices in the final decomposition will be

$$D = \operatorname{diag}(1, 1/2, d_2, d_3), \qquad L = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1/2 & 0 & 1 & 0 \\ 0 & 3/4 & \ell_{23} & 1 \end{pmatrix}.$$

- Repeat: partition

$$G^{(2)} = \left(\begin{array}{c|c} 1/8 & 0 \\ \hline 0 & 1/32 \end{array}\right),$$

identify

$$a^{(3)} = 1/8 \in \mathbb{R}, \qquad \mathbf{b}^{(3)} = \begin{pmatrix} 0 \end{pmatrix} \in \mathbb{R}^1, \qquad C^{(3)} = \begin{pmatrix} 1/32 \end{pmatrix} \in \mathbb{R}^{1\times 1}$$

compute the Schur complement of $a^{(3)}$ in $G^{(2)}$ to find

$$G^{(3)} = C^{(3)} - \frac{1}{a^{(3)}}\mathbf{b}^{(3)}(\mathbf{b}^{(3)})^\top = \begin{pmatrix} 1/32 \end{pmatrix} \in \mathbb{R}^{1\times 1},$$

record

$$d_2 = a^{(3)} = 1/8, \qquad \ell_2 = \frac{1}{a^{(3)}}\mathbf{b}^{(3)} = \begin{pmatrix} 0 \end{pmatrix},$$

and observe the matrices in the final decomposition will be

$$D = \operatorname{diag}(1, 1/2, 1/8, d_3), \qquad L = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1/2 & 0 & 1 & 0 \\ 0 & 3/4 & 0 & 1 \end{pmatrix}.$$

- Since $G^{(3)} = (1/32) \in \mathbb{R}^{1\times 1}$ is a 1×1 matrix, we trivially partition it so that its lone entry $a^{(4)}$ defines the last unknown d_3:

$$d_3 = a^{(4)} = 1/32.$$

- Return the final decomposition

$$D = \operatorname{diag}(1, 1/2, 1/8, 1/32), \qquad L = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1/2 & 0 & 1 & 0 \\ 0 & 3/4 & 0 & 1 \end{pmatrix}.$$

(iii) Verify that upon re-assembling the factorized pieces, we obtain the original Gram matrix in the monomial basis:

$$G_w^{(\mathcal{M}_3)} = LDL^\top.$$

(iv) Verify that

$$L^{-1} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -1/2 & 0 & 1 & 0 \\ 0 & -3/4 & 0 & 1 \end{pmatrix},$$

(v) Define

$$\mathbf{m}(x) = \begin{pmatrix} 1 \\ x \\ x^2 \\ x^3 \end{pmatrix},$$

and verify that

$$\boldsymbol{\omega}(x) := L^{-1}\mathbf{m}(x) \implies \begin{pmatrix} \omega_0(x) \\ \omega_1(x) \\ \omega_2(x) \\ \omega_3(x) \end{pmatrix} = \begin{pmatrix} 1 \\ x \\ x^2 - \frac{1}{2} \\ x^3 - \frac{3}{4}x \end{pmatrix}.$$

(vi) Confirm that

$$\omega_0(x) = T_0(x), \qquad \omega_1(x) = T_1(x), \qquad \omega_2(x) = \frac{1}{2}T_2(x), \qquad \omega_3(x) = \frac{1}{4}T_3(x),$$

where T_0, T_1, T_2, T_3 are the first four Chebyshev polynomials,

$$T_k(x) = \cos(k \arccos x), \qquad k = 0, 1, 2, 3.$$

In particular, conclude that our "modified" Gram-schmidt procedure developed here yields the **monic** Chebyshev polynomials when applied to the monomial basis $\mathcal{M}_{n-1}$ under the arcsine weight w_A.

3.4.3 The Orthogonal Polynomial Pipeline

At this point, we take a moment to summarize what we've learned through the first four exercises.

(i) In Exercise 1, we discovered that if $\alpha\colon\ \mathcal{X} \to [0,\infty)$ was a monotone increasing function defined on $\mathcal{X} \subseteq \mathbb{R}$ (possibly a finite or countable subset), then if V is a vector space of functions on $\mathcal{X}$ and $f, g \in V$, the Stieltjes pairing

$$\langle f, g\rangle_\alpha := \int_{\mathcal{X}} f(x)g(x)\, d\alpha(x)$$

defined a valid semi-inner product on V. We realized that we could upgrade the pairing to a *genuine* inner product — i.e., one which is **definite**, $\langle f, f\rangle_w = 0 \implies f \equiv 0 \in V$ — whenever:

  (a) α is a step function with n positive jumps: $\alpha(x_i) = 0$ for $x_i \in \mathcal{X} = \{x_1, x_2, \dots, x_n\}$. In this case, $\langle\cdot,\cdot\rangle_\alpha$ defines a valid inner product *only on* the polynomials of degree at most $n-1$, $V = \Pi_{n-1}$.

  (b) α is continuously differentiable with positive derivative $\alpha' > 0$. In this case, $\langle\cdot,\cdot\rangle_\alpha$ defines a valid inner product on the *full space* of continuous functions, $V = C(\mathcal{X})$.

In this latter case, we often choose to identify $\langle\cdot,\cdot\rangle_\alpha$ equivalently by defining

$$w(t) = \alpha'(t),$$

and instead working with the inner product

$$\langle f, g\rangle_w := \int_{\mathcal{X}} f(x)g(x)\, \underbrace{w(x)\, dx}_{d\alpha(x)}.$$

Indeed, if instead we'd first identified a positive, continuous function $w\colon\ \mathcal{X} \to (0,\infty)$, we could immediately construct $\langle\cdot,\cdot\rangle_w$ as above, implicitly understanding that it corresponds with an equivalent $\langle\cdot,\cdot\rangle_\alpha$ with $\alpha(x) = \int_{\inf(\mathcal{X})}^{x} w(t)\, dt$.

(ii) We then applied the mechanism of Exercise 1 by taking $f = T_m$ and $g = T_n$ the mth and nth Chebyshev polynomials — i.e., $T_k(x) = \cos(k \arccos x)$, where $k = 0, 1, 2, \dots$ and $x \in [-1, 1]$ — and asking whether there existed a weight function w on $(-1, 1)$ under which

$$\langle T_m, T_n\rangle_w \stackrel{?}{=} 0, \qquad m \neq n.$$

We discovered that such a function w did exist, and moreover, that (up to a normalizing constant) it was *unique*. Indeed, we imposed as convention a preference for *normalized* weights w with the property that $\int_{\mathcal{X}} w(x)\, dx = 1$, in turn lending the weights a much desired probabilistic interpretation. What we found was that the unique normalized solution rendering the Chebyshev polynomials orthogonal was the **arcsine weight**

$$w_A(x) = \frac{1}{\pi\sqrt{1-x^2}}, \qquad x \in (-1, 1).$$

The name "arcsine weight" here owes itself to the fact that the integrator α_A corresponding to $w_A = \alpha_A'$ is, of course,

$$\alpha_A(x) = \frac{1}{2} + \frac{1}{\pi}\arcsin(x), \qquad x \in [-1, 1].$$

To verify the orthogonality of the T_k under $\langle\cdot,\cdot\rangle_{w_A}$, we can identify their equivalence with our known cosine orthogonality relations from Chapter 2: on substituting $x = \cos\theta$, we find

$$\langle T_m, T_n\rangle_{w_A} = \int_{-1}^{1} T_m(x)T_n(x)\cdot\frac{dx}{\pi\sqrt{1-x^2}} = \frac{1}{\pi}\int_0^{\pi}\cos(m\theta)\cos(n\theta)\,d\theta = \begin{cases} 0, & m \neq n, \\ 1, & m = n = 0, \\ \frac{1}{2}, & m = n \geq 1. \end{cases}$$

At this point, we want to be honest about the logic here. There was something deliberately circular in this derivation. We *already knew* from the Intermission that the Chebyshev polynomials T_k were "cosines in disguise," so when we asked "can the T_k be made orthogonal under some weight w?", of course the answer was yes — we just ripped off the disguise by substituting $x = \cos\theta$, at which point the orthogonality was a property of the *cosines themselves*, inherited wholesale from Chapter 2. We did not discover the arcsine weight by any general method; we *recognized a symmetry* and *exploited it.* The substitution $x = \cos\theta$ "worked" because it simultaneously did three things we certainly cannot count on in general: it (a) turned $T_k(x)$ into $\cos(k\theta)$; it (b) absorbed the Jacobian $\sin\theta$ (from $dx = -\sin\theta d\theta$) into the weight (since the substitution gives $\sqrt{1-x^2} = \sqrt{1-\cos^2(\theta)} = \sqrt{\sin^2(\theta)} = \sin\theta$); and it (c) happened to turn into an integral whose orthogonality we already recognized from our previous work in Chapter 2.

At the time, this was all well and good because it happened to work out. It should be clear, though, that we cannot count on the strategy to save us again.

(iii) In Exercise 3, we made the observation that's been implicitly guiding us towards a more general method. Indeed, working under a continuous, strictly positive weight w, we examined the Gram matrix $G_w^{(\mathcal{M})}$ of the monomial basis $\mathcal{M}_{n-1} = \{1, x, x^2, \dots, x^{n-1}\}$ under $\langle\cdot,\cdot\rangle_w$ and discovered that its entries are **moments** — i.e., expectations of the form

$$(G_w^{(\mathcal{M})})_{jk} = \langle x^j, x^k\rangle_w = \int_{\mathcal{X}} x^{j+k} w(x)\,dx =: \mu_w^{(j+k)}.$$

We categorized $G_w^{(\mathcal{M})}$ as **Hankel** — constant along anti-diagonals — and thus asserted it was governed entirely by the list of $2n-1$ moments

$$\left\{\mu_w^{(0)}, \mu_w^{(1)}, \dots, \mu_w^{(2n-2)}\right\}.$$

The reason this matters is computational. Each entry in a Gram matrix $G_w^{(\Psi)}$ is an integral — an inner product $\langle\psi_j, \psi_k\rangle_w$ that must be evaluated. A general $n \times n$ matrix has n^2 entries; since any Gram matrix is symmetric, we need only compute the $\frac{n(n+1)}{2}$ entries on and above the diagonal. But $\frac{n(n+1)}{2}$ is still $O(n^2)$ — the number of integrals we'd need to evaluate grows *quadratically* in the number of basis functions. Under the monomial basis, however, the Hankel property collapses all $\frac{n(n+1)}{2}$ distinct entries down to $2n-1$ moments: every entry on the same anti-diagonal $j+k=r$ equals the same moment $\mu_w^{(r)}$, so we compute each moment once and fill in the matrix by lookup. The cost drops from $O(n^2)$ integrals to $O(n)$ — and, as we are about to see, the generating function will allow us to extract the entire moment sequence from a *single* computation, rather than evaluating $\int x^k w(x)\,dx$ separately for each k. And what we proved then was that *any* Gram matrix $G_w^{(\Psi)}$ that "turns out to be Hankel" *must* have $\Psi = a\mathcal{M} + b$ (in loose notation) — i.e., Hankel Grams come from monomial bases, full stop. For this reason, if we want the nice computational properties of the Hankel Gram, we're essentially forced to rely on the monomial basis $\mathcal{M}_{n-1}$.

(iv) In Exercise 4, we took the machinery one step further. Given the Hankel Gram matrix $G_w^{(\mathcal{M})}$ assembled from the moments $\{\mu_w^{(k)}\}_{k=0}^{2n-2}$, we showed that $G_w^{(\mathcal{M})}$ admitted a factorization

$$G_w^{(\mathcal{M})} = LDL^\top,$$

where L is unit lower-triangular and D is diagonal with positive entries. What makes this factorization remarkable is that it *is* Gram-Schmidt, expressed in the language of matrix algebra. At each stage r, we remove from the remaining monomials $x^{r+1}, x^{r+2}, ...$ whatever component they share with the rth orthogonal polynomial ω_r; the projection coefficients $\ell_{j,r} = \langle x^j, \omega_r \rangle_w / \|\omega_r\|_w^2$ fill in the lower triangle of L, and the squared norms $d_r = \|\omega_r\|_w^2$ occupy the diagonal of D. The factorization separates direction from scale: L and $L^\top$ encode how each monomial decomposes along the orthogonal directions, while D records how large each direction is. Inverting L reassembles the orthogonal polynomials from the monomials:

$$\boldsymbol{\omega}(x) = L^{-1}\mathbf{m}(x).$$

We verified the procedure by feeding the arcsine moments into the Hankel matrix, carrying out three passes of the factorization, and recovering the first four monic Chebyshev polynomials — the same polynomials whose orthogonality we had verified in Exercise 2, only this time, seemingly constructed *from scratch* using nothing more than the Gram matrix $G_{w_A}^{(\mathcal{M})}$ of the monomial basis $\mathcal{M}$ under the continuous, positive, normalized arcsine weight w_A.

The pipeline, then, is *real*:

$$\underbrace{\mathcal{M}_{n-1},\ w}_{\textbf{Basis \& Weight}} \xrightarrow{\text{Integrate: } \mu_w^{(r)} = \int_{\mathcal{X}} x^r w(x)\,dx} \underbrace{\mu_w^{(0)},\ \mu_w^{(1)},\ \dots,\ \mu_w^{(2n-2)}}_{\textbf{Moment sequence}}$$

$$\xrightarrow{\text{Assemble: } (G_w^{(\mathcal{M})})_{jk} = \mu_w^{(j+k)}} \underbrace{G_w^{(\mathcal{M})}}_{\textbf{Hankel matrix}} \xrightarrow{\text{Factor: } G = LDL^\top} \underbrace{\boldsymbol{\omega}(x) = L^{-1}\mathbf{m}(x)}_{\textbf{Orthogonal polynomials}}$$

What we've just demonstrated at the end of Exercise 4 is that the pipeline recovers the (monic) Chebyshev polynomials when we take $w = w_A$ the (normalized) arcsine weight, exactly as promised. Indeed, this is exciting to us for a few reasons. First of all, it is *much easier* to identify a suitable weight function w *producing* an orthogonal family of polynomials $\Omega = \{\omega_0, \omega_1, ...\}$ than it is to "identify a family of polynomials and hope to verify that they're orthogonal by finding a suitable weight." The latter is what we did in Exercise 2: we started with the Chebyshev polynomials T_k, *asked* whether there existed a weight rendering them orthogonal, and succeeded only because a specific substitution, $x = \cos\theta$, happened to reduce the question to one we'd already answered. The pipeline inverts this chain of logic entirely. We start with *any* continuous, positive, normalized weight $w\colon \mathcal{X} \to (0, \infty)$; the Hankel matrix $G_w^{(\mathcal{M})}$ and $LDL^\top$ factorization do the rest. The orthogonal polynomials are no longer *guessed* — they are *produced*.

Second, notice that the pipeline is *universal*: at no point does it depend on the specific form of w. The same three steps — integrate, assemble, factor — work for any weight satisfying the conditions of Exercise 1. We do not need to know anything about the resulting polynomials in advance; we do not need a clever substitution; we do not even need to know whether the family has a name. The machinery is entirely *mechanical*, which is to say, it is a genuine *method* — the kind of thing we confessed in (ii) that the Chebyshev computation was not.

The only obstacle is the first arrow. Integration is hard. Computing $\mu_w^{(r)} = \int_{\mathcal{X}} x^r w(x)\, dx$ one moment at a time, for each $r = 0, 1, \ldots, 2n-2$, requires us to evaluate $2n-1$ separate integrals — and for most weights of practical interest, these integrals have no elementary closed form that we can simply read off. What we want is a single object that encodes the entire moment sequence at once: something we compute *once*, and from which we can then extract $\mu_w^{(0)}, \mu_w^{(1)}, \mu_w^{(2)}, \ldots$ as needed. The object is called a **generating function**.

3.4.4 Generating Functions

Exercise 5: OGFs and PGFs

A sequence of real numbers $\{a_0, a_1, a_2, \ldots\}$ is, at the end of the day, a list. But we have spent the better part of these exercises and, indeed, this chapter, learning that lists become more useful when we embed them in *structure*. Given the structural focus of this chapter, the polynomial under a fixed basis, it is only sensible for us to begin our pursuit of "a better way to compute moments $\mu_w^{(r)}$" by presenting the following formal definition.

Definition 3.4.4 (Ordinary Generating Function (OGF))

Given a sequence of real numbers $(a_k)_{k\geq 0}$ whose members $|a_k| < \infty$ are each finite, we define its **ordinary generating function** (OGF) to be the *formal* power series

$$A(t) := \sum_{k=0}^{\infty} a_k t^k.$$

(We say "formal" here because, at this point, we do not know whether the power series $A(t)$ converges — this is, of course, a matter of the members a_k and the input t (or neither, or both, as it may happen). In any case, $A(t)$ is nothing more than a *device for organizing* the list (a_k) as the coefficients of a power series in an auxiliary variable t, with the understanding that we will "address convergence if/when it matters.")

(a) Observe that the OGF $A(t) = \sum a_k t^k$ represents the sequence (a_k) in the monomial basis $\mathcal{M} = \{1, t, t^2, \ldots\}$: the coefficient of t^k is a_k.

Consider the alternative: the infinite Lagrange basis $\mathcal{L} = \{\ell_0(t), \ell_1(t), \ell_2(t), ...\}$ at the integer nodes $t_k = k$, so that $\ell_j(k) = \mathbf{1}_{\{j=k\}}$. Under this basis, we could represent the sequence by $A^{(\mathcal{L})}(t) = \sum a_k \ell_k(t)$, and the coefficients would still be the a_k.

Now suppose we want to *multiply* two generating functions $A(t)$ and $B(t)$ and read off the coefficients of the product. Under the monomial basis, the product $t^j \cdot t^k = t^{j+k}$ sends degree-j and degree-k terms to a degree-$(j+k)$ term — a clean rule involving only the *sum* of the indices. Explain why no such rule holds for the Lagrange basis: in general, $\ell_j(t) \cdot \ell_k(t)$ is not a scalar multiple of $\ell_{j+k}(t)$, nor of any single basis function.

Conclude that the monomial basis is the natural home for generating functions precisely because its multiplication rule $t^j \cdot t^k = t^{j+k}$ converts *function-multiplication* into *index-addition*. (Compare Exercise 3(h), where the same multiplicative property was the reason the monomial Gram matrix is Hankel.)

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In the last part, we gestured towards the possibility of *multiplying* OGFs. As such, suppose that

$$A(t) = \sum_{k=0}^{\infty} a_k t^k, \qquad B(t) = \sum_{k=0}^{\infty} b_k t^k$$

are OGFs of the sequences (a_k) and (b_k), and that we define

$$C(t) := A(t) \cdot B(t), \qquad t \in \mathbb{R}.$$

We would, of course, have that

$$\begin{aligned} C(t) &= (a_0 + a_1 t + a_2 t^2 + \cdots) \cdot (b_0 + b_1 t + b_2 t^2 + \cdots) \\ &= (a_0 b_0) + (a_0 b_1 + a_1 b_0)t + (a_0 b_2 + a_1 b_1 + a_2 b_0)t^2 + \cdots, \end{aligned}$$

so that, in general, we may define

$$C(t) = \sum_{k=0}^{\infty} c_k \, t^k$$

where the kth coefficient is

$$c_k = \sum_{j=0}^{k} a_j \, b_{k-j}.$$

(b) The sequence (c_j) defined above is called the **ordinary convolution** of (a_k) and (b_k),

$$(a * b)_k := \sum_{j=0}^{k} a_j \, b_{k-j}.$$

Verify that the computation above proves the following: "If $A(t)$ and $B(t)$ are the OGFs of (a_k) and (b_k), and $C(t) = A(t) \cdot B(t)$, then $C(t)$ is the OGF of the ordinary convolution $(a * b)$."

In other words: *multiplication of OGFs $A(t)$ and $B(t)$ is convolution of sequences (a_k) and (b_k)*. This is the payoff of part (a) — the index-addition rule $t^j \cdot t^k = t^{j+k}$ is the mechanism that collects, in the coefficient of t^k, exactly those products $a_j b_{k-j}$ whose indices sum to k.

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(Probability Generating Functions (PGFs)) As an initial application demonstrating the utility of generating functions, consider the case when

$$X \sim \mathrm{Bin}(n, p)$$

with PMF

$$p_k := \mathbb{P}(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \qquad k = 0, 1, \dots, n.$$

We considered binomial random variables when constructing Bernstein's polynomials — indeed, we can recognize that $p_k \equiv b_{n,k}$ — but here, we want to consider the p_k quite deliberately as a sequence of PMF evaluations.

Indeed, the members of the sequence $(p_0, p_1, \dots, p_n)$ are finite, so the sequence has an OGF. We call this the **probability generating function** (PGF) of the random variable X:

$$G_X(z) := \sum_{k=0}^{n} p_k z^k.$$

(i) Show that

$$G_X(z) = \mathbb{E}_X[z^X].$$

(Hint: We index by $X = k$ in $\mathbb{P}(X = k)$ — compare with $G_X(z) = \sum p_k z^k$!)

(ii) Verify that

$$G_X(1) = 1.$$

Why must the PGF of any other random variable satisfy $G(1) = 1$?

(iii) For $X \sim \mathrm{Bin}(n, p)$, show that

$$G_X(z) = \left((1-p) + pz\right)^n.$$

(Hint: Recall the binomial theorem,

$$(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^k b^{n-k}.$$

)

(c) Now, let $Y \sim \mathrm{Bin}(m, p)$ be independent of X. We are often interested in computing probabilities regarding the **sum** of the random variables X and Y:

$$p_{X,Y}(k) := \mathbb{P}(X + Y = k).$$

We often prefer to recast the situation by defining a new random variable, say

$$Z := X + Y,$$

and then computing probabilities of the form above relative to the PMF of Z:

$$p_Z(k) := \mathbb{P}(Z = k).$$

Of course, we don't *know* the PMF of Z yet, but remarkably, the PGF is **distribution-determining** in this case. Indeed:

(i) Verify that the PMF of $Z = X + Y$ for independent, integer-valued X and Y is the ordinary convolution of their individual PMFs:

$$p_Z(k) = \mathbb{P}(X + Y = k) = \sum_{j=0}^{k} \mathbb{P}(X = j)\mathbb{P}(Y = k - j) = \sum_{j=0}^{k} p_X(j)p_Y(k - j).$$

(ii) Using part (b), conclude that

$$G_Z(z) = G_X(z) \cdot G_Y(z).$$

(iii) For $X \sim \text{Bin}(n, p)$ and $Y \sim \text{Bin}(m, p)$, use the closed form from part (c) to verify that

$$G_Z(z) = ((1 - p) + pz)^{n+m}.$$

(iv) Compare the forms of the PGFs of X, Y, and $Z = X + Y$:

$$G_X(z) = ((1 - p) + pz)^n, \qquad G_Y(z) = ((1 - p) + pz)^m,$$
$$G_{X+Y}(z) = ((1 - p) + pz)^{n+m}.$$

Observe that G_{X+Y} retains the form of the PGF for a single binomial random variable, G_X and G_Y. Conclude that, since the PGF determines the PMF (its coefficients *are* the PMF), conclude that

$$X + Y \sim \text{Bin}(n + m, p).$$

That is: the sum of two independent binomials $X \sim \text{Bin}(n, p)$ and $Y \sim \text{Bin}(m, p)$ with the same success probability p is again binomial — $X + Y \sim \text{Bin}(n + m, p)$ with the same success probability p and sample size $n + m$ equal to the sum of the sizes of the original X and Y.

(d) (PGF Generates Factorial Moments) We now observe that the PGF encodes more than just the values of the PMF $\mathbb{P}(X = k)$ — it also provides access to a natural family of summary statistics.

(i) Again letting $X \sim \text{Bin}(n, p)$ and $G_X(z) = \sum_{k=0}^{n} p_k z^k$, apply linearity of differentiation over the sum defining G_X to show that

$$G'_X(z) = \sum_{k=1}^{n} k p_k z^{k-1}.$$

Note that we re-indexed the sum to start at $k = 1$. Why?

(ii) Show that the evaluation of G'_x at $z = 1$ yields the mean:

$$G'_X(1) = \sum_{k=1}^{n} k p_k = \mathbb{E}_X[X].$$

(iii) Differentiate again and evaluate at $z = 1$ to show that

$$G''_X(1) = \sum_{k=2}^{n} k(k - 1)p_k = \mathbb{E}_X[X(X - 1)].$$

(iv) Now, define the rth **falling factorial** as

$$X^{(\downarrow r)} = X(X-1)(X-2)\cdots(X-(r-1)) = \frac{X!}{(X-r)!},$$

and the rth **factorial moment** of X as

$$\mu^{(\downarrow r)} = \mathbb{E}_X[X^{(\downarrow r)}] = \mathbb{E}_X\left[\frac{X!}{(X-r)!}\right].$$

Show that, in general, the rth derivative of the PGF G_X at $z=1$ is the rth factorial moment:

$$G_X^{(r)}(1) = \mu^{(\downarrow r)}.$$

(Hint: the rth derivative of z^k is $k(k-1)\cdots(k-r+1)z^{k-r} = k^{(\downarrow r)}z^{k-r}$).

(v) The factorial moments are not the quite the same as the raw moments, but they're often just as useful. Indeed, while the quantity $G''_X(1) = \mathbb{E}_X[X(X-1)]$ is *not* the second moment $\mathbb{E}_X[X^2]$, it is certainly quite close. Confirm that

$$G''_X(1) = \mathbb{E}_X[X^2] - G'_X(1),$$

and conclude we may recover the raw second moment via

$$\mathbb{E}_X[X^2] = G''_X(1) + G'_X(1).$$

‖

The PGF works because discrete random variables X with countable support sets of the form $\mathcal{X} = \{0,1,2,...\}$ come with a *natural map* from their distribution to a sequence: the PMF itself,

$$k \in \{0,1,2,...\} \mapsto p_k := \mathbb{P}(X=k).$$

The index k ranges over the nonnegative integers, the PMF evaluations p_k are entries of the sequence, and the OGF of (p_k) is the PGF $G_X(z) = \sum p_k z^k$. Crucially, this map is **distribution-determining**: the PMF specifies the distribution of X completely, in the sense that "if we know $\mathbb{P}(X=k) = \mathbb{P}(Y=k)$ for $X \sim \text{Bin}(n,p)$ and all $k \in \{0,1,2,\dots,n\}$, then we immediately know that $Y \sim \text{Bin}(n,p)$." It is not possible to find "PMFs $p_X(k) = p_Y(k)$ which agree for all $k \in \{0,1,2,\dots,n\}$, yet wind up with $X \overset{d}{\neq} Y$." In this way, no information is lost in the passage from "random variable" to "sequence."

For a continuous random variable, the situation is quite different. Supposing $X \sim f_X$ a density supported on $\mathcal{X} \subseteq R$, no such map is immediately available — the density is a *function*, and it isn't immediately clear how we may "compress the uncountably many evaluations $x \in \mathcal{X} \mapsto f_X(x)$ into a countable sequence while preserving the full structure of X."

(e) To convince yourself this is the case, consider the naive map $k \in \{0,1,2,...\} \mapsto f_X(k)$ for f_X a continuous density on $\mathcal{X} \subseteq \mathbb{R}$.

(i) Explain why this map cannot be distribution-determining — either exhibit (or argue for the existence of) two continuous densities $f \neq \tilde{f}$ on $\mathcal{X} = [a,b]$ satisfying

$$f(k) = \tilde{f}(k) \quad \text{for every integer } k \in [a,b],$$

but where

$$f \overset{d}{\neq} \tilde{f} \text{ on all of } \mathcal{X} = [a,b].$$

‖

The failure displayed here is actually the same one we encountered in Exercise 1(b): evaluating a continuous function at finitely many (or even countably many) prescribed points discards almost all of its information. The map $k \in \{0, 1, 2, ...\} \mapsto f_X(k)$ "looks at" f_X at the integers and ignores everything else — and a continuous density can be freely modified between integers without affecting any of the sampled values.

What we would like, then, is a map that's able to compress the "uncountable information" contained in a density f_X into a "countable sequence" whose members *are* distribution-determining. This might seem like a lofty ambition, but as it turns out, the tool we need is already at our disposal.

(f) Given a continuous density $f_X \colon \mathcal{X} \to (0, \infty)$, define the sequence of integrals

$$a_k = \int_{\mathcal{X}} \varphi_k(x)\, f_X(x)\, dx, \qquad k = 0, 1, 2, \dots,$$

for some family of functions $\Phi = \{\varphi_0, \varphi_1, ...\}$ defined on the common support $\mathcal{X} \subseteq \mathbb{R}$.

(i) Validate the intuition for choosing the sequence above: "unlike the pointwise evaluation $k \mapsto f_X(k)$, the integrals a_k 'see' f_X over all of $\mathcal{X}$." Argue that each a_k depends on the *entire* density f_X.

(ii) Show that if we choose $\varphi_k(x) = x^k$ the kth monomial basis function, then

$$a_k = \int_{\mathcal{X}} x^k f_X(x)\, dx = \mathbb{E}_X[X^k] =: \mu^{(k)}.$$

Conclude that the Gram matrix $G^{(\mathcal{M})}_{f_X}$ can be assembled from the sequence (a_k) by the rule

$$(G^{(\mathcal{M})}_{f_X})_{jk} = a_{j+k}.$$

Explain why no other choice of Φ guarantees this. (Hint: this follows by the logic we developed in Exercise 3.)

(g) Fill in the gaps of the proof of the following theorem.

Theorem 3.4.5 (Moment Sequences on Compact Intervals)

Let f and $\tilde{f}$ be two continuous densities on a compact interval $\mathcal{X} = [a, b]$ with the same moments:

$$\int_a^b x^k f(x)\, dx = \int_a^b x^k \tilde{f}(x)\, dx, \qquad k = 0, 1, 2, ...$$

Then f and $\tilde{f}$ determine the same distribution:

$$F(x) = \widetilde{F}(x), \qquad \forall x \in [a, b].$$

(Proof:)

(i) By linearity, show that the hypothesis "all moments agree,"

$$\int_a^b x^k f(x)\,dx = \int_a^b x^k \tilde{f}(x)\,dx, \qquad k = 0, 1, 2, \dots$$

implies

$$\int_a^b p(x) f(x)\,dx = \int_a^b p(x)\tilde{f}(x)\,dx$$

for every polynomial $p \in C([a,b])$. (Hint: Use Exercise 1 after observing that $w = f$ and $w = \tilde{f}$ are continuous, positive densities on the common support $\mathcal{X} = [a,b]$.)

(ii) Let $g\colon [a,b] \to \mathbb{R}$ be any continuous function on $\mathcal{X} = [a,b]$. Whence, by the Weierstrass approximation theorem, we can identify a sequence of polynomials (p_n) such that

$$p_n \rightrightarrows g \text{ uniformly on } \mathcal{X} = [a,b].$$

Using this, show that

$$\left| \int_a^b p_n(x) f(x)\,dx - \int_a^b g(x) f(x)\,dx \right| \le \sup_{x\in[a,b]} |p_n(x) - g(x)| \cdot \underbrace{\int_a^b f(x)\,dx}_{=1},$$

and conclude that

$$\int_a^b p_n(x) f(x)\,dx \longrightarrow \int_a^b g(x) f(x)\,dx \qquad \text{as} \qquad n \to \infty.$$

(iii) Applying the same argument to $\tilde{f}$, deduce that

$$\int_a^b g(x) f(x)\,dx = \int_a^b g(x)\tilde{f}(x)\,dx$$

for every $g \in C([a,b])$.

(iv) Fix $x_0 \in (a,b)$ and $\epsilon > 0$ small enough that

$$x_0 + \epsilon < b.$$

Construct the continuous function $g_\epsilon\, [a,b] \to [0,1]$ defined by

$$g_\epsilon(x) = \begin{cases} 1, & x \in [a, x_0], \\ 1 - \dfrac{x - x_0}{\epsilon}, & x \in [x_0, x_0 + \epsilon], \\ 0, & x \in [x_0 + \epsilon, b]. \end{cases}$$

Verify that g_ϵ is continuous (Hint: clearly it suffices to show continuity of g_ϵ at the points x_0 and $x_0 + \epsilon$), and that

$$\mathbf{1}_{[a,x_0]}(x) \le g_\epsilon(x) \le \mathbf{1}_{[a,x_0+\epsilon]}, \qquad \forall x \in [a,b],$$

where we've defined the indicator function $\mathbf{1}_{[c,d]}(x) = 1$ if $x \in [c,d]$ and 0 otherwise.

(v) Using

$$\mathbb{E}_X[\mathbf{1}_{[c,d]}(x)] = \mathbb{P}(c \le X \le d) = F_X(d) - F_X(c),$$

take expectations of the indicator bound above to find

$$\mathbb{P}(a \le X \le a + x_0) \le \mathbb{E}_X[g_\epsilon(X)] \le \mathbb{P}(a \le X \le a + x_0 + \epsilon).$$

Verify this can be rearranged to yield the chain of inequalities

$$F(x_0) \le \int_a^b g_\epsilon(x) f(x)\, dx \le F(x_0 + \epsilon),$$

and conclude analogous bounds for $\tilde{f}$:

$$\tilde{F}(x_0) \le \int_a^b g_\epsilon(x) \tilde{f}(x)\, dx \le \tilde{F}(x_0 + \epsilon).$$

(vi) Using the fact that $\int g_\epsilon f = \int g_\epsilon \tilde{f}$, show that the bounds above force

$$|F(x_0) - \tilde{F}(x_0)| \le F(x_0 + \epsilon) - F(x_0)$$

for every $\epsilon > 0$.

(vii) Taking $\epsilon \downarrow 0$ and applying right-continuity of the CDF F, conclude that

$$F(x_0) = \tilde{F}(x_0),$$

as we sought to prove.

$\|$

Remark.

To summarize, the chain of reasoning is:

$$\textbf{Moments agree} \xrightarrow{\text{linearity}} \textbf{Polynomials agree} \xrightarrow{\text{Weierstrass}} \textbf{Continuous functions agree}$$
$$\xrightarrow{\text{squeeze}} \textbf{CDFs agree.}$$

Each step uses tools from this chapter. We note that the final conclusion — "CDFs agree, therefore the distributions are identical" — is itself a theorem: two probability measures that assign the same value to every interval $(-\infty, x]$ must agree on every Borel set, a result that follows from the **Dynkin π-λ theorem** (or, equivalently, the **Carathéodory extension theorem** — we'll treat each formally later). For continuous densities on compact intervals, the conclusion is immediate by differentiation; the general statement, which requires the measure-theoretic notion of **σ-algebras**, will be developed in Chapter 4. For now, we take it on faith that the CDF determines the distribution — a fact that, in any case, is consistent with everything the reader might have encountered in a first course in probability.

(h) **(Failure of the OGF on Gaussian Moments)** In Exercise 0, we defined a **standard normal** random variable $X \sim N(0,1)$ to be one whose density is defined by

$$\phi(x) := \frac{1}{\sqrt{2\pi}} e^{-x^2/2}, \qquad x \in \mathbb{R}.$$

(i) Show that the odd moments of $X \sim N(0,1)$ vanish:

$$\mu^{(2k+1)} := \mathbb{E}_X[X^{2k+1}] = 0, \qquad k = 0, 1, 2, \ldots$$

(Hint: verify the integrand $v = -x^{2k+1}e^{-x^2/2}$ is an odd function of x.)

(ii) For the even moments, apply integration by parts with

$$u = x^{2k-1}, \qquad dv = xe^{-x^2/2}$$

to show that

$$\mu^{(2k)} = (2k-1)\mu^{(2k-2)}, \qquad k = 1, 2, 3, \ldots$$

(Hint: verify that $v = -e^{-x^2/2}$ by the chain rule, and that the boundary terms $\left[x^{2k-1}(-e^{-x^2/2})\right]_{-\infty}^{\infty}$ vanish.)

(iii) Using $\mu^{(0)} = 1$, verify the recursion

$$\mu^{(2k)} = (2k-1)(2k-3)\cdots 3 \cdot 1 =: (2k-1)!!,$$

where $(2k-1)!!$ denotes the **double factorial**.

(iv) Verify that the recursion above can be rewritten as

$$\mu^{(2k)} = \frac{(2k)!}{2^k k!}.$$

(Hint: $(2k)! = (2k)(2k-1)(2k-2)\cdots(2)(1)$. Separate the event and odd factors.)

(v) Compute the first eight moments of the standard normal. Verify that

$$\mu^{(0)} = 1, \quad \mu^{(2)} = 1, \quad \mu^{(4)} = 3, \quad \mu^{(6)} = 15, \quad \mu^{(8)} = 105,$$

and that $\mu^{(1)} = \mu^{(3)} = \mu^{(5)} = \mu^{(7)} = 0$.

(vi) **(The Gutpunch)** Now attempt to form the OGF of the moment sequence: verify we have

$$(\mu^{(k)}) \mapsto \sum_{k=0}^{\infty} \mu^{(2k)} t^{2k} = \sum_{k=0}^{\infty} \frac{(2k)!}{2^k k!} t^{2k}.$$

(vii) Apply the ratio test: show that

$$\frac{\mu^{(2k+2)}t^{2k+2}}{\mu^{(2k)}t^{2k}} = (2k+1)t^2 \overset{k\to\infty}{\longrightarrow} +\infty, \quad \text{for any fixed } t \neq 0.$$

(viii) Conclude that the OGF of the Gaussian moment sequence has radius of convergence *zero*: that is, it *diverges for every* $t \neq 0$.

Remark.

We can give a second explanation for the failure by tracing the OGF back to its integral form. Recall from part (f) that the OGF of a moment sequence arises from integrating ϕ against the **geometric kernel**: formally, we observe that

$$\sum_{k=0}^{\infty} \mu^{(k)} t^k = \sum_{k=0}^{\infty} \left(\int_{-\infty}^{\infty} x^k \phi(x)\, dx \right) t^k = \int_{-\infty}^{\infty} \left(\sum_{k=0}^{\infty} (tx)^k \right) \phi(x)\, dx = \int_{-\infty}^{\infty} \frac{\phi(x)}{1-tx}\, dx,$$

provided the interchange of sum and integral is valid (as it is for the Gaussian density ϕ). One can then show that, for any $t \neq 0$, this integral diverges: the geometric kernel $1/(1-tx)$ has a pole at $x = 1/t$, which lies inside the domain of integration $\mathbb{R}$. The density $\phi(x) = e^{-x^2/2}/\sqrt{2\pi}$ is strictly positive at $x = 1/t$, so the integrand blows up and the integral cannot converge. What makes this so devastating is that t is the input *we provided*: and the second we do, the integral seems to avoid us entirely, placing a pole at $x = 1/t$. One should also verify that this version of the proof makes it quite apparent the failure was *independent* of the Gaussian — the issue here was fundamentally one of "attempting to define the OGF on an unbounded interval." The interchange was doomed from the start.

Thus, the OGF $\sum_k \mu^{(k)} t^k$ of the Gaussian moment sequence $(\mu^{(k)})$ does not merely "fail to converge on all of $\mathbb{R}$" — it converges *nowhere*. The moments $(2k-1)!!$ grow so fast that no geometric damping t^{2k} can control them. This is the standard normal distribution — the most natural weight function in statistics — and the OGF proves to be *completely useless* for it. In the next exercise, we introduce a (likely familiar) tool that will ultimately resolve this failure.

Exercise 6: Gamma, Stirling, and Factorial Growth

Exercise 5 ended on a failure: the ordinary generating function of the Gaussian moment sequence,

$$G_Z(t) := \sum_{k=0}^{\infty} \mu^{(2k)} t^{2k} = \sum_{k=0}^{\infty} (2k-1)!! t^{2k} = \sum_{k=0}^{\infty} \frac{(2k)!}{2^k k!} t^{2k},$$

diverges *everywhere*. The moments $\mu^{(2k)} = (2k-1)!!$ grow so fast in k that no geometric damping by a fixed $t \in \mathbb{R} \mapsto t^{2k}$ can save them, and the underlying geometric kernel $1/(1-tx)$ has a pole on $\mathbb{R}$ at $x = 1/t$ for every $t \neq 0$.

In this exercise, we attempt to diagnose the structural reason for this failure in an effort to ultimately rectify it. To start, suppose we stare at the rightmost form of the Gaussian OGF coefficient:

$$\mu^{(2k)} = \frac{(2k)!}{2^k k!}.$$

(a) (Gamma function) The coefficient $\mu^{(2k)} = \frac{(2k)!}{2^k k!}$ involves factorials — to understand *why* the OGF fails, we need to understand *how fast* factorials grow. We claim that a tool that will prove useful in this pursuit is the factorial's *analytic extension*.

Indeed, suppose we define the important **Gamma function**:

$$\Gamma(s) := \int_0^{\infty} x^{s-1} e^{-x}\, dx, \qquad s > 0.$$

On its nose, $\Gamma(s)$ is nothing more than an integral.

(i) Using integration by parts with

$$u = x^{s-1}, \qquad dv = e^{-x}\,dx,$$

verify the recursion

$$\Gamma(s) = (s-1)\Gamma(s-1), \qquad s > 1.$$

(Hint: verify that the boundary terms vanish: $\lim_{x\to\infty} x^{s-1}e^{-x} = 0$ since exponential decay of e^{-x} eventually dominates x^{s-1} for fixed s as $x \to \infty$. Furthermore, the contribution at $x = 0$ vanishes for $s > 1$.)

(ii) Verify that

$$\Gamma(1) = \int_0^\infty e^{-x}\,dx = 1.$$

(iii) Applying an inductive argument to the results above, conclude that

$$\Gamma(n+1) = n!, \qquad n = 0, 1, 2, \ldots$$

(iv) Verify the following explicit formula for computing $n!$:

$$n! = \int_0^\infty x^n e^{-x}\,dx, \qquad n = 0, 1, 2, \ldots$$

(b) (Stirling's approximation) What the integral representation of $n!$ provides is thus a means for its *approximation by exponentials*.

(i) Verify that the Gamma integrand may be rearranged as

$$x^n e^{-x} := e^{\varphi_n(x)}, \qquad \varphi_n(x) := n\ln x - x.$$

(ii) Show that $\varphi_n(x)$ achieves a global maximum at

$$x^\star = n.$$

(iii) Verify that φ_n in a Taylor series about $x = n$: show that the substitution $u = x - n \to x = u + n$ gives

$$\varphi_n(n+u) = (n\ln n - n) - \frac{u^2}{2n} + O\left(\frac{u^3}{n^2}\right).$$

In particular:

- Write
$$\varphi_n(n+u) = n\ln(n+u) - (n+u),$$
- Factor n out of the logarithm to obtain
$$n\ln(n+u) = n\ln n + n\ln(1+u/n).$$
- Expand $\ln(1+v)$ where $v = u/n$:
$$\ln(1+v) = v - \frac{v^2}{2} + \frac{v^3}{3} - \cdots$$
- Substitute $v = u/n$ back and multiply by n:
$$n\ln(1+u/n) = u - \frac{u^2}{2n} + \frac{u^3}{3n^2} - \cdots$$

- Return to the original:

$$\begin{aligned}\varphi_n(n+u) &= n\ln n + n\ln(1+u/n) - (n+u)\\ &= (n\ln n) + \left(u - \frac{u^2}{2n} + \frac{u^3}{3n^2} - \cdots\right) - (n+u)\\ &= (n\ln n - n) - \frac{u^2}{2n} + \frac{u^3}{3n^2} - \cdots\\ &= (n\ln n - n) - \frac{u^2}{2n} + O\left(\frac{u^3}{n^2}\right).\end{aligned}$$

(iv) Observe that the linear term u from $n\ln(n+u) = n\ln n + u - \frac{u^2}{2n} + O(u^3/n^2)$ cancels with the $-u$ from $-(n+u)$. This cancellation is not a coincidence; verify that it *is* the statement that

$$\varphi_n'(n) = 0;$$

i.e., that $x = n$ is a critical point of φ_n. Furthermore, confirm that the surviving quadratic term $-u^2/(2n)$ reflects the curvature at the peak:

$$\varphi''(n) = -\frac{1}{n} \quad \text{at the peak} \quad x^\star = n.$$

(v) Dropping the higher-order terms, conclude that near $x^\star = n$,

$$x^n e^{-x} = e^{\varphi_n(x)} \approx e^{(n\ln n - n) - u^2/(2n)} = n^n e^{-n} \cdot e^{-u^2/2n}, \qquad u = x - n.$$

That is: near its peak, the Gaussian integrand $x^n e^{-x}$ is

$$x^n e^{-x} \approx \left(\frac{n}{e}\right)^n \cdot e^{-u^2/2n}, \qquad u = x - n.$$

and verify the following interpretation: ”near its peak, the Gamma integrand $x^n e^{-x}$ is closely approximated by *a Gaussian density* centered at $x = n$, with variance n, and peak height at $(n/e)^n$.”

(vi) Approximate $n!$ by integrating the Gaussian approximation over all of $\mathbb{R}$:

$$n! = \int_0^\infty x^n e^{-x}\,dx \approx \left(\frac{n}{e}\right)^n \int_{-\infty}^\infty e^{-u^2/2n}\,du.$$

(Hint: use the Gaussian integral from Exercise 0: under the substitution $u = \sqrt{2n}v$, verify that

$$\int_{-\infty}^\infty e^{-u^2/(2n)}\,du = \sqrt{2n} \cdot \int_{-\infty}^\infty e^{-v^2}\,dv = \sqrt{2\pi n}.$$

)

(vii) Conclude **Stirling’s approximation** for $n!$:

$$\boxed{n! \;\approx\; \sqrt{2\pi n} \cdot \left(\frac{n}{e}\right)^n}$$

(viii) We say that $n!$ is **asymptotically** a Gaussian integral. The factorial — a combinatorial object, defined by the iterative multiplications of $n \times (n-1) \times \cdots \times 1$ — more and more closely begins to resemble $\sqrt{2\pi}$ (a fairly significant constant, as far as constants go) "times $n^{(n+1/2)}e^{-n}$." When this approximation is made rigorous (see Chapter 6), one shows that the **relative error** of the approximation tends to zero:

$$\frac{n!}{\sqrt{2\pi n}(n/e)^n} \longrightarrow 1 \qquad \text{as} \qquad n \to \infty.$$

For now, we simply ask you to validate the rationale behind the conventional wisdom that "Stirling's approximation gets better for larger and larger n." (Hint: we approximated $\varphi_n(n+u)$ by its quadratic term $-u^2/(2n)$ and dropped the cubic remainder $O(u^3/n^2)$.)

(ix) Verify the quality of the approximation numerically. Compute

$$S(n) := \sqrt{2\pi n}(n/e)^n$$

for $n = 1, 5, 10$, and compare with the exact values:

n	$n!$	$S(n)$	$S(n)/n!$
1	1		
5	120		
10	3,628,800		

(The reader should find that Stirling underestimates by roughly 8% at $n = 1$, by 1.7% at $n = 5$, and by $< 1\%$ at $n = 10$.)

(c) (Diagnosing the OGF failure) We now apply Stirling to understand precisely why the Gaussian OGF diverges, and to identify a prospective fix.

(i) Using Stirling's approximation, verify that

$$k! \approx \sqrt{2\pi k}\left(\frac{k}{e}\right)^k, \qquad (2k)! \approx \sqrt{4\pi k}\left(\frac{2k}{e}\right)^k,$$

and assemble these forms to show that the Gaussian moment OGF coefficients $\mu^{(2k)}$ satisfy

$$\mu^{(2k)} = \frac{(2k)!}{2^k k!} \approx \frac{\sqrt{4\pi k}(2k/e)^{2k}}{2^k \cdot \sqrt{2\pi k}(k/e)^k} = \sqrt{2}\left(\frac{2k}{e}\right)^k.$$

(ii) When we form the OGF, the kth term is $\mu^{(2k)} \cdot t^{2k}$. Verify that substituting the Stirling form here gives the following approximation for the kth term in the Gaussian OGF G_Z:

$$\mu^{(2k)} \cdot t^{2k} \approx \sqrt{2}\left(\frac{2kt^2}{e}\right)^k.$$

(iii) With the approximation in hand, substitute it into the Gaussian OGF G_Z:

$$G_Z(t) = \sum_{k=0}^{\infty} \mu^{(2k)} t^{2k} \approx \sqrt{2}\sum_{k=0}^{\infty}\left(\frac{2kt^2}{e}\right)^k.$$

Explain why this diverges for every fixed $t \neq 0$. (Hint: the base $2kt^2/e$ grows with k, so for any fixed t, there exists a $K \in \mathbb{N}$ such that for all $k > K$, $2kt^2/e > 1$. Past this point, we are "raising a number exceeding 1 to an increasing power.")

Remark.

We say that the Gaussian moments $\mu^{(2k)}$ grow **super-exponentially**: in the denominator of the series defining $G_Z(t) = \sum_k \mu^{(2k)} t^{2k}$ the geometric factor of t^{2k} for fixed $t \in \mathbb{R}$ multiplies $t \times t \times \cdots \times t$ a total of $2k$ times. This is "where we would *hope* that setting t large in the denominator would act to dampen the growth of the moments $\mu^{(2k)}$." What we've just shown, however, is that $\mu^{(2k)}$ grows too fast for such a *fixed* damping factor to tame — it "grows faster than an exponential (of a fixed base)," whence the term *super*-exponential.

(d) (Hinting at the fix) What the analysis has revealed to us, however, is precisely *how fast* the OGF G_Z's coefficients grow — and knowing this rate of growth immediately suggests a prospective solution.

We showed in part (c) that

$$\mu^{(2k)} \cdot t^{2k} \approx \sqrt{2}\left(\frac{2kt^2}{e}\right)^k.$$

The problem here is the fact that the base $2kt^2/e$ *grows in* k, the same factor that appears in the exponent — if "we could only get rid of" the factor of k^k, we'd be able to dampen the remaining $(2t^2/e)^k$ by a geometric factor. To neutralize it, we need to divide by a sequence that *kills* the factor of $(2k/e)^k$. Indeed, what if we simply substituted a new coefficient: say,

$$\mu^{(2k)} \longmapsto \frac{\mu^{(2k)}}{(2k)!}?$$

(i) Using Stirling, we can compute

$$\frac{\mu^{(2k)}}{(2k)!} \cdot t^{2k} \approx \frac{\sqrt{2}(2k/e)^k}{\sqrt{4\pi k}(2k/e)^{2k}} \cdot t^{2k} = \frac{1}{\sqrt{2\pi k}}\left(\frac{et^2}{2k}\right)^k.$$

Verify that, for any fixed $t \in \mathbb{R}$, there exists a $K \in \mathbb{N}$ such that for all $k > K$,

$$\frac{et^2}{2k} < 1.$$

(ii) Suppose we were to define

$$M_Z(t) := \sum_{k=0}^{\infty} \frac{\mu^{(2k)}}{(2k)!} t^{2k}.$$

In light of the behavior characterized in (i), assert the convergence of $M_Z(t)$ for every $t \in \mathbb{R}$ — since the base shrinks for $k > K$, we are raising a number less than 1 to an increasing power. We say the terms of $M_Z(t)$ decay **super-geometrically** to zero. (Compare: the OGF had a growing base that eventually exeeded 1, making its growth super-exponential; dividing by $(2k)!$ replaced it with a shrinking base that eventually falls below 1, making its decay super-geometric.)

(iii) Verify the convergence concretely, using the identity

$$\frac{\mu^{(2k)}}{(2k)!} = \frac{1}{2^k k!},$$

(see Exercise 5(i)), confirm that

$$M_Z(t) = \sum_{k=0}^{\infty} \frac{t^{2k}}{2^k k!} = \sum_{k=0}^{\infty} \frac{(t^2/2)^k}{k!} = e^{t^2/2}.$$

The closed form $M_Z(t) = e^{t^2/2}$ should be surprising; it certainly is to us. Realize that we divided by $(2k)!$ for purely analytic reasons — to "neutralize a growing base" and "restore convergence." We were *not* "pursuing a closed-form": we would have been happy with mere convergence — a series that converged to *some* function, named or not. Yet, what's emerged all the same is the familiar exponential. This demands some explanation — this is the subject of the following exercise.

Exercise 7: EGFs and MGFs

We closed exercise 6 on something of a cliff-hanger: we derived the formal power series

$$M_Z(t) = \sum_{k=0}^{\infty} \frac{\mu^{(2k)}}{(2k)!} t^{2k}$$

by dividing the divergent Gaussian OGF $G_Z(t) = \sum_k \mu^{(2k)} t^{2k}$ by $(2k)!$. In turn, this simple analytic maneuver rendered $M_Z(t)$ convergent everywhere. This much was intended.

What was *unintended*, however, was for our maneuver to render a *closed-form expression* for M_Z in terms of the exponential function e^x; and this makes the fact that, indeed, we have

$$M_Z(t) = e^{t^2/2}$$

all the more remarkable.

In this exercise, we want to explore this phenomenon. We begin by formalizing the "$1/k!$-rescaled OGF" that emerged in the last exercise at several levels which, we believe, are quite complementary.

(a) (The exponential generating function (EGF)) Given a sequence (a_k) whose member are finite, $|a_k| < \infty$ for all $k = 0, 1, 2, ...$, recall we defined its *ordinary* generating function (OGF) to be the formal power series

$$G(t) := \sum_{k=0}^{\infty} a_k t^k, \qquad t \in \mathbb{R}.$$

We now define the **exponential generating function** (EGF) of the sequence (a_k) (under the same conditions) to be the "$1/k!$-rescaled" variant:

$$\hat{G}(t) := \sum_{k=0}^{\infty} \frac{a_k}{k!} t^k, \qquad t \in \mathbb{R}.$$

Before digging into the EGF, we want to motivate a new operation we'll be defining shortly.

(i) Recall from Exercise 5 that OGFs enjoy a **multiplicative property** with respect to *ordinary convolution*: if $A(t)$ and $B(t)$ are OGFs of (a_k) and (b_k), then their product is

$$A(t) \cdot B(t) = \sum_{k=0}^{\infty} (a * b)_k \, t^k;$$

Really convince yourself of the intuition behind the ordinary convolution: when we multiply the power series $A(t) = \sum a_j t^j$ and $B(t) = \sum b_m t^m$, each pair of terms $a_j t^j$ and $b_m t^m$ contributes $a_j b_m t^{j+m}$ to the product. The coefficient of t^k in $A(t) \cdot B(t)$ therefore collects *all pairs* (j, m) with $j + m = k$:

$$\underbrace{(a * b)_k}_{\text{coefficient of } t^k \text{ in } A(t) \cdot B(t)} = a_0 b_k + a_1 b_{k-1} + \cdots + a_k b_0 = \sum_{j=0}^{k} a_j b_{k-j}$$

(b) (Binomial convolution) We now ask: what is the analogous operation for EGFs?

(i) Suppose that

$$\hat{A}(t) = \sum_{k=0}^{\infty} \frac{a_k}{k!} t^k, \qquad \hat{B}(t) = \sum_{k=0}^{\infty} \frac{b_k}{k!} t^k$$

are now the EGFs of the same sequences (a_k) and (b_k). Show that multiplying $\hat{A}(t) \cdot \hat{B}(t)$ and collecting the coefficient of t^k proceeds by the same logic as part (a): each coefficient of t^k in $\hat{A}(t) \cdot \hat{B}(t)$ receives contributions by pairs $a_j/j!$ and $b_{k-j}/(k-j)!$, and collecting all such pairs gives

$$\left[\text{coefficient of } t^k \text{ in } \hat{A}(t) \cdot \hat{B}(t)\right] = \frac{a_0 b_k}{0! k!} + \frac{a_1 b_{k-1}}{1!(k-1)!} + \cdots + \frac{a_k b_0}{k! 0!} = \sum_{j=0}^{k} \frac{a_j b_{k-j}}{j!(k-j)!}.$$

(ii) Factor out $1/k!$ and verify that

$$\sum_{j=0}^{k} \frac{a_j \, b_{k-j}}{j!(k-j)!} = \frac{1}{k!} \sum_{j=0}^{k} \binom{k}{j} a_j \, b_{k-j},$$

where

$$\binom{k}{j} = \frac{k!}{j!(k-j)!}.$$

(iii) The operation that has appeared is not the ordinary convolution — it carries an extra factor of $\binom{k}{j}$. Accordingly, we define the **binomial convolution** of (a_k) and (b_k) to be

$$(a \star b)_k := \sum_{j=0}^{k} \binom{k}{j} a_j \, b_{k-j},$$

(Note: $(a \star b)_k$, with a star $\star$ — as opposed to $(a * b)_k = \sum_{j=0}^{k} a_j \, b_{k-j}$, the ordinary convolution from Exercise 5, with an asterisk $*$).

Conclude that multiplication of EGFs is stable under binomial convolution: i.e., that the product

$$\hat{A}(t) \cdot \hat{B}(t) = \sum_{k=0}^{\infty} \frac{(a \star b)_k}{k!} \, t^k,$$

where, in particular,

$$\underbrace{\frac{(a\star b)_k}{k!}}_{\text{coefficient of } t^k \text{ in } \hat{A}(t)\cdot\hat{B}(t)} = \frac{1}{k!}\left[\binom{k}{0}a_0 b_k + \binom{k}{1}a_1 b_{k-1} + \cdots + \binom{k}{k}a_0 b_k\right] = \frac{1}{k!}\sum_{j=0}^{k}\binom{k}{j}a_j\, b_{k-j}$$

(One should quickly recognize the structural parallel with what we'd identified for the ordinary convolution $(a*b)$.)

(c) (Independent moments convolve binomially) The binomial convolution $(a\star b)$ that fell out of the multiplication of the EGFs $\hat{A}(t)\cdot\hat{B}(t)$ is thus "not a mere curiosity" — it is *the operation* that moment sequences *actually undergo*. Let us make this assertion rigorous — in the following, we suppose that X and Y are continuous random variables with finite moments of all orders: $\mathbb{E}_X[X^r] := \mu_X^{(r)}$ and $\mathbb{E}_Y[Y^r] := \mu_Y^{(r)}$ exist for all $r = 0, 1, 2, \ldots$.

(i) Expand $(X+Y)^k$ via the binomial theorem and take expectations with respect to the **joint distribution** of the pair (X, Y),

$$(X,Y) \sim f_{X,Y}, \qquad f_{X,Y} \colon \mathbb{R}^2 \to [0,\infty),$$

where $f_{X,Y}(x,y) = 0$ outside the support $S_{X,Y}$ of (X,Y). In this case, we can take the expectation of the function $(X,Y) \mapsto (X+Y)^k$ under $f_{X,Y}$: show that if

$$\mu_{X+Y}^{(k)} := \mathbb{E}_{X,Y}[(X+Y)^k] = \iint_{\mathcal{S}_{X,Y}} (x+y)^k f_{X,Y}(x,y)\,dxdy,$$

we can write

$$\mathbb{E}_{X,Y}[(X+Y)^k] = \mathbb{E}_{X,Y}\left[\sum_{j=0}^{k}\binom{k}{j}X^j Y^{k-j}\right],$$

and apply linearity of expectations to show that

$$\mathbb{E}_{X,Y}[(X+Y)^k] = \sum_{j=0}^{k}\binom{k}{j}\mathbb{E}_{X,Y}[X^j Y^{k-j}].$$

(ii) The quantities

$$\mathbb{E}_{X,Y}[X^j Y^{k-j}] = \iint_{\mathcal{S}_{X,Y}} x^j y^{k-j} f_{X,Y}(x,y)\,dxdy$$

are the **mixed-moments** (or **cross-moments**) of the joint distribution $f_{X,Y}$. Explain why, without further assumptions on X and Y, this is as far as we can go with the simplification of $\mathbb{E}_{X,Y}[(X+Y)^k]$.

(iii) *Now*, suppose we further assume that X and Y are *independent* random variables: in the continuous case, this is precisely equivalent to the statement that the joint density $f_{X,Y}$ factors into a product of the **marginal densities** $f_X \cdot f_Y$ describing X and Y individually — i.e., that

$$f_{X,Y}(x,y) = f_X(x) f_Y(y), \qquad x \in \mathcal{X}, \quad y \in \mathcal{Y},$$

where we note the support set $\mathcal{S}_{X,Y}$ has factored into a **cartesian product** of the form

$$\mathcal{S}_{X,Y} = \mathcal{X} \times \mathcal{Y} \subseteq \mathbb{R}^2.$$

Show that, under independence, the mixed moments factor:

$$\mathbb{E}_{X,Y}[X^j Y^{k-j}] = \iint_{\mathcal{S}_{X,Y}} x^j y^{k-j} f_X(x) f_Y(y)\, dxdy = \left(\int_{\mathcal{X}} x^j f_X(x)\, dx\right) \left(\int_{\mathcal{Y}} y^{k-j} f_Y(y)\, dy\right).$$

(iv) Conclude: if X and Y are *independent* random variables with finite moments $\mu_X^{(k)}$ and $\mu_Y^{(k)}$ of all orders $k = 0, 1, 2, ...$, then

$$\mu_{X+Y}^{(k)} = \sum_{j=0}^{k} \binom{k}{j} \mu_X^{(j)} \mu_Y^{(k-j)} = (\mu_X \star \mu_Y)_k.$$

(d) We take a *moment* (!) to reiterate what we've just shown with regards to binomial convolution and the EGF.

- In part (b), we discovered that the $1/k!$ normalization in the EGF *produces* binomial convolution upon multiplication — the "extra factor of $\binom{k}{j}$" *fell out of the algebra*.
- In part (c), we discovered — *independently* (!!) — that the moment sequence of a sum of independent random variables *combines* by binomial convolution. Here, the "extra factor of $\binom{k}{j}$" *fell out of the binomial theorem*.
- These are *two separate facts* — the first about *power series*, the second about *probability* — and *both* produce the "same extra factor of $\binom{k}{j}$." Their coincidence, then, is *the reason* the EGF of the moment sequence has a multiplicative property under independence: if $\mu_{X+Y} = \mu_X \star \mu_Y$ (part (c)) and $\hat{A} \cdot \hat{B} = \widehat{(a \star b)}$ (part (b)), then

$$\hat{M}_{X+Y}(t) = \hat{M}_X(t) \cdot \hat{M}_Y(t).$$

Convince yourself that no other normalization achieves this. Concretely: if we had divided by some other sequence d_k instead of $k!$, the product of two such series would correspond to the operation $\sum \frac{d_k}{d_j\, d_{k-j}}\, a_j\, b_{k-j}$. For this to equal the binomial convolution $(\mu_X \star \mu_Y)_k$, we would need $d_k/(d_j\, d_{k-j}) = \binom{k}{j}$ for all j, k — in particular, verify that $d_k = k!$ is the unique solution. (Hint: set $j = 1$ to get $d_k/(d_1\, d_{k-1}) = k$, which forces $d_k = d_1^k \cdot k!$ up to a constant. Rescaling absorbs d_1.)

(e) (Recognizing the MGF.) Form the EGF of the moment sequence:

$$\hat{M}(t) := \sum_{k=0}^{\infty} \frac{\mu^{(k)}}{k!}\, t^k.$$

(i) When this power series has positive radius of convergence $R > 0$, the sum defines an analytic function and the moments are recoverable as Taylor coefficients:

$$\mu^{(k)} = \hat{M}^{(k)}(0).$$

The condition $R > 0$ here is crucial, not automatic — somewhat counterintuitively, we'll discover that all moments $\mu^{(k)}$ being finite does *not* guarantee a positive radius of convergence $R > 0$ of the EGF $\hat{M}(t)$. This apparent gap — between "finite moments" and "convergent EGF" — will be taken up in the following exercises.

(ii) Identify the sum:

$$\hat{M}(t) = \sum_{k=0}^{\infty} \frac{\mathbb{E}[X^k]}{k!}\, t^k = \mathbb{E}\left[\sum_{k=0}^{\infty} \frac{(tX)^k}{k!}\right] = \mathbb{E}[e^{tX}],$$

where the interchange of sum and expectation is justified by absolute convergence for $|t| < R$.

The object $\mathbb{E}[e^{tX}]$ is the **moment generating function** (MGF) of X, written $M_X(t)$. The MGF is undoubtedly familiar to the reader who has taken an introductory statistics course — now we know it is nothing more than the exponential generating function of the moment sequence.

(i) Using parts (b)–(d), conclude that if X and Y are independent, then

$$M_{X+Y}(t) = M_X(t) \cdot M_Y(t).$$

(f) (Discrete MGFs are PGFs) In Exercise 5, we defined the probability generating function of a discrete, integer-valued random variable $X \in \mathcal{X} \subseteq \{0, 1, 2, ...\}$ to be the formal power series

$$G_X(z) := \mathbb{E}_X[z^k] = \sum_{k=0}^{\infty} p_k z^k, \qquad p_k := \mathbb{P}(X = k).$$

We then verified that a closed-form expression for the PGF of a binomial random variable $X \sim \mathrm{Bin}(n, p)$ is given by

$$G_X(z) := ((1-p) + pz)^n,$$

and, finally, we observed that the PGF is trivially **distribution-determining**: since the coefficients of $G_X(z) = \sum p_k z^k$ *are* the probabilities $p_k = \mathbb{P}(X = k)$, knowing G_X is equivalent to knowing the full distribution of X. No information is lost in the encoding, because no compression occurred — the PMF went in, and the PMF can be read back out.

(i) What's interesting about the MGF in the discrete case is that it arises directly from the PGF at the point $z = e^t$. Indeed, for X a discrete random variable taking values in $\mathcal{X} \subseteq \{0, 1, 2, ...\}$, verify the substitution

$$M_X(t) = \mathbb{E}[e^{tX}] = \mathbb{E}[(e^t)^X] = G_X(e^t).$$

(ii) Using this equivalence, show that if $X \sim \mathrm{Bin}(n, p)$, we may use the PGF we derived previously to obtain the closed-form

$$M_X(t) = ((1-p) + pe^t)^n.$$

(iii) Explain why, in this light, the MGF of a discrete, integer-valued random variable *inherits* its distribution-determinacy from the PGF: if $M_X(t) = M_Y(t)$ for all t in a neighborhood of zero, show that $G_X(z) = G_Y(z)$ for z near 1, whence $p_k^{(X)} = p_k^{(Y)}$ for all k.

(iv) Expand $G_X(e^t) = \sum p_k \, e^{kt}$ by substituting the Taylor series of each e^{kt} and interchanging the order of summation. Verify that the coefficient of t^m in the result is $\mu^{(m)}/m!$. (The probabilities p_k, which sat at individual powers z^k in the PGF, have been *smeared* by the exponential across every power of t; the moments reassemble when we collect terms, and the $1/m!$ is manufactured by the Taylor series of e^{kt}.)

(v) Explain why this argument does not extend to continuous distributions: no PMF exists, no PGF exists, and the MGF $M_X(t) = \int e^{tx} f(x) \, dx$ must be computed directly. The distribution-determining property no longer comes for free — the moments are a compression of the density, and whether the compression is lossless is a genuine question. The next part illustrates the computation; the following exercises take up the question.

(g) (The Gaussian MGF, two ways.) In Exercise 6, we assembled $M_X(t) = e^{t^2/2}$ from its Taylor coefficients: $\mu^{(2k)}/(2k)! = 1/(2^k k!)$, summed to $e^{t^2/2}$. There is a second derivation that bypasses the moment sequence entirely and arrives at the same closed form in two lines. We encourage the reader not to be discouraged by its brevity — the derivation is genuinely short, and it is *supposed* to feel anticlimactic. The question worth sitting with afterward is not "why didn't we just do this from the start?" but rather "what does the short proof *see* that the long one didn't, and what does it *miss*?"

(i) Compute $M_X(t) = \mathbb{E}[e^{tX}]$ directly from the density $\phi(x) = e^{-x^2/2}/\sqrt{2\pi}$:

$$M_X(t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{tx - x^2/2}\, dx.$$

Complete the square: $tx - x^2/2 = -(x-t)^2/2 + t^2/2$. Factor out $e^{t^2/2}$ and evaluate the shifted Gaussian integral (Exercise 0) to recover $M_X(t) = e^{t^2/2}$.

(ii) The first route — assembling Taylor coefficients from the moment sequence — demonstrated convergence through Stirling's analysis. This second route — evaluating the integral $\mathbb{E}[e^{tX}]$ directly — reveals *why* the computation is tractable: multiplying $e^{-x^2/2}$ by the exponential kernel e^{tx} produces another Gaussian (shifted by t, same variance) times the constant $e^{t^2/2}$. The density absorbs the tilt, and the integral evaluates to $\sqrt{2\pi}$ regardless of t.

(h) (A case where $0 < R < \infty$) Now, we consider a *new* class of random variables — one which we unknowingly derived the moment sequence of in Exercise 6. Indeed, in Exercise 6(a), recall that we defined the **Gamma function**

$$\Gamma(s) := \int_0^{\infty} x^{s-1} e^{-x}\, dx.$$

(i) Suppose we define

$$f_E(x) := e^{-x}, \qquad x \in \mathcal{X} = [0, \infty).$$

Verify that $f_E \geq 0$ is continuous, nonnegative, and integrates to 1 on $\mathcal{X}$. Conclude that f_E defines a valid probability density on $[0, \infty)$.

(ii) We say a random variable $X \sim f_E$ is a **standard exponential** random variable, and we write $X \sim \text{Exp}(1)$. Using the Gamma function from Exercise 6, show that the kth moment of a standard exponential random variable is

$$\mu_E^{(k)} = \int_0^{\infty} x^k e^{-x}\, dx = \Gamma(k+1) = k!, \qquad k = 0, 1, 2, \ldots$$

Conclude that the sequence of factorials $(0!, 1!, 2!, \ldots)$, ostensibly a "pure combinatorial object," also defines the moment sequence $(\mu_X^{(0)}, \mu_X^{(1)}, \mu_X^{(2)}, \ldots)$ of a standard exponential random variable.

(iii) Form the EGF of the moment sequence $(\mu_X^{(k)})$ to show that the MGF of a standard exponential random variable is given by

$$M_E(t) = \sum_{k=0}^{\infty} \frac{\mu^{(k)}}{k!} t^k = \sum_{k=0}^{\infty} t^k = \frac{1}{1-t}, \qquad |t| < 1.$$

In particular, conclude that the MGF $M_E(t)$ has a finite radius of convergence of $R = 1$: in particular, for $t \in (-1, 1)$, the definition of M_E as a formal power series reduces to a convergent geometric series.

(iv) Perform this computation directly: by definition,

$$M_E(t) = \mathbb{E}_E[e^{tx}] = \int_{\mathcal{X}} e^{tx} f_E(x)\,dx = \int_0^\infty e^{-(1-t)x}\,dx = \frac{1}{1-t}, \qquad t < 1.$$

Here, the integral exists if and only if $1 - t > 0$; i.e., if and only if $t < 1$.

(v) Observe that both representations yield the same closed-form expression,

$$M_E(t) = \frac{1}{1-t},$$

but they *disagree on where this formula is valid*: the series requires $|t| < 1$, while the integral merely requires $t < 1$.

Consider, then, the evaluation at $t = -2$.

(vi) Verify this computation directly from the integral definition:

$$M_E(t) = \int_0^\infty e^{tx} e^{-x}\,dx = \int_0^\infty e^{-(1-t)x}\,dx = \frac{1}{1-t},$$

converging if and only if $1 - t > 0$, i.e., $t < 1$.

(vii) Both representations yield $M_E(t) = 1/(1-t)$, but they disagree on where this formula is valid: the series requires $|t| < 1$, while the integral requires only $t < 1$. Consider the evaluation at $t = -2$.

- The integral gives

$$M_E(-2) = \int_0^\infty e^{-3x}\,dx = \frac{1}{3}.$$

  This is perfectly well-defined. Making t more negative only makes the integrand $e^{-(1-t)x}$ decay *faster* on $[0, \infty)$ — the integral becomes better behaved, not worse.
- The series gives

$$\sum_{k=0}^\infty (-2)^k = 1 - 2 + 4 - 8 + 16 - \cdots,$$

  which diverges. The moments $\mu^{(k)} = k!$ are all finite, the Taylor coefficients $\mu^{(k)}/k! = 1$ are as tame as could be, yet the series cannot compute $M_E(-2)$.
- The closed-form $1/(1-t)$ that the series *recovers*, however, returns $1/(1-(-2)) = 1/3$ — which happens to be the *correct answer*. It appears that the local information provided by our moment sequence was just enough to reconstruct a function whose validity extends beyond the moments' own reach. Whether this is a coincidence — whether we are *entitled* to trust the closed-form outside the radius of convergence that produced it — is what remains to be seen.

(i) (The Gaussian and the Cauchy.) Consider the alternating sequence $a_k = (-1)^k$.

(i) Verify that its OGF is

$$\sum_{k=0}^\infty (-1)^k t^k = \frac{1}{1+t}, \qquad |t| < 1,$$

and its EGF is

$$\sum_{k=0}^\infty \frac{(-1)^k}{k!} t^k = e^{-t}, \qquad t \in \mathbb{R}.$$

(ii) Substitute $t = x^2$ and verify the following interpretation. "On the OGF side: $1/(1+x^2)$ — up to normalization, the Cauchy density, convergent only for $|x| < 1$. On the EGF side: e^{-x^2} — up to normalization, the Gaussian density, analytic on all of $\mathbb{R}$. The two parable DGPs are the OGF and EGF of the same alternating sequence, evaluated at x^2. The convergence drama of this chapter was the OGF-versus-EGF distinction all along."

(iii) Recall that the Cauchy's second moment diverges: $\mu^{(2)} = +\infty$. Without a moment sequence, the EGF cannot even be formed. Show that the MGF fares no better: in particular, verify that, if $X \sim \text{Cauchy}(0,1)$ with density

$$f_X(x) = \frac{1}{\pi(1+x^2)}, \qquad x \in \mathbb{R},$$

then

$$\mathbb{E}_X[e^{tX}] = \int_{-\infty}^{\infty} \frac{e^{tx}}{\pi(1+x^2)}\,dx$$

diverges for every $t \neq 0$. (Hint: for $t > 0$, $e^{tx}/(1+x^2) \geq e^{tx}/(2x^2)$ for $x \geq 1$.)

The hierarchy of failures across Exercises 5, 6, and 7 is now three-tiered:

- **Gaussian ($R = \infty$):** all moments finite, OGF diverges (Ex. 5), EGF converges everywhere (Ex. 6), EGF is the unique normalization matched to the convolution moments undergo (Ex. 7). The MGF $M_Z(t) = e^{t^2/2}$ exists for all $t \in \mathbb{R}$. The series and the integral agree on all of $\mathbb{R}$.
- **Exponential ($0 < R < \infty$):** all moments finite, OGF diverges, EGF converges on $|t| < 1$. The MGF $M_E(t) = 1/(1-t)$ exists for all $t < 1$ — but the moment series can only see it for $t \in (-1, 1)$. The moments built a function that extends beyond their reach.
- **Cauchy (R undefined):** moments do not exist past $k = 1$. The EGF cannot be formed. The MGF integral diverges. Every tool built across these three exercises is powerless.

For the Gaussian, the series and the integral agreed everywhere. For the Exponential, they agreed on $(-1, 1)$ but the integral extended further. In both cases, the integral $\int e^{tx} w(x)\,dx$ converged somewhere beyond $t = 0$. But what if it didn't? A distribution whose moments are all finite — so the EGF $\sum \mu^{(k)} t^k/k!$ can be written down — but whose series converges only at $t = 0$, and whose integral $\int e^{tx} w(x)\,dx$ diverges for every $t \neq 0$. The moments $\mu^{(k)}$ exist; the function they are trying to represent, however, *does not*.

Exercise 8: The Lognormal Distribution

Exercise 7 ended on a hierarchy with a gap. For the Gaussian, the moment series and the integral definition of the MGF agreed on all of $\mathbb{R}$: the series $\sum \mu^{(2k)} t^{2k}/(2k)! = e^{t^2/2}$ converges everywhere, and the integral $\int e^{tx}\phi(x)\,dx$ converges everywhere. For the Exponential, the moment series converged on $|t| < 1$, while the integral converged on the larger domain $t < 1$ — but in both cases, the MGF existed as a well-defined function in some neighborhood of $t = 0$. For the Cauchy, moments past $k = 1$ do not exist, and the integral $\mathbb{E}[e^{tY}]$ diverges for every $t \neq 0$.

This leaves a natural question: is there something between the Exponential and the Cauchy? A distribution whose moments are all finite — so the EGF $\sum \mu^{(k)} t^k/k!$ can be formally assembled — but whose series converges only at $t = 0$, and whose integral $\int e^{tx} w(x)\,dx$ diverges for every $t \neq 0$? If such a distribution exists, the moments would be individually meaningful — each $\mu^{(k)} < \infty$ is a well-defined number — but collectively powerless: the function they are trying to build would not exist.

(a) (The Derivation) We exhibit a distribution whose moments *all exist*, but whose MGF *does not exist*. Indeed, if $Z \sim N(0,1)$ is a standard normal random variable, define

$$U := e^Z.$$

First, we want to verify that U actually defines a legitimate random variable.

(i) Suppose we define the general transformation

$$g(W) = e^W, \qquad W \in \mathcal{W} = \mathbb{R}.$$

Show that g is continuous and strictly increasing on all of $\mathcal{W}$. Conclude that g is invertible, and verify that its inverse is

$$g^{-1}(V) = \ln(V), \qquad V \in \mathcal{V} = (0, \infty).$$

(i.e., show that that g and g^{-1} satisfy $g^{-1}(g(W)) = W$ and $g(g^{-1}(V)) = V$).

(ii) Show that $g(W) = e^W$ is differentiable on all of $\mathcal{W}$ with

$$g'(W) = e^W > 0, \qquad W \in \mathcal{W}.$$

From this, we can invoke the **inverse function theorem** to conclude that $g^{-1}(V) = \ln(V)$ is differentiable on $\mathcal{V} = (0, \infty)$, with

$$\frac{d}{dv} g^{-1}(V) := (g^{-1})'(V) = \frac{1}{g'(g^{-1}(V))} = \frac{1}{e^{\ln V}} = \frac{1}{V}.$$

(iii) Apply the **change-of-variables formula**: if $W \sim f_W$ is a continuous random variable and $V = g(W)$ where g is continuously differentiable and strictly monotone, then the density of V is given by

$$f_V(v) = f_W(g^{-1}(v)) \cdot \left|(g^{-1})'(v)\right|.$$

Apply this result to $W = Z \sim N(0,1)$, $g(z) = e^z$, $g^{-1}(v) = \ln v$, and $(g^{-1})'(v) = 1/v$ to show that the density of the transformation $U = e^Z$ is

$$f_U(u) = \frac{1}{u} \cdot \frac{1}{\sqrt{2\pi}} e^{-(\ln u)^2/2}, \qquad u \in \mathcal{U} = (0, \infty).$$

(iv) Verify that $f_U \geq 0$ on $\mathcal{U} = (0, \infty)$, and show that $\int_0^\infty f_U(u)\, du = 1$. (Hint: just substitute back to the case you know: take $u = e^z$ with $du = e^z dz$.)

(v) Conclude that f_U defines a valid probability density on $\mathcal{U} = (0, \infty)$.

(b) (Lognormal Moments Exist) We say that $U \sim f_U$ the density derived above is a **standard lognormal** random variable, and we write $U \sim \text{Lognormal}(0,1)$ — the reason for the name is clear from the derivation: if $U \sim \text{Lognormal}(0,1)$, then $U = e^Z$ for a standard normal random variable Z, so that "the natural log $Z := \ln U$ of a standard lognormal random variable $U \sim \text{Lognormal}(0,1)$ is standard normal, $Z = \ln U \sim N(0,1)$."

What would seem to make the transformation technique valuable in this case is the fact that we *already know* the moments of the standard normal — in particular, we might imagine that "since we know the moments of $Z \sim N(0,1)$, and since we know that $U = e^Z \sim \text{Lognormal}(0,1)$, perhaps we could use these in conjunction to obtain the moments of U 'for free', usurping our need to integrate $\mu_U^{(k)} = \int_0^\infty u^k f_U(u)\, du$ directly."

(i) Indeed, verify that the kth moment of $U = e^Z \sim \text{Lognormal}(0,1)$ where $Z \sim N(0,1)$ can be computed without ever touching the density f_U directly: show that

$$\mu_U^{(k)} := \mathbb{E}_U[U^k] = \int_0^\infty u^k f_U(u)\,du = \int_{-\infty}^\infty (e^z)^k f_Z(z)\,dz = \mathbb{E}_Z[e^{kZ}] = M_Z(k).$$

(ii) Conclude that the moments of the standard lognormal are given explicitly by

$$\mu_U^{(k)} = e^{k^2/2}, \qquad k = 0, 1, 2, \ldots$$

(c) (The Lognormal MGF Doesn't Exist) Thus, the lognormal moments $\mu_U^{(k)} = e^{k^2/2} < \infty$ of all orders $k = 0, 1, 2, \ldots$ exist and are finite. What is certainly surprising to us, then, is that they turn out to be *completely useless*.

(i) Verify that the lognormal moments grow **super-factorially**: using Stirling,

$$\ln(k!) \approx k \ln k - k,$$

while $\ln(\mu_U^{(k)}) = \ln(e^{k^2/2}) = k^2/2$. Show that $k^2/2$ **eventually dominates** $k \ln k$ — i.e., show there exists $K \in \mathbb{N}$ such that for all $k \geq K$, $k^2/2 > k \ln k$. Show that

$$\ln\left(\frac{e^{k^2/2}}{k!}\right) = \frac{k^2}{2} - k\ln k \overset{k\to\infty}{\longrightarrow} +\infty,$$

and conclude that

$$\frac{e^{k^2/2}}{k!} \overset{k\to\infty}{\longrightarrow} +\infty.$$

(ii) Form the EGF of $U \sim \text{Lognormal}(0,1)$: this is the formal power series

$$\hat{M}_U(t) = \sum_{k=0}^\infty \frac{\mu_U^{(k)}}{k!} t^k = \sum_{k=0}^\infty \frac{e^{k^2/2}}{k!} t^k.$$

Verify that while

$$\hat{M}_U(0) = 1,$$

we observe

$$\hat{M}_U(t) = +\infty, \qquad t \neq 0,$$

and conclude that the radius of convergence of the EGF $\hat{M}_U(t)$ is $R = 0$.

(d) (The Hamburger Moment Problem) We have just shown that the lognormal's moments $\mu_U^{(k)} = e^{k^2/2}$ all exist and are finite, yet the MGF $M_U(t)$ has radius of convergence $R = 0$. The MGF was a *tool* for "organizing the moment sequence $(\mu^{(k)})$ into a convergent generating function" — yet, that tool has apparently failed to make good on its promise. We want you to appreciate the discomfort that follows from this apparent contradiction, because as it turns out, it will pave the way to a more complete understanding of how distribution determinacy manifests on unbounded intervals.

Indeed, at this point, one might imagine the situation is easily rectified: "Sure, we may not be able to map the moment sequence as $\mu_U^{(k)} \mapsto M_U(t) = \sum_{k=0}^\infty \frac{\mu_U^{(k)}}{k!} t^k$ and have $M_U(t)$ converge for $t \neq 0$. But we've certainly identified that those moments *exist and are finite*: they're given by $\mu^{(k)} = e^{k^2/2} < \infty$ for each $k = 0, 1, 2, \ldots$. What matter is it, then, whether or not $M_U(t)$ has a positive radius of convergence — after all, it is *the moments themselves* that matter, is it not?"

We imagine such a skeptic is quite dismayed, then, to learn that the MGF is *the entirety* of the matter the second we leave compact intervals $[a, b]$.

(i) For some fixed $\epsilon \in [-1, 1]$, define

$$f_\epsilon(u) := f_U(u)\,[1 + \epsilon \sin(2\pi \ln u)]\,, \qquad u \in \mathcal{U} = (0, \infty),$$

where f_U is the lognormal density derived in part (a). Verify that $f_\epsilon(u) \geq 0$ for all $u \in \mathcal{U}$. (Hint: $|\sin(\cdot)| \leq 1$, $|\epsilon| \leq 1$, and f_U is already a probability density.)

(ii) Show that

$$\int_0^\infty f_\epsilon(u)\,du = 1.$$

(Hint: By linearity, it suffices to show

$$\int_0^\infty f_U(u)\sin(2\pi \ln u)\,du = 0.$$

Under the substitution $v = \ln u$, this holds if and only if

$$\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-v^2/2}\, \sin(2\pi v)\,dv = 0.$$

Confirm this by verifying the integral is over a symmetric interval about $v = 0$, and showing that the integrand is odd.)

(iii) Conclude that, for each $\epsilon \in [-1, 1]$, f_ϵ is a valid probability density on $\mathcal{U} = (0, \infty)$ with the property that

$$f_\epsilon \not\equiv f_U \qquad \text{whenever} \qquad \epsilon \neq 0.$$

(e) (The Gutpunch) We now show that each density in the family $\{f_\epsilon \colon \epsilon \in [-1, 1]\}$ shares *every moment* with the lognormal f_U.

(i) Verify that the kth moment of f_ϵ for a fixed $\epsilon \in [-1, 1]$ is given explicitly by

$$\mu_\epsilon^{(k)} = \int_0^\infty u^k f_\epsilon(u)\,du = \mu_U^{(k)} + \epsilon \underbrace{\int_0^\infty u^k f_U(u)\sin(2\pi \ln u)\,du}_{:=I_k}.$$

(ii) We claim that

$$\mu_\epsilon^{(k)} = \mu_U^{(k)}, \qquad k = 0, 1, 2, \ldots$$

Explain why it is sufficient to show that $I_k = 0$ for every nonnegative integer $k = 0, 1, 2, \ldots$ to establish this, and show that under the substitution $v = \ln u$, we have

$$I_k = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{kv - v^2/2} \sin(2\pi v)\,dv.$$

(iii) Complete the square to show that

$$I_k = \frac{e^{k^2/2}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-(v-k)^2/2} \sin(2\pi v)\,dv.$$

(iv) After substituting $w = v - k$ in the form above, apply the angle addition formula

$$\sin(\alpha \pm \beta) = \sin\alpha\cos\beta \pm \cos\alpha\sin\beta,$$

(Theorem 2.2.6, (A2)) and use $\cos(2\pi k) = 1$ and $\sin(2\pi k) = 0$, $k = 0, 1, 2, \ldots$ to show that

$$I_k = \frac{e^{k^2/2}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-w^2/2} \sin(2\pi w)\,dw.$$

(v) Verify that

$$\int_{-\infty}^{\infty} e^{-w^2/2} \sin(2\pi w)\, dw = 0$$

by establishing the integrand is odd.

(vi) Use this to confirm that

$$I_k = 0,$$

and conclude that

$$\mu_\epsilon^{(k)} = \mu_U^{(k)} = e^{k^2/2}, \qquad k = 0, 1, 2, \ldots$$

(vii) Verify that, just as $\hat{M}_U(t)$ had a radius of convergence $R = 0$, so too does

$$\hat{M}_\epsilon(t) := \sum_{k=0}^{\infty} \frac{\mu_\epsilon^{(k)}}{k!} t^k.$$

In particular, use the known lognormal moments $\mu_\epsilon^{(k)} = \mu_U^{(k)}$ to conclude that $M_\epsilon(0) = 1$, while $M_\epsilon(t) = +\infty$ for any $t \neq 0$.

(f) (Where's Weierstrass?) We have just exhibited an uncountable family of distinct probability densities $\{f_\epsilon\}_{\epsilon \in [-1,1]}$ on $\mathcal{U} = (0, \infty)$ which *all* share *every moment* with the lognormal:

$$\mu_U^{(k)} = \mu_\epsilon^{(k)} = e^{k^2/2}, \qquad k = 0, 1, 2, \ldots$$

In Exercise 5(h), we proved that this *cannot happen* on a compact support set $\mathcal{X} = [a, b]$: if two continuous densities on $[a, b]$ share all moments, they *must* determine the same distribution. The behavior we're observing here for $U \sim \text{Lognormal}(0, 1)$ on the **unbounded support** $\mathcal{U} = (0, \infty)$ is *decisively different*; something has broken in the passage from $[a, b]$ to ∞.

(i) Once again, recall the chain of reasoning from Exercise 5(h):

$$\textbf{Moments agree} \xrightarrow{\text{linearity}} \textbf{Polynomials agree}$$

$$\xrightarrow{\text{Weierstrass (?)}} \textbf{Continuous functions agree} \xrightarrow{\text{squeeze}} \textbf{CDFs agree.}$$

Validate the following intuition for *why the second arrow breaks* when we move from a compact interval $[a, b]$ to an unbounded subset of $\mathbb{R}$: "The Weierstrass approximation theorem guarantees that any continuous function on $[a, b]$ can be uniformly approximated by polynomials. On $\mathbb{R}$ (or $(0, \infty)$), this guarantee fails: polynomials $p(x)$ grow *without bound* as $|x| \to \infty$, so no polynomial can approximate a function that decays to zero as $|x| \to \infty$."

We assemble the full hierarchy of radius-of-convergence behavior:

Distribution	$\mu^{(k)}$ (growth)	$\mu^{(k)}/k!$ (growth)	**Radius** R	**Moments Dist. Determining?**
Gaussian	$(k/e)^{k/2}$ (factorial)	$\to 0$ (super-geometric decay)	∞	Yes
Exponential	$k!$ (factorial)	$\to 1$ (constant)	1	Yes
Log-normal	$e^{k^2/2}$ (super-factorial)	$\to \infty$ (super-exponential growth)	0	**No**
Cauchy	undefined	—	—	—

Exercise 9: Moment (In-)Determinacy on Unbounded Intervals

The last exercise exhibited something quite unsettling to our existing intuition; we want to briefly recall the setup. Indeed, we showed that while a standard lognormal random variable $U \sim \text{Lognormal}(0,1)$ with PDF

$$f_U(u) = \frac{1}{u} \cdot \frac{1}{\sqrt{2\pi}} e^{-(\ln u)^2/2}, \qquad u \in \mathcal{U} = (0,\infty).$$

has finite moments of all orders,

$$\mu_U^{(k)} = e^{k^2/2} < \infty, \qquad k = 0,1,2,\dots,$$

we were able to construct an *uncountable family* of *distinct densities* $\{f_\epsilon\}_{\epsilon\in[-1,1]}$ — where, for a fixed $\epsilon \in [-1,1]$,

$$f_\epsilon(x) := f_U(x)\left[1 + \epsilon \sin(2\pi \ln u)\right], \qquad u \in \mathcal{U} = (0,\infty),$$

— such that

$$\underbrace{\mathbb{E}_U[U^k]}_{U\sim\text{Lognormal}(0,1)} := \mu_U^{(k)} = \mu_\epsilon^{(k)} =: \underbrace{\mathbb{E}_\epsilon[U^k]}_{U\sim f_\epsilon}, \qquad \forall k \in \{0,1,2,\dots\}.$$

This seems quite counterintuitive, and in complete defiance of the situation we observe on compact intervals $[a,b] \subset \mathbb{R}$. We identified that the breakdown of the "**moment-determining property**" was due to the failure of the Weierstrass approximation theorem on unbounded domains: our proof of moment-determinacy on $[a,b]$ proceeded according to the chain of logic

$$\textbf{Moments agree} \xrightarrow{\text{linearity}} \textbf{Polynomials agree} \xrightarrow{\text{Weierstrass}} \textbf{Continuous functions agree}$$

$$\xrightarrow{\text{squeeze}} \textbf{CDFs agree,}$$

and since polynomials are unable to approximate continuous functions that decay to 0 as $|x| \to \infty$, the *second link breaks* when we move to the lognormal's unbounded support $\mathcal{U} = (0,\infty)$.

But there is something remiss. By the end of Exercise 8, we'd seemed to have surmised the following:

Distribution	**Radius R**	**Moments Dist. Determining?**
Gaussian	∞	*Yes!*
Exponential	1	*Yes!*
Log-normal	0	**No.**

But wait a minute — have we *proven* the first two rows? We've *certainly* proved the third row: the lognormal's moments fail to determine its distribution, as the perturbation family $\{f_\epsilon\}$ demonstrates concretely. But the first two rows are, at this point, *assertions*. Our only proof of moment-determinacy (Exercise 5(h)) relied on the Weierstrass approximation theorem, which we've now realized *fails* on unbounded domains.

In other words: the lognormal has disrupted the *whole affair*. We'd taken the moment determinacy for granted because, at the time, we had no reason to doubt its legitimacy, but we now have a *very* compelling reason to be suspicious of those "Yes!" entries in the table above.

The goal of this exercise is to *partially* restore order. We will identify a real-valued criterion — a condition on members of the moment sequence $(\mu^{(k)})$ itself — that, when satisfied, *guarantees* moment-determinacy on $\mathbb{R}$. We will verify this condition for the distributions above — importantly, we will directly observe that the condition *succeeds* for the Gaussian and Exponential, while it *fails* for the Log-normal.

We will then use this condition to establish the *true* condition we've been pursuing thus far: namely, that the MGF is distribution-determining, *even on unbounded supports*, if and only if its radius of convergence $R > 0$ is *strictly positive*.

Remark.

To bury the lede: we *will not* be proving Carleman's condition — at least, not right now. The reason for this is mostly a pedagogical one; nevertheless, it runs somewhat in contrast to the "build-it-from-scratch" mentality prefacing the rest of the book, since it means we'll *essentially* be concluding the MGF is distribution determining, *without* having proven the condition we claim to "tame the moment sequence enough to ensure this determinacy is genuine." We note, however, that we believe the verification is convincing enough in its own right to serve our purposes.

But the truly fastidious reader should never fear: we'll retroactively pursue a *direct* proof of the MGF's distribution-determinacy utilizing the material we'll develop in the following chapter. Nevertheless, the reader who has encountered the MGF uniqueness theorem in an introductory statistics textbook and believes we're being somewhat careless with our approach will *certainly* be surprised to learn that it was *never proved there, either.*

(a) (Carleman's Condition) We state the following result without proof.

Theorem 3.4.6 (Carleman's Condition (1926))

Let X be a random variable with finite moments $\mu^{(k)} < \infty$ of all orders $k = 0, 1, 2, ...$. If the *even* moments satisfy

$$\sum_{k=1}^{\infty} \left(\mu^{(2k)}\right)^{-1/2k} = +\infty,$$

then the distribution of X is *uniquely determined* by its moment sequence $(\mu^{(k)})$.

The condition is a divergence test on a series built from the even moments. If the moments grow slowly enough that the sum *diverges to* $+\infty$, the distribution of X is pinned down. If the moments grow so fast that the sum *converges to a finite number*, no such guarantee exists.

We now verify Carleman's Condition in our hierarchy.

(i) **(Gaussian)** For $Z \sim N(0, 1)$, we've shown that the even moments are given by

$$\mu^{(2k)} = (2k-1)!! = (2k-1)(2k-3)\cdots(3)(1), \qquad k = 0, 1, 2, ...$$

Using Stirling, show that

$$\left(\mu^{(2k)}\right)^{-1/2k} = ((2k-1)!!)^{-1/2k} \sim \left(\frac{e}{2k}\right)^{1/2},$$

and verify that

$$\sum_{k=1}^{\infty} (\mu^{(2k)})^{-1/2k} \sim \left(\frac{e}{2}\right)^{1/2} \sum_{k=1}^{\infty} \frac{1}{\sqrt{k}} = +\infty,$$

since the series on the right is a constant multiple of the p-series with $p = 1/2 < 1$, so the series on the left diverges by the limit comparison test. Conclude that Carleman's condition is satisfied for the Gaussian moment sequence. (Hint: Use $(2k-1)!! \approx \sqrt{2}\,(2k/e)^k$ from Exercise 6(c) and raise to the $-1/(2k)$th power.)

(ii) **(Exponential)** For $X \sim \text{Exp}(1)$, we've shown that

$$\mu^{(2k)} = (2k)!, \qquad k = 0, 1, 2, ...$$

Using Stirling, show that

$$\left(\mu^{(2k)}\right)^{-1/2k} = ((2k)!)^{-1/2k} \sim \frac{e}{2k},$$

and verify that

$$\sum_{k=1}^{\infty} (\mu^{(2k)})^{-1/2k} \sim \frac{e}{2} \sum_{k=1}^{\infty} \frac{1}{k} = +\infty,$$

since the series on the right is a constant multiple of the harmonic series. Conclude that Carleman's condition is satisfied for the Exponential moment sequence.

(iii) **(Log-normal)** For $U \sim \text{Lognormal}(0, 1)$, we've shown that

$$\mu^{(2k)} = e^{2k^2}, \qquad k = 0, 1, 2, ...$$

Verify directly (no Stirling needed) that

$$\left(e^{2k^2}\right)^{-1/2k} = e^{-k},$$

and verify that

$$\sum_{k=1}^{\infty} \left(\mu^{(2k)}\right)^{-1/2k} = \sum_{k=1}^{\infty} e^{-k} = \frac{e^{-1}}{1 - e^{-1}} = \frac{1}{e - 1} < \infty.$$

Conclude that Carleman's condition **fails** for the Lognormal moment sequence.

(iv) **(Cauchy)** Verify that Carleman's condition *fails trivially* for the Cauchy — the moment sequence doesn't exist, so there's nothing to sum.

(b) **(MGF-Determinacy on $\mathbb{R}$)** We're now ready to prove the key link.

Theorem 3.4.7 (MGF Convergence Implies Carleman)

If the moment generating function $M_X(t) = \sum_{k=0}^{\infty} \mu^{(k)} t^k / k!$ has positive radius of convergence $R > 0$, then Carleman's condition is satisfied:

$$\sum_{k=1}^{\infty} \left(\mu^{(2k)}\right)^{-1/(2k)} = +\infty.$$

In particular, by Carleman's theorem, the distribution of X is uniquely determined by its moments.

(i) Suppose $M_X(t)$ converges at some $t_0 \neq 0$. Explain why the terms of a convergent series must be bounded: there exists $C > 0$ such that

$$\frac{\mu^{(k)} |t_0|^k}{k!} \leq C, \qquad k = 0, 1, 2, ...$$

(ii) Rearrange to obtain a bound on the kth moment:

$$\mu^{(k)} \leq C \cdot \frac{k!}{|t_0|^k}, \qquad k = 0, 1, 2, \ldots$$

In particular, for the even moments of current interest, we have that

$$\mu^{(2k)} \leq C \cdot \frac{(2k)!}{|t_0|^2 k}.$$

(iii) Using Stirling's approximation to $((2k)!)^{-1/2k} \sim e/2k$ that we obtained earlier (see Carleman's Exponential case), raise each side of above to the $-1/(2k)$th power: making sure to flip the inequality, we have

$$\left(\mu^{(2k)}\right)^{-1/(2k)} \geq \left(C \cdot \frac{(2k)!}{|t_0|^{2k}}\right)^{-1/2k} = C^{-1/2k} \cdot \frac{|t_0|}{\left((2k)!\right)^{1/2k}} \sim \frac{e|t_0|}{2k},$$

since $C^{-1/2k} \to 1$ as $k \to \infty$.

(iv) Therefore: if we set the constant $A = e|t_0|/2$,

$$\sum_{k=1}^{\infty} \left(\mu^{(2k)}\right)^{-1/(2k)} \geq A \sum_{k=1}^{\infty} \frac{1}{k} = +\infty.$$

(v) Conclude that a positive radius of convergence $R > 0$ of the MGF $M_X(t)$ is is sufficient to establish Carleman's condition, and by Carleman's theorem, that the moment sequence $(\mu^{(k)})$ of X determines its distribution.

(c) (The completed hierarchy.) Assemble the full picture:

Dist.	$\boldsymbol{\mu^{(k)}}$ **(growth)**	$\boldsymbol{\mu^{(k)}/k!}$ **(behavior)**	**Radius** R	**Carleman?**	**Moments Det. Dist.?**
Gaussian	$(k/e)^{k/2}$ (factorial)	$\to 0$ (super-geom. decay)	∞	$\sum k^{-1/2} = \infty$	Yes
Exponential	$k!$ (factorial)	$\to 1$ (constant)	1	$\sum k^{-1} = \infty$	Yes
Log-normal	$e^{k^2/2}$ (super-factorial)	$\to \infty$ (super-exp. growth)	0	$\sum e^{-k} < \infty$	**No**
Cauchy	undefined	—	—	—	—

We summarize the chain of implications established in this exercise as follows:

$$\underbrace{R > 0}_{\text{MGF converges}} \xrightarrow{\text{bounded terms}} \underbrace{\mu^{(2k)} \leq C \cdot \frac{(2k)!}{|t_0|^{2k}}}_{\text{factorial growth bound}}$$

$$\xrightarrow{\text{Stirling (Ex. 6)}} \underbrace{(\mu^{(2k)})^{-1/(2k)} \geq \frac{A}{k}}_{\text{harmonic lower bound}} \xrightarrow{\text{Harmonic series } A\sum 1/k \text{ diverges}} \underbrace{\text{Carleman diverges}}_{\text{condition satisfied}}$$

$$\xrightarrow{\text{Carleman's theorem}} \underbrace{\text{moments determine distribution.}}_{\text{uniqueness on } \mathbb{R}}$$

The first arrow is the boundedness of terms in a convergent series. The second is Stirling (Exercise 6). The third is the harmonic series. The fourth is Carleman's theorem, which we have taken on faith. Each link except the last uses tools the reader built in this chapter.

The Cauchy remains untouched. Its moments do not exist past $k = 1$; there is no moment sequence, no EGF, no MGF, no Carleman sum to evaluate. Every tool we have built across Exercises 5 through 9 is powerless against it. Yet the Cauchy is a perfectly legitimate distribution — it has a density, a CDF, and a definite structure. To reach it, we need a tool that encodes distributional information *without passing through the moment sequence*. The construction of such a tool is the subject of the next chapter.

Remark.

The quantity $(\mu^{(2k)})^{1/(2k)} = (\mathbb{E}[X^{2k}])^{1/(2k)}$ is, informally, the L^{2k} norm $\|X\|_{2k}$ of the random variable X — a measure of the distribution's "effective reach" at resolution $2k$. (We have not formally defined L^p norms on random variables; this will come in due course.) In this language, Carleman's sum is $\sum 1/\|X\|_{2k}$, and the condition asks how fast $\|X\|_{2k}$ grows in k.

For the Gaussian, $\|X\|_{2k} \sim \sqrt{k}$; for the Exponential, $\|X\|_{2k} \sim k$; for the log-normal, $\|X\|_{2k} \sim e^k$. The two distributions whose moments determine them have polynomial growth; the one whose moments don't has exponential growth. This is not a coincidence — it is the Weierstrass story, replaying one level of abstraction higher.

On $[a, b]$, the polynomial test functions $\{1, x, x^2, ...\}$ approximate every continuous function (Weierstrass), and moments determine the distribution. On $\mathbb{R}$, the individual monomials x^k blow up at infinity and can no longer approximate anything on their own. But we are not forced to use them one at a time. If we take *all* the monomials and average them — not equally (since we know that would diverge), but with the weights $1/k!$ that decay fast enough to keep the sum finite —

$$1 + tx + \frac{(tx)^2}{2!} + \frac{(tx)^3}{3!} + \cdots = e^{tx},$$

what emerges is a single function that works on all of $\mathbb{R}$: bounded in one direction, growing in the other, never oscillating, never blowing up in both directions at once. The monomials are all still present — differentiating k times at $t = 0$ peels off the kth moment, recovering the polynomial information term by term. The averaging did not destroy the polynomial content; it *organized* it into a form that survives on unbounded domains.

The weights $1/k!$ are not arbitrary. They are the Taylor coefficients of e^t, the function whose growth rate Stirling characterized in Exercise 6: $k! \approx \sqrt{2\pi k}\,(k/e)^k$. The constant $e \approx 2.718$ — defined in Chapter 1 as $\lim_{n\to\infty}(1+1/n)^n$ — is the base rate of factorial growth, and the $1/k!$ damping is calibrated to exactly this rate. If e were larger, the damping would be too aggressive, killing the polynomial contributions before they could contribute. If e were smaller, the damping would be too gentle, and the sum would diverge — as the OGF demonstrated. The exponential e^{tx} is the *minimal controlled average* of the monomial basis that converges to a useful test function on $\mathbb{R}$.

Carleman's condition, in this light, asks: is the distribution's effective reach growing *slowly enough* that the averaged monomials can keep up? Polynomial reach ($\|X\|_{2k} \sim k^a$) means yes — the monomials, resummed as e^{tx}, *can* track the tails. Exponential reach ($\|X\|_{2k} \sim e^k$) means no — the tails are expanding at the very rate the averaging was designed to handle, and perturbations like $\sin(2\pi \ln u)$ live in the gaps the averaged monomials e^{tx} *cannot* close.

Chapter 1, as it turns out, wasn't a mere digression. The definition $e^x = \lim_{n\to\infty}(1+x/n)^n$ is a sequence of polynomials converging to a transcendental function — the archetype of the "polynomial sequences converging to transcendental functions" that this book has been studying since its first page. The passage from Weierstrass on $[a,b]$ to the MGF on $\mathbb{R}$ is: stop using polynomials individually and start using them collectively, averaged by $1/k!$, and what emerges is e^{tx} — the same function that Chapter 1 defined, that Exercise 6 studied via Stirling, that Exercise 7 recognized as the MGF kernel, and that now certifies distribution-determinacy on $\mathbb{R}$.

Would you believe it, then, if we said we hadn't even scratched the surface with e? It really is the gift that keeps on giving!

3.4.5 Reclaiming the Pipeline

We just spent quite a bit of time on a digression into generating functions that, while informative, might have the reader questioning whether or not we've lost the plot. "I thought we were discussing polynomials," we lament — "*why* have we just spent four exercises taking up the generating function mantle?"

There were, of course, many good reasons for it. First of all, it is clear at this point that we really needed MGFs as a *classification tool* for unbounded distributions — the four-tier hierarchy of Exercise 9 (Gaussian, Exponential, Log-normal, Cauchy) would have been invisible without the generating function machinery to diagnose it. But second, and perhaps more importantly for our current purposes, we recall the pipeline we established at the end of Exercise 4:

$$\underbrace{\mathcal{M}_{n-1},\ w}_{\textbf{Basis, Weight}} \xrightarrow{\text{Integrate } 2k-1 \text{ times: } \mu_w^{(r)}=\int_{\mathcal{X}} x^r w(x)\,dx} \underbrace{\mu_w^{(0)},\ \mu_w^{(1)},\ \dots,\ \mu_w^{(2n-2)}}_{\textbf{Moment sequence}}$$

$$\xrightarrow{\text{Assemble: } (G_w^{(\mathcal{M})})_{jk}=\mu_w^{(j+k)}} \underbrace{G_w^{(\mathcal{M})}}_{\textbf{Hankel matrix}} \xrightarrow{\text{Factor: } G=LDL^\top} \underbrace{\boldsymbol{\omega}(x) = L^{-1}\mathbf{m}(x)}_{\textbf{Orthogonal polynomials}}$$

At the time, the first arrow here was an obstacle. What we said then and echo now is that, indeed, *integration is hard* — computing $\mu_w^{(r)} = \int_{\mathcal{X}} x^r w(x)\,dx$ one moment at a time, for each $r = 0, 1, 2, \dots, 2n-2$, required evaluating $2n-1$ separate integrals. And, though the examples we've curated here somewhat deliberately seem to suggest otherwise — $Z \sim N(0,1)$ with $\mu_Z^{(2k)} = (2k-1)!!$, $X \sim \text{Exp}(1)$ with $\mu_X^{(k)} = k!$, and $U \sim \text{Log-normal}(0,1)$ with $\mu_U^{(k)} = e^{k^2/2}$ — there *really isn't* a "faster way" to obtain the moments of a *general* random variable directly. We'd rather avoid this laborious integration if at all possible.

What the generating function exercises have furnished is precisely a tool to dissolve this bottleneck. The MGF

$$M_w(t) = \mathbb{E}_w[e^{tX}] = \int_{\mathcal{X}} e^{tx} w(x)\, dx$$

compresses the entire moment sequence into a *single* exponential integral, from which (as we showed in Exercise 7) the moments are recoverable as Taylor coefficients:

$$M_w^{(r)}(0) = \mu_w^{(r)}, \qquad r = 0, 1, 2, \ldots$$

In practice, we often do even better: if the integral $\int e^{tx} w(x)\, dx$ evaluates to a closed-form expression $M_w(t)$ that we recognize — an exponential, a power, a rational function — then the Taylor expansion reads off the moments *by inspection*, without differentiating at all. This is what happened for the Gaussian: we identified $M_Z(t) = e^{t^2/2}$, expanded $e^{t^2/2} = \sum_k (t^2/2)^k/k!$, and read off $\mu^{(2k)} = (2k)!/(2^k k!)$ in a single line. The process "is equivalent to differentiation," but we need not "formally differentiate" if we can *infer* how the process would play out.

What should be clear is that, if "symbolic integration is *quite hard*," then "symbolic differentiation is *relatively easy*," and "inspection is *just trivial*." Indeed, the situation is quite apparent for the Gaussian integrand $f(x) = e^{-x^2}$, whose integral on $\mathbb{R}$ required a full exercise's worth of derivation, but whose derivative is just

$$\frac{d}{dt} e^{-x^2} = -2xe^{-x^2}, \qquad x \in \mathbb{R}.$$

But even "easy" has its limits. In principle, $\mu_w^{(r)} = M_w^{(r)}(0)$ is "just the rth derivative" — but we can imagine how the exponential factor in $\mathbb{E}_w[e^{tX}]$ might impart something of a *compounding multiplicative structure* as we continue to differentiate M_w. For instance, while the first derivative of the Gaussian MGF $M_w(t) = e^{t^2/2}$ is $te^{t^2/2}$, the second is $(1+t^2)e^{t^2/2}$ — which requires us to apply the product rule $(f+g)' = f'g + g'f$. By, say, the sixth derivative $\frac{d^6}{dt^6} e^{t^2}$, then, the product rule will have generated a polynomial prefactor of degree 6 whose coefficients require genuine, nontrivial bookkeeping to track. To be clear, this is almost certainly "still easier than integrating $\int x^k e^{-x^2/2}\, dx$ six times for $k = 1, 2, \ldots, 6$"; but it *certainly* "isn't free," either.

That is why we should always keep an eye out, since *inspection* sidesteps these concerns entirely. Indeed, by somewhat cleverly recognizing $e^{t^2/2} = \sum_k (t^2/2)^k/k!$, we're able to read off $\mu^{(2k)}/(2k)! = 1/(2^k k!)$ for *every* k at once, with no differentiation at all. When a closed-form MGF admits a Taylor expansion we recognize, inspection is not merely convenient — it is the method of choice.

We thus revise the pipeline accordingly:

$$\underbrace{w}_{\textbf{Weight}} \xrightarrow{\text{Integrate once: } M_w(t) = \int e^{tx}\, w(x)\, dx} \underbrace{M_w(t)}_{\textbf{MGF}} \xrightarrow{\text{Inspect/Differentiate}} \underbrace{\mu_w^{(0)}, \mu_w^{(1)}, \ldots, \mu_w^{(2n-2)}}_{\textbf{Moments}}$$

$$\xrightarrow{\text{Assemble}} \underbrace{G_w^{(\mathcal{M})}}_{\textbf{Hankel}} \xrightarrow{\text{Factor}} \underbrace{LDL^\top}_{\textbf{Gram-Schmidt}} \xrightarrow{\text{Invert } L \text{ and Apply}} \underbrace{\boldsymbol{\omega} = L^{-1}\mathbf{m}}_{\textbf{Orthogonal polynomials}}$$

The time has come for us to see how it plays out in action.

Exercise 10: The Continuous Family Pipeline

In this exercise, we run the updated pipeline on *three continuous weight functions* — in turn, we will produce *three classical families of orthogonal polynomials*. In each case, the recipe is the same:

- **Compute** the MGF:
$$M_w(t) = \int_{\mathcal{X}} e^{tx}\, w(x)\, dx.$$
- **Verify** that the radius of convergence $R > 0$ of M_w is positive. (If instead $R = 0$, *stop*; the moment sequence $\mu_w^{(k)}$ grows too quickly, and the pipeline cannot proceed.)
- **Inspect/Differentiate** the closed-form expression for $M_w(t)$ to obtain the moment sequence
$$\mu_w^{(0)} = M_w(0), \qquad \mu_w^{(1)} = M_w'(0), \qquad \mu_w^{(2)} = M_w''(0), \qquad \dots, \qquad \mu^{(k)} = M_w^{(k)}(0).$$
- **Assemble** the Hankel Gram matrix
$$G_w^{(\mathcal{M})} = \begin{pmatrix} \mu_w^{(0)} & \mu_w^{(1)} & \mu_w^{(2)} & \cdots & \mu_w^{(n-1)} \\ \mu_w^{(1)} & \mu_w^{(2)} & \mu_w^{(3)} & \cdots & \mu_w^{(n)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \mu_w^{(n-1)} & \mu_w^{(n)} & \mu_w^{(n+1)} & \cdots & \mu_w^{(2n-2)} \end{pmatrix}.$$
- **Factor** $G_w^{(\mathcal{M})}$ as
$$G_w^{(\mathcal{M})} = LDL^\top.$$
- **Invert** L **and Apply** to obtain
$$L^{-1}\mathbf{m}(x) = \boldsymbol{\omega}(x).$$
- **Conclude** that $\boldsymbol{\omega}$ is an orthogonal family of polynomials under the continuous weight w:
$$\langle \omega_j, \omega_k \rangle_w = 0, \qquad j \neq k.$$

(a) (Uniform $\to$ Legendre) As our first application, consider the weight function
$$w_U(x) := \frac{1}{2}, \qquad x \in \mathcal{X} = [-1, 1].$$

(i) Verify that $w_U(x) > 0$ on $\mathcal{X}$ (where $w_U(x) = 0$ otherwise), and that $\int_{\mathcal{X}} w_U(x)\, dx = 1$. Conclude that w_U defines a valid probability density function on $\mathcal{X}$.

(ii) Whenever $U \sim w_U$ is a random variable distributed according to the PDF w_U, we say that U is a **Uniform**$[-1, 1]$ random variable and write
$$U \sim \text{Uniform}[-1, 1].$$

Compute the MGF of a random variable $U \sim \text{Uniform}[-1, 1]$ directly: in particular, verify that $M_U(0) = 1$ (why is this automatic from Exercise 5?) and that
$$M_U(t) = \int_{\mathcal{X}} e^{tx} \cdot w_U(x)\, dx = \frac{1}{2}\int_{-1}^{1} e^{tx}\, dx = \frac{e^t - e^{-t}}{2t}, \qquad t \neq 0.$$

Conclude that $M_U(t)$ is defined for all $t \in \mathbb{R}$, and that its radius of convergence is $R = \infty$.

(iii) **(Hyperbolic Trig Functions)** The closed-form expression for M_U can be simplified a bit further.

1. Verify that if we define the **hyperbolic sine** and **hyperbolic cosine** functions

$$\sinh(t) := \frac{e^t - e^{-t}}{2}, \qquad \cosh(t) := \frac{e^t + e^{-t}}{2}, \qquad t \in \mathbb{R},$$

then we have

$$\sinh^2(t) = \frac{e^{2t} + e^{-2t} - 2}{4}, \qquad \cosh^2(t) = \frac{e^{2t} + e^{-2t} + 2}{4},$$

and conclude that

$$\cosh^2(t) - \sinh^2(t) = 1, \qquad \forall t \in \mathbb{R}.$$

Compare with the analogous identity $\cos^2(t) + \sin^2(t) = 1$ (with a $+$!) for all $t \in \mathbb{R}$.

2. Verify that

$$\sinh(-t) = -\sinh(t), \qquad \cosh(-t) = \cosh(t), \qquad \forall t \in \mathbb{R}.$$

Conclude that $\sinh(\cdot)$ is odd, while $\cosh(\cdot)$ is even. Compare with the analogous even/odd properties for $\sin(\cdot)$ and $\cos(\cdot)$ and validate their symmetry.

3. Verify the decomposition

$$e^t = \cosh(t) + \sinh(t).$$

Conclude that $\sinh(t)$ and $\cosh(t)$ respectively comprise the odd and even parts of the exponential function e^t.

4. Verify that the hyperbolic trigonometric derivatives **cycle with period** 2:

$$\frac{d}{dt}\sinh(t) = \cosh(t), \qquad \frac{d}{dt}\cosh(t) = \sinh(t).$$

Compare with the analogous trigonometric derivatives, which cycle with period 4:

$$\frac{d}{dt}\sin(t) = \cos(t), \qquad \frac{d}{dt}\cos(t) = -\sin(t), \qquad \frac{d}{dt}(-\sin(t)) = -\cos(t), \qquad \frac{d}{dt}(-\cos(t)) = \sin$$

5. Using the derivative cycles above, compute the Taylor series $\mathcal{T}[\cdot]$ of sinh and cosh about $t = 0$: we have

$$\sinh(0) = 0, \qquad \cosh(0) = 1,$$

and the derivatives cycle with period 2 (always positive), we have

$$\mathcal{T}[\sinh(t)] = \sum_{k=0}^{\infty} \frac{\sinh^{(k)}(0)}{k!} t^k = t + \frac{t^3}{3!} + \frac{t^5}{5!} + \cdots = \sum_{k=0}^{\infty} \frac{t^{2k+1}}{(2k+1)!}, \qquad t \in \mathbb{R},$$

and

$$\mathcal{T}[\cosh(t)] = \sum_{k=0}^{\infty} \frac{\cosh^{(k)}(0)}{k!} t^k = 1 + \frac{t^2}{2!} + \frac{t^4}{4!} + \cdots = \sum_{k=0}^{\infty} \frac{t^{2k}}{(2k)!}, \qquad t \in \mathbb{R}.$$

Compare with the analogous trigonometric Taylor series, where the period-4 cycle introduces alternating signs:

$$\mathcal{T}[\sin(t)] = t - \frac{t^3}{3!} + \frac{t^5}{5!} - \cdots = \sum_{k=0}^{\infty} \frac{(-1)^k t^{2k+1}}{(2k+1)!},$$

and

$$\mathcal{T}[\cos(t)] = 1 - \frac{t^2}{2!} + \frac{t^4}{4!} - \cdots = \sum_{k=0}^{\infty} \frac{(-1)^k t^{2k}}{(2k)!}.$$

6. We could have derived these hyperbolic trigonometric Taylor series another way: directly from the definitions of sinh and cosh as exponentials in (1). Validate the following derivation:

$$\begin{aligned}\mathcal{T}[\sinh(t)] = \mathcal{T}\left[\frac{e^t - e^{-t}}{2}\right] &= \frac{1}{2}\left(\mathcal{T}\left[e^t\right] - \mathcal{T}\left[e^{-t}\right]\right) \\ &= \frac{1}{2}\left(\sum_{k=0}^{\infty} \frac{t^k}{k!} - \sum_{k=0}^{\infty} \frac{(-t)^k}{k!}\right) \\ &= \frac{1}{2}\sum_{k=0}^{\infty} \frac{t^k - (-t)^k}{k!} \\ &= \frac{1}{2}\sum_{k=0}^{\infty} \frac{2t^{2k+1}}{(2k+1)!} \\ &= \sum_{k=0}^{\infty} \frac{t^{2k+1}}{(2k+1)!}.\end{aligned}$$

Obtain an analogous derivation for cosh.

7. In light of the competing cyclic behaviors of the hyperbolic and ordinary trigonometric functions under differentiation, attempt to offer an intuitive explanation for why the hyperbolic sinh and cosh are natural solutions to

$$y''(t) = +y(t), \qquad t \geq 0,$$

while the ordinary sin and cos are natural solutions to

$$y''(t) = -y(t), \qquad t \geq 0.$$

Explain any correspondences you observe with the basic form of the ODE we encountered in chapter 1: the exponential function exp is a natural solution to

$$y'(t) = y(t), \qquad t \geq 0.$$

(Hint: Consider the cyclic behavior demonstrated in (3): interpret $y(t)$ as the position of a particle and $y''(t)$ as its acceleration. The first ODE says "the acceleration points in the same direction as the displacement" — a particle displaced to the right accelerates further to the right, producing *runaway growth*. The second says "the acceleration points opposite to the displacement" — a particle displaced to the right is pulled back to the left, overshoots, and is pulled right again, producing *oscillation*.)

(iv) Returning to the task at hand: verify the MGF from part (ii) is

$$M_U(t) = \frac{e^t - e^{-t}}{2t} = \frac{\sinh(t)}{t}, \qquad t \neq 0.$$

(v) Using the Taylor Expansion derived in part (iii), verify that

$$M_U(t) = \frac{1}{t}\sum_{k=0}^{\infty} \frac{t^{2k+1}}{(2k+1)!} = \sum_{k=0}^{\infty} \frac{1}{(2k+1)!} t^{2k}.$$

(vi) Compare with the general form:

$$M(t) = \sum_{k=0}^{\infty} \frac{\mu^{(k)}}{k!} t^k.$$

Conclude by inspection of the form of M_U that the odd moments of $U \sim \text{Uniform}[-1,1]$ vanish ($\mu^{(2k+1)} = 0$ — why must this hold for any symmetric weight about $x = 0$?) and that its even moments may be deduced via

$$\frac{\mu^{(2k)}}{(2k)!} = \frac{1}{(2k+1)!} \implies \mu^{(2k)} = \frac{1}{2k+1}, \qquad k = 0, 1, 2, \ldots$$

(vii) Verify that the first 6 moments of a $U \sim \text{Uniform}[-1,1]$ random variable are

$$\mu^{(0)} = 1, \qquad \mu^{(2)} = 1/3, \qquad \mu^{(4)} = 1/5, \qquad \mu^{(6)} = 1/7,$$

while $\mu^{(1)} = \mu^{(3)} = \mu^{(5)} = 0$.

(viii) Assemble the 4×4 Hankel Gram matrix from $\mu^{(0)}, \mu^{(1)}, \ldots, \mu^{(6)}$:

$$\mathcal{G}_{w_U}^{(\mathcal{M}_3)} = \begin{pmatrix} 1 & 0 & 1/3 & 0 \\ 0 & 1/3 & 0 & 1/5 \\ 1/3 & 0 & 1/5 & 0 \\ 0 & 1/5 & 0 & 1/7 \end{pmatrix}.$$

Using the Schur complement procedure developed in Exercise 4, factor

$$G_{w_U}^{(\mathcal{M}_3)} = LDL^\top,$$

and verify that

$$L = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1/3 & 0 & 1 & 0 \\ 0 & 3/5 & 0 & 1 \end{pmatrix}, \qquad D = \operatorname{diag}\left(1, \frac{1}{3}, \frac{4}{45}, \frac{4}{175}\right).$$

Why is it immediately clear by inspection that L is invertible? Indeed, verify

$$L^{-1} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -1/3 & 0 & 1 & 0 \\ 0 & -3/5 & 0 & 1 \end{pmatrix}.$$

(ix) Compute

$$\boldsymbol{\omega}_U = L^{-1}\mathbf{m}, \qquad \mathbf{m} = \begin{pmatrix} 1 \\ x \\ x^2 \\ x^3 \end{pmatrix}.$$

Conclude that the entries of $\boldsymbol{\omega}_U$ are the monic polynomials

$$\omega_0(x) = 1, \qquad \omega_1(x) = x, \qquad \omega_2(x) = x^2 - \frac{1}{3}, \qquad \omega_3(x) = x^3 - \frac{3}{5}x,$$

and verify that ω_k, $k = 0, 1, 2, 3$ truly are orthogonal under the Stieltjes pairing induced by the uniform weight w_U. (Hint: whenever j is odd and k is even or vice-versa, that $\langle \omega_j, \omega_k \rangle_{w_U} = 0$ would seem to follow immediately by the oddness of the resulting integrand. On the other hand, ensuring that

$$\langle \omega_1, \omega_3 \rangle_{w_U} = \frac{1}{2} \int_{-1}^{1} \left(x^4 - \frac{3}{5}x^2 \right) dx = \frac{1}{2} \left[\frac{x^5}{5} - \frac{x^3}{5} \right]_{-1}^{1} = 0$$

would seem to be a much more delicate balancing act for our pipeline.)

(x) We've just witnessed our humble pipeline produce its first family of orthogonal polynomials in real time. Indeed, define the first four **Legendre polynomials** $P_n(x)$ to be

$$P_0(x) = 1, \qquad P_1(x) = x, \qquad P_2(x) = \tfrac{1}{2}(3x^2 - 1), \qquad P_3(x) = \tfrac{1}{2}(5x^3 - 3x).$$

Verify that, under this normalization, $P_n(1) = 1$ for each $n = 0, 1, 2, 3$.

(xi) Confirm that our **monic Legendre polynomials** ω_n are scalar multiples of the P_n: observe that $\omega_0 \equiv P_0$ and $\omega_1 \equiv P_1$ directly, while

$$\omega_2(x) = \tfrac{2}{3}P_2(x), \qquad \omega_3(x) = \tfrac{3}{5}P_3(x).$$

(Of course, it would follow from our observation above that, unlike the P_n at $x = 1$, our ω_n are unstable: $\omega_0(1) = \omega_1(1) = 1$, while $\omega_2(1) \neq 1$ and $\omega_3(1) \neq 1$. On the other hand, our ω_n are *monic* — the P_n aren't. This is the only difference between the ω_n we found, and the "conventionally normalized" Legendre polynomials P_n; behaviorally, *each characterizes* the *same polynomial*.)

Remark.

The uniform weight $w_U = 1/2$ on $[-1, 1]$ is perhaps the simplest probability density one could write down — it assigns equal mass everywhere. Indeed, in Chapter 2, we encountered this weight as the **uniform Stieltjes integrator**: when $U \sim \text{Uniform}[-1, 1]$, the differential $d\alpha_U(x) = \frac{1}{2}dx$ is just a constant rescaling of the ordinary differential dx, and the Stieltjes integral $\int_{-1}^{1} f(x)\, d\alpha_U(x) = \frac{1}{2}\int_{-1}^{1} f(x)\, dx$ reduces to the Riemann integral up to normalization. In that chapter, we observed that "there is no internal bias on behalf of the strip-producing mechanism $d\alpha_U$" — every infinitesimal strip dx receives equal probability mass.

The Legendre polynomials, then, are the orthogonal polynomials for the *unbiased* inner product — the one that treats every part of the domain identically. In Exercise

2, the arcsine weight $w_A(x) = 1/(\pi\sqrt{1-x^2})$ introduced a dramatic bias toward the boundary of $[-1, 1]$, and the Chebyshev polynomials emerged with roots clustering near the endpoints. The uniform weight does no such thing, and the Legendre roots spread more evenly across the interval. Two weights on the same domain, two geometries, two polynomial families — and the only difference is what the weight decides to care about.

(b) **(Gaussian → Hermite)** Now consider the standard normal weight on the unbounded support set $\mathcal{X} = \mathbb{R}$:

$$w_Z(x) := \frac{1}{\sqrt{2\pi}} e^{-x^2/2}, \qquad x \in \mathbb{R}.$$

(i) Verify that w_Z defines a valid probability density on $\mathbb{R}$: $w_Z(x) > 0$ for all $x \in \mathbb{R}$, and $\int_{\mathcal{X}} w_Z(x)\,dx = 1$. We write $Z \sim N(0, 1)$.

(ii) In Exercise 7(g), we derived the MGF of Z by completing the square in the Gaussian integral. Verify that $M_Z(t) = e^{t^2/2}$ is defined for all $t \in \mathbb{R}$, and in particular, that the radius of convergence of M_Z is $R = \infty$. Conclude that the Gaussian moment sequence is distribution determining.

(iii) We've derived closed-form expressions for the standard normal moments: for the even and odd moments respectively, these are

$$\mu^{(2k)} = (2k-1)!!, \qquad \mu^{(2k+1)} = 0.$$

Verify the moments by inspection of M_Z: substituting $u = t^2/2$ into $e^u = \sum_{k=0}^{\infty} u^k/k!$,

$$M_Z(t) = \sum_{k=0}^{\infty} \frac{1}{k!}\left(\frac{t^2}{2}\right)^k = \sum_{k=0}^{\infty} \frac{t^{2k}}{2^k k!}.$$

Deduce that the odd moments vanish: $\mu^{(2k+1)} = 0$, $k = 0, 1, 2, \ldots$ Then, comparing with $M(t) = \sum_{k=0}^{\infty} \frac{\mu^{(k)} t^k}{k!}$, read off the even moments:

$$\frac{\mu^{(2k)}}{(2k)!} = \frac{1}{2^k} k!, \qquad \text{i.e.,} \qquad \mu^{(2k)} = \frac{(2k)!}{2^k k!} = (2k-1)!!.$$

(iv) Verify the first 7 moments of $Z \sim N(0, 1)$ are

$$\mu^{(0)} = 1, \qquad \mu^{(2)} = 1, \qquad \mu^{(4)} = 3, \qquad \mu^{(6)} = 15,$$

while $\mu^{(1)} = \mu^{(3)} = \mu^{(5)} = 0$.

(v) Assemble the 4×4 Hankel Gram matrix $G_{w_Z}^{(\mathcal{M})}$ from $\mu^{(0)}, \mu^{(1)}, \ldots, \mu^{(6)}$:

$$G_{w_Z}^{(\mathcal{M}_3)} = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 3 \\ 1 & 0 & 3 & 0 \\ 0 & 3 & 0 & 15 \end{pmatrix}.$$

Using the Schur complement procedure of Exercise 4, factor $G_{w_Z}^{(\mathcal{M}_3)} = LDL^\top$, and verify that

$$L = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 3 & 0 & 1 \end{pmatrix}, \qquad D = \mathrm{diag}(1, 1, 2, 6).$$

Verify that

$$L^{-1} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & -3 & 0 & 1 \end{pmatrix}.$$

(vi) Compute

$$\boldsymbol{\omega}_Z = L^{-1}\mathbf{m}, \qquad \mathbf{m} = \begin{pmatrix} 1 \\ x \\ x^2 \\ x^3 \end{pmatrix}.$$

Conclude that the entries of $\boldsymbol{\omega}_Z$ are the monic polynomials

$$\omega_0(x) = 1, \qquad \omega_1(x) = x, \qquad \omega_2(x) = x^2 - 1, \qquad \omega_3(x) = x^3 - 3x.$$

(vii) Verify the orthogonality of members of $\boldsymbol{\omega}$ under $\langle \cdot, \cdot \rangle_{w_Z}$ directly for the least obvious cases: show that

$$\langle \omega_2, \omega_0 \rangle_{w_Z} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} (x^2 - 1) e^{-x^2/2}\, dx = \mu^{(2)} - 1 = 0,$$

and that

$$\langle \omega_1, \omega_3 \rangle_{w_Z} = \int_{-\infty}^{\infty} (x^4 - 3x^2) e^{-x^2/2}\, dx = \mu^{(4)} - 3\mu^{(2)} = 3 - 3(1) = 0.$$

Why are the other cases immediately obvious by symmetry? (Compare with the previous part.)

(viii) Verify the diagonal entries of D: confirm that

$$\|\omega_0\|^2_{w_Z} = 1, \qquad \|\omega_1\|^2_{w_Z} = 1, \qquad \|\omega_2\|^2_{w_Z} = 2, \qquad \|\omega_3\|^2_{w_Z} = 6,$$

and in particular, observe that

$$d_r = r!, \qquad r = 0, 1, 2, 3.$$

This pattern persists: $\|\omega_r\|^2_{w_Z} = r!$ for all $r = 0, 1, 2, \ldots$. (The reader may wish to prove this in general!)
(Hint: For instance,

$$\|\omega_1\|^2_{w_Z} = \int_{-\infty}^{\infty} x^2 e^{-x^2/2}\, dx = \mu_Z^{(2)} = 1,$$

and

$$\|\omega_2\|^2_{w_Z} = \int_{-\infty}^{\infty} (x^4 - 2x^2 + 1) e^{-x^2/2}\, dx = \mu_Z^{(4)} - 2\mu_Z^{(2)} + 1 = 3 - 2 + 1 = 2.$$

Verify ω_0 and ω_3 similarly.)

(ix) The polynomials $\omega_0, \omega_1, \omega_2, \omega_3$ are the first four **probabilist's monic Hermite polynomials**, often written

$$\mathrm{He}_j(x) := \omega_j(x), \qquad j = 0, 1, 2, \ldots$$

In particular, then, the first four monic Hermite polynomials are

$$\mathrm{He}_0(x) = 1, \qquad \mathrm{He}_1(x) = x, \qquad \mathrm{He}_2(x) = x^2 - 1, \qquad \mathrm{He}_3(x) = x^3 - 3x.$$

Verify these are equivalent to the first four **physicist's non-monic Hermite polynomials**,

$$H_0(x) = 1, \qquad H_1(x) = 2x, \qquad H_2(x) = 4x^2 - 2, \qquad H_3(x) = 8x^3 - 12x$$

by the rescaling

$$H_n(x) = (\sqrt{2})^n \, \mathrm{He}_n(x\sqrt{2}), \qquad n = 0, 1, 2, 3.$$

(x) **(Location-scale invariance)** Recognize that the substitution $x \mapsto x\sqrt{2}$ appearing in the rescaling above is the same one from Exercise 0(g): it passes between two members of the Gaussian scale-family, $e^{-x^2}/\sqrt{\pi}$ (variance 1/2) and $e^{-z^2/2}/\sqrt{2\pi}$ (variance 1). What we're encouraged to realize, then, is that if we are in possession of *some particular* Gaussian's orthogonal polynomials, we may recover the orthogonal polynomials of *any other* member of the Gaussian scale-family without rerunning the pipeline. Indeed, the equivalence extends even to the most general class of Gaussians, the location-scale family.

To see this explicitly, suppose $\{\omega_0, \omega_1, \ldots\}$ are the monic orthogonal polynomials under a weight w on $\mathcal{X}$.

- Under the affine substitution $x = (y - \mu)/\sigma$ with $\sigma > 0$ and $\mu \in \mathbb{R}$, verify that

$$\int_{\mathcal{X}} \omega_j(x)\, \omega_k(x)\, w(x)\, dx = 0 \qquad \Longrightarrow \qquad \int_{\tilde{\mathcal{X}}} \omega_j\!\left(\tfrac{y-\mu}{\sigma}\right) \omega_k\!\left(\tfrac{y-\mu}{\sigma}\right) w\!\left(\tfrac{y-\mu}{\sigma}\right) dy = 0,$$

  since the substitution merely relabels the variable of integration (and contributes a factor of σ that is positive and thus cannot produce a zero).
- Define

$$\tilde{\omega}_j(y) := \sigma^j\, \omega_j\!\left(\frac{y-\mu}{\sigma}\right).$$

  Verify that $\tilde{\omega}_j$ is monic of degree j. (Hint: verify that the leading term of $\omega_j(x) = x^j + \cdots$ becomes

$$\sigma^j \cdot \left(\frac{y-\mu}{\sigma}\right)^j \quad + \quad \cdots \quad = \quad (y-\mu)^j \quad + \quad \cdots \quad = \quad y^j \quad + \quad \cdots$$

)
- Conclude: the monic orthogonal polynomials for the location-scale weight $\tilde{w}(y) \propto w\!\left(\frac{y-\mu}{\sigma}\right)$ are

$$\tilde{\omega}_j(y) = \sigma^j\, \omega_j\!\left(\frac{y-\mu}{\sigma}\right).$$

  Thus, if we are in possession of some Gaussian's orthogonal polynomials, we are in possession of any other Gaussian's orthogonal polynomials.

- Confirm: for the Gaussian location-scale family $N(\mu, \sigma^2)$, the monic orthogonal polynomials are

$$\tilde{\omega}_j(y) = \sigma^j \ \mathrm{He}_j\left(\frac{y-\mu}{\sigma}\right).$$

The physicist's Hermite polynomials correspond to $\mu = 0$, $\sigma = 1/\sqrt{2}$ — verify this recovers our familiar

$$H_n(x) = (\sqrt{2})^n \ \mathrm{He}_n(x\sqrt{2}).$$

The upshot: location-scale families capture the class of *linear transformations* of a "standard" random variable. These are the *mildest possible* transformations of a weight — they change the center (via μ) and the spread (via σ) but preserve the shape. A Gaussian shifted by μ and stretched by σ is still a Gaussian — and its orthogonal polynomials are still Hermite, shifted and stretched by the same μ and σ. No information is created or destroyed; the polynomials adjust their argument to match the new coordinates. This is why the two Hermite conventions are not competing definitions but the *same family* viewed through the two natural choices of Gaussian scale from Exercise 0.

(c) (Exponential $\to$ Laguerre) We now leave the realm of symmetric distributions. Consider the standard exponential weight on $\mathcal{X} = [0, \infty)$:

$$w_E(x) := e^{-x}, \qquad x \in [0, \infty).$$

(i) Confirm: In Exercise 7(h), we verified that w_E defines a valid probability density on $[0, \infty)$ and ultimately derived the MGF

$$M_E(t) := \frac{1}{1-t}, \qquad t < 1.$$

In Exercise 9(a), we confirmed that Carleman's condition $\sum_{k=1}^{\infty} \left(\mu^{(2k)}\right)^{-1/2k} = +\infty$, was satisfied for the exponential moment sequence, which were identified in Exercise 7(h) to be

$$\mu_E^{(k)} = \int_0^{\infty} x^k e^{-x} \, dx = \Gamma(k+1) = k!, \qquad k = 0, 1, 2, ...$$

At this point, we may even verify our moment sequence via differentiation of the MGF M_E: confirm that

$$\frac{d^k}{dt^k} M_E(t) = k! \cdot (1-t)^{-(k+1)} \qquad \Longrightarrow \qquad M_E^{(k)}(0) \equiv \mu_E^{(k)} = k!, \qquad k = 0, 1, 2, ...$$

As such, the pipeline may proceed.

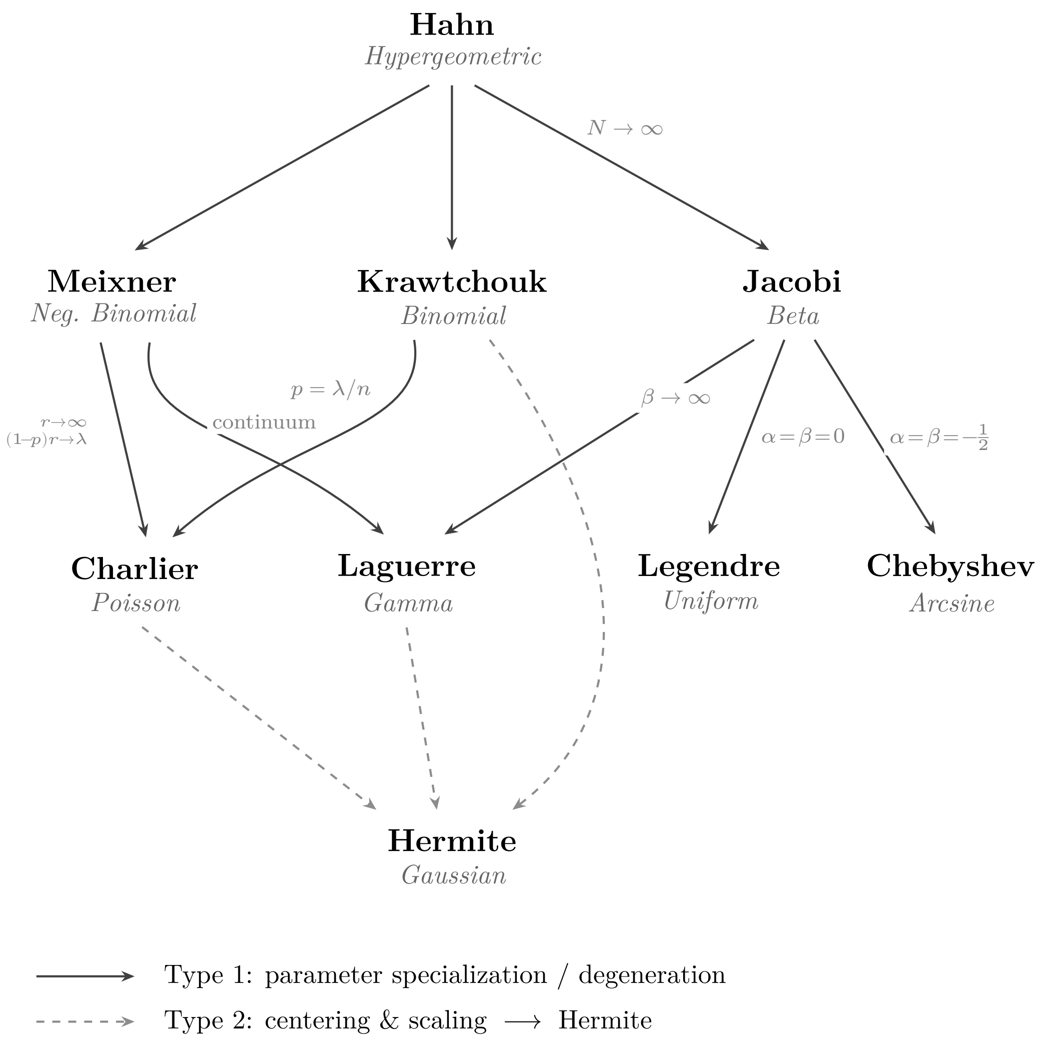
(ii) Compute the first 7 moments of a standard exponential random variable — verify

$$\mu_E^{(0)} = 1, \qquad \mu_E^{(1)} = 1, \qquad \mu_E^{(2)} = 2, \qquad \mu_E^{(3)} = 6,$$
$$\mu_E^{(4)} = 24, \qquad \mu_E^{(5)} = 120, \qquad \mu_E^{(6)} = 720,$$

and assemble the 4×4 Hankel Gram matrix:

$$G_{w_E}^{(\mathcal{M}_e)} = \begin{pmatrix} 1 & 1 & 2 & 6 \\ 1 & 2 & 6 & 24 \\ 2 & 6 & 24 & 120 \\ 6 & 24 & 120 & 720 \end{pmatrix}.$$

(iii) Observe: unlike the Chebyshev, Legendre, and Hermite cases, this Gram matrix has *no zero entries*.



The local Askey scheme for the polynomial families constructed in this chapter. Each node is a family of orthogonal polynomials, labeled by the distribution to which it is attached via the Stieltjes pairing. Solid arrows denote parameter specialization or degeneration (Type 1); dashed arrows denote centering and scaling (Type 2). Every dashed arrow terminates at Hermite.